

The structure and interpretation of (pro)nominal expressions in Spanish

by

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Abstract

According to the DP hypothesis, the merger of a determiner and a noun yields a determiner phrase (DP) rather than a noun phrase (*nP*). Focusing on Spanish, I defend the DP hypothesis but reject the notion that argumenthood is contingent upon a DP layer. Instead, I maintain that arguments can be as small as *nP* provided that they are c-commanded by a verb or a preposition, in which case the variables that they introduce are bound through a last-resort operation of existential closure. I then consider the ways in which functional projections above *nP* shape the denotation of nominal arguments and adopt Borer’s (2005) view that a division phrase (DivP) is responsible for countability. Nonplural indefinite expressions can either have mass or singular readings depending on the determiner (*mucho tomate* ‘much tomato’ versus *un tomate* ‘a tomato’), whereas nonplural definite expressions can have both mass and singular readings no matter the determiner (*el/este/mi tomate* ‘the/this/my tomato’). I attribute the systematic ambiguity of nonplural definite expressions to the absence of DivP in their structure and the maximality operator that is a component of definiteness.

Throughout the thesis, I explore the formal representation of ϕ -features and how their semantic behaviour is conditioned by their syntactic position. I propose that gender features have consequences for interpretation when they are hosted by Panagiotidis’s (2019) animacy phrase (AnimP) but not by *nP* and extend this argument to number features on DivP versus *nP*, thereby eliminating the need for “interpretable” and “uninterpretable” versions of the same feature. As for pronouns, it is not immediately apparent where they are generated in Spanish because they are not mutually exclusive with the definite article (*nosotros los cocineros* ‘we the chefs’). After upholding Höhn’s (2016, 2017) claim that pronouns originate in a dedicated person phrase (π P) in languages that use the definite article in pronoun-noun constructions, I address the issue of the third-person gap (**ellos los cocineros* ‘they the chefs’) and argue that third-person pronouns are forms that realize π , D, Div, Anim, and *n*, a configuration that prevents other material from occupying these same heads.

In memory of Paula Kempchinsky, who set me on this journey.

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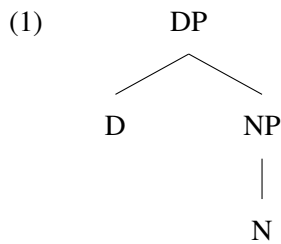
List of Abbreviations

1	first person
2	second person
3	third person
ACC	accusative
AUTH	author
COND	conditional
DAT	dative
DOM	differential object marking
F	feminine
FUT	future
HON	honorific
IMP	imperative
IMPF	imperfect
M	masculine
PART	participant
PL	plural
PRET	preterite
PTCP	participle
SBJV	subjunctive
SG	singular

Chapter 1

Introduction

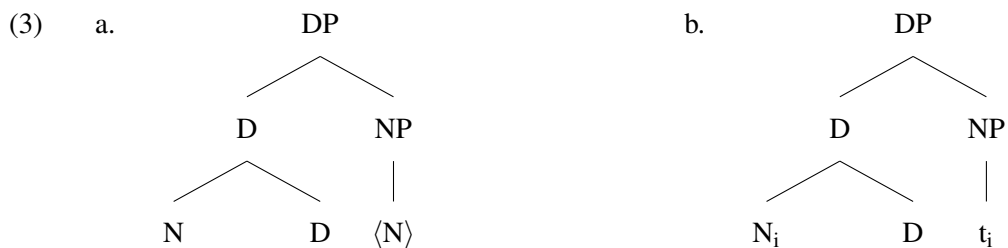
According to the line of investigation that has come to be known as the DP hypothesis, determiners take nominal expressions as their complements (Brame 1981, 1982; Fukui 1986; Fukui & Speas 1986; Hellan 1986; Abney 1987; Horrocks & Stavrou 1987; Szabolcsi 1987; Eguren 1989; Mallén 1989; Felix 1990). This view is shown in (1) and reflects Grimshaw's (2000) proposal that lexical items may be dominated by functional structure in what constitutes their extended projection.



In this thesis, which focuses on Spanish, I explore the question of whether there can be arguments that lack DP, as in (2a), or arguments that lack NP, as in (2b).



Moreover, I consider the possibility that some elements may originate in N and raise to D, movement that is represented by an unpronounced copy in (3a) and by a trace in (3b).



In section 1.1 of this chapter, I present the strong and weak versions of the DP hypothesis and use bare nouns to illustrate the differences between them. In section 1.2, I discuss the notion that pronouns are a type of determiner and show that the empirical landscape is especially complex in Spanish because of constructions like *nosotros los cocineros* ‘we the chefs’. In section 1.3, I introduce the theoretical framework of Distributed Morphology, and in section 1.4, I address the topic of gender in Distributed Morphology together with issues that arise when translating syntactic structures to logical form. Finally, in section 1.5, I identify the four research questions that form the basis of chapters 2 through 5 and summarize the major contributions of these chapters.

1.1 The DP hypothesis and bare nouns

In work that assumes the DP hypothesis, the role that D plays in argumenthood has been the subject of extensive debate. For some authors, D alone has the ability to convert NPs into arguments, even in languages that lack article systems (Stowell 1989, 1991; Longobardi 1994; Szabolcsi 1994; Benedicto 1998; Progovac 1998; Crisma 1999; Zamparelli 2000; Longobardi 2001; Borer 2005; Longobardi 2005, 2008). For other authors, D is one path to argumenthood, and there may be others depending on the language or the construction under study (Masullo 1992, 1996; Chierchia 1998b; Li 1998; Dayal 2004; Dobrovie-Sorin et al. 2005; Danon 2006; Pereltsvaig 2006; de Swart & Zwarts 2009; Espinal 2010; Dayal 2011a,b; Espinal & McNally 2011). It is therefore important to distinguish between the stronger version of the DP hypothesis in (4a) and the weaker one in (4b).

- (4) a. Only D can license an NP in argument position.
- b. D can license an NP in argument position.

In this section, I evaluate how each version of the DP hypothesis fares in capturing bare nouns in Spanish, which have a restricted distribution and obligatory existential readings.

As shown in (5), bare nouns can occur postverbally but not preverbally in Spanish (Lapesa 1974, Sánchez de Zavala 1976, Suñer 1982, Contreras 1986, Lois 1987, Eguren 1989, Delfitto & Schroten 1991, Masullo 1992, Bosque 1996, Brugè & Brugger 1996, Contreras 1996, Laca 1996, Masullo 1996, Olarrea 1998, Zubizarreta 1998, Bosque 1999, Laca 1999, Rigau 1999, Dobrovie-Sorin & Laca 2003, Dobrovie-Sorin et al. 2005, Stark 2008, Laca 2013, Fábregas 2018).

- (5) a. En la mañana llegaron cocineros.
 in the morning arrived.3PL chefs
 ‘Chefs arrived in the morning.’
- b. *Cocineros llegaron en la mañana.
 chefs arrived.3PL in the morning
 Intended: ‘Chefs arrived in the morning.’

Only the strong DP hypothesis commits to a phonologically null D in (5a), which is represented by the $\emptyset$ symbol in (6).

- (6) En la mañana llegaron [DP Ø [NP cocineros]].
in the morning arrived.3PL chefs
'Chefs arrived in the morning.'

The weak DP hypothesis is compatible with the notion of a covert D, but it also allows for the possibility that the bare noun is an NP, as in (7), one that is licensed as an argument by some other means, such as by the verb itself.

- (7) En la mañana llegaron [NP cocineros].
in the morning arrived.3PL chefs
'Chefs arrived in the morning.'

Not only must the strong DP hypothesis posit a null D in (5a), but it must also explain why this element cannot license the bare preverbal noun in (5b). If bare nouns are DPs, as in (6), it is not immediately apparent why the configuration in (8) should be ungrammatical.

- (8) *_{[DP ∅} _[NP cocineros]] llegaron en la mañana.
 chefs arrived.3PL in the morning
 Intended: ‘Chefs arrived in the morning.’

In response to this issue, Longobardi (1994, 2000, 2001, 2005, 2008) parameterizes the strong DP hypothesis and maintains that null D needs to be properly governed, or c-commanded by a lexical head, in Romance languages like Spanish. Even if one subscribes to the weak DP hypothesis, this same logic could be extended to (9) since the verb does not c-command the NP.

- (9) *_[NP Cocineros] llegaron en la mañana.
chefs arrived.3PL in the morning
Intended: ‘Chefs arrived in the morning.’

The weak DP hypothesis does not explicitly state that bare nouns are NPs rather than DPs, however, so I concentrate on null D itself and how this element might be represented at logical form.

I start with the semantic type of NPs in Spanish, which are unambiguously property-denoting, or type $\langle e, t \rangle$, expressions because they cannot refer to kinds as bare arguments, even in postverbal position (Laca 1990, 1996, 1999; Dobrovie-Sorin & Laca 2003; McNally 2004; Bleam 2005; Dobrovie-Sorin et al. 2005; Dobrovie-Sorin 2009; Borik & Espinal 2012; Laca 2013; Borik & Espinal 2015). As illustrated in (10), the mass noun *arroz* ‘rice’ cannot denote the kind, or species, *Oryza sativa* if it is not preceded by the definite article.

- (10) En el futuro podría extinguirse *(el) arroz.
in the future could extinguish.3 the rice
'Rice could become extinct in the future.'

As predicates, NPs introduce variables at logical form that must be bound, and in the case of bare nouns, the question is whether null D serves as the binder.

Bare nouns can only be interpreted existentially in Spanish, but if one were to endow null D with the ability to supply existential force, like the indefinite article, it would be necessary to explain why bare nouns exhibit different semantic behaviour than indefinite nouns do (Garrido Medina 1991; Bosque 1996; Laca 1996, 1999; Gutiérrez-Rexach 2001; Dobrovie-Sorin & Laca 2003; McNally 2004; Bleam 2005; Laca 2013; Fábregas 2018). In (11), for instance, only the indefinite noun *unos ratones* ‘some mice’ can take wide scope with respect to the universal quantifier *todos* ‘all’ and give rise to the reading where every chef saw the same mice.

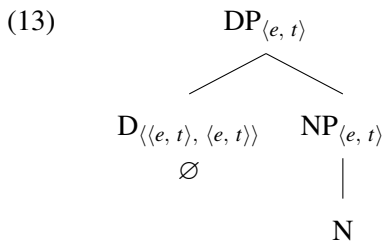
- (11) a. Todos los cocineros vieron unos ratones. $\forall > \exists$ or $\exists > \forall$
 all the chefs saw.3PL some mice
 ‘All the chefs saw some mice.’
 b. Todos los cocineros vieron ratones. $\forall > \exists$ only
 all the chefs saw.3PL mice
 ‘All the chefs saw mice.’

In other words, it cannot be the case that null D existentially binds the variables that are introduced by NPs, suggesting that the binder is external to the noun phrase.

Longobardi (2000, 2001, 2005, 2008) attributes the existential readings of bare postverbal nouns in Italian to Diesing’s (1992) rule of VP-level existential closure, though no formal implementation is provided. I attempt to do so now based on the standard definition of the existential operator ($\exists$) in (12), according to which this element applies to one property-denoting expression and then another before returning a truth value (Partee 1987, Chierchia 1998b, Dayal 2004).

$$(12) \quad \exists := \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \exists x_e [P(x) \wedge Q(x)]$$

Bare nouns must be of type $\langle e, t \rangle$ to belong to the domain of this operator, and because NPs are already of type $\langle e, t \rangle$ in Spanish, the only way to preserve null D would be to represent this head as a vacuous identity function, as in (13).

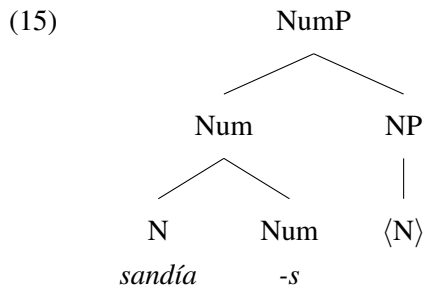


If one concedes that null D does not make a semantic contribution, the need for a DP layer is reduced to a purely syntactic constraint, one that violates structural economy since NPs themselves are of the correct type to be targeted by existential closure. In chapter 2, I identify further issues with the strong DP hypothesis and pursue an analysis of bare nouns where the verb licenses them directly through a process known as semantic incorporation.

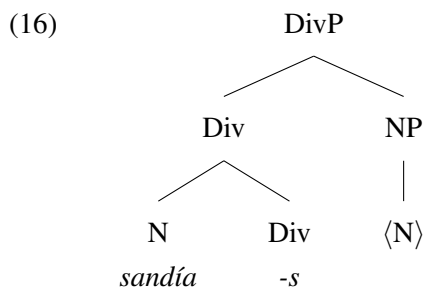
So far, I have characterized mass nouns and plural nouns as NPs, and though their distributions overlap in many ways, they diverge in others. For example, the mass noun in (14a) does not tolerate the indefinite article, but the plural noun in (14b) does.

- (14) a. Julia trajo (#una) sal de la cocina.
 Julia brought.3SG a salt from the kitchen
 ‘Julia brought salt from the kitchen.’
 b. Julia trajo (unas) sandías de la cocina.
 Julia brought.3SG some watermelons from the kitchen
 ‘Julia brought (some) watermelons from the kitchen.’

Since Ritter (1991, 1992) first proposed a number phrase (NumP), plural morphology is commonly assigned to this functional layer, as in the structure of *sandías* ‘watermelons’ in (15).



In this thesis, I use Borer’s (2005) division phrase (DivP) in place of NumP, as in (16), because this label more transparently reflects the role that number marking plays in establishing countability.



If nonplural *sandía* ‘watermelon’ is an NP, like nonplural *sal* ‘salt’, one might expect it to have a mass reading as a bare argument. This prediction is borne out in (17), where *sandía* can only be interpreted as watermelon substance.

- (17) Julia trajo sandía de la cocina.
 Julia brought.3SG watermelon from the kitchen
 ‘Julia brought watermelon from the kitchen.’

In chapter 3, I conduct a thorough investigation of the mass-count distinction in Spanish and develop a novel account of singularity based on the different semantic behaviour of indefinite and definite

determiners. In (18a), *una sandía* ‘a watermelon’ can only have a singular reading, but in (18b), *la sandía* ‘the watermelon’ can have a mass or singular reading depending on the context.

- (18) a. Julia trajo una sandía de la cocina.
 Julia brought.3SG a watermelon from the kitchen
 ‘Julia brought a watermelon from the kitchen.’
- b. Julia trajo la sandía de la cocina.
 Julia brought.3SG the watermelon from the kitchen
 ‘Julia brought the watermelon from the kitchen.’

As it turns out, nonplural definite descriptions like *la sandía* are systematically ambiguous between mass and singular denotation, a phenomenon that I attribute to two factors: the lack of DivP in their structure and the maximality operator that is encoded by definite determiners.

1.2 Pronouns as determiners

In Spanish, the third-person pronouns in (19) are morphologically similar to the forms of the definite article in (20), as is the case in other Romance languages, and this resemblance provides empirical support for Postal’s (1966) claim that pronouns are a type of determiner (Bello 1847, Greenberg 1978, Harris 1991, Schrotten 1992, Cardinaletti 1994, Zwarts 1994, Uriagereka 1995, F. Roca 1996, Luján 2000, Kayne 2003, Fábregas 2018).

- | | | | | | | | | |
|------|----|----|----|------|----|-------|----|--------|
| (19) | a. | él | b. | ella | c. | ellos | d. | ellas |
| | | he | | she | | they | | they.F |
-
- | | | | | | | | | |
|------|----|-----|----|-------|----|--------|----|----------|
| (20) | a. | el | b. | la | c. | los | d. | las |
| | | the | | the.F | | the.PL | | the.F.PL |

One of the ways in which Spanish differs from many languages, including some closely related ones, is that none of the pronouns in (21) can be directly followed by a noun.

- (21) a. *Nosotros cocineros somos altos.
 we chefs are.1PL tall
 Intended: ‘We chefs are tall.’
- b. *Vosotros cocineros sois altos.
 you.PL chefs are.2PL tall
 Intended: ‘You (informal) chefs are tall.’
- c. *Ustedes cocineros son altos.
 you.PL chefs are.3PL tall
 Intended: ‘You chefs are tall.’

To express the equivalent of *we chefs* and *you chefs* in Spanish, the definite article must be inserted after the pronoun, as in (22), even though there is no prosodic break (Villa-García 2010; Choi 2013, 2014a,b; Höhn 2016, 2017, 2020).

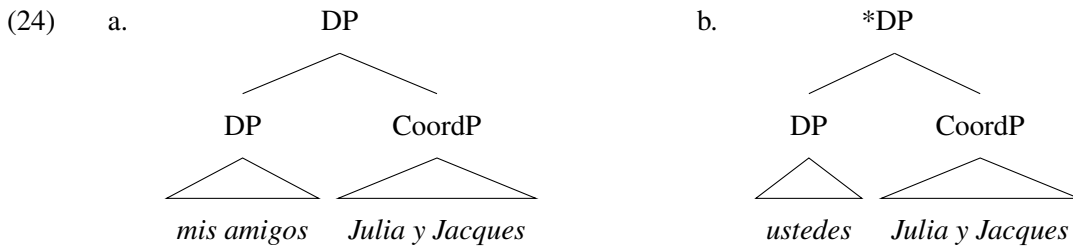
- (22) a. Nosotros *(los) cocineros somos altos.
 we the chefs are.1PL tall
 ‘We chefs are tall.’
- b. Vosotros *(los) cocineros sois altos.
 you.PL the chefs are.2PL tall
 ‘You (informal) chefs are tall.’
- c. Ustedes *(los) cocineros son altos.
 you.PL the chefs are.3PL tall
 ‘You chefs are tall.’

One possibility is that constructions like *nosotros los cocineros* ‘we the chefs’ consist of a pronoun and a definite description in apposition, and another is that they consist of a single extended nominal projection that contains two determiners, similar to *todos los cocineros* ‘all the chefs’.

There are at least two reasons to suspect that pronoun-noun constructions in Spanish are not instances of apposition, the first of which relates to substitution of the definite description component. As Lawrenz (1993) and Höhn (2016, 2017, 2020) observe in English and German, appositive configurations permit coordinated names as the second constituent, but pronoun-noun constructions do not. This contrast is also found in Spanish, as illustrated in (23), where I use *ustedes* ‘you all’ to maintain the same verb form and establish a minimal pair.

- (23) a. Mis amigos Julia y Jacques planearon el menú.
 my friends Julia and Jacques planned.3PL the menu
 ‘My friends Julia and Jacques planned the menu.’
- b. *Ustedes Julia y Jacques planearon el menú.
 you.PL Julia and Jacques planned.3PL the menu
 Literal: ‘You Julia and Jacques planned the menu.’

Assuming that *mis amigos Julia y Jacques* ‘my friends Julia and Jacques’ has a structure like that in (24a), it is unclear why **ustedes Julia y Jacques* ‘you Julia and Jacques’ could not have the structure in (24b) if *ustedes* constitutes its own DP.

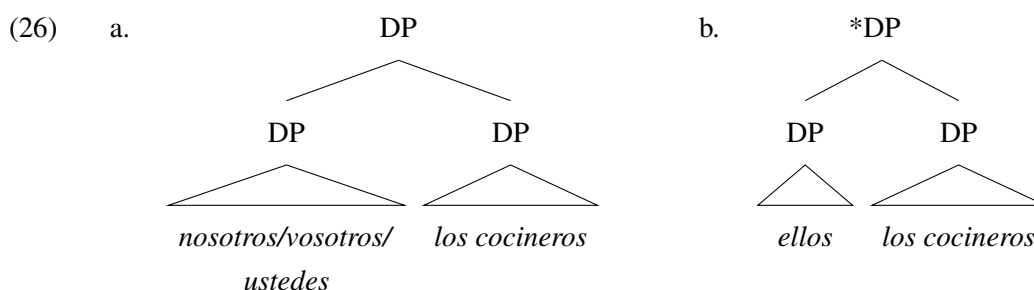


The second reason to discard apposition as a viable analysis of pronoun-noun constructions in Spanish concerns the “third-person gap”, or the inability of third-person pronouns to appear in syntactic environments that other pronouns can (Postal 1966; Abney 1987; Cardinaletti 1994; Lyons 1999; Déchaine & Wiltschko 2002; Giusti 2002; Panagiotidis 2002; Roehrs 2005; Bernstein 2008a,b;

Höhn 2016, 2017; Ackema & Neeleman 2018; Höhn 2020). As indicated in (25), *ellos* ‘they’ is the only plural pronoun that cannot occur in the same prosodic unit as a definite description.

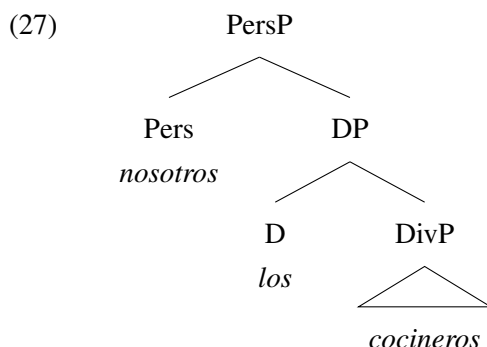
- (25) a. Jacques quiere hablar con {nosotros / vosotros / ustedes / ellos}.
 Jacques wants speak with us you.PL you.PL them
 ‘Jacques wants to speak with us/you (informal)/you/them.’
- b. Jacques quiere hablar con {nosotros / vosotros / ustedes / *ellos} los cocineros.
 Jacques wants speak with us you.PL you.PL them the chefs
 ‘Jacques wants to speak with us/you (informal)/you chefs.’

If *nosotros/vosotros/ustedes los cocineros* ‘we/you (informal)/you the chefs’ all have the appositive configuration in (26a), the question is why **ellos los cocineros* ‘they the chefs’ could not have the same configuration in (26b).

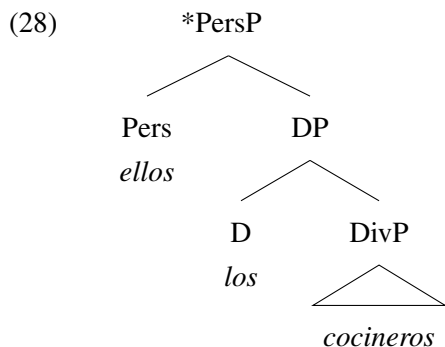


I now consider the alternative possibility, which is that pronoun-noun constructions in Spanish involve a single extended nominal projection.

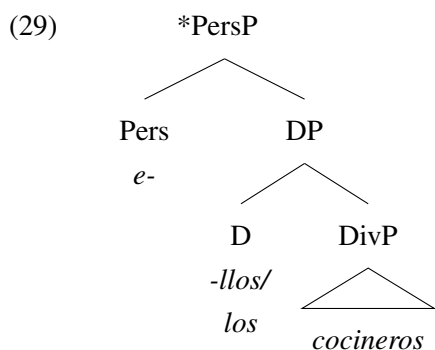
According to Höhn (2016, 2017), person features are hosted by a dedicated person phrase (PersP) in languages like Spanish that require the definite article in pronoun-noun constructions, as shown with *nosotros los cocineros* ‘we the chefs’ in (27).



This proposal explains why pronouns are not in complementary distribution with the definite article in Spanish, but it does not account for the ungrammaticality of **ellos los cocineros* ‘they the chefs’. After all, it is not evident what prevents *ellos* ‘they’ from occupying the Pers head in (28).

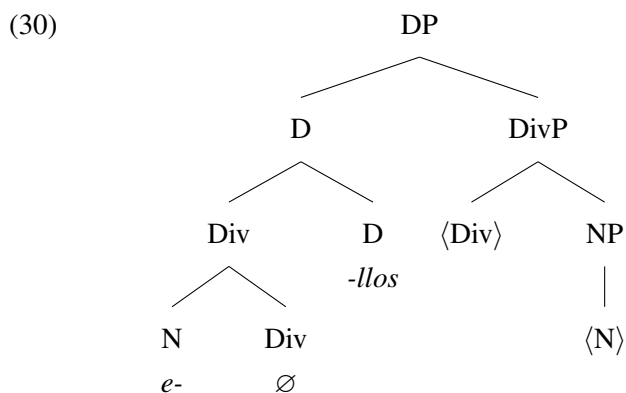


Considering that *ellos* seems to include the definite article *los* ‘the’, one could split *ellos* into *e-* in Pers and *-llos* in D, as in (29), and attribute the problem with **ellos los cocineros* to the impossibility of D being realized by more than one exponent.



The weakness of this analysis is that it suggests that **ellos cocineros* ‘they chefs’ should be grammatical, yet it is not.

To rule out **ellos los cocineros* ‘they the chefs’ together with **ellos cocineros* ‘they chefs’, one could invoke Zwarts’s (1994) claim that third-person pronouns are generated in N and raise to D, in which case both heads would be spelled out by the pronoun itself. One implementation of this view is given in (30), where I maintain *-llos* in D and position the initial *e-* segment in N to derive the correct morpheme order.



This approach faces the challenge of identifying a plausible trigger for N-to-D movement, which is otherwise unattested in Spanish. For example, names do not raise to D when this position is available, as shown in (31) and discussed at length in chapter 2.

- (31) a. Vino el viejo García.
 came.3SG the old García
 ‘Old García came.’
- b. *Vino García viejo.
 came.3SG García old
 Intended: ‘Old García came.’

In chapter 4, I explore the possibility that the *-o* and *-s* of *ellos* ‘they’ are not gender and number concord on D but rather realizations of N and Div, respectively, just like the *-o* and *-s* of *cocineros* ‘chefs’. Like Zwarts’s proposal, this solution explains why *ellos* and *cocineros* are mutually exclusive, but it does so without resorting to N-to-D movement.

1.3 Distributed Morphology

In this thesis, I adopt the theoretical framework of Distributed Morphology as conceptualized in Figure 1.1 (Halle & Marantz 1993, 1994; Marantz 1996; Embick 1997; Halle 1997; Marantz 1997; Harley & Noyer 1999, 2000; Embick & Noyer 2001; Marantz 2001; Arad 2003, 2005; Embick & Noyer 2007; Embick & Marantz 2008; Embick 2010; Marantz 2013).

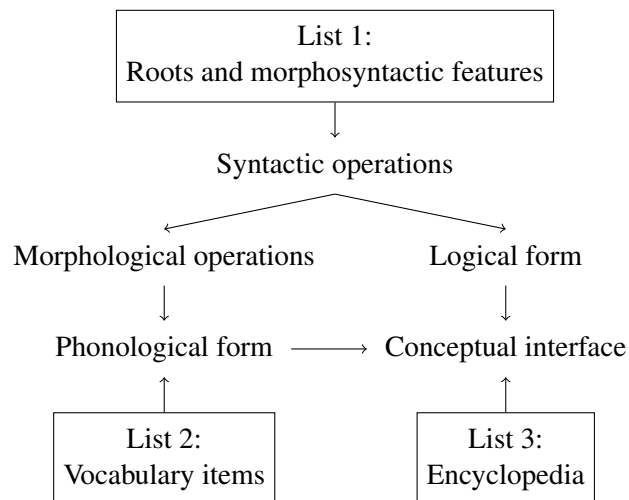


Figure 1.1: Distributed Morphology

Instead of a traditional lexicon, or a predetermined inventory of “words”, all that exists prior to syntax is List 1, which consists of roots and morphosyntactic features. Using only these elements, syntactic operations assemble structures that are transferred to morphological operations and logical form along two independent paths.

One of the tenets of Distributed Morphology is the late insertion of phonological material, or the vocabulary items of List 2. To demonstrate this process, I use the pronominal clitics of Spanish and the minimum number of privative features that are needed to differentiate them: the person features [AUTH(OR)] and [PARTICIPANT)], the number feature [PL(URAL)], the case features [DAT(IVE)] and [ACC(USATIVE)], and the gender feature [F(EMININE)]. In (32), I represent the target of insertion as φ to simplify the discussion but note that clitics can be decomposed into a stem, a class marker, and a plural suffix, following Harris (1995).

- (32)
- a. /nos/ $\leftrightarrow \varphi_{[\pi:AUTH, \#:PL]}$
 - b. /me/ $\leftrightarrow \varphi_{[\pi:AUTH]}$
 - c. /os/ $\leftrightarrow \varphi_{[\pi:PART, \#:PL]}$
 - d. /te/ $\leftrightarrow \varphi_{[\pi:PART]}$
 - e. /les/ $\leftrightarrow \varphi_{[K:DAT, \#:PL]}$
 - f. /las/ $\leftrightarrow \varphi_{[\#:PL, \gamma:F]}$
 - g. /los/ $\leftrightarrow \varphi_{[\#:PL]}$
 - h. /le/ $\leftrightarrow \varphi_{[K:DAT]}$
 - i. /la/ $\leftrightarrow \varphi_{[\gamma:F]}$
 - j. /lo/ $\leftrightarrow \varphi_{[K:ACC]}$
 - k. /se/ $\leftrightarrow \varphi$

The notion that some vocabulary items are less specified than others is fundamental to Distributed Morphology, and syncretism is captured through Halle's (1997: 428) Subset Principle in (33).

- (33) The phonological exponent of a Vocabulary item is inserted into a morpheme in the terminal string if the item matches all or a subset of the grammatical features specified in the terminal morpheme. Insertion does not take place if the Vocabulary item contains features not present in the morpheme. Where several Vocabulary items meet the conditions for insertion, the item matching the greatest number of features specified in the terminal morpheme must be chosen.

Since first-person and second-person clitics do not overtly express gender or case distinctions, these features are not included in the vocabulary items in (32a) through (32d). The examples in (34) confirm that *nos* 'us' is inserted regardless of whether the stressed pronoun that the clitic corresponds to is masculine or feminine or serves as the direct or indirect object.

- (34)
- a. Julia nos saludó a {nosotros / nosotras}.
 Julia 1PL greeted.3SG DOM us us.F
 'Julia greeted us.'
 - b. Julia nos dio la receta a {nosotros / nosotras}.
 Julia 1PL gave.3SG the recipe to us us.F.PL
 'Julia gave the recipe to us.'

Third-person clitics do, on the other hand, inflect for gender and case, so it is necessary to specify these features in at least some of the vocabulary items in (32e) through (32k). In (35), I use plural third-person forms to illustrate the relevant differences.

- (35) a. Jacques los saludó a ellos.
 Jacques 3PL.ACC greeted.3SG DOM them
 ‘Jacques greeted them.’
- b. Jacques las saludó a ellas.
 Jacques 3PL.F.ACC greeted.3SG DOM them.F
 ‘Jacques greeted them.’
- c. Jacques les dio la receta a {ellos / ellas}.
 Jacques 3PL.DAT gave.3SG the recipe to them them.F
 ‘Jacques gave the recipe to them.’

As the Subset Principle states, vocabulary items can be underspecified with respect to the environment in which they are inserted, but they can never be overspecified.

Another tenet of Distributed Morphology is that morphological operations can manipulate syntactic structures on their way to phonological form, meaning that features that are present at phonological form may differ from those that are present at logical form. One phenomenon in Spanish that has been attributed to a postsyntactic morphological operation is the “spurious *se* rule”, to use Perlmutter’s (1970, 1971) term. As shown in (36) and (37), third-person dative clitics take the form *se* rather than *le* or *les* when they precede third-person accusative clitics.

- (36) a. Julia le entregó el manuscrito a la editora.
 Julia 3SG.DAT submitted.3SG the manuscript to the editor
 ‘Julia submitted the manuscript to the editor.’
- b. Julia {se / *le} lo entregó.
 Julia 3 3SG.DAT 3SG.ACC submitted.3SG
 ‘Julia submitted it to her.’
- (37) a. Julia les entregó el manuscrito a los editores.
 Julia 3PL.DAT submitted.3SG the manuscript to the editor
 ‘Julia submitted the manuscript to the editors.’
- b. Julia {se / *les} lo entregó.
 Julia 3 3PL.DAT 3SG.ACC submitted.3SG
 ‘Julia submitted it to them.’

To account for this constraint, Bonet i Alsina (1991) proposes that dative clitics lose their case feature in the context of accusative clitics through postsyntactic “impoverishment”, an analysis that has been widely embraced in the generative literature (Halle & Marantz 1994, Harris 1995, Heap 2002, Nevins 2007, Cuervo 2013, Arregi & Nevins 2018). I provide one way of formalizing this process in (38) using the same privative case features as before.

- (38) $\varphi_{[K:DAT]} \rightarrow \varphi / \text{---} \varphi_{[K:ACC]}$

According to this rule, a $\varnothing$ head that is valued as dative loses this feature in the context of a $\varnothing$ head that is valued as accusative, reducing the possible candidates for insertion to *se*, or the vocabulary item in (32k) above.

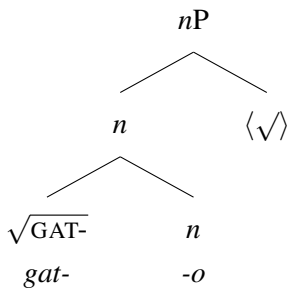
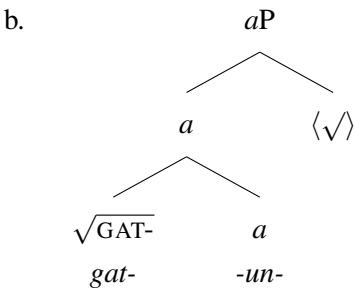
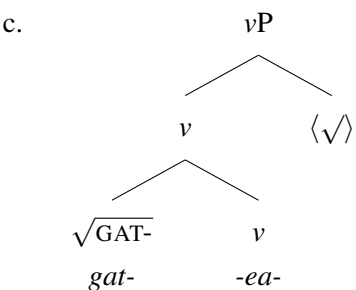
I close this section by discussing List 3 in Distributed Morphology, which is known as the encyclopedia and constitutes the storehouse of extralinguistic information. The relevance of the encyclopedia is evident from the first stanza of Lewis Carroll’s poem “Jabberwocky” as annotated by Borer (2005: vi) in (39).

- (39) ’Twas [A brillig], and the [NP [A slithy] toves]
 Did [V gyre] and [V gimble] in the [N wabe]:
 All [A mimsy] were the [N borogoves],
 And the [NP [A/N mome] raths] [V outgrabe].

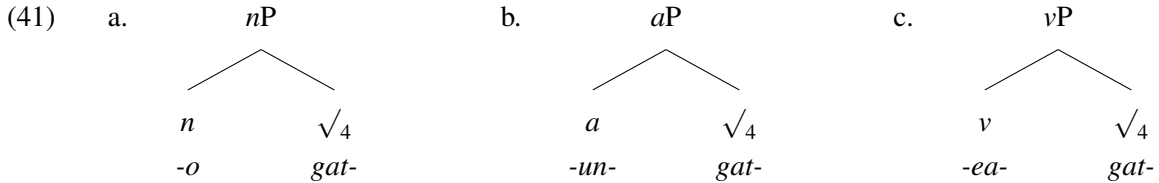
The syntactic category of every lexical item other than *mome* is clear, yet the poem is incomprehensible because the invented roots do not have corresponding entries in the encyclopedia to supply their “meaning”. In the case of *slithy*, for instance, it is evident that this word functions as an adjective, but it is impossible to determine the property that it expresses.

1.4 Lexical decomposition and gender

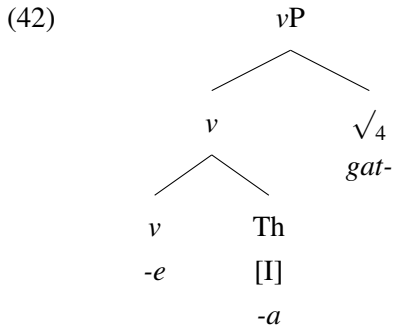
I assume, following Marantz (1996, 1997), that roots lack a syntactic category and must be provided one before they can be pronounced or interpreted. As shown in (40), the noun *gato* ‘cat’, the stem of the adjective *gatuno* ‘catlike’, and the stem of the verb *gatear* ‘to crawl’ can be decomposed into a common root, $\sqrt{\text{GAT-}}$, and a nominalizer (*n*), adjectivizer (*a*), and verbalizer (*v*), respectively.

- (40) a. 
- b. 
- c. 

The fact that **gat* never occurs in isolation supports the claim that roots must be categorized. In more recent work on lexical decomposition, roots have been modelled as numerical indices to reflect the arbitrary connection between the form that they take in a given syntactic environment and the interpretation that they have in that environment (Acquaviva 2009, Pfau 2009, Saab 2010, Acquaviva & Panagiotidis 2012, Acquaviva 2014, Harley 2014, Panagiotidis 2014). I demonstrate this approach in (41) by replacing $\sqrt{\text{GAT-}}$ with the randomly chosen index $\sqrt{4}$.

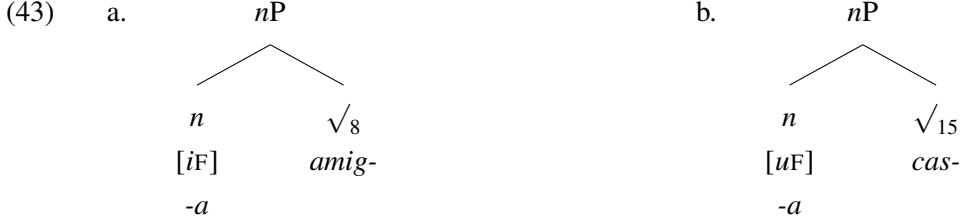


Here and in the remainder of this section, I abstract away from head movement to facilitate the discussion. As for theme vowels and class markers, many authors who assume Distributed Morphology consider dedicated theme nodes to be inserted postsyntactically, and some include class features on these nodes (Oltra-Massuet 1999, Embick & Halle 2005, Oltra-Massuet & Arregi 2005, Embick & Noyer 2007, Embick 2010, Lampitelli 2014, Kramer 2015). For example, the verb stem *gatea-* ‘crawl’ can be represented as in (42), where a theme node that bears class feature [I] is added to v during morphological operations and is spelled out as *-a*.



The analyses that I develop in this thesis are amenable to this view but do not hinge on adopting it, so I do not include theme nodes in morphosyntactic structures and instead situate theme vowels and class markers under the relevant syntactic head. I do, however, take a stance on the formal representation of gender.

The nouns *amiga* ‘(female) friend’ and *casa* ‘house’ are both feminine in Spanish, but this gender specification only has semantic relevance in the case of *amiga*. One way of implementing this distinction is through “interpretable” versus “uninterpretable” versions of a feminine feature, such as $[iF]$ and $[uF]$, that are hosted by n (Matushansky 2013; Kramer 2014, 2015; Anagnostopoulou 2017; Smith 2017; Wurmbrand 2017a,b; Panagiotidis 2019; Smith 2021; Steriopo 2024; Adamson 2025; Adamson & Anagnostopoulou to appear). According to the syntactic structures in (43), *amiga* and *casa* are both grammatically feminine, but only *amiga* expresses femaleness.

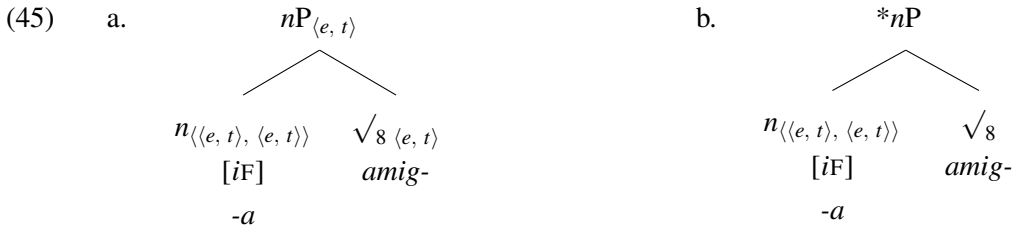


Feminine gender that has repercussions at logical form is often assigned the presuppositional semantics in (44), where I use [F] rather than [iF] because $[i\gamma]$ - $[u\gamma]$ notation is not universally employed (Percus 2011, Matushansky 2013, Merchant 2014, Spathas & Sudo 2020, Sudo & Spathas 2020).

$$(44) \quad \llbracket [F] \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Female}(x) . P(x)$$

This feature takes a property-denoting expression and returns a function that first checks that for every object x , x is female. If this condition is met, the function then maps x to the truth value 1 if and only if x has property P .

If n is the sole locus of gender features, the root of *amiga* ‘(female) friend’ must acquire its own semantic type through nominalization, as in (45a), or else composition cannot take place between n and the root, as in (45b).



The problem, however, is that the interpretable structure in (45a) violates Embick & Marantz’s (2008: 6) categorization assumption: “roots cannot appear (cannot be pronounced or interpreted) without being *categorized*; they are categorized by merging syntactically with category-defining functional heads”. More specifically, it would not be possible to calculate the denotation of this structure without providing the root with its own lexical entry, as in (46), even though it has not yet combined with n to become a noun.

$$(46) \quad \llbracket \sqrt{8} \rrbracket = \lambda x_e . \text{Friend}(x)$$

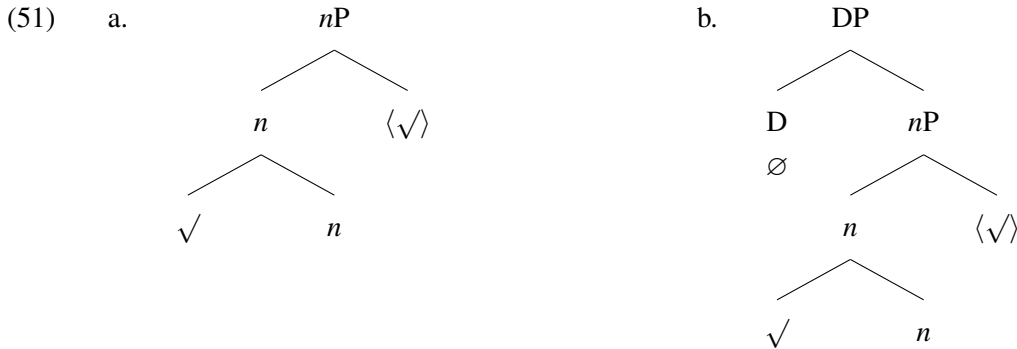
One way to avoid this situation would be to assign interpretable gender features to a functional projection that dominates nP , such as Picallo’s (1991) gender phrase (GenP). Based on Harley & Ritter’s (2002) feature geometry, where masculine and feminine nodes are dependents of an animate node, I opt to use Panagiotidis’s (2019) animacy phrase (AnimP) instead. As illustrated in (47), roots do not require their own semantic type in an approach where gender features that have consequences for logical form are hosted by a head above n .

- (50) a. Vi (*a) la casa.
saw.1SG DOM the house
'I saw the house.'
- b. Vi *(a) la víctima.
saw.1SG DOM the victim
'I saw the victim.'

In chapter 5, I develop an even more nuanced account of gender and animacy in Spanish, one where nouns and pronouns are organized into six classes based on their syntactic and semantic behaviour: (i) animate masculine nouns and pronouns that presuppose maleness, (ii) animate feminine nouns and pronouns that presuppose femaleness, (iii) animate masculine nouns and pronouns that do not presuppose maleness, (iv) animate feminine nouns that do not presuppose femaleness, (v) inanimate masculine nouns, and (vi) inanimate feminine nouns. I then formally differentiate each class through the presence or absence of AnimP in the structure, privative [M] and [F] features, and the location of these features on Anim or *n*.

1.5 Research questions and contributions

This thesis is organized around four research questions that stem from the unresolved issues in Spanish that I have discussed in this introductory chapter. The first research question is whether arguments can be as small as *n*Ps in this language, as in (51a), or whether they must contain an unpronounced DP layer, as in (51b).



In chapter 2, I demonstrate that Longobardi's (1994, 2000, 2001, 2005, 2008) null D approach faces a variety of empirical and theoretical problems, as does Contreras's (1996) null Q approach. I then combine insights from the crosslinguistic literature on semantic incorporation and Benedicto's (1998) view of existential closure to argue that verbs themselves license bare *n*Ps and DivPs in their c-command domain. I formalize the analysis in event semantics and propose a type-shifting operation that enables verbs and prepositions to existentially bind the variables that are introduced by bare nouns. Next, I explore the contrast that Jaeggli (1982: 59) observes in (52), where the bare indirect object *mujeres* 'women' cannot be doubled by a dative clitic.

- (52) a. Regalaré todos mis libros a mujeres.
 gift.FUT.1SG all my books to women
 ‘I will give all my books to women.’
- b. *Les regalaré todos mis libros a mujeres.
 3PL.DAT gift.FUT.1SG all my books to women
 Intended: ‘I will give all my books to women.’

In Demonte’s (1995) and Cuervo’s (2003a,b, 2008, 2010a,b) accounts of the dative alternation in Spanish, *a* ‘to’ is a genuine preposition in (52a) but a marker of dative case in (52b). Accordingly, existential closure is available as a rescue operation in the first example but not in the second. Lastly, I consider several environments that disallow bare postverbal nouns, such as the psychological predicate in (53a), the inchoative construction in (53b), and the passive sentence in (53c).

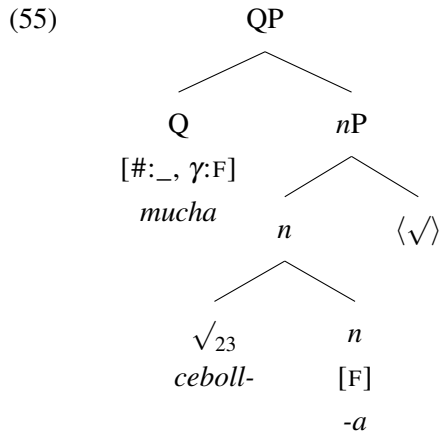
- (53) a. Julia odia *(el) cilantro.
 Julia hates the cilantro
 ‘Julia hates cilantro.’
- b. Se abrieron *(las) ventanas.
 3 opened.3PL the windows
 ‘The windows opened.’
- c. Fueron vendidos *(los) terrenos.
 were.3PL sold the lands
 ‘The lands were sold.’

In each case, I identify a semantic explanation for the obligatory determiner and conclude that these types of data do not undermine my broader analysis of bare nouns.

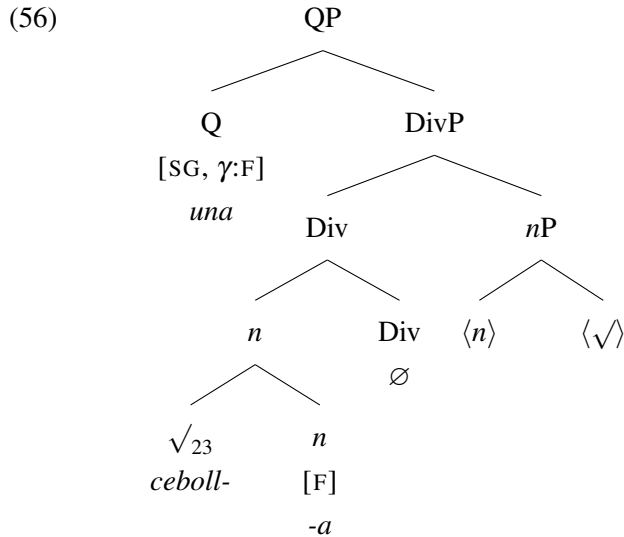
The second research question relates to the mass-count distinction and the facts in (54), where *mucha cebolla* ‘much onion’ only permits a mass reading, *una cebolla* ‘an onion’ only permits a singular reading, and *la cebolla* ‘the onion’ is ambiguous between a mass and a singular reading.

- (54) a. Jacques puso mucha cebolla en el caldo.
 Jacques put.3SG much onion in the broth
 ‘Jacques put a lot of onion in the broth.’
- b. Jacques puso una cebolla en el caldo.
 Jacques put.3SG an onion in the broth
 ‘Jacques put an onion in the broth.’
- c. Jacques puso la cebolla en el caldo.
 Jacques put.3SG the onion in the broth
 ‘Jacques put the onion in the broth.’

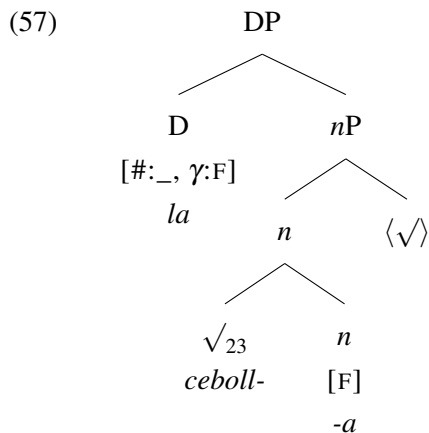
In chapter 3, I advocate for Borer’s (2005) view that nominal expressions that are interpreted as mass lack DivP, as illustrated by the morphosyntactic structure of *mucha cebolla* in (55), where Q combines directly with the *nP*.



To derive the singular reading of *una cebolla*, I propose that *una* spells out a Q head that bears a privative [SG] feature, as in (56), the presence of which conditions the null realization of Div.



Next, I argue that the ambiguous interpretation of *la cebolla* is due to the absence of DivP in (57) and the maximality operator that is part of the semantic value of the definite article.

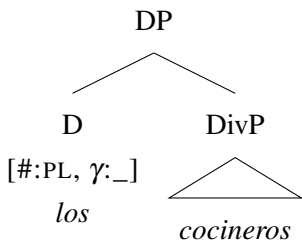
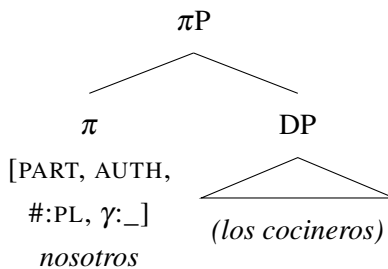


In essence, *la cebolla* denotes a unique maximal object that is onion, and it is entirely dependent on the context whether this object corresponds to a totality of onion substance or to a whole onion. I then address bare number-neutral objects of verbs of possession, such as *carro* ‘car’ in *tengo carro* ‘I have a car/cars’, and demonstrate how Espinal & McNally’s (2011) account of this phenomenon can be integrated into the view of argument structure that I adopt in this thesis. At the end of chapter 3, I consider several types of form-meaning mismatches in Spanish and argue that lexical plurals like *alicates* ‘pliers’ and pluralia tantum like *enseres* ‘belongings, equipment’ have their [PL] feature on *n* rather than on Div, in which case this feature still triggers plural agreement and concord but is disregarded at logical form. Finally, I determine that group nouns like *ejército* ‘army’ and aggregate nouns like *ganado* ‘livestock’ cannot be subsumed under a general analysis of the mass-count distinction and warrant their own treatment.

The third research question concerns pronouns and their base-generation position in Spanish. As shown in (58), at least some pronouns can appear in the same prosodic unit as the definite article, suggesting that they do not originate in D.

- (58) a. Nosotros los cocineros trabajamos día y noche.
 we the chefs work.1PL day and night
 ‘We chefs work day and night.’
 b. Vosotros los cocineros trabajáis día y noche.
 you.PL the chefs work.2PL day and night
 ‘You (informal) chefs work day and night.’
 c. Ustedes los cocineros trabajan día y noche.
 you.PL the chefs work.3PL day and night
 ‘You chefs work day and night.’

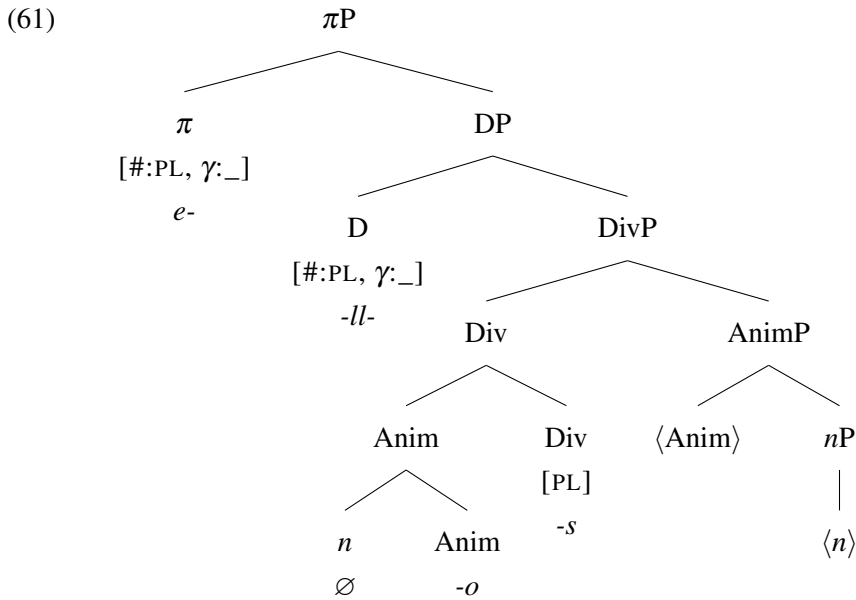
In chapter 4, I use a variety of diagnostics to explore whether constructions like *nosotros los cocineros* ‘we the chefs’ consist of a pronoun and a definite description in apposition and ultimately rule out this possibility. Instead, I argue in favour of Höhn’s (2016, 2017) claim that person features belong to their own syntactic projection in Spanish, which I label as π P, and defend the view that only pronominal expressions are specified for person (Bianchi 2006; Ackema & Neeleman 2013b, 2018; Sichel & Wiltschko 2021; Toosarvandani 2023; Sichel & Toosarvandani to appear). In this approach, a definite description like *los cocineros* ‘the chefs’ is a personless DP, as in (59a), and a pronoun like *nosotros* ‘we’ is a π P whose DP component can be overt or covert, as in (59b).

- (59) a. 
- b. 

Next, I consider the third-person gap in Spanish, or the inability of *ellos* ‘they’ to precede the definite description in (60).

- (60) (*Ellos) los cocineros trabajan día y noche.
 they the chefs work.3PL day and night
 ‘The chefs work day and night.’

To explain this restriction, I decompose *ellos* into a third-person marker (*e-*) in π , a definiteness marker (*-ll-*) in D, a class marker (*-o*) in Anim, and a plural marker (*-s*) in Div, as indicated in (61).



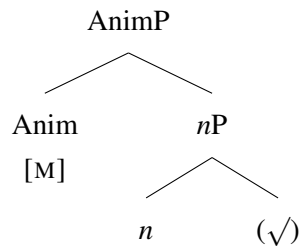
If D, *n*, Anim, and Div are all spelled out by the pronoun itself, it follows that these heads cannot host other material, hence the ungrammaticality of both **ellos los cocineros* ‘they the chefs’ and **ellos cocineros* ‘they chefs’. Finally, I observe that the third-person gap extends to pronouns that are modified by restrictive relative clauses, as in (62).

- (62) a. Solo nosotros los que tenemos visa podemos estudiar en Francia.
 only we the that have.1PL visa can.1PL study in France
 ‘Only we who have a visa can study in France.’
 b. Solo vosotros los que tenéis visa podéis estudiar en Francia.
 only you.PL the that have.2PL visa can.2PL study in France
 ‘Only you all (informal) who have a visa can study in France.’
 c. Solo ustedes los que tienen visa pueden estudiar en Francia.
 only you.PL the that have.3PL visa can.3PL study in France
 ‘Only you all who have a visa can study in France.’
 d. Solo (*ellos) los que tienen visa pueden estudiar en Francia.
 only they the that have.3PL visa can.3PL study in France
 ‘Only the ones who have a visa can study in France.’

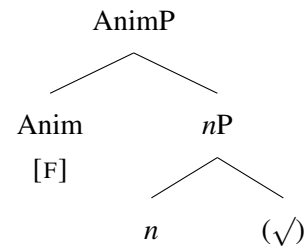
Based on this constraint, I posit that the realization of a third-person pronoun is contingent upon the strict adjacency of π , D, and $n + \text{Anim}$ in singular forms and of π , D, and $n + \text{Anim} + \text{Div}$ in plural forms. Restrictive relative clauses merge within the domain of modification below DP and, in so doing, interrupt the adjacency of D and $n + \text{Anim}$ (+ Div), preventing the structure from being spelled out as a third-person pronoun.

The fourth research question is whether ϕ -features have the same syntactic representation and make the same semantic contribution in nouns and pronouns, and in chapter 5, I maintain that all features that are common to both types of expressions do. Starting with gender and animacy, I argue that the six configurations of projections and features in (63) capture the behaviour of virtually all nouns and pronouns in Spanish.

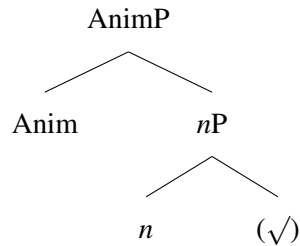
- (63) a. Animate masculine (pro)nouns
that presuppose maleness



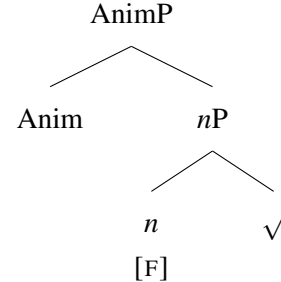
- b. Animate feminine (pro)nouns
that presuppose femaleness



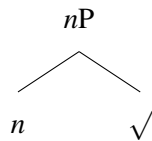
- c. Animate masculine nouns



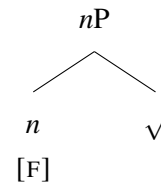
- d. Animate feminine nouns



- e. Inanimate masculine nouns



- f. Inanimate feminine nouns

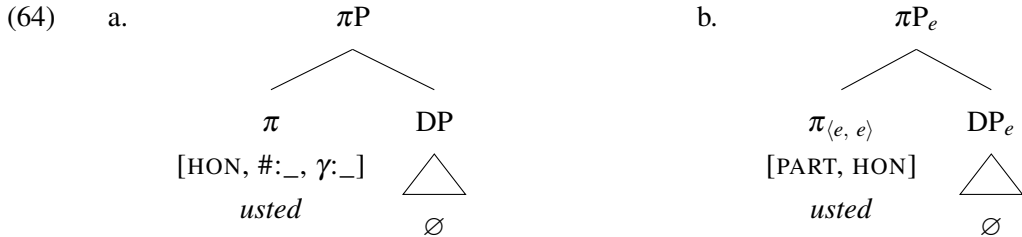


I later account for the three-way contrast that is found with *él* ‘he’, *ella* ‘she’, and the “neuter” pronoun *ello* ‘it’ by assigning *él* the AnimP component in (63a), *ella* the AnimP component in (63b), and *ello* a rootless version of the nP component in (63e).

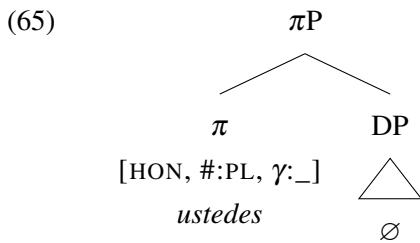
Next, I take up the issue of number in plural pronouns that have heterogeneous extensions. For instance, *nosotros* ‘we’ does not denote a plurality of speakers but rather the speaker and at

least one nonspeaker, meaning that *nosotros* does not have the same relation to *yo* ‘I’ that a plural noun like *cocineros* ‘chefs’ has to *cocinero* ‘chef’. I demonstrate that the heterogeneity of *nosotros*, *vosotros* ‘you all (informal)’, and *ustedes* ‘you all’ stems from the semantics of their person features and not to a different type or locus of number. Concretely, *nosotros* bears an [AUTH] feature that presupposes that the speaker is one of its referents, and *vosotros* and *ustedes* bear a [PART] feature that presupposes that the addressee is one of their referents.

In the final section of chapter 5, I explore the syntactic identity of *usted* ‘you (formal)’ and *ustedes* and determine on the basis of Person-Case Constraint effects that both are second person. To explain why these pronouns appear with third-person verb forms and clitics, I propose that they are generated with an [HON(ORIFIC)] feature on π that triggers the deletion of their [PART] feature through a morphological rule of impoverishment. In the analysis that I develop, *usted* has the morphosyntactic structure in (64a), which lacks [PART], and the interpretable structure in (64b), which retains [PART].

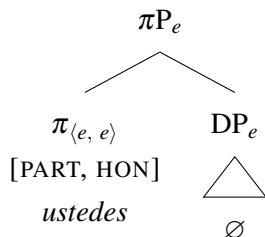


As for *ustedes*, this pronoun has the morphosyntactic structure in (65), where π exhibits plural concord due to the presence of DivP immediately below DP.



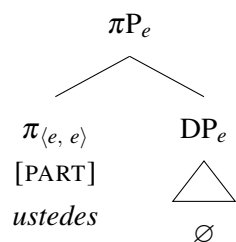
Lastly, the interpretable structure of *ustedes* depends on the variety of Spanish under consideration. In Peninsular Spanish, where this pronoun conveys formality, I maintain [HON] on π , as in (66).

(66) Peninsular Spanish



In Latin American Spanish, where this pronoun does not convey formality, I argue that [HON] is eliminated at logical form in the context of DivP. The result of this impoverishment process is the interpretable structure of *ustedes* in (67).

(67) Latin American Spanish



To conclude, gender and number features are the same in nouns and pronouns, while person and honorific features are exclusive to pronouns.

Chapter 2

Bare nouns and existential closure

In Spanish and many other Romance languages, nouns can only appear without a determiner as postverbal arguments or as the objects of prepositions. For some authors, bare nouns in Romance are licensed by a covert determiner that must itself be licensed by a verb or preposition (Contreras 1986; Longobardi 1994; Contreras 1996; Longobardi 2000, 2001, 2005, 2008). For other authors, bare nouns in Romance are licensed directly by verbs and prepositions through a process known as semantic incorporation (Dobrovie-Sorin & Laca 2003, McNally 2004, Blears 2005, Dobrovie-Sorin et al. 2005, Dobrovie-Sorin 2009, Espinal 2010, Espinal & McNally 2011, Laca 2013). In this chapter, I explore the predictions of both analyses and conclude that semantic incorporation is more promising, but only if it is constrained to rule out bare preverbal nouns. To this end, I propose a mechanism of existential closure that raises the semantic type of verbs and prepositions so that they can combine with property-denoting arguments in their c-command domain. Next, I investigate bare indirect objects and their intolerance of clitic doubling, as exemplified by *Julia* (*les) *regaló juguetes a niños* ‘Julia gave toys to children’, and to capture this phenomenon, I appeal to the view that *a* ‘to’ is only an authentic preposition in the absence of a dative clitic (Demonte 1995; Cuervo 2003a,b, 2008, 2010a,b). Finally, I explore a set of contexts that disallow bare nouns even postverbally and determine that semantic rather than syntactic factors are at play.

In section 2.1, I present the distribution of bare mass and plural nouns in Spanish and establish the empirical focus of this chapter. In section 2.2, I evaluate the null determiner approach to bare nouns in Romance and show that it faces numerous syntactic and semantic problems, many of which have not been discussed in the literature. In section 2.3, I develop a novel analysis of existential closure in event semantics where a single type-shifting rule enables verbs and prepositions to bind the variables that are introduced by bare nouns at logical form. In section 2.4, I examine the dative alternation in Spanish and build on Cuervo’s (2003a,b, 2008, 2010a,b) account of double-object constructions to explain why they tolerate bare direct objects but not bare indirect objects. In section 2.5, I argue that the various restrictions on bare postverbal nouns that are found with psychological predicates, causative and inchoative verbs, and constructions that involve adjectival elements are due to a variety of semantic considerations. In section 2.6, I conclude and identify some unresolved questions that pertain to the mass-count distinction and are taken up in chapter 3.

2.1 Bare mass and plural nouns

Throughout this thesis, I use the terms “mass noun”, “count noun”, “plural noun”, and “singular noun” as descriptive shorthand for nominalized roots that occur in a given syntactic environment. In work on English, Allan (1980) remarks that count readings are only possible if a noun bears number marking, as in *salads*, or is preceded by a determiner, as in *a salad*. Spanish largely patterns the same in that *ensalada* ‘salad’ is interpreted as a substance in (1a), *ensaladas* ‘salads’ is interpreted as a plurality of salads in (1b), and *una ensalada* ‘a salad’ is interpreted as a single salad in (1c).

- (1) a. Julia preparó ensalada.
 Julia prepared.3SG salad
 ‘Julia prepared salad.’
 b. Julia preparó ensaladas.
 Julia prepared.3SG salads
 ‘Julia prepared salads.’
 c. Julia preparó una ensalada.
 Julia prepared.3SG a salad
 ‘Julia prepared a salad.’

One difference between English and Spanish relates to the semantic behaviour of bare nonplural nouns that appear as the objects of verbs of possession, broadly construed. In an English sentence like *#Julia has cat*, *cat* can only have an improbable mass reading, but in the Spanish equivalent of this sentence in (2), *gato* ‘cat’ has a number-neutral reading (Dobrovie-Sorin et al. 2005, Espinal 2010, Espinal & Mateu 2011, Espinal & McNally 2011).

- (2) Julia tiene gato.
 Julia has cat
 ‘Julia has a cat/cats.’

In (3), where I have replaced the verb of possession with a verb of perception, *gato* is no longer number neutral and must be introduced by a determiner.

- (3) a. #Julia ve gato.
 Julia sees cat
 Literal: ‘Julia sees cat.’
 b. Julia ve {un / el} gato.
 Julia sees a the cat
 ‘Julia sees a/the cat.’

In this chapter, I focus on bare nonplural nouns that can only be interpreted as mass, and in section 3.4, I return to the phenomenon of number neutrality.

It has long been observed in Spanish that unmodified mass and plural nouns do not require a determiner as postverbal arguments (Lapesa 1974, Sánchez de Zavala 1976, Suñer 1982, Contreras

1986, Lois 1987, Eguren 1989, Delfitto & Schroten 1991, Masullo 1992, Bosque 1996, Brugè & Brugger 1996, Contreras 1996, Laca 1996, Masullo 1996, Olarrea 1998, Zubizarreta 1998, Bosque 1999, Laca 1999, Rigau 1999, Dobrovie-Sorin & Laca 2003, Dobrovie-Sorin et al. 2005, Stark 2008, Laca 2013, Fábregas 2018). In (4), the mass noun *sopa* ‘soup’ and the plural noun *aperitivos* ‘appetizers’ occur as bare postverbal subjects of an unaccusative verb, and in (5), these same nouns occur as bare objects of a transitive verb.

- | | |
|--|---|
| <p>(4) a. Quedaba sopa.
 remained.IMPF soup
 ‘Soup remained.’</p> <p> b. Quedaban aperitivos.
 remained.IMPF.3PL appetizers
 ‘Appetizers remained.’</p> | <p>(5) a. Pidieron sopa.
 ordered.3PL soup
 ‘They ordered soup.’</p> <p> b. Pidieron aperitivos.
 ordered.3PL appetizers
 ‘They ordered appetizers.’</p> |
|--|---|

Bare nouns have existential readings in Spanish, and in the majority of cases, they must be adjacent to the verb. For instance, the adverb *ayer* ‘yesterday’ cannot intervene between the verb and its object in (6) but rather must precede the verb or follow the object.

- (6) (Ayer) pidieron (*ayer) sopa (ayer).
 yesterday ordered.3PL yesterday soup yesterday
 ‘(Yesterday) they ordered soup (yesterday).’

Judgments may differ with respect to the placement of adverbs of frequency, as shown by the contrast between the example in (7a), which Lois (1987: 95) considers to be grammatical, and the example in (7b), which Masullo (1992: 284) considers to be ungrammatical.¹

- (7) a. María escribe siempre cartas.
 María writes always letters
 ‘María always writes letters.’
- b. *Juan lee siempre poesía.
 Juan reads always poetry
 Intended: ‘Juan always reads poetry.’

The account of bare nouns that I develop in section 2.3 requires them to be c-commanded by the verb but not necessarily adjacent to it, so it can accommodate an intervening adverb for those speakers who accept this word order.

Turning to unergative verbs, there is consensus that bare postverbal subjects are enabled by locative inversion, as illustrated in (8) with the verb *trabajar* ‘to work’ (Suñer 1982, Torrego 1989, Rigau 1999, Dobrovie-Sorin et al. 2005).

- (8) En el restaurante trabajaban cocineros.
 in the restaurant worked.IMPF.3PL chefs
 ‘Chefs were working in the restaurant.’

¹Whenever cited data lack glosses or translations, I include the missing details as appropriate.

There is, however, disagreement regarding the possibility of bare subjects when the locative expression is postverbal as well. For example, Torrego (1989: 258) finds the construction in (9a) to be unacceptable, whereas Dobrovie-Sorin et al. (2005: 131) find the nearly identical construction in (9b) to be acceptable.

- (9) a. *Juegan niños en este parque.
 play.3PL children in this park
 Intended: ‘Children are playing in this park.’
 b. Merodeaban leones en la selva.
 prowled.IMPF.3PL lions in the jungle
 ‘Lions were prowling in the jungle.’

There is reason to suspect that the form of the verb plays a role in these judgments since the intended episodic reading, where the bare noun is interpreted existentially, is easier to access in the imperfect tense than it is in the simple present tense. In section 2.3, I use the sentence in (9b) to demonstrate how my analysis of bare nouns captures postverbal subjects that are agentive.

Unless a speaker manipulates the prosody of an utterance, unmodified mass and plural nouns are prohibited as bare preverbal subjects in Spanish (Suñer 1982, Contreras 1986, Eguren 1989, Masullo 1992, Bosque 1996, Contreras 1996, Laca 1996, Zubizarreta 1998, Dobrovie-Sorin & Laca 2003, Dobrovie-Sorin et al. 2005, Laca 2013, Fábregas 2018). This restriction is shown in (10) with the unaccusative verb *llegar* ‘to arrive’ and in (11) with the transitive verb *mejorar* ‘to improve’.

- (10) a. *Queso llegó de Francia.
 cheese arrived.3SG from France
 Intended: ‘Cheese arrived from France.’
 b. *Caracoles llegaron de Francia.
 snails arrived.3PL from France
 Intended: ‘Snails arrived from France.’
 (11) a. *Queso mejoró la receta.
 cheese improved.3SG the recipe
 Intended: ‘Cheese improved the recipe.’
 b. *Caracoles mejoraron la receta.
 snails improved.3PL the recipe
 Intended: ‘Snails improved the recipe.’

If, on the other hand, bare nouns are focused, as in (12), or topicalized, as in (13), they can occur preverbally (Suñer 1982, Contreras 1986, Eguren 1989, Masullo 1992, Bosque 1996, Brugè & Brugger 1996, Contreras 1996, Laca 1996, Olarra 1998, Zubizarreta 1998, Gutiérrez-Rexach & Silva-Villar 1999, Laca 1999, Dobrovie-Sorin & Laca 2003, Laca 2013, Fábregas 2018).

- (12) a. QUESO llegó de Francia, no caracoles.
 cheese arrived.3SG from France not snails
 ‘Cheese arrived from France, not snails.’

- b. CARACOLES mejoraron la receta, no queso.
snails improved.3PL the recipe not cheese
'Snails improved the recipe, not cheese.'
- (13) ¿Qué quedaba después de la cena?
what remained.IMPF after of the dinner
'What remained after the dinner?'
- a. Queso quedaba, pero muy poco.
cheese remained.IMPF but very little
'Cheese remained, but very little.'
 - b. Caracoles quedaban, pero muy pocos.
snails remained.IMPF.3PL but very few
'Snails remained, but very few.'

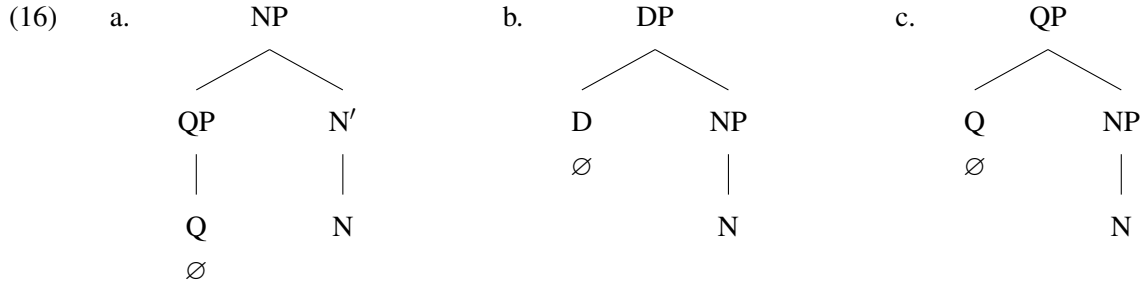
There are other means of licensing bare preverbal subjects in Spanish, including modification and coordination (Lapesa 1974, Sánchez de Zavala 1976, Suñer 1982, Lois 1987, Eguren 1989, Delfitto & Schrotten 1991, Masullo 1992, Bosque 1996, Brugè & Brugger 1996, Dobrovie-Sorin & Laca 2003, Laca 2013, Fábregas 2018). Lapesa's (1974: 299) examples in (14) involve a variety of modification strategies, and Suñer's (1982: 222) in (15) illustrate that coordinated bare nouns tend to constitute parts of a whole.

- (14) a. Hombres de calidad estudian el asunto.
men of quality study.3PL the matter
'Men of quality are studying the matter.'
- b. Voces prestigiosas consiguieron frenar los desmanes.
voices prestigious managed.3PL stop the outrage
'Prestigious voices managed to stop the outrage.'
- c. Cosas como ésta desaniman a cualquiera.
things like this discourage.3PL DOM anyone
'Things like this discourage anyone.'
- (15) a. Hombres y mujeres aceptaron mi opinión.
men and women accepted.3PL my opinion
'Men and women accepted my opinion.'
- b. Niños y adultos respetaban sus palabras.
children and adults respected.IMPF.3PL their words
'Children and adults respected their words.'
- c. Autos, ómnibus y camiones fueron paralizados por la inundación.
cars buses and trucks were.3PL paralyzed by the flood
'Cars, buses, and trucks were at a standstill because of the flood.'

I concentrate on bare postverbal nouns in this chapter and defer the analysis of grammatical bare preverbal nouns to future research. In the next section, I explore whether there may be empirical or theoretical support for the claim that bare nouns are introduced by a silent determiner in Spanish.

2.2 Null determiners and compositionality

Early accounts of bare nouns in Romance converge on the notion that they are licensed as arguments by some type of phonologically null determiner whose distribution is constrained by Chomsky's (1981) Empty Category Principle. Contreras (1986) first proposed that bare nouns are NPs that host an unpronounced quantifier phrase (QP) in their specifier, as in (16a), Longobardi (1994) then argued that bare nouns are covert DPs, as in (16b), and Contreras (1996) later combined aspects of both approaches to model bare nouns as covert QPs, as in (16c).



In all three cases, the putative null determiner is subject to lexical government, meaning that it must be c-commanded by a lexical head like V, which is the case in (17a) but not in (17b).

- (17) a. Aparecieron ∅ leones en la sabana.
 appeared.3PL lions on the savannah
 'Lions appeared on the savannah.'
- b. *∅ leones aparecieron en la sabana.
 lions appeared.3PL on the savannah
 Intended: 'Lions appeared on the savannah.'

Longobardi's analysis is, moreover, illustrative of the strong DP hypothesis, according to which DPs alone can be arguments (Stowell 1989, 1991; Szabolcsi 1994; Benedicto 1998; Progovac 1998; Crisma 1999; Zamparelli 2000; Borer 2005). This view remains influential in work on Romance but has proven controversial as a crosslinguistic generalization.

To derive the existential readings of bare nouns in Romance, Chierchia (1998b) resorts to null D and posits that this head is the locus of the "down" operator ($\cap$) in (18), where I use Dayal's (2004: 399) definition for clarity.²

$$(18) \quad \cap := \lambda P_{\langle s, \langle e, t \rangle \rangle} . \lambda w_s . \iota x_e [P_s(x)]$$

In an intensional semantics, this operator takes a property-denoting expression and a world argument and returns the unique maximal object that has property P in the world of evaluation. The resulting

²The counterpart of $\cap$ is the "up" operator ($\cup$) in (i), which takes a kind-denoting expression and returns a property-denoting one (Dayal 2004: 399).

$$(i) \quad \cup := \lambda k_{\langle s, e \rangle} . \lambda x_e . x \leq k_s$$

DP can then be targeted by the rule of Derived Kind Predication in (19), which applies to kind-denoting expressions and supplies existential quantification over the realizations of the kind.

(19) *Derived Kind Predication (DKP):*

If P applies to objects and k denotes a kind, then

$$P(k) = \exists x_e [\cup k(x) \wedge P(x)]$$

The issue with this analysis is that there is no evidence to suggest that bare nouns can denote kinds in Spanish, meaning that they are not eligible for Derived Kind Predication. As discussed in section 1.1, and as indicated in (20), nouns that serve as arguments of kind-selecting predicates like *extinguirse* ‘to become extinct’ require the definite article even in postverbal position (Laca 1990, 1996, 1999; Dobrovie-Sorin & Laca 2003; McNally 2004; Bleam 2005; Dobrovie-Sorin et al. 2005; Dobrovie-Sorin 2009; Borik & Espinal 2012; Laca 2013; Borik & Espinal 2015).

- (20) Se extinguieron *(los) mamuts.
 3 extinguished.3PL the mammoths
 ‘Mammoths became extinct.’

In response to this obstacle, Chierchia (1998b: 385) asserts that modified nouns can be the subject of *essere estinto* ‘to be extinct’ in Italian based on examples like (21a), but Longobardi (2001: 343) and Zamparelli (2002: 323) dispute this characterization and remark that the similar examples in (21b) and (21c) are ungrammatical.

- (21) a. Insegnanti davvero dediti nella scuola di oggi sono quasi estinti.
 teachers really devoted in.the school of today are.3PL nearly extinct
 ‘Really devoted teachers in today’s school are nearly extinct.’
 b. *Elefanti di colore bianco sono estinti.
 elephants of colour white are.3PL extinct
 Intended: ‘White-coloured elephants have become extinct.’
 c. *Tigri dai denti a sciabola sono estinte.
 tigers from.the teeth to sabre are.3PL extinct
 Intended: ‘Sabre-toothed tigers are extinct.’

Independently of the debate in Italian, there is no motivation to assign null D the semantics of $\cap$ in Spanish. In the next part of this section, I identify two more empirical challenges in this language for the strong DP hypothesis, one that relates to extraction of material from within the noun phrase and another that relates to the lack of n -to-D movement.

I begin with the phenomenon of extraction and a contrast that many authors have noted between indefinite and definite noun phrases (Campos 1988, Mallén 1991, Masullo 1992, Martín 1993, Masullo 1996, Gutiérrez-Rexach & Silva-Villar 1999, Ticio 2005). As illustrated by the *wh*-questions in (22) and (23), it is possible to extract the complement of *varios libros* ‘several books’, which is indefinite, but not the complement of *los libros* ‘the books’, which is definite.

- (22) a. Compraron varios libros de Julia Child.
bought.3PL several books of Julia Child
'They bought several books by Julia Child.'
- b. ¿De qué autor compraron varios libros?
of what author bought.3PL several books
Literal: 'By what author did they buy several books?'
- (23) a. Compraron los libros de Julia Child.
bought.3PL the books of Julia Child
'They bought the books by Julia Child.'
- b. *¿De qué autor compraron los libros?
of what author bought.3PL the books
Literal: 'By what author did they buy the books?'

One might expect bare nouns to pattern like indefinite rather than definite nouns when it comes to extraction of their complements, and this prediction is borne out in (24).

- (24) a. Compraron libros de Julia Child.
bought.3PL books of Julia Child
'They bought books by Julia Child.'
- b. ¿De qué autor compraron libros?
of what author bought.3PL books
Literal: 'By what author did they buy books?'

These facts undermine the null D account of bare nouns, but they do not necessarily favour the null Q account either since *libros* 'books' could be a DivP instead of a covert QP.

The third problem that the strong DP hypothesis faces concerns the distribution of names, which are another type of *nP*. As shown in (25), names can be preceded by the same determiners and modifiers that nouns can, confirming that they are generated in the lexical layer of the noun phrase.³

- (25) a. La amable Julia me saludó.
the kind Julia 1SG greeted.3SG
'Kind Julia greeted me.'
- b. La amable cocinera me saludó.
the kind chef 1SG greeted.3SG
'The kind chef greeted me.'

A major difference between names and nouns is that names can appear as bare preverbal arguments, as in (26), raising questions about their internal structure.

- (26) Julia me saludó.
Julia 1SG greeted.3SG
'Julia greeted me.'

³See Jambrović 2024 for arguments that names contain two nominalizers, one that establishes the name itself and another that converts it into a predicate.

If null D must be c-commanded by a lexical head in Spanish, it cannot be the case that *Julia* in (26) has the syntactic representation in (27).

- (27) [DP $\emptyset$ [_{NP} *Julia*]] me saludó.
 Julia 1SG greeted.3SG
 ‘Julia greeted me.’

To address this issue in Italian, Longobardi (1994, 2001, 2005, 2008) proposes that names raise to D whenever this head is available as a landing site, as indicated in (28).

- (28) [DP *Julia* [_{NP} $\langle$ *Julia* $\rangle$]] me saludó.
 Julia 1SG greeted.3SG
 ‘Julia greeted me.’

As a result of this process, D acquires phonological material and becomes exempt from the Empty Category Principle.

In support of the *n*-to-D analysis of names, Longobardi (1994: 623) offers the paradigm in (29), which illustrates that *Gianni* can precede or follow the possessive determiner *mio* ‘my’ if the definite article is present but necessarily precedes *mio* if the definite article is absent, suggesting that *Gianni* raises to occupy D.

- (29) a. Il mio Gianni ha finalmente telefonato.
 the my Gianni has finally called
 ‘My Gianni finally called.’
 b. *Mio Gianni ha finalmente telefonato.
 my Gianni has finally called
 Intended: ‘My Gianni finally called.’
 c. Gianni mio ha finalmente telefonato.
 Gianni my has finally called
 ‘My Gianni finally called.’
 d. Il Gianni mio ha finalmente telefonato.
 the Gianni my has finally called
 ‘My Gianni finally called.’

As it turns out, there is no evidence of *n*-to-D movement in Spanish, refuting Longobardi’s claim that Italian is representative of the Romance languages in general.

The definite article is mutually exclusive with prenominal possessive determiners in Spanish, as revealed by *mi Juan* ‘my Juan’ versus **el mi Juan* ‘the my Juan’, so the only way to determine whether names raise to D is through modifiers that can occur prenominally.⁴ As shown in (30), *Juan* can follow the adjective *viejo* ‘old’, confirming that the name originates in *n*, but *Juan* cannot raise above the domain of modification to D.

⁴In Brazilian Portuguese, where the definite article and prenominal possessive determiners are not mutually exclusive, names must remain in situ. For Schmitt & Munn (2002: 195), there is a contrast in grammaticality between *o meu Pedro telefonou* ‘my Pedro called’ and **Pedro meu telefonou* ‘my Pedro called’.

- (30) a. El viejo Juan por fin llamó.
the old Juan finally called.3SG
'Old Juan finally called.'
- b. *Juan viejo por fin llamó.
old Juan finally called.3SG
Intended: 'Old Juan finally called.'

The same observation extends to the place name *Roma* 'Rome' in (31) and the temporal names *jueves* 'Thursday' and *mayo* 'May' in (32), which are direct translations of Longobardi's (1994: 624–625) examples (30a,c) and (31a,c).

- (31) a. La antigua Roma fue la ciudad más importante del Mediterráneo.
the ancient Rome was the city more important of.the Mediterranean
'Ancient Rome was the most important city of the Mediterranean.'
- b. *Roma antigua fue la ciudad más importante del Mediterráneo.
Rome ancient was the city more important of.the Mediterranean
Intended: 'Ancient Rome was the most important city of the Mediterranean.'
- (32) a. El pasado {jueves / mayo} fue un {día / mes} terrible.
the last Thursday May was a day month terrible
'Last Thursday/May was a terrible day/month.'
- b. *{Jueves / Mayo} pasado fue un {día / mes} terrible.
Thursday May last was a day month terrible
Intended: 'Last Thursday/May was a terrible day/month.'

If names do not move to D in Spanish, it seems that null D need not be c-commanded by a lexical head after all. One way to grapple with this conclusion would be to posit an unconstrained null D for bare names, which are referential expressions, and a constrained null Q for bare nouns, which are not. Beyond the stipulatory nature of this proposal, the semantic behaviour of bare nouns rules out null Q as a viable option.

As I first mentioned in section 1.1 and now demonstrate through the broader set of facts in (33) through (35), indefinite nouns like *unos leones* 'some lions' and bare nouns like *leones* 'lions' both have existential readings, but only indefinite nouns can take wide scope with respect to other semantic operators (Garrido Medina 1991; Bosque 1996; Laca 1996, 1999; Gutiérrez-Rexach 2001; Dobrovie-Sorin & Laca 2003; McNally 2004; Bleam 2005; Laca 2013; Fábregas 2018).⁵

- (33) a. Todos vieron unos leones. $\forall > \exists$ or $\exists > \forall$
all saw.3PL some lions
'Everyone saw some lions.'
- b. Todos vieron leones. $\forall > \exists$ only
all saw.3PL lions
'Everyone saw lions.'

⁵In English, Carlson (1977, 1980) first noted that bare plural nouns are semantically inert in these types of environments and many others.

- (34) a. Deben encontrar unos leones. $\square > \exists$ or $\exists > \square$
 must.3PL find some lions
 ‘They must find some lions.’
- b. Deben encontrar leones. $\square > \exists$ only
 must.3PL find lions
 ‘They must find lions.’
- (35) a. Buscan unos leones. $\wedge > \exists$ or $\exists > \wedge$
 seek.3PL some lions
 ‘They are looking for some lions.’
- b. Buscan leones. $\wedge > \exists$ only
 seek.3PL lions
 ‘They are looking for lions.’

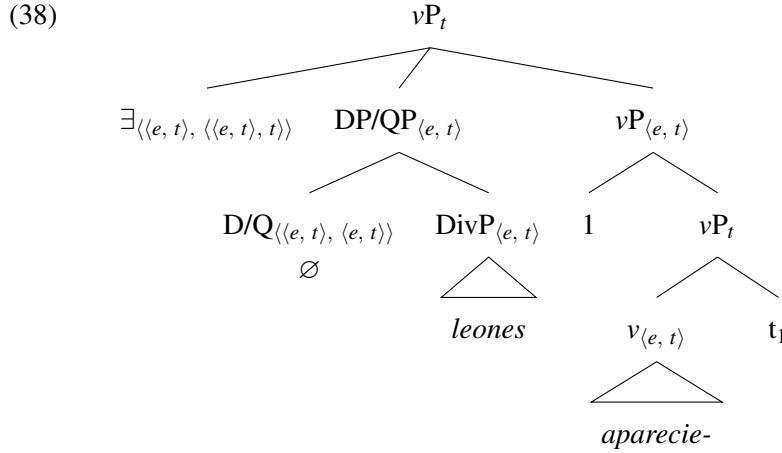
That is, only *unos leones* permits the reading where there are particular lions. Further evidence that bare nouns obligatorily take narrow scope is found in the morphological expression of mood in Spanish. If the direct objects in (35) are modified by a restrictive relative clause, as in (36), the bare noun *leones* is incompatible with an indicative form of the embedded verb, confirming that it does not allow for a specific reading (Jaeggli 1982; Kliffer 1984; Bosque 1996; Brugè & Brugger 1996; Laca 1996, 1999; Rigau 1999; Dobrovie-Sorin & Laca 2003; Bleam 2005; Fábregas 2018).

- (36) a. Buscan unos leones que {tengan / tienen} melena. $\wedge > \exists$ or $\exists > \wedge$
 seek.3PL some lions that have.SBJV.3PL have.3PL mane
 ‘They are looking for some lions that have manes.’
- b. Buscan leones que {tengan / *tienen} melena. $\wedge > \exists$ only
 seek.3PL lions that have.SBJV.3PL have.3PL mane
 ‘They are looking for lions that have manes.’

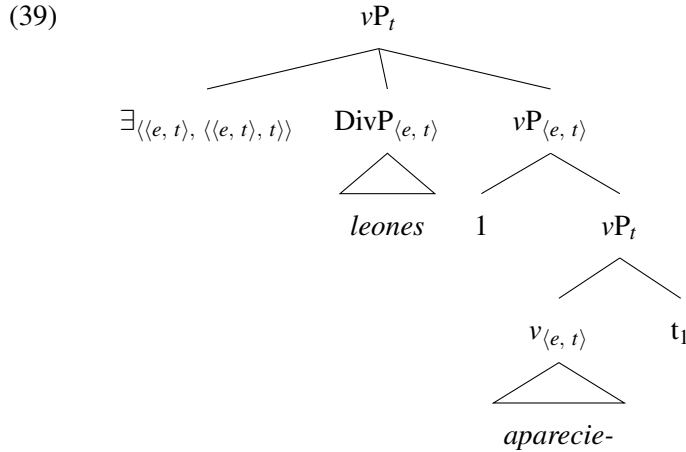
If bare nouns cannot undergo quantifier raising, there is no reason to model them as covert QPs. Still, the question remains how they come to have existential interpretations if existential force is not introduced within the noun phrase itself.

To account for the existential readings of bare nouns in Italian, Longobardi (2000, 2001, 2005, 2008) invokes Diesing’s (1992) view of existential closure, where an existential operator ($\exists$) is adjoined to VP at logical form. Since $\exists$ is external to the noun phase in this approach, it follows that bare nouns only take narrow scope. At the same time, Longobardi does not provide interpretable structures or otherwise identify the semantic contribution of the null determiner that supposedly licenses bare nouns as arguments. Using the unaccusative construction in (37), and abstracting away from tense, I seek to do so in (38) in a way that is consistent with the tripartite configuration that Diesing adopts from Heim (1988).

- (37) Aparecieron leones.
 appeared.3PL lions
 ‘Lions appeared.’



Regardless of whether one considers the bare noun to be a DP or a QP, the semantic type of $\exists$ requires both of its arguments to be of type $\langle e, t \rangle$, which in turn requires the null D or Q to be a vacuous identity function of type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$. Since this head does not have a phonological realization or any bearing on logical form, structural economy dictates that it should be eliminated altogether, as in (39).⁶



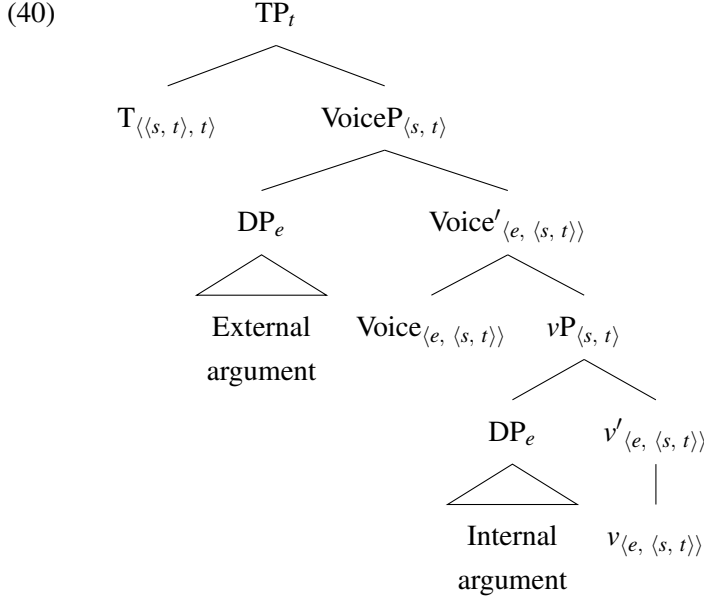
This solution represents a step in the right direction but is overly simplistic in terms of argument structure and overly complex in terms of ternary branching and movement, issues that I seek to resolve in the next section.

2.3 Existential closure in event semantics

In the interest of incorporating tense into the picture, I develop an account of existential closure in event semantics, where verbs require an entity (e) as well as an event (s) to return a truth value (t)

⁶Independently of the phenomenon under study, structural economy is often the deciding factor between competing analyses (Chomsky 1981, Chierchia 1998b, Cardinaletti & Starke 1999, Schlenker 2005, Dayal 2011a, Katzir 2011, Patel-Grosz & Grosz 2017).

and are therefore minimally of type $\langle e, \langle s, t \rangle \rangle$ rather than $\langle e, t \rangle$. Furthermore, I assume Kratzer's (1996) view of argument structure in (40), according to which internal arguments merge as specifiers of vP , or VP , and external arguments merge as specifiers of VoiceP .



In this system, Voice and vP combine through the rule of event identification in (41), which I have adapted to reflect the notation used elsewhere in this thesis (Kratzer 1996: 122).

(41) Event Identification

$$P_{\langle e, \langle s, t \rangle \rangle} Q_{\langle e, t \rangle} \rightarrow R_{\langle e, \langle s, t \rangle \rangle} \\ \lambda x_e . \lambda e_s . P(x)(e) \wedge Q(e)$$

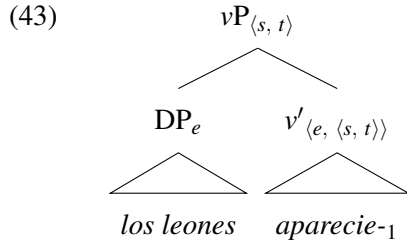
As for indirect objects, I represent these arguments as complements of v in cases where they are introduced by a preposition, a topic that I return to in section 2.4.

Instead of adjoining $\exists$ to vP to derive the existential readings of bare nouns, as in Diesing's (1992) approach, many authors resort to a process known as semantic incorporation, where the verb itself provides existential force (Van Geenhoven 1995; Dobrovie-Sorin 1997; Van Geenhoven 1998, 2000; Cohen & Erteschik-Shir 2002; Dobrovie-Sorin & Laca 2003; Farkas & de Swart 2003; McNally 2004; Bleam 2005; Van Geenhoven & McNally 2005; Dobrovie-Sorin et al. 2005; Dobrovie-Sorin 2009; de Swart & Zwarts 2009; Espinal 2010; Dayal 2011b; Espinal & McNally 2011; Laca 2013; Collins 2019; Sağ 2019, 2022).⁷ For instance, Van Geenhoven (1995, 1998) claims that verbs come in two guises, one for object-denoting, or type e , arguments and another for property-denoting, or type $\langle e, t \rangle$, arguments. In (42), I use the stem of the unaccusative verb *aparecer* 'to appear' to illustrate this proposal in event semantics.

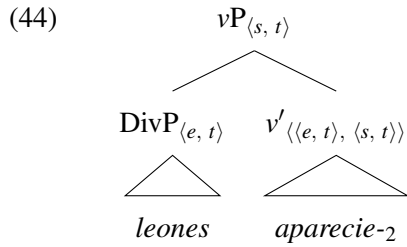
⁷In early theoretical work on Spanish, it was argued that bare nouns undergo syntactic incorporation and become part of the verbal complex (Lois 1987; Delfitto & Schrotten 1991; Masullo 1992, 1996). This view has largely been abandoned in favour of semantic incorporation.

- (42) a. $\llbracket \text{aparece-}_1 \rrbracket = \lambda x_e . \lambda e_s . \text{Appear}(x)(e)$
 b. $\llbracket \text{aparece-}_2 \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda e_s . \exists x_e [P(x) \wedge \text{Appear}(x)(e)]$

The simpler version of *aparece-* in (42a) can combine with a referential expression like *los leones* ‘the lions’, as in (43).



The more complex version of *aparece-* in (42b) can combine with a bare noun like *leones* ‘lions’ and existentially bind its variable, as in (44),



The major disadvantage of Van Geenhoven’s analysis is the redundancy that stems from positing two lexical entries for the relevant verbs.

In a different take on existential closure, Benedicto (1998: 54) maintains a single guise for all verbs and argues that they can only existentially bind variables that lie within their c-command domain, as stated in (45).⁸

- (45) $\exists$ -with-*e* (The ‘Existential with the Event’)
- Existential Closure is introduced with the event argument of the predicate.
 - The scope of $\exists$ is its c-command domain.

This view captures the obligatory postverbal position of bare nouns in Spanish without confining existential closure to *vP* or any other layer of the clause. One reason to favour Benedicto’s proposal over Diesing’s (1992) relates to interrogative sentences that have bare subjects, such as the polar question in (46).

- (46) ¿Merodeaban leones en la selva?
 prowled.IMPF.3PL lions in the jungle
 ‘Were lions prowling in the jungle?’

⁸Although Benedicto subscribes to strong DP hypothesis, neither claim in (45) depends on this assumption.

If one considers *leones* ‘lions’ to occupy the specifier of TP in this example, its variable could be bound through $\exists$ -with-*e* but not through vP -level existential closure. The question is how to translate Benedicto’s $\exists$ -with-*e* account to logical form.

One possibility is to characterize $\exists$ -with-*e* as a last-resort mechanism that raises the semantic type of verbs so that they can combine with property-denoting arguments, an approach that has precedent in the literature (Cohen & Erteschik-Shir 2002, McNally 2004, Laca 2013). In essence, the objective is to derive the more complex guise of the verb stem *aparece*- ‘appear’ in (42b), repeated in (47b), from its simpler guise in (42a), repeated in (47a).

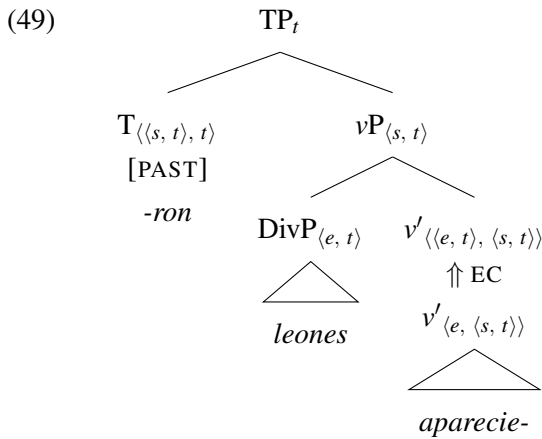
- (47) a. $\llbracket \text{aparece-1} \rrbracket = \lambda x_e . \lambda e_s . \text{Appear}(x)(e)$
 b. $\llbracket \text{aparece-2} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda e_s . \exists x_e [P(x) \wedge \text{Appear}(x)(e)]$

Not only does the proposed definition of existential closure (EC) in (48) achieve this goal, but it also enables prepositions to apply to property-denoting objects, as I demonstrate in section 2.4.

- (48) $\text{EC} := \lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda Q_{\langle e, t \rangle} . \lambda e_s . \exists x_e [Q(x) \wedge P(x)(e)]$

If, as Benedicto (1998) argues, existential closure is delimited by the c-command domain of the verb, the type-shifting operation in (48) is all that is needed to capture the interpretation of bare nouns in Spanish.

To illustrate the mechanics of the analysis, I start with the interpretable structure of the unaccusative construction *aparecieron leones* ‘lions appeared’ in (49), where I omit VoiceP because the subject is generated as an internal argument and assume, like Kratzer (1996), that heads are interpreted in situ.



The verb ending *-ron* in T c-commands the bare DivP in the specifier of vP , so this configuration is eligible for existential closure. The truth conditions of this sentence can be calculated in three steps, each of which involves the substitution of one or more lexical entries, abbreviated as “LE”, followed by β -reduction, or simplification, abbreviated as “BR”. First, existential closure (EC) lifts the type of *aparecie*- in (50).

$$\begin{aligned}
(50) \quad & \text{EC}(\llbracket \text{aparecie-} \rrbracket) \\
&= [\lambda P_{\langle e, \langle s, t \rangle \rangle} \cdot \lambda Q_{\langle e, t \rangle} \cdot \lambda e_s \cdot \exists x_e [Q(x) \wedge P(x)(e)]] (\lambda y_e \cdot \lambda e'_s \cdot [\text{Appear}(y)(e')]) \quad \text{LE} \\
&= \lambda Q_{\langle e, t \rangle} \cdot \lambda e_s \cdot \exists x_e [Q(x) \wedge \text{Appear}(x)(e)] \quad \text{BR}
\end{aligned}$$

Second, the type-shifted verb stem applies to the DivP *leones* ‘lions’ in (51).⁹

$$\begin{aligned}
(51) \quad & \llbracket \uparrow \text{aparecie-} \rrbracket (\llbracket \text{leones} \rrbracket) \\
&= [\lambda Q_{\langle e, t \rangle} \cdot \lambda e_s \cdot \exists x_e [Q(x) \wedge \text{Appear}(x)(e)]] (\lambda y_e \cdot * \text{Lion}(y)) \quad \text{LE} \\
&= \lambda e_s \cdot \exists x_e [* \text{Lion}(x) \wedge \text{Appear}(x)(e)] \quad \text{BR}
\end{aligned}$$

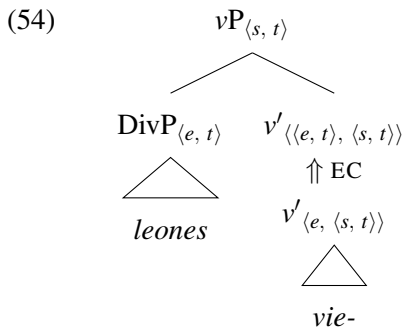
Third, the preterite tense suffix *-ron*, which has the denotation that Kratzer suggests for $T_{[\text{PAST}]}$, applies to the vP *leones aparecie-* ‘lions appear’ in (52).

$$\begin{aligned}
(52) \quad & \llbracket \text{-ron} \rrbracket (\llbracket \text{leones aparecie-} \rrbracket) \\
&= [\lambda P_{\langle s, t \rangle} \cdot \exists e'_s [\text{Past}(e') \wedge P(e')]] (\lambda e_s \cdot \exists x_e [* \text{Lion}(x) \wedge \text{Appear}(x)(e)]) \quad \text{LE} \\
&= 1 \text{ iff } \exists e'_s [\text{Past}(e') \wedge \exists x_e [* \text{Lion}(x) \wedge \text{Appear}(x)(e')]] \quad \text{BR}
\end{aligned}$$

That is, *aparecieron leones* is true if and only there is a past event e' in which there was an object x that consisted of lions and x appeared in e' .

Transitive constructions have external arguments, but they are no different from unaccusative constructions at the level of vP in Kratzer’s (1996) system. For instance, the vP of the sentence in (53) has the interpretable structure in (54).

- (53) Vieron leones.
saw.3PL lions
‘They saw lions.’

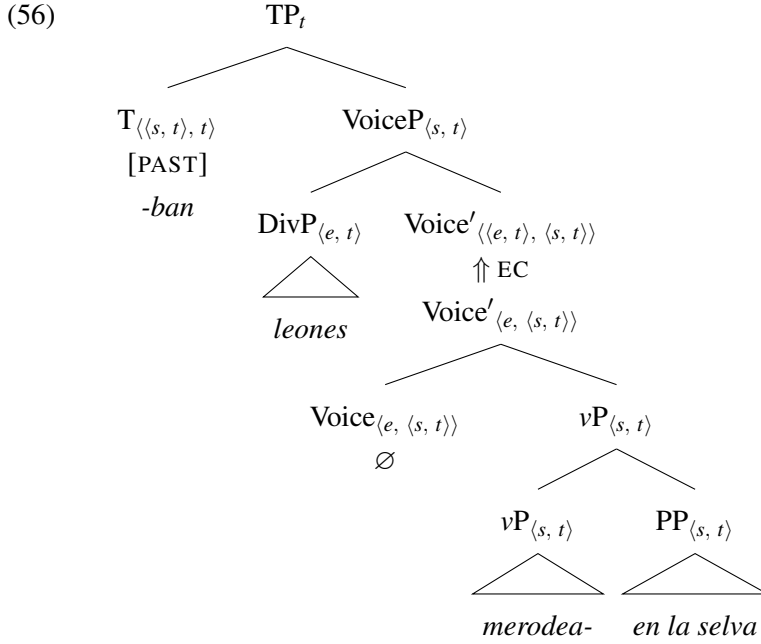


Here as well, EC raises the type of the verb stem *vie-* ‘see’ so that it can combine with its property-denoting internal argument *leones* ‘lions’.

As for unergative constructions, I propose that Dobrovie-Sorin et al.’s (2005: 131) example in (9b), repeated in (55), has the interpretable structure in (56), where the external argument occupies the specifier of VoiceP and the PP adjoins to vP.

⁹See section 3.1 for discussion of Link’s (1983) star operator (*), which represents plurality.

- (55) Merodeaban leones en la selva.
prowled.IMPF.3PL lions in the jungle
'Lions were prowling in the jungle.'



First, the vP *merodea-* 'prowl' and the PP *en la selva* 'in the jungle', which are both expressions of type $\langle s, t \rangle$, conjoin in (57).

$$(57) \quad \llbracket \text{merodea- en la selva} \rrbracket = \lambda e_s . [\text{Prowl}(e) \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e)]$$

Second, Voice and vP combine through Kratzer's (1996) rule of event identification in (41) above, yielding the predicate in (58).

$$(58) \quad \llbracket \emptyset \text{ merodea- en la selva} \rrbracket = \lambda y_e . \lambda e_s . [\text{Prowl}(e) \wedge \text{AGENT}(y)(e) \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e)]$$

Third, EC lifts the type of Voice' *merodea- en la selva* 'prowl in the jungle' in (59).

$$\begin{aligned}
 (59) \quad & \text{EC}(\llbracket \emptyset \text{ merodea- en la selva} \rrbracket) \\
 &= [\lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge P(z)(e')]] (\lambda y_e . \lambda e_s . [\text{Prowl}(e) \wedge \text{AGENT}(y)(e) \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e)]) \quad \text{LE} \\
 &= \lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge [\text{Prowl}(e') \wedge \text{AGENT}(z)(e') \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e')]] \quad \text{BR}
 \end{aligned}$$

Fourth, the type-shifted verb stem applies to the DivP *leones* 'lions' in (60).

$$\begin{aligned}
 (60) \quad & \llbracket \uparrow \emptyset \text{ merodea-} \rrbracket (\llbracket \text{leones} \rrbracket) \\
 &= [\lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge [\text{Prowl}(e') \wedge \text{AGENT}(z)(e') \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e')]]] (\lambda y_e . * \text{Lion}(y)) \quad \text{LE} \\
 &= \lambda e'_s . \exists z_e [* \text{Lion}(z) \wedge [\text{Prowl}(e') \wedge \text{AGENT}(z)(e') \wedge \text{In}(\iota x_e \text{ Jungle}(x))(e')]] \quad \text{BR}
 \end{aligned}$$

Fifth, the imperfect tense suffix *-ban* applies to the entire VoiceP *leones merodea- en la selva* ‘lions prowl in the jungle’ in (61).

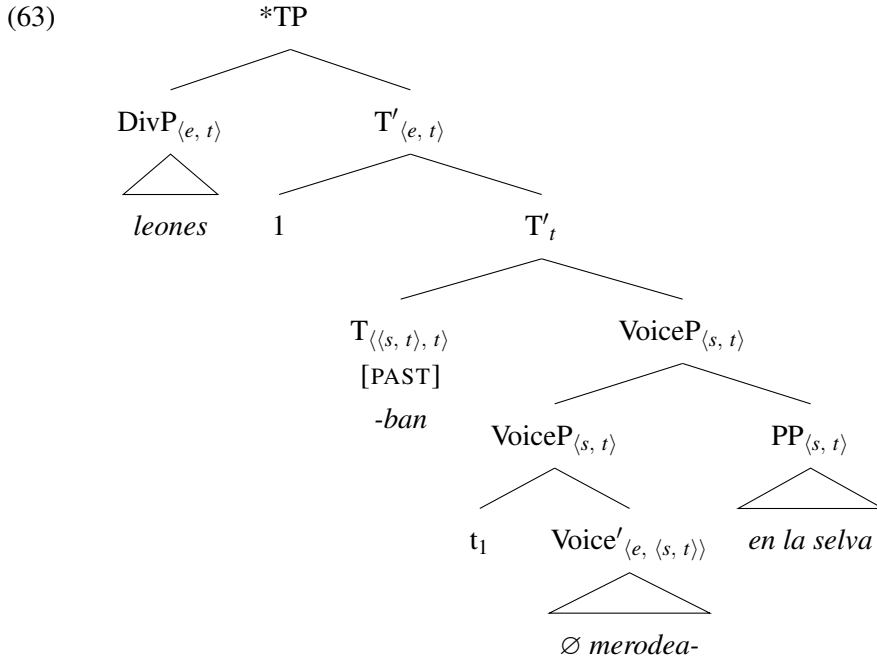
- (61) $\llbracket \text{-ban} \rrbracket (\llbracket \text{leones } \emptyset \text{ merodea- en la selva} \rrbracket)$
 $= [\lambda P_{\langle s, t \rangle} . \exists e_s [\text{Past}(e) \wedge P(e)] (\lambda e'_s . \exists z_e [* \text{Lion}(z) \wedge [\text{Prowl}(e') \wedge$ LE
 $\text{AGENT}(z)(e') \wedge \text{In}(\iota x_e \text{Jungle}(x))(e')]]]$
 $= 1 \text{ iff } \exists e_s [\text{Past}(e) \wedge \exists z_e [* \text{Lion}(z) \wedge [\text{Prowl}(e) \wedge \text{AGENT}(z)(e) \wedge$ BR
 $\text{In}(\iota x_e \text{Jungle}(x))(e)]]]$

In prose, *merodeaban leones en la selva* ‘lions were prowling in the jungle’ is true if and only if there is a past event e of prowling in which there was an object z that consisted of lions, z was the agent of e , and e took place in the unique object x that is a jungle.

The final issue to explore is how the analysis that I have developed rules out bare preverbal subjects, such as *leones* ‘lions’ in (62).

- (62) **Leones merodeaban en la selva.*
 lions prowled.IMP.F.3PL in the jungle
 Intended: ‘Lions were prowling in the jungle.’

In this case, I assume that the external argument has moved to the specifier of TP, leaving behind a trace and triggering Heim & Kratzer’s (1998) rule of predicate abstraction, as indicated in (63).



In its derived position, the subject *leones* is not c-commanded by any part of the verb, so existential closure cannot rescue the structure. Nor would predicate modification solve the problem since the combination of DivP and T' would result in an expression of the same type, $\langle e, t \rangle$, not a truth value.

In summary, I have modelled existential closure as a last-resort operation that resolves type mismatches between verbs and property-denoting arguments that lie within their c-command domain. As in Contreras's (1986, 1996) null Q approach and Longobardi's (1994, 2000, 2001, 2005, 2008) null D approach, I attribute the limited distribution of bare nouns in Spanish to a constraint that is based on hierarchical relations, but I consider the relevant component of the grammar to be logical form rather than syntactic operations. The next step is to evaluate how the proposal fares in capturing bare objects of prepositions, including those that serve as indirect objects.

2.4 Bare objects of prepositions

It is generally recognized that prepositions, like verbs, can license bare nouns in Spanish, as shown in (64) with examples by Suñer (1982: 216) and Lois (1987: 66), respectively.

- (64) a. La cosecha fue destruida por langostas.
 the harvest was destroyed by locusts
 'The harvest was destroyed by locusts.'
- b. Ayer salí con amigos.
 yesterday went.out.1SG with friends
 'I went out with friends yesterday.'

Whether prepositions uniformly have this ability has been called into question by Brugè & Brugger (1996: 19), for whom the bare objects in (64) cannot be substituted by those in (65).

- (65) a. La cosecha fue destruida por { *mujeres / *enemigos / *terroristas }.
 the harvest was destroyed by women enemies terrorists
 Intended: 'The harvest was destroyed by women/enemies/terrorists.'
- b. Ayer salí con { *porteros / *pobres / *marineros }.
 yesterday went.out.1SG with doorkeepers poor sailors
 Intended: 'I went out with doorkeepers/the poor/sailors yesterday.'

One reason to venture that the issue is pragmatic rather than syntactic or semantic in nature is that all of the bare nouns in (66) are felicitous.

- (66) La cosecha fue destruida por { conejos / hongos / saltamontes }.
 the harvest was destroyed. by rabbits fungi grasshoppers
 'The harvest was destroyed by rabbits/fungi/grasshoppers.'

Moreover, the use of any bare noun with *salir con* 'to go out with' can be improved by providing sufficient context, as in (67).

- (67) Parece más divertido salir con marineros que con porteros.
 seems more fun go.out with sailors than with doorkeepers
 'It seems more fun to go out with sailors than with doorkeepers.'

These additional data suggest that prepositions can indeed license bare nouns productively, but there is one context that warrants careful examination.

The first point to establish is that indefinite and definite indirect objects can both be doubled by dative clitics in Spanish, as indicated in (68) with the verb *enviar* ‘to send’ (Roldán 1971; Strozer 1976; Silva-Corvalán 1981; Jaeggli 1982, 1986; Suñer 1988, 1989; F. Roca 1992; Fernández Soriano 1993; Demonte 1995; F. Roca 1996; Parodi 1998; Fernández Soriano 1999; Bleam 2003; Cuervo 2003a,b, 2008, 2010a,b; Gabriel & Rinke 2010; Fischer et al. 2019).¹⁰

- (68) a. Jacques (les) envió recetas a varios amigos.
 Jacques 3PL.DAT sent.3SG recipes to some friends
 ‘Jacques sent recipes to some friends.’
 b. Jacques (les) envió recetas a sus amigos.
 Jacques 3PL.DAT sent.3SG recipes to his friends
 ‘Jacques sent recipes to his friends.’

Curiously, bare indirect objects are incompatible with clitic doubling, a restriction that Jaeggli (1982: 59) illustrates with the facts in (69).

- (69) a. Regalaré todos mis libros a mujeres.
 gift.FUT.1SG all my books to women
 ‘I will give all my books to women.’
 b. *Les regalaré todos mis libros a mujeres.
 3PL.DAT gift.FUT.1SG all my books to women
 Intended: ‘I will give all my books to women.’

The same constraint is found in (70), where the preposition varies according to the absence versus presence of the dative clitic (Masullo 1992: 278).

- (70) a. Recogieron ropa para mendigos.
 collected.3PL clothes for beggars
 ‘They collected clothes for beggars.’
 b. *Les recogieron ropa a mendigos.
 3PL.DAT collected.3PL clothes DAT beggars
 Intended: ‘They collected clothes for beggars.’

This alternation is indicative of the dative shift in Spanish and is vital to understanding the complementary distribution of bare indirect objects and dative clitics.

Demonte (1995), Bleam (2003), and Cuervo (2003a,b, 2008, 2010a,b) all maintain that a subset of the constructions that involve dative clitics are equivalent to double-object constructions in English, such as *Jacques sent his friends recipes*, even though the word order does not change in Spanish. In (71), for instance, the direct object precedes the indirect object in both sentences, but the version that contains a dative clitic is ungrammatical (Cuervo 2003a: 53–54).

¹⁰I section 4.1, I present clitic doubling in greater detail and discuss the parallels between this phenomenon and object-verb agreement.

- (71) a. Valeria mostró el maestro a sí mismo.
Valeria showed.3SG the teacher to himself
'Valeria showed the teacher to himself.'
- b. *Valeria le mostró el maestro a sí mismo.
Valeria 3SG.DAT showed.3SG the teacher DAT himself
Approximate: 'Valeria showed himself the teacher.'

These facts suggest that *el maestro* 'the teacher' c-commands the reflexive pronoun *sí mismo* 'himself' in (71a) but not in (71b), hence the different translations of these examples. The minimal pair in (72) exhibits the same hierarchical asymmetry in that the direct object *cada cuadro* 'every painting' can only take scope over the indirect object *un museo* 'a museum', which is the reading that permits the inclusion of *distinto* 'different', if there is no clitic doubling (Cuervo 2003a: 61).

- (72) a. Andrés mandó cada cuadro a un museo (distinto).
Andrés sent.3SG each painting to a museum different
'Andrés sent each painting to a (different) museum.'
- b. Andrés le mandó cada cuadro a un museo (#distinto).
Andrés 3SG.DAT sent.3SG each painting DAT a museum different
'Andrés sent a museum each painting.'

I consider the ditransitive constructions in (71a) and (72a) to have the vP component in (73a) and, following Cuervo, the double-object constructions in (71b) and (72b) to have that in (73b).

- (73) a. Ditransitive construction
-
- ```

graph TD
 vP --> DP1[DP]
 vP --> v_prime[v']
 DP1 --> Direct[Direct object]
 v_prime --> v[v]
 v_prime --> PP[PP]
 PP --> P[P]
 P --> a[a]
 PP --> DP2[DP]
 DP2 --> Indirect[Indirect object]

```
- b. Double-object construction
- 
- ```

graph TD
    vP --> v[v]
    vP --> ApplP[ApplP]
    ApplP --> a_plus_DP[a + DP]
    a_plus_DP --> Indirect[Indirect object]
    ApplP --> Appl_prime[Appl']
    Appl_prime --> Appl[Appl]
    Appl --> le[le]
    Appl_prime --> DP[DP]
    DP --> Direct[Direct object]
  
```

In (73a) the direct object c-commands the indirect object, and each argument is introduced by a separate head. In (73b), the indirect object c-commands the direct object, and both arguments are introduced by an applicative head (Appl) that assigns dative case to its specifier. This head then raises to become part of the verbal complex, where it is spelled out as a dative clitic, and the direct object raises to the specifier of vP, the same position that it occupies in (73a). The essential point

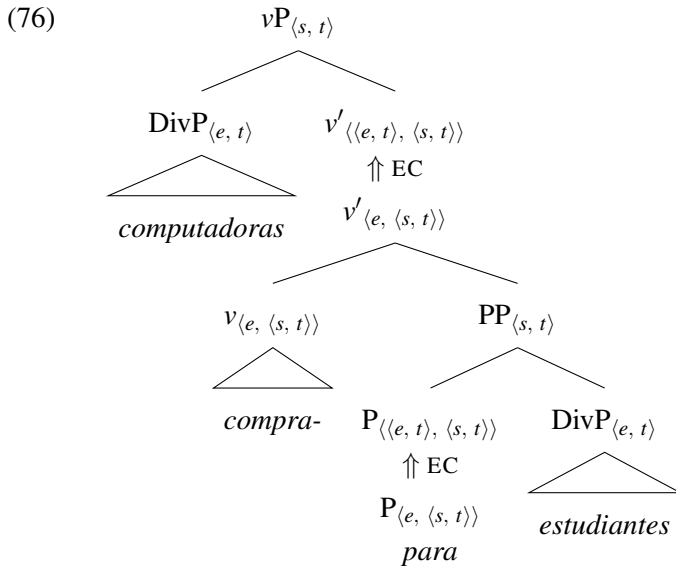
for the discussion of bare nouns is that Cuervo, like Demonte, analyzes *a* as a marker of dative case in double-object constructions.¹¹ If *a* is not a legitimate preposition in these structures, there is no reason to expect that it could license a bare argument.

In event semantics, prepositions are of the same semantic type as verbs, $\langle e, \langle s, t \rangle \rangle$, and thus are eligible for the existential closure operation that I proposed in (48), repeated in (74).

$$(74) \quad EC := \lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda Q_{\langle e, t \rangle} . \lambda e_s . \exists x_e [Q(x) \wedge P(x)(e)]$$

To illustrate how this mechanism works with both direct and indirect objects, I use Masullo's (1996: 186) ditransitive example in (75), whose *vP* has the interpretable structure in (76).

- (75) Compraron computadoras para estudiantes.
 bought.3PL computers for students
 'They bought computers for students.'



The denotation of this structure can be computed in five steps. First, EC raises the type of the preposition *para* ‘for’ in (77).

$$\begin{aligned}
 (77) \quad EC(\llbracket para \rrbracket) &= [\lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda Q_{\langle e, t \rangle} . \lambda e_s . \exists x_e [Q(x) \wedge P(x)(e)]] (\lambda y_e . \lambda e'_s . For(y)(e')) && \text{LE} \\
 &= \lambda Q_{\langle e, t \rangle} . \lambda e_s . \exists x_e [Q(x) \wedge For(x)(e)] && \text{BR}
 \end{aligned}$$

Second, the type-shifted preposition applies to the DivP *estudiantes* ‘students’ in (78).

$$\begin{aligned}
 (78) \quad \llbracket \uparrow para \rrbracket (\llbracket estudiantes \rrbracket) &= [\lambda Q_{\langle e, t \rangle} . \lambda e_s . \exists x_e [Q(x) \wedge For(x)(e)]] (\lambda y_e . *Student(y)) && \text{LE} \\
 &= \lambda e_s . \exists x_e [*Student(x) \wedge For(x)(e)] && \text{BR}
 \end{aligned}$$

¹¹ Similarly, Kempchinsky (1992: 699) uses the term “dative *a*”, and Masullo (1992: 24) states that *a* is not an “overt content preposition” when a dative clitic is present.

Third, the verb stem *compra-* ‘buy’ and the PP *para estudiantes* ‘for students’ combine through event identification in (79).

$$(79) \quad \llbracket \text{compra- para estudiantes} \rrbracket = \lambda y_e . \lambda e_s . [\text{Buy}(y)(e) \wedge \exists x_e [*Student(x) \wedge \text{For}(x)(e)]]$$

Fourth, EC lifts the type of the predicate *compra- para estudiantes* ‘buy for students’ in (80).

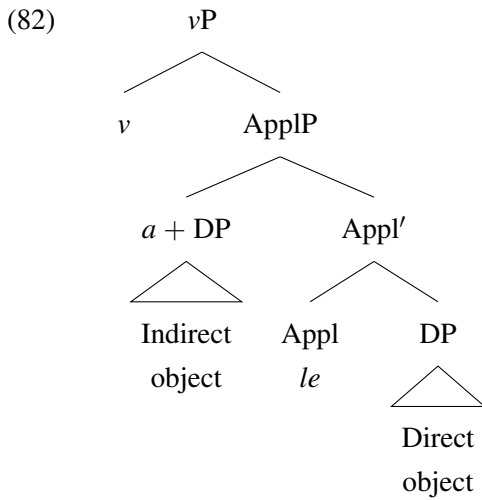
$$(80) \quad \begin{aligned} & \text{EC}(\llbracket \text{compra- para estudiantes} \rrbracket) \\ &= [\lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge P(z)(e')]] (\lambda y_e . \lambda e_s . [\text{Buy}(y)(e) \wedge \exists x_e [*Student(x) \wedge \text{For}(x)(e)]]]) \quad \text{LE} \\ &= \lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge [\text{Buy}(z)(e') \wedge \exists x_e [*Student(x) \wedge \text{For}(x)(e')]]] \quad \text{BR} \end{aligned}$$

Fifth, the type-shifted predicate applies to the DivP *computadoras* ‘computers’ in (81).

$$(81) \quad \begin{aligned} & \llbracket \uparrow \text{compra- para estudiantes} \rrbracket (\llbracket \text{computadoras} \rrbracket) \\ &= [\lambda Q_{\langle e, t \rangle} . \lambda e'_s . \exists z_e [Q(z) \wedge [\text{Buy}(z)(e') \wedge \exists x_e [*Student(x) \wedge \text{For}(x)(e')]]]] (\lambda y_e . *Computer(y)) \quad \text{LE} \\ &= \lambda e'_s . \exists z_e [*Computer(z) \wedge [\text{Buy}(z)(e') \wedge \exists x_e [*Student(x) \wedge \text{For}(x)(e')]]] \quad \text{BR} \end{aligned}$$

The resulting vP denotes a function that maps every event e' to the truth value 1 if and only if there is an object z that consists of computers, z is bought in e' , there is an object x that consists of students, and e' is for x . Voice would then merge to introduce the external argument, and $T_{[PAST]}$ would existentially bind e' , anchoring the event in the past.

If double-object constructions in Spanish have the structure that Cuervo (2003a,b, 2008, 2010a,b) proposes in (73b), repeated in (82), one might ask why they permit bare direct objects but not bare indirect objects, as indicated in (83).

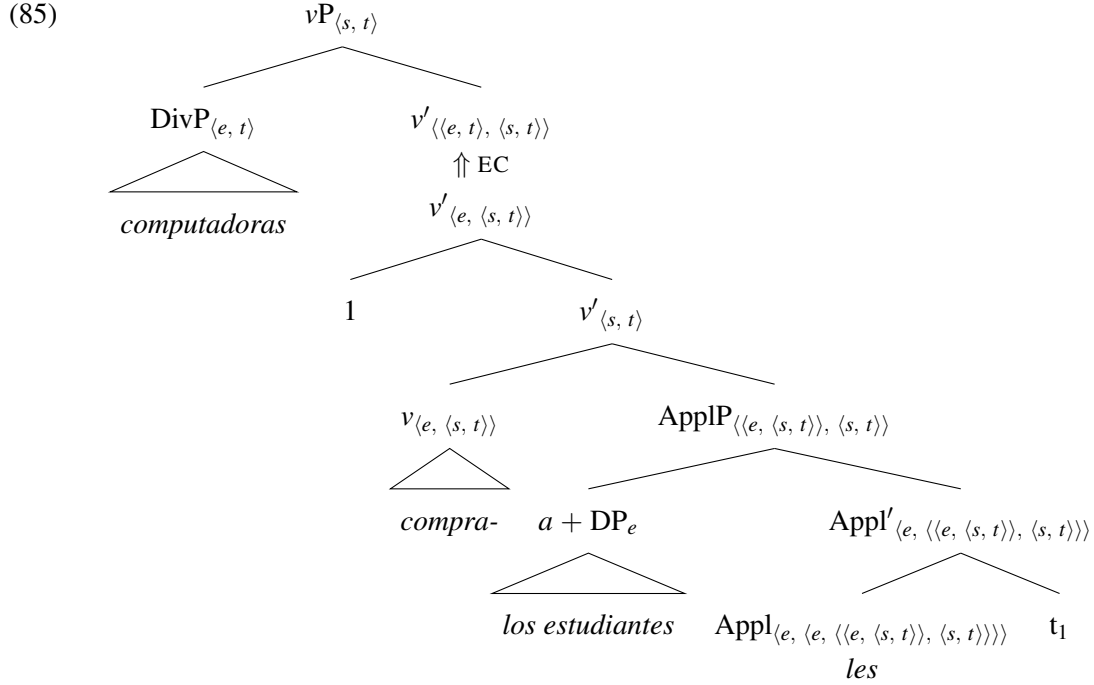


- (83) Les compraron computadoras a *(los) estudiantes.
 3PL.DAT bought.3PL computers DAT the students
 ‘They bought computers for the students.’

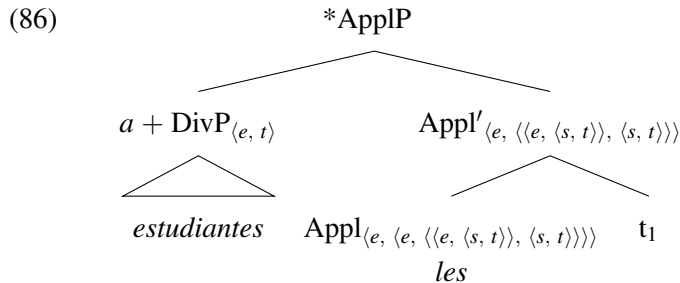
Pylkkänen (2008: 18) assigns recipient applicatives the semantic value in (84), according to which they are ternary predicates of type $\langle e, \langle e, \langle \langle e, \langle s, t \rangle \rangle, \langle s, t \rangle \rangle \rangle \rangle$ that first apply to the direct object, then to the indirect object, and finally to the verb itself.

$$(84) \quad \lambda x_e . \lambda y_e . \lambda P_{\langle e, \langle s, t \rangle \rangle} . \lambda e_s . P(e, x) \wedge \text{THEME}(e, x) \wedge \text{TO-THE-POSSESSION}(x, y)$$

Consequently, the vP of the sentence in (83) has the interpretable structure in (85), where the direct object occupies the specifier of vP and has its variable bound through existential closure.



From this configuration, it is evident why the indirect object cannot be a bare noun: Appl' is not of the appropriate semantic type to be targeted by existential closure, so it cannot combine with the property-denoting expression *estudiantes* 'students' in (86).



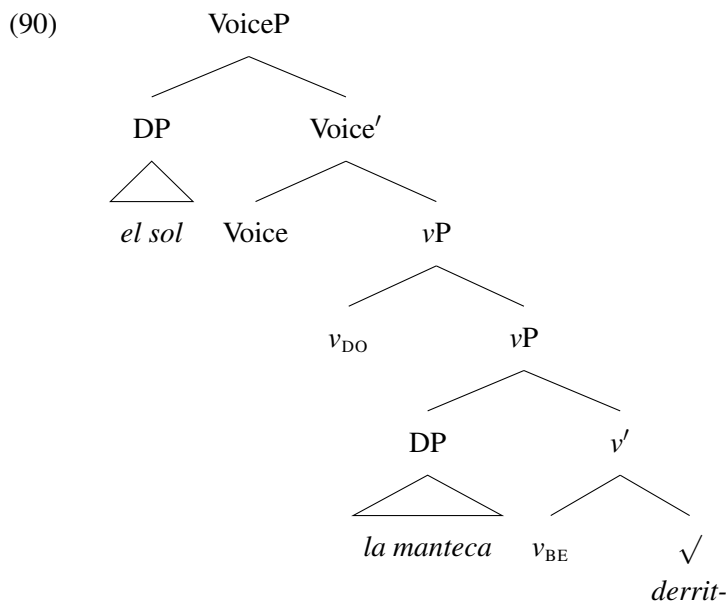
In conclusion, double-object constructions in Spanish do not permit bare indirect objects because *a* is a marker of dative case, not a genuine preposition, and dative clitics are not eligible to have their type raised through existential closure.

2.5 Restrictions on bare postverbal nouns

In this section, I discuss several constructions where bare postverbal nouns are disallowed in Spanish and how to best account for these facts. Four of the relevant environments are illustrated in (87) through (89), which include small clauses, causative and inchoative constructions, and psychological predicates (Cuervo 2003a: 114–115, 166–167).¹²

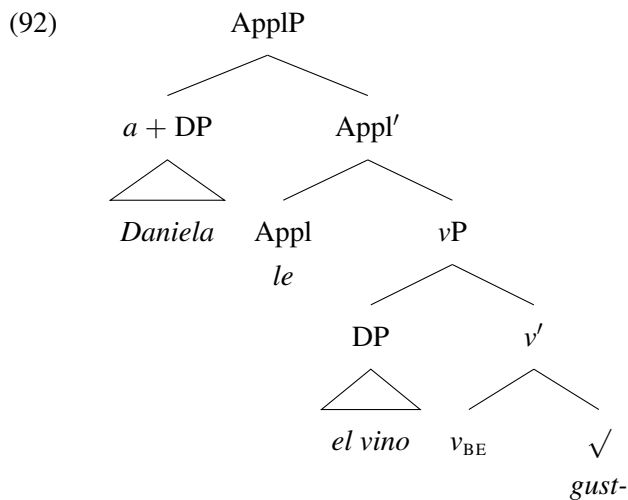
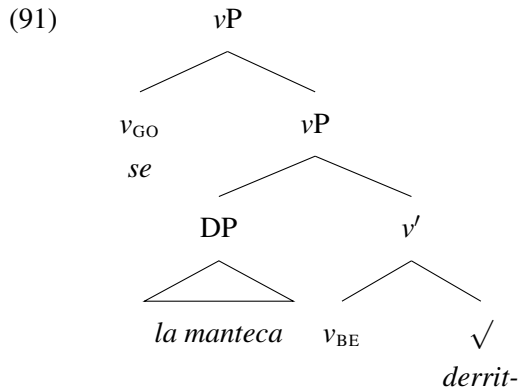
- (87) Tu amiga consideraba interesantes *(las) películas.
 your friend considered.IMPF interesting the movies
 ‘Your friend used to consider (the) movies to be interesting.’
- (88) a. El sol derretió *(la) manteca.
 the sun melted.3SG the butter
 ‘The sun melted the butter.’
 b. Se derretió *(la) manteca.
 3 melted.3SG the butter
 ‘The butter melted.’
- (89) A Daniela le gusta *(el) vino.
 DAT Daniela 3SG.DAT pleases the wine
 ‘Daniela likes wine.’

For Cuervo (2003a, 2008, 2010b, 2014, 2015), the reason that these postverbal nouns must all be introduced by a determiner is that they occupy “inner” subject positions. According to the structure that Cuervo (2003a: 112) proposes for causatives, for instance, the sentence in (88a) has the VoiceP in (90), where *la manteca* ‘the butter’ is the subject of the lower vP.



¹²The example in (88b) is by Masullo (1992: 272), and many more authors have identified the need for a determiner in these contexts (Contreras 1986, Lois 1987, Laca 1990, Bosque 1996, Contreras 1996, Laca 1996, Masullo 1996, Laca 1999, Dobrovie-Sorin & Laca 2003, Bleam 2005, Dobrovie-Sorin et al. 2005, Dobrovie-Sorin 2009).

Since Cuervo (2003a: 133, 165) considers inchoatives and psychological predicates that involve a dative argument to have the same lower *vP* as causatives, *la manteca* in (88b) and *el vino* ‘the wine’ in (89) are both inner subjects as well, as represented in (91) and (92), respectively.



Based on these structures, Cuervo (2003a: 116) argues that Suñer's (1982: 209) Naked Noun Constraint in (93) can be made even more general and proposes the amended version in (94).

(93) Naked Noun Constraint

An unmodified common noun in preverbal position cannot be the surface subject of a sentence under conditions of normal stress and intonation.

(94) The Naked Noun Constraint Revised

An unmodified common noun cannot be the subject of a predicate under conditions of normal stress and intonation.

I do not find a uniform analysis viable for both empirical and theoretical reasons and instead claim that the need for a determiner stems from distinct semantic factors in each case.

I begin with psychological predicates because the restriction on bare nouns is not limited to dative constructions, as shown in (95) with the verbs *amar* ‘to love’ and *odiar* ‘to hate’.

- (95) a. Julia ama *(los) gatos.
 Julia loves the cats
 ‘Julia loves cats.’
- b. Jacques odia *(los) gatos.
 Jacques hates the cats
 ‘Jacques hates cats.’

In these sentences, *los gatos* ‘the cats’ is a direct object, not an inner subject, yet the definite article remains obligatory. Given that bare nouns can only be interpreted existentially in Spanish and that the intended reading of the examples in (95) is generic rather than existential, it is unsurprising that this determiner cannot be omitted. I argue that the dative construction in (96) disallows a bare postverbal noun for this same reason, not because *los gatos* is an inner subject.

- (96) A Julia le encantan *(los) gatos.
 DAT Julia 3SG.DAT enchant.3PL the cats
 ‘Julia loves cats.’

In support of this point, the definite article cannot be replaced by the indefinite article, which lexicalizes the existential operator in Spanish.

- (97) a. #Julia ama unos gatos.
 Julia loves some cats
 Intended: ‘Julia loves some cats.’
- b. #A Julia le encantan unos gatos.
 DAT Julia 3SG.DAT enchant.3PL some cats
 Intended: ‘Julia loves some cats.’

It is, nevertheless, possible to use an indefinite determiner like *algunos* ‘some’ or *ciertos* ‘certain’, as in (98), which convey specificity and enable two readings, one where there are particular cats that Julia loves and another where there are particular subkinds of cats that Julia loves.

- (98) a. Julia ama algunos gatos.
 Julia loves some cats
 ‘Julia loves some cats (Geoffrey, Minette, le Petit Prince).’
 ‘Julia loves some cats (Birmans, Chartreux, Ragdolls).’
- b. A Julia le encantan ciertos gatos.
 DAT Julia 3SG.DAT enchant.3PL certain cats
 ‘Julia loves certain cats (Geoffrey, Minette, le Petit Prince).’
 ‘Julia loves certain cats (Birmans, Chartreux, Ragdolls).’

In short, psychological predicates cannot license bare nouns because there is a fundamental incompatibility between these predicates and existential closure.

Another potential challenge for the view that inner subjects require a determiner is found with verbs like *quemar(se)* ‘to burn’ that participate in the causative-inchoative alternation. These verbs

do, in fact, permit bare postverbal nouns as long as the construction in question does not entail a final result, as revealed by the imperfect versus preterite versions of the sentence in (99).¹³

- (99) a. Se quemaban casas en toda la región.
 3 burned.IMPF.3PL houses in all the region
 ‘Houses were burning all over the region.’
 b. #Se quemaron casas en toda la región.
 3 burned.3PL houses in all the region
 Literal: ‘Houses burned all over the region.’

Tense is but one factor that plays a role in whether a final result has been attained. The four-way contrast in (100) indicates that agentivity is relevant as well (Cuervo 2003a, 2008, 2015).

- (100) a. El incendiario quemó las casas.
 the arsonist burned.3SG the houses
 ‘The arsonist burned the houses.’ (Activity or causative)
 b. El incendiario quemó casas.
 the arsonist burned.3SG houses
 ‘The arsonist burned houses.’ (Activity)
 c. El incendio quemó las casas.
 the fire burned.3SG the houses
 ‘The fire burned the houses.’ (Causative)
 d. #El incendio quemó casas.
 the fire burned.3SG houses
 Literal: ‘The fire burned houses.’ (Causative)

The use of a nonagentive subject in (100c) and (100d) rules out the possibility of an activity reading and confirms that the causative guise of *quemar*, like the inchoative guise of *quemarse* in (99b), disallows a bare postverbal noun. This point is underscored by the fact that the sentence in (100b) becomes ungrammatical if one adds a modifier like *en un instante* ‘in an instant’, as in (101), that blocks the activity interpretation and imposes the causative interpretation.

- (101) #El incendiario quemó casas en un instante.
 the arsonist burned.3SG houses in an instant
 Literal: ‘The arsonist burned houses in an instant.’ (Causative)

One might ask how transitive *quemar* behaves in the imperfect tense, and the data in (102) suggest that the causative reading is far more difficult, if not impossible, to access.

- (102) a. El incendiario quemaba las casas.
 the arsonist burned.IMPF the houses
 ‘The arsonist was burning the houses.’ (Activity)

¹³The example in (99b) is felicitous if the intended reading is passive (‘houses were burned all over the region’), which is orthogonal to the discussion.

- b. El incendiario quemaba casas.
 the arsonist burned.IMPF houses
 'The arsonist was burning houses.' (Activity)

Since activities demand agentive subjects, the expectation is that *el incendiario* 'the arsonist' cannot be replaced by *el incendio* 'the fire' even in the first example, a prediction that is borne out in (103).

- (103) a. #El incendio quemaba las casas.
 the fire burned.IMPF the houses
 Literal: 'The fire was burning the houses.' (Activity)
- b. #El incendio quemaba casas.
 the fire burned.IMPF houses
 Literal: 'The fire was burning houses.' (Activity)

The generalization that emerges from the data in (99) through (103) is that nouns must be introduced by a determiner in constructions that entail a final result, and to test for this entailment, one must take both tense and agentivity into account.

Lastly, small clauses are but one of many environments that involve adjectival elements and prohibit bare nouns. For instance, the copular construction in (104), the absolute construction in (105), the periphrastic passive construction in (106), and the resultative construction in (107) all require the noun to be preceded by a determiner (Suñer 1982, Contreras 1986, Lois 1987, Masullo 1992, Contreras 1996, Laca 1996, Masullo 1996, Laca 1999).

- (104) Aquí son mansas *(las) ardillas.
 here are.3PL tame the squirrels
 'The squirrels are tame here.'
- (105) Caídas *(las) hojas, ya no se veía el camino.
 fallen the leaves no longer 3 saw.IMPF the path
 Approximate: 'The leaves having fallen, the path could no longer be seen.'
- (106) Fueron vendidas *(las) flores.
 were.3PL sold the flowers
 'The flowers were sold.'
- (107) Ya tienen escritas *(las) novelas.
 already have.3PL written the novels
 'They already have the novels written.'

The sentences in (105) through (107) contain participles that agree in gender and number with the adjacent noun, unlike their counterparts in (108) through (110), and it is telling that the determiner can be omitted in the absence of agreement.

- (108) Habiendo caído (las) hojas, ya no se veía el camino.
 having fallen the leaves no longer 3 saw.IMPF the path
 Approximate: '(The) leaves having fallen, the path could no longer be seen.'

- (109) Se han vendido (las) flores.
 3 have.3PL sold the flowers
 Approximate: ‘(The) flowers have been sold.’
- (110) Ya han escrito (las) novelas.
 already have.3PL written the novels
 ‘They have already written (the) novels.’

In other words, verbal participles can license bare nouns, but adjectival participles cannot, a difference that could be attributed to their semantic types. If verbal participles are of type $\langle e, \langle s, t \rangle \rangle$, like verbs, and adjectival participles are of type $\langle e, t \rangle$, like adjectives, it follows that only verbal participles can have their type raised by the rule of existential closure that I defined in (48) above. This hypothesis covers more empirical ground than one based on inner subjects, but it must be evaluated by investigating each of the constructions in (104) through (110) separately, a task that is beyond the scope of this chapter.

In summary, the restrictions on bare postverbal nouns that I have addressed in this section are not problematic for the view of existential closure that I developed in section 2.3, but they do seem to evade a uniform syntactic analysis. Psychological predicates have generic rather than existential readings, so there is no reason to expect that they would license bare nouns through existential closure. Causative and inchoative constructions entail a final result, and this interpretation requires a determiner. Finally, adjectives and adjectival participles are not of the correct semantic type to be targeted by existential closure and therefore cannot license bare nouns in small clauses or copular, absolute, periphrastic passive, and resultative constructions.

2.6 Conclusion

In this chapter, I presented the general distribution of bare mass and plural nouns in Spanish and explored the question of whether they contain an unpronounced determiner layer, such as DP or QP. I maintained that there is neither empirical nor theoretical support for this approach and showed that the only means of translating a phonologically null D or Q to logical form in a way that captures the semantic behaviour of bare nouns in Spanish would be to represent these heads as vacuous identity functions. In response to these concerns, I eliminated covert determiners altogether and proposed a last-resort operation of existential closure that enables verbs and prepositions, both of which are of type $\langle e, \langle s, t \rangle \rangle$, to combine with property-denoting expressions that they c-command. Next, I demonstrated that the impossibility of clitic doubling with bare indirect objects is predicted by Demonte (1995) and Cuervo’s (2003a,b, 2008, 2010a,b) view that *a* is a marker of dative case in double-object constructions. I then considered a variety of environments that disallow bare postverbal nouns and concluded that determiners are needed for disparate semantic reasons.

One issue that has yet to be explained is why nouns that are typically described as singular cannot appear in the same contexts that mass and plural nouns can. In (111), for example, *nieve* ‘snow’ and *gatos* ‘cats’ are felicitous in object position, but *gato* ‘cat’ is not.

- (111) a. Jacques vio nieve afuera.
 Jacques saw.3SG snow outside
 ‘Jacques saw snow outside.’
- b. Jacques vio gatos afuera.
 Jacques saw.3SG cats outside
 ‘Jacques saw cats outside.’
- c. #Jacques vio gato afuera.
 Jacques saw.3SG cat outside
 Literal: ‘Jacques saw cat outside.’

In chapter 3, I argue that singularity is derived through additional structure above DivP and that this phenomenon plays out differently in indefinite and definite expressions.

Chapter 3

The mass-count distinction

In chapter 2, I rejected the claim that bare mass nouns and plural nouns are DPs in Spanish, and in this chapter, I model these arguments as *n*Ps and DivPs, respectively.¹ Like Borer (2005), I pursue an analysis where countability is a structural rather than lexical notion, but I employ a different theoretical framework and explicitly identify the semantic contribution of the relevant functional projections. I begin by engaging with two debates: first, whether *n*Ps denote properties of objects or properties of kinds in Spanish, and second, whether DivPs include or exclude atomic objects in their extension. Next, I consider the role that indefinite and definite determiners play in interpretation and observe that only indefinite determiners display sensitivity to the mass-count distinction. I then demonstrate that exclusively singular interpretation is contingent upon an indefinite determiner in Spanish and argue that singularity is contributed by QP instead of DivP. As for “singular” definite expressions like *el aguacate* ‘the avocado’, I show that they are ambiguous between mass and singular denotation and attribute this phenomenon to the maximality operator that is encoded by definite determiners. Lastly, I address the number neutrality of bare nonplural objects in certain constructions, such as *tengo gato* ‘I have a cat/cats’, along with other cases of form-meaning mismatches that relate to number.

In section 3.1, I develop a system where *n*Ps denote properties of unindividuated objects and DivPs denote properties of individuated objects, both of which can be represented as complete join semilattices. In section 3.2, I propose that singularity is introduced at the level of QP by a select few indefinite determiners in Spanish, such as *cada* ‘every’, *un* ‘a’, and *ningún* ‘no’. In section 3.3, I posit that the ambiguous interpretation of *el aguacate* ‘the avocado’ and other nonplural definite expressions is due to the lack of DivP in their structure and the maximality operator in D. In section 3.4, I discuss some of the environments where bare nonplural objects have number-neutral readings and illustrate how Espinal & McNally’s (2011) account of these constructions can be implemented in the view of argument structure that I adopted in chapter 2. In section 3.5, I propose morphosyntactic and interpretable structures for lexical plurals and pluralia tantum and explore other types of semantically unpredictable nouns, including group nouns and aggregate nouns. In section 3.6, I summarize the major contributions of this chapter and conclude.

¹As mentioned in section 2.1, the terms “mass noun”, “count noun”, “plural noun”, and “singular noun” describe instances of nominalized roots whose interpretation is conditioned by the syntactic context in which they appear.

3.1 The semantics of *nP* and *DivP*

Bare nouns cannot refer to kinds in Spanish, a constraint that I first discussed in section 1.1 and considered at length in section 2.2. As shown with the kind-selecting predicate *extinguirse* ‘to become extinct’ in (1), the mass noun *arroz* ‘rice’ and the plural noun *papas* ‘potatoes’ must be introduced by the definite article even when they occur postverbally (Laca 1990, 1996, 1999; Dobrovie-Sorin & Laca 2003; McNally 2004; Bleam 2005; Dobrovie-Sorin et al. 2005; Dobrovie-Sorin 2009; Borik & Espinal 2012; Laca 2013; Borik & Espinal 2015).

- (1) a. En el futuro podría extinguirse *(el) arroz.
 in the future could extinguish.3 the rice
 ‘Rice could become extinct in the future.’
 b. En el futuro podrían extinguirse *(las) papas.
 in the future could.3PL extinguish.3 the potatoes
 ‘Potatoes could become extinct in the future.’

It must therefore be the case that nouns are property-denoting, or type $\langle e, t \rangle$, expressions in Spanish, and in this section, I endorse the view that *nPs* denote properties of objects. After presenting my analysis of mass nouns and plural nouns, I address another possibility, which is that *nPs* denote properties of kinds (Espinal 2010; Espinal & McNally 2011; Borik & Espinal 2012, 2015). I find this account to be untenable on both empirical and theoretical grounds because it takes mass nouns to have the same structural representation as singular nouns.

Ever since Carlson’s (1977, 1980) work on bare plural nouns in English, the domain of entities, or individuals, is widely regarded to consist of objects and kinds, neither of which are necessarily atomic. For this reason, the claim that *nPs* denote properties of objects must not be conflated with the much stronger claim that *nPs* denote properties of atomic objects. For example, the sentence in (2) contains the bare *nP* *arroz* ‘rice’ and does not convey any sense of atomicity.

- (2) Julia comió arroz.
 Julia ate.3SG rice
 ‘Julia ate rice.’

Distributed Morphology, the theoretical framework that I assume in this thesis, rejects the traditional notion of a lexicon, meaning that there cannot be preexisting lists of mass nouns or count nouns in a given language. Instead, this opposition must be derived through pieces of structure, features, or some combination of the two.

In the approach to the mass-count distinction that I develop in this chapter, I adopt Borer’s (2005) proposal that count interpretation stems from a division phrase (*DivP*), which is realized by plural morphology in languages like English and Spanish.² Based on the examples in (3), Borer

²There is a large body of crosslinguistic research that explores the connection between number marking and countability (Greenberg 1972; Sanches & Slobin 1973; Sharvy 1978; Allan 1980; Doetjes 1996; Chierchia 1998a,b; Bouchard 2002; Déprez 2005; Cowper & Hall 2009; Chierchia 2010; Borik & Espinal 2012; Cowper & Hall 2012; Mathieu 2012; Pelletier 2012; Mathieu 2014; Borik & Espinal 2015; Alexiadou 2019; Borik & Espinal 2020; Fábregas 2024).

(2005: 120) claims that $DivP$ does not establish atoms because these sentences do not necessarily express that a single apple was entirely eaten, a single house was entirely built, or a single circle was entirely drawn.

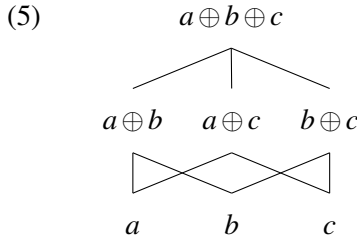
- (3) a. Kim ate apples this afternoon (for an hour).
 b. Pat built houses (all summer).
 c. My kid sister drew circles (all morning).

There is a confounding factor, however, which is the activity reading that is made salient by the parenthetical material. As Borer (2005: 129) acknowledges in a footnote, the telic predicate in (4) does not readily permit an interpretation where Jane did not complete a single house.

- (4) Jane built houses in a month.

In the absence of conclusive evidence to suggest otherwise, I maintain that $DivP$ is indeed responsible for atomicity and now formalize this view.

Following Link (1983), the extension of a mass or plural noun can be modelled as a complete join semilattice, as in (5), and following Dali & Mathieu (2020, 2021), it is $DivP$ that determines whether the semilattice includes an atomic layer.



In a “mass” semilattice, a , b , and c have proper parts, whereas in a “plural” semilattice, they are atomic. To derive this contrast, I propose that $DivP$ conditions the semantics of nP in a process that can be construed as a type of contextual allosemy (Anagnostopoulou & Samioti 2013, Marantz 2013, Anagnostopoulou & Samioti 2014, Harley 2014, Wood 2015, Myler 2016, Wood & Marantz 2017, Harðarson 2021, Wood 2023). Assuming Champollion & Krifka’s (2016: 377) definition of an atomic object in (6), nPs have the count interpretation in (7a) in structures that contain $DivP$ and the elsewhere mass interpretation in (7b) in structures that lack $DivP$.

$$(6) \quad \text{ATOM}(x) := \neg \exists y [y < x]$$

- (7) a. $\llbracket nP \rrbracket = \lambda x_e : P(x) \wedge \text{ATOM}(x) . P(x) / \text{--- } DivP$
 b. $\llbracket nP \rrbracket = \lambda x_e . P(x)$

In both cases, P is supplied by the nominalized root. Div itself has the lexical entry in (8a), where Link’s star operator ($*$) represents the algebraic closure of P , and since this head only combines with nPs that have the denotation in (7a), $DivPs$ have the denotation in (8b).

- (8) a. $\llbracket \text{Div} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e . *P(x)$
 b. $\llbracket \text{DivP} \rrbracket = \lambda x_e . *P(x)$, defined only if $P(x)$ and $\text{ATOM}(x)$

To use the plural guise of the semilattice in (5) for the purposes of illustration, the extension of an *nP* that is dominated by *DivP* would be the set of atoms $\{a, b, c\}$, and the extension of this *DivP* would be the set of atoms and their sums $\{a, b, c, a \oplus b, a \oplus c, b \oplus c, a \oplus b \oplus c\}$.

The noun *soga* ‘rope’ is known for its flexibility in terms of the mass-count distinction, as shown in (9), and is well suited to demonstrate how the system works.

- (9) a. Trajeron sogā.
 brought.3PL rope
 ‘They brought rope.’
 b. Trajeron sogas.
 brought.3PL ropes
 ‘They brought ropes.’

In the first sentence, *soga* is a bare *nP* that has the interpretable structure and semantic value in (10), which corresponds to a semilattice that lacks an atomic layer.³

- (10) a. $nP_{\langle e, t \rangle}$
 $\swarrow \quad \searrow$
 $n \quad \sqrt{2}$
 $[F] \quad sog-$
 $-a$
 b. $\llbracket \text{soga} \rrbracket = \lambda x_e . \text{Rope}(x)$

In the second sentence, *sogas* ‘ropes’ is a bare *DivP* that has the interpretable structure and semantic value in (11), which corresponds to a semilattice that contains an atomic layer.

- (11) a. $\text{DivP}_{\langle e, t \rangle}$
 $\swarrow \quad \searrow$
 $\text{Div}_{\langle \langle e, t \rangle, \langle e, t \rangle \rangle} \quad nP_{\langle e, t \rangle}$
 $[PL] \quad \swarrow \quad \searrow$
 $-s \quad n \quad \sqrt{3}$
 $\quad [F] \quad sog-$
 $\quad -a$
 b. $\llbracket \text{sogas} \rrbracket = \lambda x_e . * \text{Rope}(x)$, defined only if $\text{Rope}(x)$ and $\text{ATOM}(x)$

³See sections 1.4 and 5.2 for the view that roots do not have a semantic type independently of their categorizer and that gender features on *n* are visible but ignored at logical form. Here and throughout the thesis, I assume that heads are interpreted in situ.

I include a [PL] feature on Div in (11a) for two reasons. First, in section 3.2, I argue that [PL] is the target of a postsyntactic rule of impoverishment in singular constructions. Second, in section 3.5, I differentiate between [PL] on Div, where this feature has consequences for interpretation, and [PL] on *n*, where it does not, to account for canonical plurals like *sogas* versus lexical plurals like *alicates* ‘pliers’ and pluralia tantum like *bártulos* ‘gear’.

In the previous discussion, I have tacitly subscribed to the inclusive view of plurality, according to which the extension of a plural noun like *papas* ‘potatoes’ includes atomic ropes as well as sums of ropes (Krifka 1989, Schwarzschild 1996, Van Geenhoven 2000, Farkas & de Swart 2003, Krifka 2003, Sauerland 2003, Sauerland et al. 2005, Spector 2007, Zweig 2009, Chierchia 2010, Farkas & de Swart 2010, Grimm 2013, Mathieu 2014). There is empirical motivation for this characterization of plurality in downward-entailing contexts, where a given expression can be replaced by a more specific one without fundamentally changing the interpretation. For instance, *papas* ‘potatoes’ can be replaced by *una o más papas* ‘one or more potatoes’ in the negative, interrogative, and modal environments in (12).

- (12) a. No comí {papas / una o más papas}.
not ate.1SG potatoes one or more potatoes
‘I did not eat potatoes/one or more potatoes.’
- b. ¿Compraste {papas / una o más papas}?
bought.2SG potatoes one or more potatoes
‘Did you buy potatoes/one or more potatoes?’
- c. Jacques necesita {papas / una o más papas}.
Jacques needs potatoes one or more potatoes
‘Jacques needs potatoes/one or more potatoes.’

As for sentences like *comí papas* ‘I ate potatoes’ where a one-or-more reading of *papas* is unavailable, it has been argued that pragmatic factors are at play (Krifka 1989, Sauerland 2003, Sauerland et al. 2005, Spector 2007, Zweig 2009). Since this phenomenon is not particular to any one language, I do not attempt to resolve it in this chapter.

Recently, some authors have claimed that nouns in Spanish denote properties of kinds rather than properties of objects, a possibility that is illustrated in (13) with *arroz* ‘rice’ (Espinal 2010; Espinal & McNally 2011; Borik & Espinal 2012, 2015).⁴

$$(13) \quad \llbracket \text{arroz} \rrbracket = \lambda x_e^k. \text{Rice}(x^k)$$

That is, *arroz* denotes a function that maps every kind x^k to the truth value 1 if and only if x^k is rice. One conceptual issue with this approach is that kinds are typically considered to be unique entities, like the referents of proper names (Langford 1949; Carlson 1977, 1980; Heyer 1985; Krifka 1995; Krifka et al. 1995; Chierchia 1998a,b; Doron 2003). From this perspective, it seems that the extension of any noun would be a singleton set.

⁴The existence of expressions that denote properties of kinds has been raised in work on other languages as well (Lumsden 1988; Krifka 1995; Dayal 2004; McNally & Boleda 2004; Dobrovie-Sorin & Pires de Oliveira 2008, 2010).

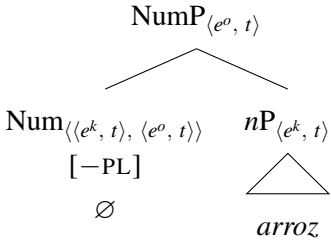
An even more pressing concern is how to derive the existential readings of bare mass nouns, such as *arroz* ‘rice’ in (2) above, repeated in (14), if *nPs* denote properties of kinds.

- (14) Julia comió arroz.
 Julia ate.3SG rice
 ‘Julia ate rice.’

Borik & Espinal (2015) recognize this challenge and suggest that such occurrences of mass nouns contain a phonologically null NumP layer that is valued as [−PL], the same as in singular nouns. According to the lexical entry that these authors propose for Num_[−PL] in (15), this head takes an expression that denotes a property of kinds and returns a function that maps every object x^o to the truth value 1 if and only if there is a kind x^k that has property P , x^o is a realization of x^k , and x^o is atomic (Borik & Espinal 2015: 188).

$$(15) \quad \llbracket \text{Num}_{[-\text{PL}]} \rrbracket = \lambda P_{\langle e^k, t \rangle} . \lambda x_e^o . \exists x_e^k [P(x^k) \wedge R(x^o, x^k) \wedge x^o \in \text{At}]$$

In (16), I provide the interpretable structure and semantic value of *arroz* as a singular NumP, where this expression denotes a function that maps every object x^o to the truth value 1 if and only if there is a kind x^k that is rice, x^o is a realization of x^k , and x^o is atomic.

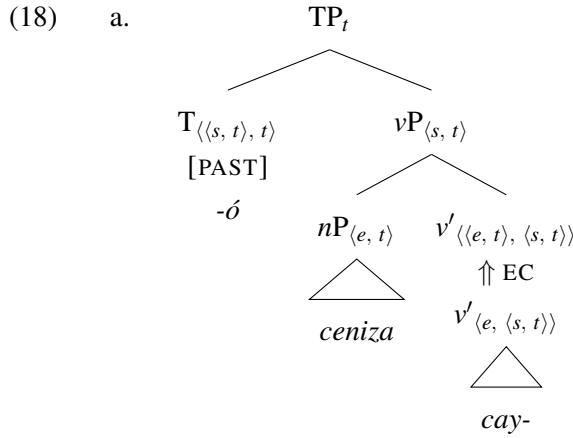
- (16) a. 
- b. $\llbracket \text{arroz} \rrbracket = \lambda x_e^o . \exists x_e^k [\text{Rice}(x^k) \wedge R(x^o, x^k) \wedge x^o \in \text{At}]$

The problem is that bare mass nouns do not have atomic readings, and the lack of empirical support for NumP in their structure jeopardizes the claim that nouns denote properties of kinds in Spanish.

I end this section by demonstrating how my analysis captures the interpretation of bare *nPs* and *DivPs* in the clause, such as *ceniza* ‘ash’ in (17a) and *cenizas* ‘ashes’ in (17b).

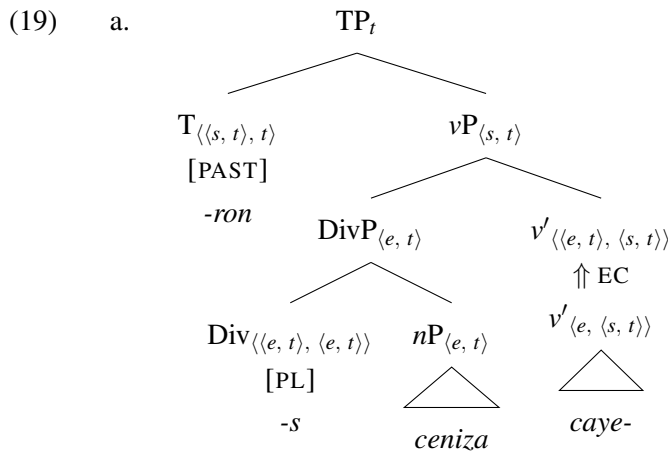
- (17) a. Cayó ceniza.
 fell.3SG ash
 ‘Ash fell.’
- b. Cayeron cenizas.
 fell.3PL ashes
 ‘Ashes fell.’

In section 2.3, I argued that the variables that bare nouns introduce at logical form are bound through a last-resort operation of existential closure (EC). For instance, *cayó ceniza* ‘ash fell’ in (17a) has the interpretable structure and truth conditions in (18).



- b. $\llbracket \text{cayó ceniza} \rrbracket = 1 \text{ iff } \exists e_s [\text{Past}(e) \wedge \exists x_e [\text{Ash}(x) \wedge \text{Fall}(x)(e)]]$

That is, *cayó ceniza* ‘ash fell’ is true if and only if there is a past event e in which there was an object x that consisted of ash substance and x fell in e . Next, the sentence *cayeron cenizas* ‘ashes fell’ in (17b) has the interpretable structure and truth conditions in (19), where DivP triggers the presupposition that the ashes are individuated.



- b. $\llbracket \text{cayeron cenizas} \rrbracket = 1 \text{ iff } \exists e_s [\text{Past}(e) \wedge \exists x_e [* \text{Ash}(x) \wedge \text{Fall}(x)(e)]]$,
defined only if $\text{Ash}(x)$ and $\text{ATOM}(x)$

In this case, *cayeron cenizas* is true if and only if there is a past event e in which there was a sum of objects x that consisted of atomic ashes and x fell in e . With this system in place, where nPs and $DivPs$ both denote properties of objects, the next issue to explore is how structure above the level of $DivP$ affects the interpretation of the noun phrase.

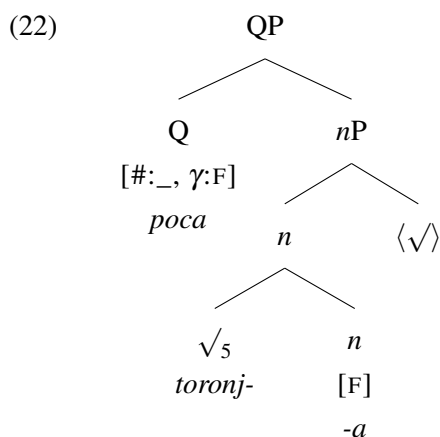
3.2 *Ē plūribus ūnum*

The objective of this section is to consider the morphosyntactic and semantic behaviour of indefinite determiners. In this endeavour, I start with two that fully inflect for gender and number: *mucho*/

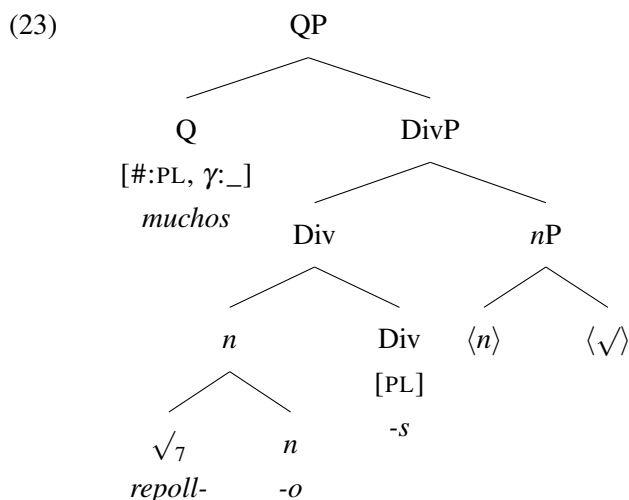
mucha/muchos/muchas ‘much, many’ and *poco/poca/pocos/pocas* ‘little, few’. When these determiners appear with nonplural nouns, as in (20), the interpretation is mass, and when they appear with plural nouns, as in (21), the interpretation is count.

- | | |
|--|--|
| (20) a. Usaron mucho repollo.
used.3PL much cabbage
‘They used a lot of cabbage.’
b. Usaron poca toronja.
used.3PL little grapefruit
‘They used little grapefruit.’ | (21) a. Usaron muchos repollos.
used.3PL many cabbages
‘They used many cabbages.’
b. Usaron pocas toronjas.
used.3PL few grapefruits
‘They used few grapefruits.’ |
|--|--|

For all determiners, I assume that feminine concord is triggered by a [F] feature that is present on a lower head and that plural concord is triggered by a [PL] feature. In the morphosyntactic structure of *poca toronja* ‘little grapefruit’ in (22), where I use γ for gender concord and # for number concord, there is no plural marking on *poca* ‘little’ in Q because # remains unvalued in the absence of $\text{Div}_{[\text{PL}]}$.



It is likewise possible for γ to remain unvalued, as in the morphosyntactic structure of *muchos repollos* ‘many cabbages’ in (23), a nominal expression that lacks a gender feature on *n*.



The configuration in (22) is representative of indefinite expressions that have mass readings, and the configuration in (23) is representative of indefinite expressions that have plural readings.

Indefinite expressions that are interpreted as singular are morphologically indistinguishable from those that are interpreted as mass, as illustrated by the interchangeability of *un/una* ‘a’ and *mucho/mucha* ‘much’ in (24).

- (24) a. Necesitan {un / mucho} repollo.
 need.3PL a much cabbage
 ‘They need a/a lot of cabbage.’
- b. Necesitan {una / mucha} toronja.
 need.3PL a much grapefruit
 ‘They need a/a lot of grapefruit.’

The indefinite article evolved from the numeral *ūnus* ‘one’ in Latin, but there are at least two reasons to suspect that, synchronically, it does not occupy the same structural position as numerals. First, as the contrast in (25) reveals, the indefinite article is mutually exclusive with the definite article, but numerals are not (Bernstein 1993; Fábregas 2018, 2024).

- (25) a. *el un repollo
 the one cabbage
 Intended: ‘the one cabbage’
- b. los dos repollos
 the two cabbages
 ‘the two cabbages’

Second, as shown in (26), the indefinite article can be used together with a numeral to convey a sense of approximation, but two numerals cannot be used together (Bello 1847, Fábregas 2024).

- (26) a. unas tres toronjas
 some three grapefruits
 ‘three or so grapefruits’
- b. *dos tres toronjas
 two three grapefruits
 Literal: ‘two three grapefruits’

In the interest of theoretical parsimony, I assign the indefinite article in Spanish to Q rather than to Numeral or another lower head, and this same element is a component of *algún/alguna/ algunos/ algunas* ‘a, some’ and *ningún/ninguna/ ningunos/ningunas* ‘no’ as well.⁵ Even a prototypical mass noun like *cerveza* ‘beer’ has a singular reading when it is preceded by either of these determiners, as in (27), an effect that can be described as the “universal packager” if *cerveza* is interpreted as a quantity of beer and as the “universal sorter” if *cerveza* is interpreted as a type of beer (Gleason 1965, Bunt 1985, Bach 1986, Jackendoff 1991, Landman 1991).

⁵For the semantic differences between *un* and *algún*, see Gutiérrez-Rexach 2001, López Palma 2007, and Martí 2008.

- (27) a. ¿Hay alguna cerveza que te guste?
 has a beer that 2SG pleases.SBJV
 ‘Is there a beer that you like?’
 b. Ninguna cerveza vale la pena.
 no beer is.worth the trouble
 ‘No beer is worth the trouble.’

There are only five other indefinite determiners that yield exclusively singular readings of nouns in Spanish, all of which follow in (28).

- (28) a. {Cada / Todo} ingrediente resultó necesario.
 every every ingredient resulted.3SG necessary
 ‘Every ingredient turned out to be necessary.’
 b. ¿{Qué / Cuál} ingrediente buscas?
 what which ingredient seek.2SG
 ‘Which ingredient are you looking for?’
 c. Cualquier ingrediente sería mejor que nada.
 whichever ingredient be.COND better than nothing
 ‘Any ingredient would be better than nothing.’

I now turn to the morphosyntactic analysis of these indefinite determiners along with their plural counterparts and then address their semantics.

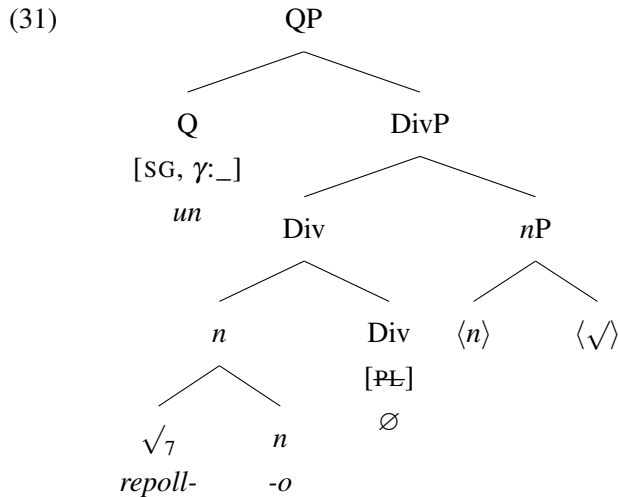
Assuming that division, or individuation, is a prerequisite for countability, it must be the case that the indefinite expressions in (29) contain DivP in their structure even if this projection is not phonologically realized as the plural suffix.

- (29) a. {un / algún / ningún / cada / todo / qué / cuál / cualquier} repollo
 a some no every every what which whichever cabbage
 ‘a/some/no/every/every/which/which/whichever cabbage’
 b. {una / alguna / ninguna / cada / toda / qué / cuál / cualquier} toronja
 a some no every every what which whichever grapefruit
 ‘a/some/no/every/every/which/which/whichever grapefruit’

In recent work on the Germanic and Romance languages, I argue that singularity is introduced at the level of QP, not DivP, and consider indefinite determiners like those in (29) to spell out Q heads that bear a [SG] feature (Jambrović 2023). I now propose that this feature has two effects in different components of the grammar: during syntactic operations, [SG] impedes plural concord on Q, and during morphological operations, [SG] conditions the deletion of [PL] on Div through the impoverishment rule in (30).

- (30) Div_[PL] → Div / ___ Q_[SG]

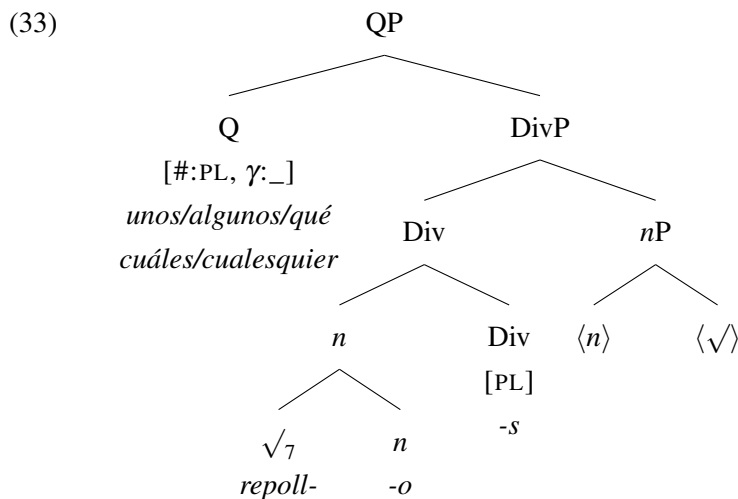
This rule eliminates the context of insertion for the plural suffix because it reduces Div_[PL] to Div, as in the morphosyntactic structure of *un repollo* ‘a cabbage’ in (31), where this head is unpronounced.



This configuration is shared by all singular indefinite nouns, including those like *una cerveza* ‘a (quantity or type of) beer’ that have a universal packager or sorter reading.

Turning to the plural indefinite expressions in (32), I assign them the morphosyntactic structure in (33), which is the same as that of *muchos repollos* ‘many cabbages’ in (23) above.

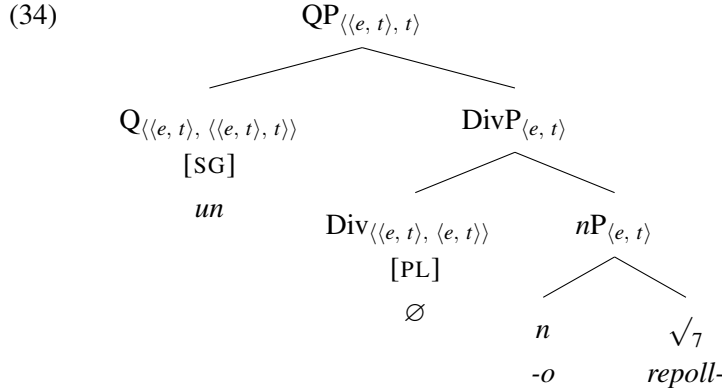
- (32) {unos / algunos / qué / cuáles / cualesquier} repollos
 some some what which whichever cabbages
 ‘some/some/which/which/whichever cabbages’



The interrogative determiner *qué* ‘what’ does not inflect for number, unlike *cuál/cuáles* ‘which’, so I propose that *qué* is a vocabulary item that is underspecified for number and can be inserted into $Q_{[SG]}$ or $Q_{[#:PL]}$.⁶ When *qué* realizes $Q_{[SG]}$, it behaves identically to *cuál* in that it exerts a universal packager or sorter effect on mass nouns, as in *qué cerveza* ‘which (quantity or type of) beer’.

⁶There are two other indefinite determiners whose plural forms have a limited distribution in Spanish. The first, *ningunos/ningunas* ‘no’, is only found with lexical plurals, as indicated by the contrast between *ningunas tijeras* ‘no scissors’ and **ningunas toronjas* ‘no grapefruits’. The second, *todos/todas* ‘all’, must be followed by a definite description: *todas *(las) toronjas* ‘all the grapefruits’.

A key principle of Distributed Morphology is that postsyntactic rules of impoverishment, such as the one that I defined in (30) above, have consequences for phonological form but not for logical form. In my analysis, singular indefinite expressions like *un repollo* ‘a cabbage’ include a DivP layer whose [PL] feature is transferred to logical form independently of its fate during the morphological component of the grammar. For example, compare the morphosyntactic structure of *un repollo* in (31), where Div has lost its [PL] feature, with the corresponding interpretable structure in (34), where Div retains this feature.



The same observation extends to any set of morphosyntactic and interpretable structures in which QP projects and is valued as [SG].

I model the semantic contribution of [SG] as $|x| = 1$, which states that x has a cardinality of one, and to illustrate some of the interpretations that $\text{Q}_{[\text{SG}]}$ can have, I incorporate $|x| = 1$ into the standard treatment of universal quantification (*cada/todo/toda* ‘every’), existential quantification (*un/una* ‘a’), and negative quantification (*ningún/ninguna* ‘no’) in (35).

- (35)
- a. $\llbracket \textit{cada/todo/toda} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \forall x_e [P(x) \wedge |x| = 1 \rightarrow Q(x)]$
 - b. $\llbracket \textit{un/una} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \exists x_e [P(x) \wedge |x| = 1 \wedge Q(x)]$
 - c. $\llbracket \textit{ningún/ninguna} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \neg \exists x_e [P(x) \wedge |x| = 1 \wedge Q(x)]$

It is now possible to calculate the semantic value of *un repollo* according to its interpretable structure in (34), where the nP has the count denotation in (7a) above due to the presence of DivP. In (36) and all other step-by-step computations in this thesis, I abbreviate the lexical entries as “LE” and β -reduction, or simplification, as “BR”.

- (36)
- $$\begin{array}{ll}
 \llbracket \textit{un} \rrbracket (\llbracket \emptyset \rrbracket (\llbracket \textit{repollo} \rrbracket)) & \\
 = \llbracket \textit{un} \rrbracket ([\lambda P_{\langle e, t \rangle} . \lambda x_e . *P(x)](\lambda y_e : \text{Cabbage}(y) \wedge \text{ATOM}(y) . \text{Cabbage}(y))) & \text{LE} \\
 = \llbracket \textit{un} \rrbracket (\lambda x_e . *\text{Cabbage}(x)), & \text{BR} \\
 \quad \text{defined only if } \text{Cabbage}(x) \text{ and } \text{ATOM}(x) & \\
 = [\lambda P_{\langle e, t \rangle} . \lambda Q_{\langle e, t \rangle} . \exists y_e [P(y) \wedge |y| = 1 \wedge Q(y)]](\lambda x_e . *\text{Cabbage}(x)), & \text{LE} \\
 \quad \text{defined only if } \text{Cabbage}(x) \text{ and } \text{ATOM}(x) &
 \end{array}$$

$$= \lambda Q_{\langle e, t \rangle} . \exists y_e [*Cabbage(y) \wedge |y| = 1 \wedge Q(y)], \quad \text{BR}$$

defined only if *Cabbage*(*y*) and *ATOM*(*y*)

The singular indefinite article takes the extension of *DivP*, which consists of atomic cabbages and their sums, and excludes the sums, resulting in a generalized quantifier that applies to a property-denoting expression and returns a truth value if and only if there is one cabbage with property *Q*.

One last issue to consider is the use of plural marking in (37), examples that are based on similar facts in English (Krifka 1989, 2003; Sauerland 2003; Borer 2005).

- (37) a. Jacques compró cero toronjas.
 Jacques bought.3SG zero grapefruits
 ‘Jacques bought zero grapefruits.’
- b. Jacques compró cero punto uno toronjas.
 Jacques bought.3SG zero point one grapefruits
 ‘Jacques bought zero point one grapefruits.’
- c. Jacques compró uno punto cero toronjas.
 Jacques bought.3SG one point zero grapefruits
 ‘Jacques bought one point zero grapefruits.’

There is no motivation to represent *cero* ‘zero’ as a head or phrase that bears a [SG] feature since this numeral does not presuppose singularity, so it is unsurprising that the plural form *toronjas* ‘grapefruits’ appears in (37a). The situation is more puzzling in (37b) and (37c), however, because *uno* ‘one’ does presuppose singularity. Curiously, *uno* does not exhibit feminine concord in decimal expressions even though it must do so as part of a cardinal numeral, as shown in (38).

- (38) Jacques compró treinta y {una / *un} toronjas.
 Jacques bought.3SG thirty and one grapefruits
 ‘Jacques bought thirty-one grapefruits.’

Another telling difference relates to the class marker *-o* when *uno* precedes a masculine noun. In the decimal *cero punto uno* ‘zero point one’ in (39a), *-o* must be included, but in the cardinal numeral *treinta y un* ‘thirty-one’ in (39b), *-o* must be omitted.

- (39) a. Jacques compró cero punto {uno / *un} repollos.
 Jacques bought.3SG zero point one one cabbages
 ‘Jacques bought zero point one cabbages.’
- b. Jacques compró treinta y {un / *uno} repollos.
 Jacques bought.3SG thirty and one one cabbages
 ‘Jacques bought thirty-one cabbages.’

These data suggest that the internal structure of decimals is complex to such a degree that all morphosyntactic interactions between *uno* and the noun are blocked. Just as *uno* does not condition the form of the nouns in (37b), (37c), or (39a), neither do these nouns condition the form of *uno*.

To summarize the analysis, *mucho repollo* ‘much cabbage’ contains QP but not DivP, whereas *un repollo* ‘a cabbage’ contains both QP and DivP. In *un repollo* and other singular indefinite expressions, Q bears a [SG] feature that triggers the deletion of [PL] on Div and results in the null realization of this head. In plural indefinite expressions, Q does not have its own number specification and instead displays number concord with Div, which retains its [PL] feature, so *muchos repollos* ‘many cabbages’ and *unos repollos* ‘some cabbages’ are structurally identical.

3.3 Maximality and its effects

Having developed a theory of indefinite expressions, I now turn to definite expressions and focus in particular on those that lack plural marking. For Borer (2005), definite descriptions that are interpreted as mass have the structure in (40a), where DivP is absent, and definite descriptions that are interpreted as count have the structure in (40b), where DivP is present.⁷

- (40) a. [DP [#P [NP]]] Mass configuration
 b. [DP [#P [DivP [NP]]]] Count configuration

One immediate concern is how to map the definite article onto the pieces of structure in (40). Borer’s solution is to endow this determiner with the ability to merge unpronounced copies in heads other than D, as schematized in (41) with *el repollo* ‘the cabbage’, a referential expression that permits a mass and a count reading.

- (41) a. [DP *el* [#P *el* [NP *repollo*]]] Mass configuration
 b. [DP *el* [#P *el* [DivP *el* [NP *repollo*]]]] Count configuration

Beyond this proliferation of unpronounced material, one might ask whether there is empirical rationale to assign the same form to more than one structure.

A fundamental difference between indefinite and definite determiners is that only the former have distributions that are influenced by the mass-count distinction. For instance, a quantifier like *mucho* ‘much’ is generally infelicitous when used with a prototypical count noun like *carro* ‘car’ (*#mucho carro* ‘much car’), and the indefinite article is generally infelicitous when used with a prototypical mass noun like *hielo* ‘ice’ (*#un hielo* ‘an ice’). In contrast, the definite article readily occurs with both nouns: *el carro* ‘the car’ and *el hielo* ‘the ice’. Moreover, it can be shown that definite determiners give rise to a systematic ambiguity between mass and singular denotation when they combine with nonplural nouns (Jambrović 2023). Without additional context, it is impossible to tell whether the nonplural definite expressions in (42) refer to substances or to atomic objects.

- (42) a. Julia encontró {el / ese / su} repollo.
 cabbage found.3SG the that her cabbage
 ‘Julia found the/that/her cabbage.’

⁷In Borer’s exo-skeletal approach, #P is a functional projection that is akin to QP.

- b. Jacques encontró {la / esa / su} toronja.
 Jacques found.3SG the that his grapefruit
 ‘Jacques found the/that/his grapefruit.’

This ambiguity is made more salient by the adjectives in (43), the first of which corresponds to the mass reading and the second of which corresponds to the singular reading.

- (43) a. Julia encontró el repollo que estaba {rallado / entero}.
 Julia found.3SG the cabbage that was shredded whole
 ‘Julia found the cabbage that was shredded/whole.’
 b. Jacques encontró {esa / su} toronja, que estaba {machacada / intacta}.
 Jacques found.3SG that his grapefruit that was crushed intact
 ‘Jacques found that/his grapefruit, which was crushed/intact.’

Before presenting my account of this phenomenon, I identify three more pieces of evidence that support this characterization of nonplural definite expressions.

First, even human-denoting nouns can have mass interpretation when they are introduced by a definite determiner, a fact that Stephen Sondheim exploits in the lyrics to “A Little Priest” from *Sweeney Todd*. In the final line of the abridged verse in (44), *the priest* denotes a totality of priest substance, not an individual priest.

- (44) Mrs. Lovett: Here we are, now! Hot out of the oven.
 Sweeney Todd: What is that?
 Mrs. Lovett: It’s priest. Have a little priest.
 ...
 Sweeney Todd: Haven’t you got poet, or something like that?
 Mrs. Lovett: No, you see, the trouble with poet is how do you know it’s deceased?
 Try the priest!

If one devises a similar context in Spanish, as in (45), both *el cura* ‘the priest’ and *el poeta* ‘the poet’ favour a mass reading.

- (45) ¿Vas a probar el cura? Es mejor que el poeta.
 go.2SG to try the priest is better than the poet
 ‘Are you going to try the priest? It is better than the poet.’

Second, pronouns behave like other nonplural definite determiners in that they are indifferent to mass versus singular denotation. In (46a), *él* ‘it’ and *ella* ‘it’ are coreferential with mass occurrences of *repollo* ‘cabbage’ and *toronja* ‘grapefruit’, and in (46b), these pronouns are coreferential with singular occurrences of the same nouns.

- (46) a. Compraron {repollo / toronja} sin saber qué hacer con {él / ella}.
 bought.3PL cabbage grapefruit without know what do with it it
 ‘They bought cabbage/grapefruit without knowing what to do with it.’

- b. Compraron {un repollo / una toronja} sin saber qué hacer con {él / ella}.
 bought.3PL a cabbage a grapefruit without know what do with it it
 ‘They bought a cabbage/grapefruit without knowing what to do with it.’

Third, I build on Sharvy’s (1980) observation that the referent of a nonplural definite expression can occupy multiple points in space, as illustrated in (47).

- (47) Limpiaron {el repollo / la toronja} que estaba por todo el piso.
 cleaned.3PL the cabbage the grapefruit that was through all the floor
 ‘They cleaned up the cabbage/grapefruit that was all over the floor.’

As it turns out, there are even contexts where these expressions can denote entities that consist partially of substances and partially of atomic objects. If the sentence in (48) is uttered by an employee at a grocery store, for example, *el repollo* ‘the cabbage’ could refer to the total inventory of cabbage substance and whole cabbages just as *la toronja* ‘the grapefruit’ could refer to the total inventory of grapefruit substance and whole grapefruits.

- (48) Hay una retirada de producto y tenemos que desechar {el repollo / la toronja}.
 has a recall of product and have.1PL that throw.out the cabbage the grapefruit
 ‘There is a product recall, and we have to throw out the cabbage/grapefruit.’

Based on the data in (42) through (48), I conclude that nonplural definite expressions are consistently vague with respect to mass versus singular interpretation.

To unify the analysis of nonplural and plural definite expressions, such as *el repollo* ‘the cabbage’ and *los repollos* ‘the cabbages’, many authors posit that a maximality operator (MAX) is a component of definite determiners (Montague 1973a; Sharvy 1980; Link 1983, Krifka 1989; Landman 1989a,b; Chierchia 1998a,b; Zamparelli 1998, 2002; Bale & Barner 2009; C. Gillon 2009; Heim 2011; Champollion & Krifka 2016). I consider this operator to be the reason that *el repollo* can denote a totality of cabbage substance as well as an atomic cabbage and base my proposal on Heim’s (2011: 998) definition of maximalized properties in (49).

$$(49) \quad \text{MAX}(P) := \lambda x_e . P(x) \wedge \neg \exists y_e [P(y) \wedge x < y]$$

In prose, a maximalized property is a function that maps every object x to the truth value 1 if and only if x has property P and there is no object y such that y has property P and x is a proper part of y . Focusing on the definite article in English, Heim (2011: 998) assigns $\text{MAX}(P)$ to both its presupposed and asserted content in (50).⁸

⁸To incorporate domain restriction (C) into the semantic value of the definite article, one could follow C. Gillon (2009) and add $C(x)$ to the assertion, as in (i).

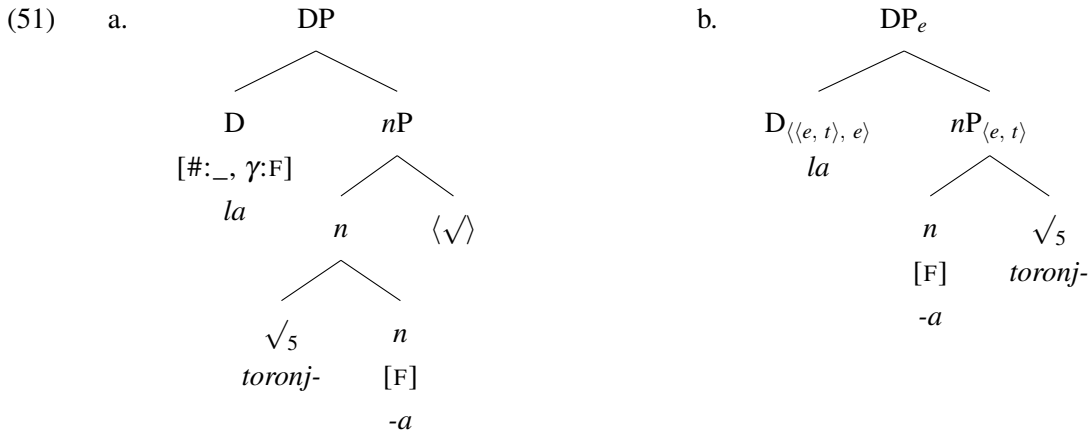
(i) $\llbracket \text{the} \rrbracket = \lambda P_{\langle e, t \rangle} : \exists x_e \forall y_e [\text{MAX}(P)(y) \leftrightarrow x = y] . \iota x_e [\text{MAX}(P)(x) \wedge C(x)]$

For other approaches to domain restriction, see Westerståhl 1984, von Stechow 1994, Martí Martínez 2003, Giannakidou 2004, and Schwarz 2009.

$$(50) \quad \llbracket \text{the} \rrbracket = \lambda P_{\langle e, t \rangle} : \exists x_e \forall y_e [\text{MAX}(P)(y) \leftrightarrow x = y] . \iota x_e \text{MAX}(P)(x)$$

The definite article first checks that a unique maximal object exists and then returns that object. More precisely, it presupposes that there is an object x such that every object y is a maximal object that has property P if and only if x is identical to y . If this condition is met, this determiner yields the unique maximal object that has property P .

Given that nonplural definite descriptions can have mass readings no matter the nominalized root, as revealed by *el cura* ‘the priest’ and *el poeta* ‘the poet’ in (45) above, I have argued in previous work that these expressions uniformly lack DivP (Jambrović 2023). To keep the discussion from becoming too gruesome, I illustrate the mechanics of the proposal using *la toronja* ‘the grapefruit’, which has the morphosyntactic and interpretable structures in (51).

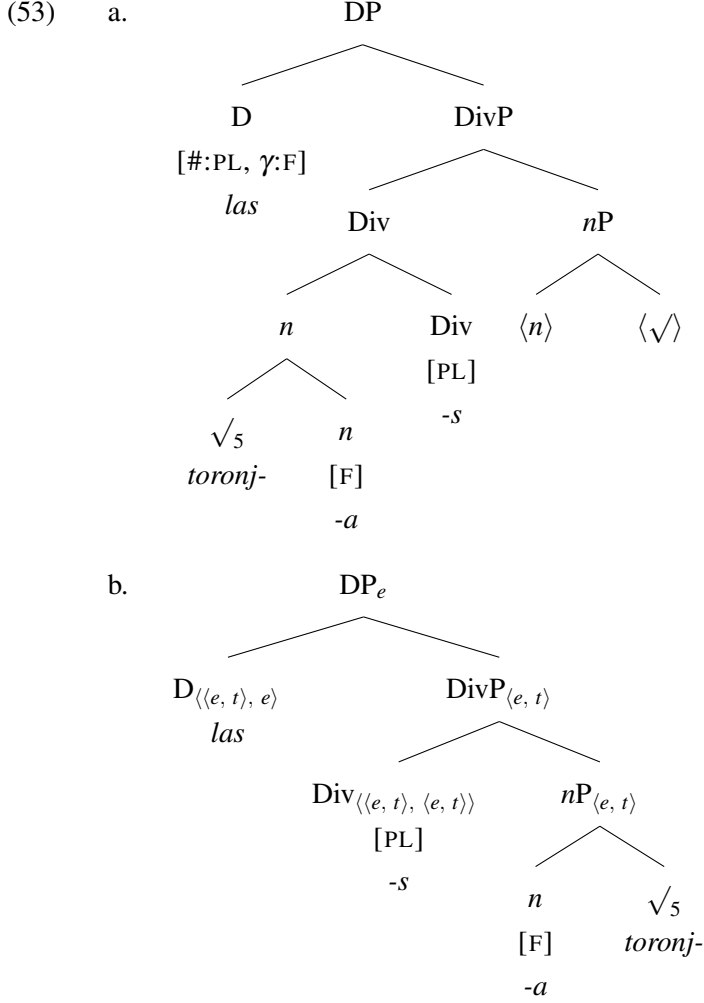


Since DivP is not present to condition the interpretation of the nP , *toronja* ‘grapefruit’ has the elsewhere semantic value in (7b) above, which is reflected in (52).

$$\begin{aligned}
 (52) \quad & \llbracket \text{la} \rrbracket(\llbracket \text{toronja} \rrbracket) \\
 &= [\lambda P_{\langle e, t \rangle} : \exists x_e \forall y_e [\text{MAX}(P)(y) \leftrightarrow x = y] . \iota x_e \text{MAX}(P)(x)](\lambda z_e . \text{Grapefruit}(z)) \quad \text{LE} \\
 &= \iota x_e \text{MAX}(\text{Grapefruit})(x), \quad \text{BR} \\
 &\quad \text{defined only if } \exists x_e \forall y_e [\text{MAX}(\text{Grapefruit})(y) \leftrightarrow x = y]
 \end{aligned}$$

Accordingly, *la toronja* denotes a unique maximal object x that is grapefruit provided that such an object exists. Depending on the context of utterance, this unique maximal object could be a totality of grapefruit substance or an entire grapefruit. By this same reasoning, the referent of *el cura* could be a totality of priest substance, as in *probaron el cura* ‘they tried the priest’, or an entire priest, as in *vieron al cura* ‘they saw the priest’.

With the exception of lexical plurals and pluralia tantum, which I address in section 3.5, plural definite expressions do contain DivP. For instance, *las toronjas* ‘the grapefruits’ has the morphosyntactic structure in (53a), where the definite article has the form *las* ‘the’ rather than *la* ‘the’ due to number concord on D, along with the interpretable structure in (53b).



In the environment of DivP, the *nP* *toronja* ‘grapefruit’ has the count denotation in (7a) above, so the calculation of *las toronjas* in (54) begins with a presupposition of atomicity that the calculation of *la toronja* ‘the grapefruit’ in (52) lacks.

$$\begin{aligned}
 (54) \quad & \llbracket \text{las} \rrbracket (\llbracket -s \rrbracket (\llbracket \text{toronja} \rrbracket)) \\
 &= \llbracket \text{las} \rrbracket ([\lambda P_{\langle e, t \rangle} . \lambda x_e . *P(x)](\lambda y_e : \text{Grapefruit}(x) \wedge \text{ATOM}(y) . \\
 &\quad \text{Grapefruit}(y))) \quad \text{LE} \\
 &= \llbracket \text{las} \rrbracket (\lambda x_e . * \text{Grapefruit}(x)), \quad \text{BR} \\
 &\quad \text{defined only if Grapefruit}(x) \text{ and } \text{ATOM}(x) \\
 &= [\lambda P_{\langle e, t \rangle} : \exists y_e \forall z_e [\text{MAX}(P)(z) \leftrightarrow y = z] . \iota y_e \text{MAX}(P)(y)](\lambda x_e . \\
 &\quad * \text{Grapefruit}(x)), \quad \text{LE} \\
 &\quad \text{defined only if Grapefruit}(x) \text{ and } \text{ATOM}(x) \\
 &= \iota y_e \text{MAX}(* \text{Grapefruit})(y), \quad \text{BR} \\
 &\quad \text{defined only if Grapefruit}(y), \text{ATOM}(y), \\
 &\quad \text{and } \exists y_e \forall z_e [\text{MAX}(* \text{Grapefruit})(z) \leftrightarrow y = z]
 \end{aligned}$$

In other words, *las toronjas* denotes a unique maximal sum of objects y that consists of grapefruits provided that the grapefruits are atomic and such a unique maximal sum of objects exists.

As I first discussed in section 3.1, definite descriptions can refer to kinds as well as objects in Spanish, so it is essential to formally differentiate the kind-denoting use of *la toronja* ‘the grapefruit’ in (55a) from the object-denoting use in (55b).

- (55) a. La toronja podría extinguirse.
 the grapefruit could extinguish.³
 ‘(The) grapefruit could become extinct.’
- b. La toronja podría caerse.
 the grapefruit could fall.³
 ‘The grapefruit could fall.’

To this end, all that is needed is to apply Montague’s (1973b) intensional operator ($\hat{}$) to the semantic value of *la toronja* in (52), as in (56), for this expression to refer to the species *Citrus × paradisi* rather than a totality of grapefruit substance or a single grapefruit.

- (56) $\llbracket \text{la toronja} \rrbracket = \hat{\iota}x_e \text{MAX}(\text{Grapefruit})(x)$,
 defined only if $\exists x_e \forall y_e [\text{MAX}(\text{Grapefruit})(y) \leftrightarrow x = y]$

As Chierchia (1998b) remarks, $\hat{\iota}$ is semantically equivalent to the “down” operator ($\cap$), which this author uses to derive kinds from predicates in languages like English. Plural definite descriptions like *las toronjas* ‘the grapefruits’ can also denote kinds, as in (57a), or objects, as in (57b).⁹

- (57) a. Las toronjas podrían extinguirse.
 the grapefruits could.3PL extinguish.³
 ‘Grapefruits could become extinct.’
- b. Las toronjas podrían caerse.
 the grapefruits could.3PL fall.³
 ‘The grapefruits could fall.’

To account for the kind-referring occurrence of *las toronjas*, I employ the same intensionalization process as before and add $\hat{}$ to the denotation of this expression in (54), as in (58).

- (58) $\llbracket \text{las toronjas} \rrbracket = \hat{\iota}x_e \text{MAX}(*\text{Grapefruit})(x)$,
 defined only if $\text{Grapefruit}(x)$, $\text{ATOM}(x)$,
 and $\exists x_e \forall y_e [\text{MAX}(*\text{Grapefruit})(y) \leftrightarrow x = y]$

In both (56) and (58), kind denotation is achieved through a single covert operation in Spanish, one that has precedent in the literature.

To review the findings of the last two sections, *mucha toronja* ‘much grapefruit’ is unambiguously mass, *una toronja* ‘a grapefruit’ is unambiguously singular, and *la toronja* ‘the grapefruit’ can

⁹See Jambrović 2023 for a structure-based explanation of the contrast between *(the) grapefruit could become extinct* and *(*the) grapefruits could become extinct* in English and other Germanic languages.

be mass or singular. In all three cases, the denotation of the noun phrase is determined by its internal structure: *mucha toronja* is a QP that lacks DivP, *una toronja* is a QP that bears a [SG] feature and contains DivP, and *la toronja* is a DP that lacks DivP but has a maximality operator in D that is responsible for its vague interpretation.

3.4 Number neutrality via modification

As a general rule, bare nonplural objects of verbs have mass readings in Spanish, hence the obligatory determiner in (59).

- (59) Julia alimenta #(un) gato.
 Julia feed.3SG a cat
 ‘Julia is feeding a cat.’

There are, nevertheless, some constructions where bare nonplural objects can have count readings, which usually involve a verb of possession, an intensional verb, or a light verb (Sánchez de Zavala 1976, Wonder 1990, Garrido Medina 1991, Masullo 1992, Bosque 1996, Masullo 1996, Laca 1999, Dobrovie-Sorin et al. 2005, Espinal 2010, Espinal & Mateu 2011, Espinal & McNally 2011).¹⁰ In (60), I provide an example of each type of predicate.

- (60) a. Julia tiene gato.
 Julia has cat
 Approximate: ‘Julia has a cat.’
 b. Julia busca casa.
 Julia seeks house
 Approximate: ‘Julia is looking for a house.’
 c. Julia hizo copia de su pasaporte.
 Julia made.3SG copy of her passport
 Approximate: ‘Julia made a copy of her passport.’

As Espinal (2010) and Espinal & McNally (2011) remark, the interpretation of bare nonplural count objects in Catalan and Spanish is more accurately described as number neutral rather than singular. In support of this view, the sentences in (60) permit continuations of the sort in (61), confirming that the nouns in question can have plural readings as well.

- (61) a. Julia tiene gato: uno joven y uno más viejo.
 Julia has cat a young and a more old
 Approximate: ‘Julia has cats: a young one and an older one.’
 b. Julia busca casa: una en la ciudad y una en el campo.
 Julia seeks house one in the city and one in the country
 Approximate: ‘Julia is looking for houses: one in the city and one in the country.’

¹⁰Many authors consider possession to be a semantic component of intensional verbs and light verbs (Masullo 1992, Bosque 1996, Masullo 1996, Fodor & Lepore 1998, Borthen 2003, Harley 2004, Dobrovie-Sorin et al. 2005, Espinal 2010, Espinal & Mateu 2011, Espinal & McNally 2011, Harves & Kayne 2012).

- c. Julia hizo copia de su pasaporte: una para ella y una para su esposo.
 Julia made.3SG copy of her passport one for her and one for her spouse
 Approximate: ‘Julia made copies of her passport: one for her and one for her spouse.’

Crosslinguistically, bare nonplural count objects often exhibit number neutrality (Van Geenhoven 1995, 1998; Farkas & de Swart 2003; Dayal 2011b; Modarresi 2014; Borik & Gehrke 2015; Dayal 2015; Krifka & Modarresi 2016). In this section, I present an overview of the debates that surround this topic in Spanish and then discuss Espinal & McNally’s (2011) analysis that bare nonplural count objects behave semantically as modifiers of the verb.

The first point to clarify is that the possibility of bare number-neutral objects in Spanish does not undermine my claim in section 3.2 that exclusively singular readings of nouns result from a limited number of indefinite determiners, including the indefinite article. In (62), for example, only *un carro* ‘a car’ is necessarily interpreted as singular.

- (62) a. Jacques tiene carro.
 Jacques has car
 Approximate: ‘Jacques has a car/cars.’
 b. Jacques tiene un carro.
 Jacques has a car
 ‘Jacques has a car.’

There is, nonetheless, at least one variety of Spanish where the empirical landscape seems to be more complex. For Rinaldi (2018: 57–58), the bare nonplural object in (63) has a singular rather than number-neutral reading in Rioplatense Spanish.

- (63) Finalmente encontré casa. #Una en Buenos Aires y una en La Plata.
 finally found.1SG house one in Buenos Aires and one in La Plata
 Approximate: ‘I finally found a house. #One in Buenos Aires and one in La Plata.’

In (61b) above, I followed Espinal & McNally (2011) in using an intensional verb in the present tense, and even speakers of Rioplatense Spanish find the number-neutral interpretation of *casa* ‘house’ in *Julia busca casa* ‘Julia is looking for a house/houses’ to be available. Tense could therefore play a role in Oggiani’s (2020: 231) judgment that *estrenamos obra* ‘we debuted a play’ in (64), a construction that is possible in Rioplatense Spanish, only allows for a singular reading.

- (64) Este año estrenamos obra. #Una en vacaciones de julio y la otra en vacaciones de primavera.
 this year debuted.1PL play one in vacations of July and the other in vacations of spring
 Approximate: ‘This year we debuted a play. #One during the July holidays and the other one during the spring holidays.’

There is, moreover, a pragmatic consideration that could explain why *buscamos casa* ‘we are looking for a house/houses’ can be interpreted as number neutral while *estrenamos obra* ‘we debuted a

play’ cannot: it is far more common to simultaneously possess more than one object of the same type than it is to simultaneously present more than one object of the same type. In light of these potentially confounding factors, further investigation is needed to evaluate whether the data in (63) and (64) are indeed problematic for the generalization that bare nonplural count objects have number-neutral readings in Spanish.

Another point of contention in research on bare nonplural count objects is whether to classify them as discourse opaque or transparent. According to Espinal & McNally (2011), these objects are discourse opaque because they cannot serve as the antecedents of pronouns, be modified by nonrestrictive relative clauses, or control implicit subjects, behaviour that differentiates them from canonical arguments. These authors use Catalan to illustrate the relevant phenomena but state that the judgments hold in Spanish as well, and in (65), I provide direct translations of their examples (16a), (18a), and (19a) for reference (Espinal & McNally 2011: 94–96).

- (65) a. Hoy lleva falda. #Se la regalamos el año pasado.
 today wears skirt 3 3SG.F.ACC gifted.1PL the year past
 Literal: ‘Today she is wearing skirt. #We gave it to her last year.’
- b. *Por fin hemos encontrado piso, que empezaremos a renovar muy pronto.
 finally have.1PL found apartment that begin.FUT.1PL to renovate very soon
 Literal: ‘We have finally found apartment, which we will begin to renovate very soon.’
- c. *Busco piso para PRO ser compartido.
 seek.1SG apartment for be shared
 Literal: ‘I am looking for apartment to be shared.’

Rinaldi (2018: 47–52) disputes this characterization of bare nonplural count objects and finds all three sentences in (66) to be grammatical in Rioplatense Spanish, suggesting that *casa* ‘house’, *departamento* ‘apartment’, and *niñera* ‘nanny’ are discourse transparent.

- (66) a. Encontré casa y ya la compré.
 found.1SG house and already 3SG.F.ACC bought.1SG
 Approximate: ‘I found a house and have already bought it.’
- b. Encontré departamento, que voy a renovar pronto para poder venderlo.
 found.1SG apartment that go.1SG to renovate soon for can sell.3SG.ACC
 Approximate: ‘I found an apartment, which I will renovate soon to be able to sell.’
- c. Busco niñera responsable para PRO cuidar a mis hijos.
 seek.1SG nanny responsible for care DOM my children
 Approximate: ‘I am looking for a responsible nanny to care for my children.’

On the other hand, Oggiani (2020: 248) reports that bare nonplural count objects of verbs of creation do not permit nonrestrictive modification in Rioplatense Spanish, as shown in (67).

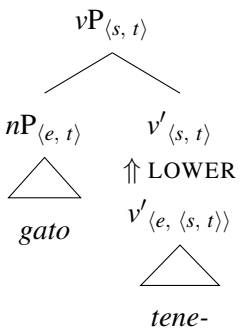
- (67) *Estoy escribiendo monografía, que quiero entregar a fin de año.
 am writing monograph that want.1SG submit at end of year
 Literal: ‘I am writing monograph, which I want to submit at the end of the year.’

This example merits attention because there is consensus in the literature that the ability to host a nonrestrictive relative clause is a reliable indicator of discourse transparency (Thorne 1972; Karttunen 1976; McCawley 1988; Potts 2002, 2005). In (68), I use the relative pronoun *quien* ‘who’ rather than *que* ‘that’ so that the relative clause is unambiguously nonrestrictive, and speakers of Rioplatense Spanish prefer the inclusion of the indefinite article.

- (68) Julia tiene #(una) pareja, quien es el amor de su vida.
 Julia has a partner who is the love of her life
 ‘Julia has a partner, who is the love of her life.’

It seems to be the case that the discourse transparency of bare nonplural count objects is degraded in at least one way, supporting Espinal & McNally’s (2011) claim that these objects are syntactic but not semantic arguments.

To derive the number neutrality of bare nonplural count objects, several authors have proposed that they are interpreted as modifiers of the verb (Dobrovie-Sorin et al. 2005, Espinal 2010, Espinal & McNally 2011). From this perspective, a sentence like *tenemos gato* ‘we have a cat/cats’ can also be translated as ‘we cat-have’ to better reflect the number-neutral reading ‘we have one or more cats’. In other words, *cat* in the predicate *cat-have* is like *cat* in the compound *cat tree*, where this modifier can be paraphrased as ‘for one or more cats’. As far as the technical implementation, Espinal & McNally (2011) posit two rules at logical form, one that lowers the semantic type of HAVE-predicates from $\langle e, \langle s, t \rangle \rangle$ to $\langle s, t \rangle$ and another that allows the type-shifted verb to combine with a direct object of type $\langle e, t \rangle$, the outcome of which is an expression of type $\langle s, t \rangle$. In Kratzer’s (1996) view of argument structure, where direct objects merge as specifiers of the verb, the vP of *tenemos gato* can be assigned the interpretable structure in (69a) and the semantic value in (69b).

- (69) a. 
- b. $\llbracket \text{gato tene-} \rrbracket = \lambda e_s . [\text{Cat}(\theta(e)) \wedge \text{Have}(e)]$

If, following Espinal & McNally (2011: 112), θ represents an “implicit role function”, this vP denotes a function that maps every event e to the truth value 1 if and only if e is an event of cat-having. In this way, the syntactic argument *gato* ‘cat’ comes to modify the verb stem *tene-* ‘have’, resulting in the predicate *gato tene-* ‘cat-have’.

The purpose of Espinal & McNally’s (2011) implicit role function, or θ , is to capture the fact that bare nonplural count objects are only possible in Catalan and Spanish if they can be understood

to characterize the subject in some relevant way. These authors note that a sentence like *en Joan té juguina* ‘Joan has a toy’ in Catalan, which translates to *Juan tiene juguete* ‘Juan has a toy’ in Spanish, would be infelicitous as an out-of-the-blue utterance but acceptable in a context where the relevance of having a toy is apparent, such as a preschool activity. Bare mass objects are not subject to the same pragmatic factors because they can combine with the verb through the more general mechanism of existential closure (EC) that I proposed in section 2.3. For instance, the *vP* of *tenemos cerveza* ‘we have beer’ has the interpretable structure and denotation in (70).

- (70) a.
- b. $\llbracket \text{cerveza tene-} \rrbracket = \lambda e_s . \exists x_e [\text{Beer}(x) \wedge \text{Have}(x)(e)]$

If *tenemos gato* ‘we have a cat/cats’ had the *vP* in (69a) instead of the one in (70a), *gato* ‘cat’ would have an infelicitous mass reading.

The goal of this section has not been to settle the debate regarding bare nonplural count objects in Spanish but rather to highlight some of the semantic and pragmatic issues that warrant further study. I find Espinal & McNally’s (2011) view that these objects function syntactically as arguments but semantically as modifiers to be the most compelling account to date, one that can serve as the basis for future research that seeks to explain the differences in interpretation that have been reported in Rioplatense Spanish and other varieties.

3.5 Form-meaning mismatches

In Spanish, as in many languages, there are nouns whose interpretation is not predictable from their form or the syntactic environment in which they appear, and the question is whether these lexical items challenge the approach to the mass-count distinction that I developed in sections 3.1 through 3.3. I start the discussion with nouns that bear plural marking but do not behave semantically like canonical plurals, such as *alicates* ‘pliers’ and *bártulos* ‘gear’, and then to turn nouns that lack plural marking but behave semantically like canonical plurals to at least some degree, such as *comité* ‘committee’ and *equipaje* ‘luggage’.

I reserve the term “lexical plural” for nouns in Spanish whose plural forms can have the same interpretation as their nonplural forms, such as *alicate/alicates* ‘pliers’, *calzoncillo/calzoncillos* ‘underwear’, *pantalón/pantalones* ‘pants’, and *tijera/tijeras* ‘scissors’, all of which denote properties of symmetrical objects. In (71), *la tijera* ‘the scissors’ and *las tijeras* ‘the scissors’ can both re-

fer to a single pair of scissors, demonstrating that pluralization does not necessarily change the interpretation of this noun.

- (71) Jacques perdió {la tijera / las tijeras} que había comprado.
 Jacques lost.3SG the scissor the scissors that had.IMPF bought
 ‘Jacques lost the scissors that he had bought.’

As for nouns like *bártulos* ‘gear’ and *entrañas* ‘entrails’ that only have plural forms, I use the term “pluralia tantum”. One difference between lexical plurals and pluralia tantum relates to their tolerance of numerals, as shown in (72) versus (73).

- | | |
|--|---|
| <p>(72) a. los dos pantalones
 the two pants
 ‘the two pants’</p> <p> b. las tres tijeras
 the three scissors
 ‘the three scissors’</p> | <p>(73) a. los (*dos) bártulos
 the two gear
 ‘the gear’</p> <p> b. las (*tres) entrañas
 the three entrails
 ‘the entrails’</p> |
|--|---|

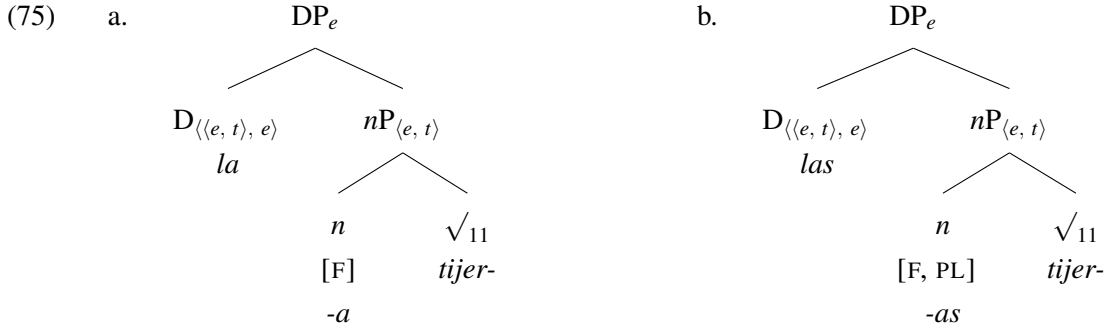
This contrast suggests that *pantalones* and *tijeras* can be canonical plurals as well as lexical plurals, and in their guise as canonical plurals, these nouns must contain DivP because they are countable. The question is how to capture their guise as lexical plurals and whether pluralia tantum can be subsumed under the same analysis.

Many authors who work on noncanonical plurals in Distributed Morphology consider them to have a [PL] feature on *n* rather than on Div or Num (Lecarme 2002, Acquaviva 2008, Lowenstamm 2008, Acquaviva 2009, Kramer 2009, Alexiadou 2011, Mathieu 2014, Kramer 2016, Punske & Jackson 2017, Dali & Mathieu 2021, Smith 2021, Alexiadou 2024).¹¹ In (74), the proposed morphosyntactic structures of *la tijera* ‘the scissors’ and of *las tijeras* ‘the scissors’ in its guise as a lexical plural, number features on *n* participate in nominal concord just as gender features do.

- | | |
|--|--|
| <p>(74) a.</p> <div style="margin-left: 40px;"> <p>DP</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>D</p> <p>[#:_, γ:F]</p> <p>la</p> </div> <div style="text-align: center;"> <p>nP</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>n</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>√₁₁</p> <p>tijer-</p> </div> <div style="text-align: center;"> <p>n</p> <p>[F]</p> <p>-a</p> </div> </div> </div> </div> </div> </div> </div> | <p>b.</p> <div style="margin-left: 40px;"> <p>DP</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>D</p> <p>[#:PL, γ:F]</p> <p>las</p> </div> <div style="text-align: center;"> <p>nP</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>n</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>√₁₁</p> <p>tijer-</p> </div> <div style="text-align: center;"> <p>n</p> <p>[F, PL]</p> <p>-as</p> </div> </div> </div> </div> </div> </div> </div> |
|--|--|

¹¹The literature on lexical plurals and pluralia tantum is not limited to Distributed Morphology, nor do all who adopt this framework endorse this particular solution (McCawley 1975, Booij 1996, Corbett 2000, Acquaviva 2004, Ojeda 2005, Wiltschko 2008, Arregi & Nevins 2014, Fábregas 2024).

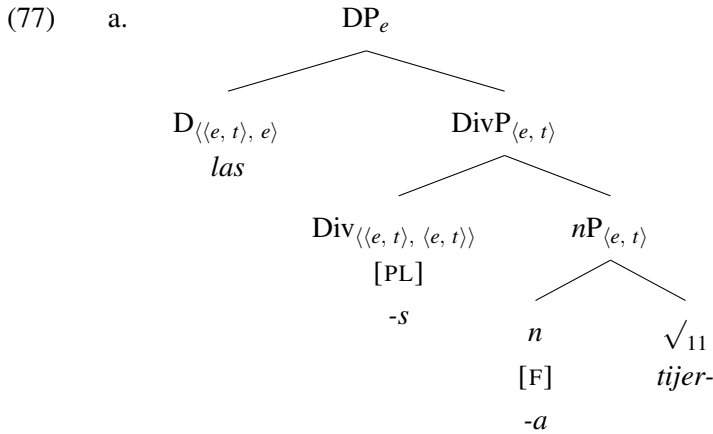
In sections 1.4 and 5.2, I maintain that gender features on n are visible but disregarded at logical form, and I extend the same argument to number features on this head. The interpretable structures of *la tijera* and *las tijeras* in (75) both have the semantic value in (76), according to which they denote a unique maximal object that is scissors.



$$(76) \quad \llbracket \text{la(s) tijera(s)} \rrbracket = \iota x_e \text{MAX}(\text{Scissors})(x),$$

defined only if $\exists x_e \forall y_e [\text{MAX}(\text{Scissors})(y) \leftrightarrow x = y]$

In its guise as a canonical plural, *las tijeras* has the interpretable structure in (77a), where DivP projects, and the semantic value in (77b), according to which this expression denotes a unique maximal sum of objects that consists of scissors.



b. $\llbracket \text{las tijeras} \rrbracket = \iota x_e \text{MAX}(*\text{Scissors})(x),$
 defined only if $\text{Scissors}(x)$, $\text{ATOM}(x)$,
 and $\exists x_e \forall y_e [\text{MAX}(*\text{Scissors})(y) \leftrightarrow x = y]$

Pluralia tantum reject numerals and thus lack DivP, so a definite description like *los bártulos* ‘the gear’ only maps onto the morphosyntactic structure in (74b) and the interpretable structure in (75b). As indicated in (78), *los bártulos* denotes a unique maximal object that is gear.

$$(78) \quad \llbracket \text{los bártulos} \rrbracket = \iota x_e \text{MAX}(\text{Gear})(x)$$

defined only if $\exists x_e \forall y_e [\text{MAX}(\text{Gear})(y) \leftrightarrow x = y]$

Landman 1989a,b; Barker 1992; B. Gillon 1992; Schwarzschild 1996; B. Gillon 1999; Rothstein 1999; Zucchi & White 2001; Rothstein 2004, 2010, 2017).

- (81) a. Se combinaron dos comités para hacer un solo comité.
 3 combined.3PL two committees for make a single committee
 ‘Two committees were combined to make a single committee.’
- b. Se combinaron dos muros para hacer un solo muro.
 3 combined.3PL two walls for make a single wall
 ‘Two walls were combined to make a single wall.’
- c. #Se combinaron dos gatos para hacer un solo gato.
 3 combined.3PL two cats for make a single cat
 Literal: ‘Two cats were combined to make a single cat.’
- (82) a. Se dividió un comité para hacer dos comités.
 3 divided.3SG a committee for make two committees
 ‘A committee was divided to make two committees.’
- b. Se dividió un muro para hacer dos muros.
 3 divided.3SG a wall for make two walls
 ‘A wall was divided to make two walls.’
- c. #Se dividió un gato para hacer dos gatos.
 3 divided.3SG a cat for make two cats
 Literal: ‘A cat was divided to make two cats.’

Considering that walls are physically homogeneous objects, it is unsurprising that *muro* is felicitous in both contexts, and the same logic applies to all of the following count nouns, among many others: *bosque* ‘forest’, *cadena* ‘chain’, *estanque* ‘pond’, *mosaico* ‘mosaic’, *prado* ‘field’, *rama* ‘branch’, *soga* ‘rope’, *tubo* ‘tube’, and *valla* ‘fence’. What is less clear is why *comité*, a group noun, can occur in cumulative and divisive environments.

One defining characteristic of group nouns is that they can take a prepositional phrase complement that embeds a plural noun and identifies the objects that constitute the group, as shown in (83) with *comité* ‘committee’ and *colección* ‘collection’ (Barker 1992).¹²

- (83) a. un comité de {cocineros / expertos / padres}
 a committee of chefs experts parents
 ‘a committee of chefs/experts/parents’
- b. una colección de {discos / estampillas / monedas}
 a collection of records stamps coins
 ‘a collection of records/stamps/coins’

Even when these nouns appear without a complement, as in (84), they are nevertheless interpreted as a committee of people and a collection of things, respectively.

¹²See Pearson 2011 for evidence that group nouns can be subdivided into two classes: *committee*-type nouns and *collection*-type nouns.

- (84) a. Tenemos que formar un comité.
 have.1PL that form a committee
 ‘We have to form a committee.’
 b. Vamos a empezar una colección.
 go.1PL to start a collection
 ‘We are going to start a collection.’

It is likely the case that group nouns that lack an explicit argument still take an implicit one, such as the plural versions of Kayne’s (2004, 2008) light nouns PERSON and THING in (85).

- (85) a. Tenemos que formar un comité ~~de PERSONAS~~.
 have.1PL that form a committee of PEOPLE
 ‘We have to form a committee.’
 b. Vamos a empezar una colección ~~de COSAS~~.
 go.1PL to start a collection of THINGS
 ‘We are going to start a collection.’

These silent plural nouns could be the source of the cumulative and divisive reference of *comité* and *colección*, but I leave a comprehensive exploration of this possibility to future investigation.

Finally, there are aggregate nouns like *equipaje* ‘luggage’ that do not bear plural marking yet denote properties of atomic objects and their sums (B. Gillon 1992; Chierchia 1998a,b; B. Gillon 1999; Rothstein 1999, 2004; Barner & Snedeker 2005; Bale & Barner 2009; Chierchia 2010; Joosten 2010; Rothstein 2010; Cowper & Hall 2012; De Belder 2013; Alexiadou 2015; Grimm & Levin 2017; Rothstein 2017; Smith 2021). In a series of experiments with English-speaking participants, Barner & Snedeker (2005) demonstrate that aggregate nouns are evaluated based on cardinality, just as plural nouns are, and not based on volume or weight, as mass nouns would be. Given the premises in (86a) and (86b), speakers of Spanish would judge the inference in (86c) to be valid because Julia has more pieces of luggage than Jacques even though her luggage weighs less collectively.

- (86) a. Julia tiene tres maletas y cada una pesa 10 kilogramos.
 Julia has three suitcases and each one weighs 10 kilograms
 ‘Julia has three suitcases, and each one weighs 10 kilograms.’
 b. Jacques tiene dos maletas y cada una pesa 20 kilogramos.
 Jacques has two suitcases and each one weighs 20 kilograms
 ‘Jacques has two suitcases, and each one weighs 20 kilograms.’
 c. Julia tiene más equipaje que Jacques.
 Julia has more luggage than Jacques
 ‘Julia has more luggage than Jacques.’

One aspect of aggregate nouns that does not receive enough attention in the literature, particular in semantics, is their morphology.

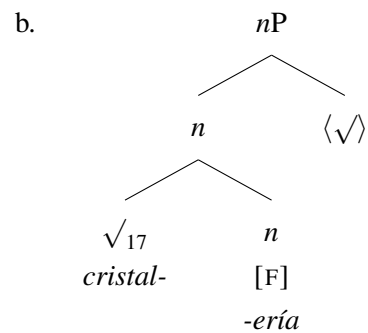
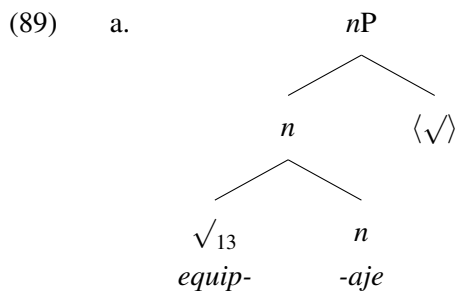
In Spanish, the vast majority of aggregate nouns are polymorphemic. For example, *equipaje* ‘luggage’ decomposes into the root *equip-* ‘equip(ment)’ and the suffix *-aje* ‘-age’, and *cristalería*

‘glassware’ decomposes into the root *cristal-* ‘crystal’ and the suffix *-ería* ‘-ery’. In (87) and (88), I provide six more aggregate nouns of each type.

- (87) a. andami-aje
 ‘scaffolding’
 b. cortin-aje
 ‘drapery’
 c. foll-aje
 ‘foliage’
 d. plum-aje
 ‘plumage’
 e. rop-aje
 ‘robes’
 f. vend-aje
 ‘bandaging’

- (88) a. artill-ería
 ‘artillery’
 b. cubert-ería
 ‘cutlery’
 c. estant-ería
 ‘shelving’
 d. lenc-ería
 ‘lingerie’
 e. mantel-ería
 ‘table linens’
 f. sill-ería
 ‘chair set’

Nouns that end in *-aje* and *-ería* in Spanish are more tolerant of pluralization than their English counterparts would suggest. In fact, one can find numerous results in the Corpus de Referencia del Español Actual for *equipajes* ‘luggages’, *cristalerías* ‘glasswares’, and the plural form of every other noun that is listed in (87) and (88). De Belder (2013) posits that aggregative suffixes in Dutch are realizations of *n*, and the same analysis could be extended to *-aje* and *-ería* in Spanish, as illustrated with *equipaje* and *cristalería* in (89).



To differentiate *equipaje* from *equipo* ‘equipment’ and *cristalería* from *cristal* ‘crystal, glass’, one could include an aggregative feature on *n*, such as [AGG], and propose a lexical entry for $n_{[AGG]}$ that would account for its collectivizing effect.

Several other aggregate nouns are derived from verbs in Spanish, in which case one can appeal to Grimm & Levin’s (2017) view that these nouns encode an associated event that serves to identify the disparate objects in their extensions. For instance, *correo* ‘mail’ contains the stem of the verb *correr* ‘to run’, and the extension of this noun can include letters, packages, postcards, and virtually any other object that has passed or will pass through a postal service. Accordingly, what determines

whether an object is *correo* is not any sort of inherent property but rather its association with an event of mailing. Similarly, the extension of *cambio* ‘change’, a noun that comes from the verb *cambiar* ‘to change’, can include any object that is given in return for an overpayment. The aggregate nouns in (90) are all participial in form and thus have verbal origins as well.

- (90)
- a. calz-ado
shoe-PTCP
‘footwear’
 - b. col-ada
filter-PTCP
‘laundry’
 - c. gan-ado
earn-PTCP
‘livestock’

Any object that is worn on the feet can be considered *calzado* ‘footwear’, just as any object that is washed as if it were a garment can be considered *colada* ‘laundry’. In the case of *ganado* ‘livestock’, the objects in its extension must be living animals, but what counts as *ganado* can vary greatly from one culture or region to another.

Monomorphemic aggregate nouns are rare in Spanish, but there are at least four. The first two, *fauna* ‘fauna’ and *flora* ‘flora’, are scientific terms that are derived from the names of Roman goddesses, so encyclopedic knowledge could be involved in their interpretation. The second two, *ropa* ‘clothing’ and *tráfico* ‘traffic’, are amenable to Grimm & Levin’s (2017) event-based approach since the objects in their extensions are arguably distinguished by their function and not by their inherent properties. After all, any material that is worn on the body can be regarded as *ropa*, and any objects that are in motion on the same pathway can be regarded as *tráfico*, suggesting that both nouns introduce an associated event at logical form.

In summary, none of the form-meaning mismatches that I have discussed in this section compromise the analysis of the mass-count distinction that I have advanced in this chapter. The non-canonical plurality of lexical plurals and pluralia tantum stems from *n* hosting [PL], in which case this feature participates in number agreement and concord but is ignored at logical form. The cumulative and divisive reference of group nouns is likely due to an implicit argument that contains an unpronounced plural light noun, such as PEOPLE or THINGS. Lastly, the unique behaviour of aggregate nouns is less perplexing if one factors in their morphological complexity and the possible semantic ramifications that aggregative suffixes and verbal bases may have.

3.6 Conclusion

Inspired by Borer (2005), I have characterized the mass-count distinction as a structural rather than lexical phenomenon, one where plurality, singularity, and maximality are all contributed by functional projections that dominate nominalized roots. I began this chapter by arguing that *n*Ps denote

properties of unindividuated objects and that DivPs denote properties of individuated objects. Consequently, bare *n*Ps are interpreted as mass, and bare DivPs are interpreted as plural. For a noun to have an exclusively singular reading, it must be preceded by an indefinite determiner, such as *un* ‘a’, which spells out a Q head that bears a [SG] feature, and at logical form, $Q_{[SG]}$ takes the extension of DivP and excludes the sums. To capture the null realization of Div in singular indefinite expressions like *un repollo* ‘a cabbage’, I proposed that $Q_{[SG]}$ triggers a postsyntactic rule of impoverishment that deletes [PL] on Div, eliminating the context of insertion for the plural suffix. As for nonplural definite descriptions like *el repollo* ‘the cabbage’, I showed that they are systematically ambiguous between mass and singular denotation and attributed this effect to the absence of DivP in their structure and the maximality operator that is introduced by definite determiners. I then considered bare nonplural count objects like *gato* ‘cat’ in *tengo gato* ‘I have cats/a cat’ and concurred with Espinal & McNally (2011) that these objects are best analyzed as modifiers of the verb. I closed the chapter by addressing several cases of form-meaning mismatches that relate to lexical plurals, pluralia tantum, group nouns, and aggregate nouns and maintained that their idiosyncrasies do not undermine my general approach to the mass-count distinction.

Chapter 4

The base-generation position of pronouns

This chapter focuses on stressed pronouns in Spanish and the question of where they are generated in the noun phrase. Pronouns can directly precede nouns in many languages, including English (*we chefs*) and Italian (*noi cuochi* ‘we chefs’), which suggests that they originate in D and take a nominal complement. The closest equivalent of this construction in Spanish is *nosotros los cocineros* ‘we the chefs’, where the pronoun occurs in the same prosodic unit as a definite description. Assuming that *nosotros* ‘we’ and *los* ‘the’ cannot both occupy D, one could assign *nosotros* to its own DP and posit that *nosotros los cocineros* constitutes two referential expressions in apposition, similar to *mis amigos los cocineros* ‘my friends the chefs’. One of the many problems with this approach is that it does not explain why the third-person pronoun *ellos* ‘they’ cannot appear in this same environment: **ellos los cocineros* ‘they the chefs’. Instead of analyzing *nosotros los cocineros* in terms of apposition, I endorse Höhn’s (2016, 2017) view that first-person and second-person pronouns in Spanish originate in a dedicated person phrase (PersP or π P) that dominates DP. This proposal captures the cooccurrence of the pronoun and the definite article in *nosotros los cocineros*, but it does not rule out **ellos los cocineros*. To do so, I argue that *ellos* ‘they’ and other third-person pronouns realize all of the major heads in the noun phrase, a configuration that does not leave room for the definite article, a noun, or any other material.

In section 4.1, I introduce the distinction between stressed pronouns and pronominal clitics in Spanish and discuss some of the parallels that have been observed between clitic doubling and object-verb agreement. In section 4.2, I explore, and ultimately reject, the possibility that stressed pronouns are generated in *n* and raise to D in Spanish. In section 4.3, I present numerous differences between *nosotros los cocineros* ‘we the chefs’ and known instances of apposition and conclude that *nosotros los cocineros* must consist of a single extended nominal projection, one that contains two determiners. In section 4.4, I review work that advances a formal separation of person and definiteness in Spanish and defend the claim that only pronominal expressions are specified for person. In section 4.5, I consider the internal structure of third-person pronouns and propose that the *-os* of *ellos* ‘they’ is genuine nominal morphology, like the *-os* of *cocineros* ‘chefs’, and not gender and number concord on D, like the *-os* of *los* ‘the’. In section 4.6, I review the main findings of this chapter and identify some pending issues, all of which are resolved in chapter 5.

4.1 The pronominal system of Spanish

Pronouns in Spanish are generally recognized to have two forms, one that is stressed and appears in argument position and another that is unstressed and must be adjacent to the verbal complex. The etymological paradigm of stressed pronouns is given in Table 4.1.¹

		Singular	Plural
First person	Nominative	<i>yo</i>	<i>nosotros/nosotras</i>
	Oblique	<i>mí</i>	
Second person	Nominative	<i>tú, vos</i>	<i>vosotros/vosotras</i>
	Oblique	<i>ti, vos</i>	
Third person		<i>él/ella/ello</i>	<i>ellos/ellas</i>
Honorific		<i>usted</i>	<i>ustedes</i>

Table 4.1: The stressed pronouns of Spanish

The classification of *usted* ‘you (formal)’ and *ustedes* ‘you all (formal only in Spain)’ is controversial due to their mixed properties. These pronouns refer to the addressee, either alone or as part of a group of individuals, but they both occur with third-person verb forms in (1).

- (1) a. *Usted* {*es* / **eres*} amable.
 you is are.2SG kind
 ‘You (formal) are kind.’
 b. *Ustedes* {*son* / **sois*} amables.
 you.PL are.3PL are.2PL kind
 ‘You all are kind.’

Throughout this chapter, I demonstrate that *usted* and *ustedes* are not subject to the syntactic restrictions that third-person pronouns are, and in section 5.5, I argue on the basis of the Person-Case Constraint that these pronouns originate with a [PARTICIPANT] feature in the syntax.

The etymological paradigm of unstressed pronouns, or clitics, follows in Table 4.2, where the third-person accusative forms are the only ones that inflect for gender.²

		Singular	Plural
First person		<i>me</i>	<i>nos</i>
Second person		<i>te</i>	<i>os</i>
Third person	Accusative	<i>lo/la</i>	<i>los/las</i>
	Dative	<i>le</i>	<i>les</i>
	Reflexive		<i>se</i>

Table 4.2: The pronominal clitics of Spanish

¹The singular pronoun *vos* ‘you (informal)’ is used in many regions of Latin America, whereas the plural pronouns *vosotros/vosotras* ‘you all (informal)’ are exclusive to Spain (Páez Urdaneta 1981, Carricaburo 1997, Fontanella de Weinberg 1999, Benavides 2003). The oblique forms *mí* ‘me’ and *ti* ‘you (informal)’ occur after prepositions and after *a* in its guises as a differential object marker and a dative case marker.

²I abstract away from the phenomena known as *leísmo*, *laísmo*, and *loísmo* since this chapter is primarily about stressed pronouns (Fernández-Ordóñez 1993, Klein-Andreu 1993, Lapesa 1993, Bleam 1999a, Fernández-Ordóñez 1999, Heap 2002, Gómez Seibane 2012).

A comparison of Table 4.1 and Table 4.2 reveals two more differences between stressed pronouns and clitics. First, there are no honorific clitics, and second, case distinctions in the stressed series are limited to singular first-person and second-person forms, while case distinctions in the unstressed series are limited to third-person forms.

Subject pronouns can be omitted in Spanish, as indicated in (2), hence its classification as a pro-drop language (Chomsky 1981, Jaeggli 1982, Montalbetti 1984, Jaeggli & Safir 1989, Fernández Soriano 1999, Luján 1999, Suñer 1999, Camacho 2013).

- (2) {Ellos / Ø} viven en Nueva Inglaterra.
 they live.3PL in New England
 ‘They live in New England.’

If the referent is animate, as in (3), pro-drop is typically optional, but if the referent is inanimate, as in (4), pro-drop is obligatory (Strozer 1976, Luján 1980, Jaeggli 1982, Schroten 1992, Fernández Soriano 1993, Cardinaletti & Starke 1999, Corver & Delfitto 1999, Fernández Soriano 1999).

- | | |
|--|---|
| <p>(3) a. ¿Dónde está Julia?
 where is Julia
 ‘Where is Julia?’</p> <p>b. (Ella) está en la cocina.
 she is in the kitchen
 ‘She is in the kitchen.’</p> | <p>(4) a. ¿Dónde está la escoba?
 where is the broom
 ‘Where is the broom?’</p> <p>b. (#Ella) está en la cocina.
 it is in the kitchen
 ‘It is in the kitchen.’</p> |
|--|---|

Pro-drop is possible in object position as well, as shown in (5a) with *él* ‘him’ as a direct object and in (5b) with *él* as an indirect object (Roldán 1971, Strozer 1976, Jaeggli 1982, Montalbetti 1984, Torrego 1995, Fernández Soriano 1999, Luján 1999, Suñer 1999).

- (5) a. Julia lo llamó {a él / Ø}.
 Julia 3SG.ACC called.3SG DOM him
 ‘Julia called him.’
- b. Julia le escribió una carta {a él / Ø}.
 Julia 3SG.DAT wrote.3SG a letter DAT him
 ‘Julia wrote him a letter.’

These examples also serve to illustrate clitic doubling, or the use of a clitic together with a nominal or pronominal expression in argument position: *Julia lo llamó a él* ‘Julia called him’.

In all varieties of Spanish, clitic doubling is obligatory with pronominal objects, as in (6), and either obligatory or optional with nonpronominal indirect objects, as in (7), depending on the construction (Roldán 1971; Strozer 1976; Silva-Corvalán 1981; Jaeggli 1982, 1986; Suñer 1988, 1989; F. Roca 1992; Fernández Soriano 1993; Demonte 1995; F. Roca 1996; Parodi 1998; Fernández Soriano 1999; Luján 1999; Suñer 1999; Bleam 2005; Di Tullio & Zdrojewski 2006; Leonetti 2008; Gabriel & Rinke 2010; Gómez Seibane 2012; Di Tullio et al. 2019; Fischer et al. 2019).³

³See section 2.4 for extensive discussion of the dative alternation in Spanish.

- (6) a. Jacques *(la) reconoció a ella.
 Jacques 3SG.F.ACC recognized.3SG DOM her
 ‘Jacques recognized her.’
- b. Jacques *(le) habló a ella.
 Jacques 3SG.DAT spoke.3SG to her
 ‘Jacques spoke to her.’
- (7) a. Jacques *(le) compró un libro a Julia.
 Jacques 3SG.DAT bought.3SG a book DAT Julia
 ‘Jacques bought Julia a book.’
- b. Jacques le mandó un libro a Julia.
 Jacques 3SG.DAT sent.3SG a book DAT Julia
 ‘Jacques sent Julia a book.’
- c. Jacques mandó un libro a Julia.
 Jacques sent.3SG a book to Julia
 ‘Jacques sent a book to Julia.’

In Rioplatense Spanish, nonpronominal direct objects that are definite or specific can also be doubled by a clitic (Roldán 1971; Strozer 1976; Silva-Corvalán 1981; Jaeggli 1982, 1986; Suñer 1988, 1989; F. Roca 1992; Fernández Soriano 1993; F. Roca 1996; Parodi 1998; Fernández Soriano 1999; Cuervo 2003a; Di Tullio & Zdrojewski 2006; Leonetti 2008; Gómez Seibane 2012; Di Tullio et al. 2019; Fischer et al. 2019). This possibility is shown in (8a) with a proper name and in (8b) with a differentially marked indefinite noun.

- (8) a. Jacques (la) reconoció a Julia.
 Jacques 3SG.F.ACC recognized.3SG DOM Julia
 ‘Jacques recognized Julia.’
- b. Jacques (la) reconoció a una colega.
 Jacques 3SG.F.ACC recognized.3SG DOM a colleague
 ‘Jacques recognized a (female) colleague.’

In Andean Spanish, even inanimate direct objects permit clitic doubling, though not necessarily by a form that reflects the same gender and number features (Pozzi-Escot 1972, Godenzzi 1986, Luján 1987, Caravedo 1996, Escobar 2000, Klee & Caravedo 2005, Zdrojewski & Sánchez 2014, Mayer 2017). As Luján (1987: 115) observes regarding the data in (9), the default third-person clitic *lo* ‘it’ is used instead of what would be the agreeing form in brackets.

- (9) a. Lo [La] botó su pistola.
 3SG.ACC 3SG.F.ACC hurled.3SG his pistol
 ‘He hurled his pistol.’
- b. Se lo [los] llevó caramelos.
 3 3SG.ACC 3PL.ACC took.3SG sweets
 ‘She took away sweets.’

To summarize the discussion so far, not only is clitic doubling required with pronominal arguments in Spanish, but it is also pervasive with nominal arguments in some varieties.

In light of the myriad ways in which clitics can appear together with direct and indirect objects in Spanish, it has been suggested that they are exponents of object-verb agreement (Lenz 1920; Heger 1966; Llorente & Mondéjar 1974; Givón 1976; Silva-Corvalán 1981; Wonder 1990). Among generativists, a major point of debate is the base-generation position of clitics. For some authors, clitics originate as part of the verb itself, usually as a head that adjoins to V or *v* depending on the theoretical framework (Strozer 1976; Rivas 1977; Jaeggli 1982; Borer 1984; Jaeggli 1986; Suñer 1988, 1999; Baker & Kramer 2018). For other authors, clitics are a type of inflectional morphology that spell out a functional projection in the verbal domain, and many consider accusative and dative clitics to have distinct structural representations (Saltarelli 1990, F. Roca 1992, Franco 1993, Torrego 1995, F. Roca 1996, Uriagereka 1995, Sportiche 1996, Uriagereka 1996, Parodi 1998, Torrego 1998, Bleam 1999a, Franco 2000, Cuervo 2003a, Fernández-Rubiera 2009, Ormazabal & Romero 2013; Arregi & Nevins 2018). What is common to all accounts is the notion that clitics are generated in a different position than stressed pronouns.

The parallel between subject-verb agreement and clitic doubling is clear from the annotated examples in (10), where I represent the obligatory verb endings and clitics as probes and the optional stressed pronouns in argument position as their goals.

- (10) a. (Él) *(la) _____ reconoci-*(ó) (a ella).
 he 3SG.F.ACC recognize-PRET.3SG DOM her
 'He recognized her.'
- b. (Él) *(le) _____ habl-*(ó) (a ella).
 he 3SG.DAT speak-PRET.3SG DAT her
 'He spoke to her.'

There is, moreover, typological support for the perspective that clitics are undergoing a process of grammaticalization in Spanish. Building on insights by Moravcsik (1974), Givón (1976: 152), argues that languages develop agreement systems in accordance with the four hierarchies in (11).⁴

- (11) a. Human > Nonhuman
b. Definite > Indefinite
c. More involved participant > Less involved participant (Agent > Dative > Accusative)
d. First person > Second Person > Third person

⁴Similar views are expressed by Hawkinson & Hyman (1974), Duranti & Ochs (1979), and Franco (1993, 2000). The hierarchy in (11b) may explain the following contrast that Ormazabal & Romero (2013: 306) identify. In general, *todos* 'all' must be doubled by clitic when it occurs in object position, as in **(los) vimos todos* 'we saw them all'. When *todos* 'refers to an unspecified set of objects', however, the clitic can be omitted, as in *tengo el primero, el segundo, el tercero...tengo todos* 'I have the first one, the second one, the third one...I have (them) all'.

The first three hierarchies predict the evolution of clitic doubling in Spanish as described by Fischer et al. (2019: 60) in Figure 4.1, which I have minimally adapted for clarity and reasons of space.

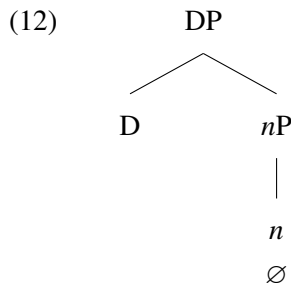
Stage	Status	Language or variety
Stage I	No clitic doubling	Latin
Stage II	Optional doubling with pronominal objects	Old Spanish
Stage III	Obligatory doubling with pronominal objects, optional doubling with [+ANIM, +DEF, +SPEC] indirect nominal objects	(Early) Modern Spanish
Stage IV	Obligatory doubling with pronominal objects, obligatory doubling with indirect nominal objects, optional doubling with [+ANIM, +DEF, +SPEC] direct nominal objects	Judeo-Spanish, Rioplatense Spanish
Stage V	Generalized doubling (with all objects, even inanimates)	Andean Spanish

Figure 4.1: Fischer et al.'s (2019) clitic doubling cycle

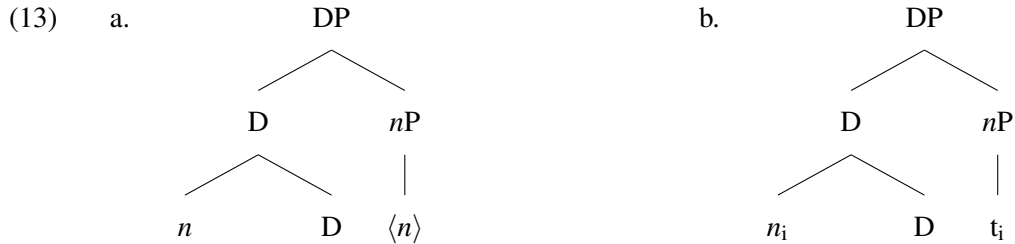
Concretely, these hierarchies are reflected in the emergence of clitic doubling with pronominal objects in Old Spanish, with definite and specific animate indirect objects in (Early) Modern Spanish, and with definite and specific animate direct objects in Judeo-Spanish and Rioplatense Spanish. As for the fourth hierarchy, the question is whether clitic doubling is attested with first-person or second-person pronouns at an earlier stage than it is with third-person pronouns, a topic that I leave for future study. The essential point is that pronominal clitics must be analyzed separately from stressed pronouns, which are the focus of this chapter.

4.2 Stressed pronouns and prenominal elements

A core issue in work on pronouns is the source of their nominality, or what makes them pro-nouns (Zwarts 1994, Panagiotidis 2002). The most common solution is to represent those that occur in argument position as determiners that combine with an elided nominal constituent, as in (12), though some accounts are more explicit than others regarding the formal details (Postal 1966; Evans 1977; Bennett 1978; Cooper 1979; Evans 1980; Heim 1990; Uriagereka 1995; Noguchi 1997; Cardinaletti & Starke 1999; Corver & Delfitto 1999; Elbourne 2001; Déchaine & Wiltschko 2002; Giusti 2002; Panagiotidis 2002, 2003; Elbourne 2005; Roehrs 2005; Sauerland 2007, 2008b; Elbourne 2013; Höhn 2016, 2017; Patel-Grosz & Grosz 2017; Höhn 2020; Dékány 2021).



A less common solution is to represent pronouns as nominal elements that raise to D, as in (13), in which case there is either an unpronounced copy of *n* or a trace in this position (Cardinaletti 1994; Progovac 1998; Rauh 2002, 2004; Bernstein 2008b; Longobardi 2008; Conrod 2019).

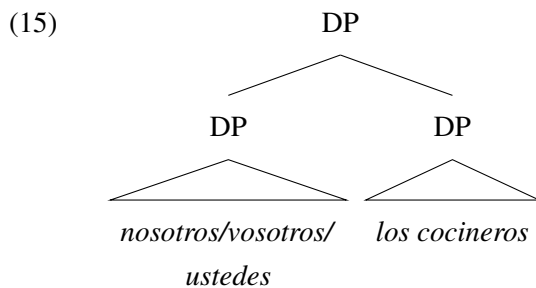


A third possibility would be to follow Zwarts (1994) in assigning first-person and second-person pronouns to the structure in (12) and third-person pronouns to one of the structures in (13) to explain the contrast between *we/you chefs* and **they chefs* in languages like English. If *they* originates in *n*, no other material could occupy this head even if the pronoun moves to D.

The situation is more complex in Spanish because the counterparts of *we chefs* and *you chefs* in (14) require the definite article to intervene between the pronoun and the noun (Villa-García 2010; Choi 2013, 2014a,b; Höhn 2016, 2017, 2020).

- (14) a. Nosotros *(los) cocineros somos artistas y científicos.
 we the chefs are.1PL artists and scientists
 ‘We chefs are artists and scientists.’
- b. Vosotros *(los) cocineros sois artistas y científicos.
 you.PL the chefs are.2PL artists and scientists
 ‘You (informal) chefs are artists and scientists.’
- c. Ustedes *(los) cocineros son artistas y científicos.
 you.PL the chefs are.3PL artists and scientists
 ‘You chefs are artists and scientists.’

To capture the ungrammaticality of **nosotros/vosotros/ustedes cocineros* ‘we/you (informal)/you chefs’, one could argue that all pronouns in Spanish are generated in *n* and raise to D. If *n* is never available in pronominal structures in this language, it could be that apposition is the only way for pronouns to occur together with nouns, as shown in (15).



In this section, I consider whether there is any evidence that pronouns originate in *n* in Spanish, and in section 4.3, I explore the possibility that constructions like *nosotros los cocineros* ‘we the chefs’ consist of two DPs in apposition.

If pronouns in Spanish were to begin their grammatical lives as nouns, one might expect to find at least some distributional overlap between these two types of expressions, such as pronouns that are preceded by a determiner. The only recurring example in the literature involves the combination of the indefinite article and a third-person pronoun to convey the notion of maleness or femaleness, as in *un él* ‘a he’ and *una ella* ‘a she’ (Cardinaletti 1994, Noguchi 1997, Siewierska 2004, Conrod 2019). These collocations are rather odd in out-of-the-blue utterances, so their relevance to a theory of pronouns is debatable. To inform someone about the sex of a domestic animal, for instance, the sentence in (16a) would be more natural than that in (16b).

- (16) a. Su mascota es {macho / hembra}.
 their pet is male female
 ‘Their pet is male/female.’
 b. #Su mascota es {un él / una ella}.
 their pet is a he a she
 ‘Their pet is a he/she.’

Where *un él* and *una ella* tend to be used is in exchanges of the sort in (17), where the first speaker uses a feminine pronoun to refer to a male cat and is corrected by the second speaker.

- (17) Context: the first speaker is pointing at the second speaker’s cat.
 a. Ella es muy hermosa.
 she is very beautiful
 ‘She is very beautiful.’
 b. Es un él, no una ella.
 is a he not a she
 ‘It is a he, not a she.’

Further indication that *macho* ‘male’ and *hembra* ‘female’ are not readily substituted by third-person pronouns lies in the indefinite environments in (18), the definite environments in (19), and the plural environments in (20).

- (18) a. Cada {macho / *él} lleva un collar amarillo.
 every male he wears a collar yellow.
 ‘Every male is wearing a yellow collar.’
 b. Ninguna {hembra / *ella} está lista para ser adoptada.
 no female she is ready for be adopted
 ‘No female is ready to be adopted.’
 (19) a. El {macho / *él} lleva un collar amarillo.
 the male he wears a collar yellow
 ‘The male is wearing a yellow collar.’

- b. Esta {hembra / *ella} no está lista para ser adoptada.
 this female she not is ready for be adopted
 ‘This female is not ready to be adopted.’
- (20) En esta camada hay dos {machos / *ellos} y tres {hembras / *ellas}.
 in this litter has two males them and three females them.F
 ‘In this litter, there are two males and three females.’

Based on the facts in (16) through (20), it seems more accurate to characterize *un él* and *una ella* as instances of pronouns that have been converted into nouns for the purpose of expressing maleness or femaleness (Noguchi 1997, Siewierska 2004, Bernstein 2008a, Cowper & Hall 2009, De Belder & van Craenenbroeck 2015).⁵ After all, functional items are frequently repurposed as lexical items in Spanish and other languages: *el quién*, *el dónde* y *el cómo* ‘the who, the where, and the how’. For this reason, I conclude that *un él* and *una ella* are uninformative when it comes to the base-generation position of pronouns.

Just as pronouns do not productively follow determiners in Spanish, they do not productively follow modifiers either. At the time of writing, there are no results in the Corpus de Referencia del Español Actual (Reference Corpus of Contemporary Spanish) for any sequence of the evaluative adjectives *amable* ‘kind’, *estúpido* ‘stupid’, *inteligente* ‘smart’, *listo* ‘clever’, *perezoso* ‘lazy’, or *tonto* ‘silly’ and a pronoun.⁶ There is one result for *afortunado yo* ‘lucky me’, which is provided in (21a), and four results for *pobre yo* ‘poor me’, one of which is provided in (21b).

- (21) a. Pero apreciaban —afortunado yo— que la experiencia ofrece siempre
 but appreciated.IMPF.3PL lucky I that the experience offers always
 algo que contar a los demás.
 something that tell to the others
 ‘But they appreciated —lucky me— that experience always offers something that one
 can tell to other people.’ (La rusa by Juan Luis Cebrián, 1986, page 40)
- b. Me quiere, eso no lo dudo. Pero me dejará de querer, de eso
 1SG loves that not 3SG.ACC doubt.1SG but 1SG leave.FUT.3SG of love of that
 estoy seguro. Entonces, ¡pobre yo!
 am sure so poor I
 ‘She loves me, I do not doubt it. But she will stop loving me, of that I am sure. So,
 poor me!’ (El caso 117.720 by Carmen Naranjo, 1987, page 118)

As these excerpts suggest, and as the data in (22) confirm, adjective-pronoun constructions cannot serve as arguments regardless of whether they appear with a first-person or third-person verb form or are introduced by the definite article.

⁵In English, one also finds compounds like *he-goat* and *she-wolf* in which the pronouns *he* and *she* perform a similar function (Déchaine & Wiltschko 2002, Cowper & Hall 2009, Conrod 2019).

⁶All possible forms of each adjective were considered along with the nominative and oblique forms of pronouns that inflect for case: *tonto yo/tonta yo/tonto mí/tonta mí* ‘silly me’. It should be noted that Sigmund Freud’s terms *das Ich* ‘the ego’ and *das Über-Ich* ‘the superego’ are often translated as *el yo* and *el superyó*, respectively, where *yo* ‘ego’ is unambiguously a noun because it has descriptive content. I have therefore excluded all results where this usage of *yo* is modified in some way, such as *el otro yo* ‘the alter ego’.

- (22) a. *Pobre yo {estoy / está} enfermo de nuevo.
 poor I am is sick of new
 Intended: ‘Poor me is sick again.’
- b. *El pobre yo {estoy / está} enfermo de nuevo.
 the poor I am is sick of new
 Intended: ‘Poor me is sick again.’

The collocation *pobre de mí* ‘poor me’, where the preposition *de* ‘of’ triggers the use of the oblique form of the pronoun, has the same distribution as *pobre yo*. As shown in (23), *pobre de mí* can occur in vocative position but not in argument position.

- (23) a. Pobre de mí, estoy enfermo de nuevo.
 poor of me am sick of new
 ‘Poor me, I am sick again.’
- b. *Pobre de mí {estoy / está} enfermo de nuevo.
 poor of me am is sick of new
 Intended: ‘Poor me is sick again.’
- c. *El pobre de mí {estoy / está} enfermo de nuevo.
 the poor of me am is sick of new
 Intended: ‘Poor me is sick again.’

It is, on the other hand, possible to use modified first-person and second-person pronouns for rhetorical effect, as in (24), but only with a third-person verb form.⁷

- (24) a. El nuevo yo {es / *soy} más comprensivo.
 the new I is am more understanding
 ‘The new me is more understanding.’
- b. La tú de esa época {parecía / *parecías} menos preocupada.
 the you of that time seemed.IMPF seemed.IMPF.2SG less worried
 ‘The you (informal) of that time seemed less worried.’

The lack of canonical agreement with *yo* ‘I’ and *tú* ‘you (informal)’ in these examples can be attributed to pronoun-to-noun conversion, a possibility that is supported by the following contrast, where the unmodified pronouns in (25a) inflect for case and the modified pronouns in (25b) do not.

- (25) a. No saben qué pensar de {mí / ti}.
 not know.3PL what think of me you
 ‘They do not know what to think of me/you (informal).’
- b. No saben qué pensar del nuevo {yo / tú}.
 not know.3PL what think of.the new I you
 ‘They do not know what to think of the new me/you (informal).’

⁷In an *n*-to-D approach, one could explain the agreement facts in (24) by arguing that pronouns do not have person features in *n* and only acquire them when they raise to D. The problem is that person features could then not be used to differentiate *yo* ‘I’ and *tú* ‘you (informal)’ as vocabulary items when they remain in *n*. One could, in principle, resort to roots for this purpose, but this solution would be conceptually similar to one that posits conversion.

In summary, the few adjective-pronoun constructions that are attested in Spanish are either confined to vocative position or behave syntactically as if they were nouns, suggesting that they are occurrences of functional items that are being used as lexical items, the same conclusion that I drew regarding *un él* ‘a he’ and *una ella* ‘a she’. There is, consequently, no empirical motivation for the claim that pronouns originate in the lexical layer of the noun phrase.

4.3 Pronoun-noun constructions

Having ruled out the possibility that pronouns are generated in *n* in Spanish, I now investigate the pronoun-noun constructions in (26), where each pronoun appears in the same prosodic unit as a definite description (Villa-García 2010; Choi 2013, 2014a,b; Höhn 2016, 2017, 2020).⁸

- (26) a. Nosotros los cocineros tenemos muchas destrezas.
 we the chefs have.1PL many skills
 ‘We chefs have many skills.’
 b. Vosotros los cocineros tenéis muchas destrezas.
 you.PL the chefs have.2PL many skills
 ‘You (informal) chefs have many skills.’
 c. Ustedes los cocineros tienen muchas destrezas.
 you.PL the chefs have.3PL many skills
 ‘You chefs have many skills.’

The first question to explore is whether these constructions are fundamentally different from those in (27), where the intervening prosodic break is indicative of apposition.

- (27) a. Nosotros, los cocineros, tenemos muchas destrezas.
 we the chefs have.1PL many skills
 ‘We, the chefs, have many skills.’
 b. Vosotros, los cocineros, tenéis muchas destrezas.
 you.PL the chefs have.2PL many skills
 ‘You all (informal), the chefs, have many skills.’
 c. Ustedes, los cocineros, tienen muchas destrezas.
 you.PL the chefs have.3PL many skills
 ‘You all, the chefs, have many skills.’

One reason to suspect that the prosodic break is structurally relevant is that it enables the inclusion of the universal quantifier *todos* ‘all’, which would otherwise be ungrammatical, as shown in (28).

- (28) a. Nosotros, todos los cocineros, tenemos muchas destrezas.
 we all the chefs have.1PL many skills
 ‘We, all the chefs, have many skills.’

⁸I use the term “pronoun-noun construction” in place of the more precise term “pronoun-article-noun construction” solely for convenience.

- b. *Nosotros todos los cocineros tenemos muchas destrezas.
 we all the chefs have.1PL many skills
 Literal: ‘We all the chefs have many skills.’

Another way in which prosody and quantification expose structural differences relates to scope (Pesetsky 1978). In (29a), the partitive expression *algunos de* ‘some of’ only takes scope over the pronoun, but in (29b), *algunos de* takes scope over the pronoun and the definite description.

- (29) a. [Algunos de [nosotros]], los cocineros, tenemos muchas destrezas.
 some of us the chefs have.1PL many skills
 ‘Some of us, the chefs, have many skills.’
 b. [Algunos de [nosotros los cocineros]] tenemos muchas destrezas.
 some of us the chefs have.1PL many skills
 ‘Some of us chefs have many skills.’

The first example conveys that only some of the referents of *nosotros* ‘we, us’ are chefs, while the second conveys that all of the referents are chefs, a contrast that has repercussions for the more complex partitive expression *algunos de...otros de* ‘some of...others of’ (Pesetsky 1978). For instance, the sentence in (30) is coherent because the scope of both quantifiers is limited to the pronoun itself, yielding an interpretation where some of the referents of *nosotros* are chefs and others are servers.

- (30) [Algunos de [nosotros]], los cocineros, estamos preocupados por [otros de [nosotros]],
 some of us the chefs are.1PL concerned for others of us
 los meseros.
 the servers
 ‘Some of us, the chefs, are concerned about others of us, the servers.’

The sentence in (31), however, is incoherent because the scope of both quantifiers extends to the definite descriptions, implying that all of the referents of *nosotros* are chefs as well as servers.

- (31) #[Algunos de [nosotros los cocineros]] estamos preocupados por [otros de [nosotros los
 some of us the chefs are.1PL concerned for others of us the
 meseros]].
 servers
 Approximate: ‘Some of us chefs are concerned about others of us servers.’

Based on these syntactic and semantic factors, it is apparent that *nosotros los cocineros* ‘we the chefs’ must be analyzed differently than *nosotros, los cocineros* ‘we, the chefs’.

The next possibility to consider is whether pronoun-noun constructions in Spanish pattern like cases of apposition that lack a prosodic break, such as *mis amigos los cocineros* ‘my friends the chefs’. As mentioned in section 1.2, one informative diagnostic involves replacing the definite description component with another type of constituent (Lawrenz 1993; Höhn 2016, 2017, 2020). As illustrated in (32), *mis amigos* ‘my friends’ can immediately precede *los cocineros* ‘the chefs’ and also the coordinated names *Julia y Jacques* ‘Julia and Jacques’.

- (32) a. Mis amigos los cocineros trabajan muchas horas.
my friends the chefs work.3PL many hours
'My friends the chefs work many hours.'
- b. Mis amigos Julia y Jacques trabajan muchas horas.
my friends Julia and Jacques work.3PL many hours
'My friends Julia and Jacques work many hours.'

In (33), on the other hand, *ustedes* ‘you all’ can only precede *los cocineros*, revealing that the syntactic relation between the pronoun and the definite description is not one of apposition.

- (33) a. Ustedes los cocineros trabajan muchas horas.
you.PL the chefs work.3PL many hours
'You chefs work many hours.'
- b. *Ustedes Julia y Jacques trabajan muchas horas.
you.PL Julia and Jacques work.3PL many hours
Literal: 'You Julia and Jacques work many hours.'

This finding is bolstered by the facts in (34), where *mis amigos* can be directly followed by coordinated singular definite descriptions but *ustedes* cannot.

- (34) a. Mis amigos el cocinero y el mesero trabajan muchas horas.
my friends the chef and the server work.3PL many hours
'My friends the chef and the server work many hours.'
- b. *Ustedes el cocinero y el mesero trabajan muchas horas.
you.PL the chef and the server work.3PL many hours
Literal: 'You the chef and the server work many hours.'

If *ustedes los cocineros* ‘you the chefs’ has the structure in (35a), or any other type of appositive structure, one would expect that in (35b) to be grammatical as well.

- (35) a.
 DP
 / \
 DP DP
 / \ / \
 / \ / \
 usted los
 cocineros
- b.
 *DP
 / \
 DP CoordP
 / \ / \
 / \ / \
 usted Julia y Jacques/
 el cocinero y el mesero

Still, it is possible for pronouns to be followed by coordinated plural definite descriptions, as in (36), a phenomenon that has not been previously identified in the literature.

- (36) Ustedes los cocineros y los meseros trabajan muchas horas.
you.PL the chefs and the servers work.3PL many hours
'You chefs and servers work many hours.'

It may seem surprising that *ustedes los cocineros y los meseros* ‘you the chefs and the servers’ in (36) is grammatical when **ustedes el cocinero y el mesero* ‘you the chef and the server’ in (34b) is not, but there is an explanation for this contrast.

The universal quantifier *todos* ‘all’, which must be followed by a definite description in Spanish, exhibits the same behaviour as *ustedes* ‘you all’ in (37).

- (37) a. Todos los cocineros y los meseros trabajan muchas horas.
 all the chefs and the servers work.3PL many hours
 ‘All the chefs and the servers work many hours.’
 b. *Todos el cocinero y el mesero trabajan muchas horas.
 all the chef and the server work.3PL many hours
 Literal: ‘All the chef and the server work many hours.’

Another way in which *todos* and *ustedes* pattern together is that both determiners can be repeated before *los meseros* ‘the servers’ in (38).

- (38) a. Todos los cocineros y todos los meseros trabajan muchas horas.
 all the chefs and all the servers work.3PL many hours
 ‘All the chefs and all the servers work many hours.’
 b. Ustedes los cocineros y ustedes los meseros trabajan muchas horas.
 you.PL the chefs and you.PL the servers work.3PL many hours
 ‘You chefs and you servers work many hours.’

The subjects in (38) can be bracketed as in (39), where *todos los cocineros* ‘all the chefs’ and *ustedes los cocineros* ‘you the chefs’ constitute the first conjunct, but not as in (40), where *todos* and *ustedes* are in apposition to a set of coordinated expressions.⁹

- (39) a. [CoordP [QP todos los cocineros] [y [QP todos los meseros]]]
 b. [CoordP [π P ustedes los cocineros] [y [π P ustedes los meseros]]]
 (40) a. *[[QP todos] [CoordP [DP los cocineros] [y [QP todos los meseros]]]]
 b. *[[π P ustedes] [CoordP [DP los cocineros] [y [π P ustedes los meseros]]]]

As for *todos los cocineros y los meseros* ‘all the chefs and the servers’ and *ustedes los cocineros y los meseros* ‘you the chefs and the servers’, one could bracket them as in (41), where the second instance of *todos* and *ustedes* is unpronounced, or as in (42), where the two definite descriptions are first coordinated and then embedded under a single occurrence of *todos* or *ustedes*.

- (41) a. [CoordP [QP todos los cocineros] [y [QP $\emptyset$ los meseros]]]
 b. [CoordP [π P ustedes los cocineros] [y [π P $\emptyset$ los meseros]]]
 (42) a. [QP todos [CoordP [DP los cocineros] [y [DP los meseros]]]]
 b. [DP ustedes [CoordP [DP los cocineros] [y [DP los meseros]]]]

⁹In anticipation of the proposal that I develop in section 4.4, I use π P for constituents that are headed by *ustedes*.

What does not seem tenable is to assign *todos los cocineros y los meseros* the structure in (43), where the determiner *todos* and the coordinated definite descriptions *los cocineros y los meseros* ‘the chefs and the servers’ are in apposition.

- (43) *[[QP todos] [CoordP [DP los cocineros] [y [DP los meseros]]]]

In light of the other similarities between *todos* and *ustedes*, I rule out the bracketing of *ustedes los cocineros y los meseros* in (44) on the same grounds.

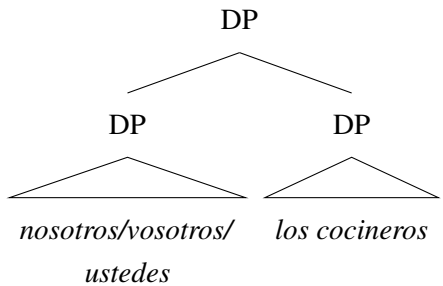
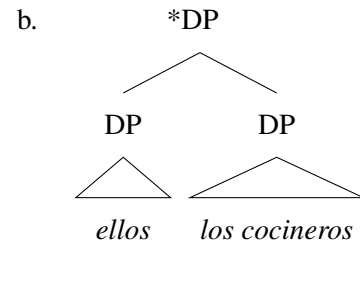
- (44) *[[_πP ustedes] [CoordP [DP los cocineros] [y [DP los meseros]]]]

Independently of whether one represents *ustedes los cocineros y los meseros* as in (41b) or (42b), the pronoun belongs to the same extended nominal projection as the first definite description, so neither configuration can be taken as support for apposition.

A third reason to discard apposition as a viable analysis of pronoun-noun constructions in Spanish relates to the paradigmatic gap in (45).

- (45) a. Julia quiere reunirse con {nosotros / vosotros / ustedes / *ellos} los cocineros.
 Julia wants meet.3 with us you.PL you.PL them the chefs
 ‘Julia wants to meet with us/you (informal)/you chefs.’
 b. Julia quiere reunirse con {nosotras / vosotras / ustedes / *ellas} las cocineras.
 Julia wants meet.3 with us.F you.F.PL you.PL them.F the chefs
 ‘Julia wants to meet with us/you (informal)/you chefs.’

This constraint is known as the “third-person gap” and is found in many languages, including those like English that do not use the definite article in pronoun-noun constructions: *we/you/*they chefs* (Postal 1966; Abney 1987; Cardinaletti 1994; Lyons 1999; Déchaine & Wiltschko 2002; Giusti 2002; Panagiotidis 2002; Roehrs 2005; Bernstein 2008a,b; Höhn 2016, 2017; Ackema & Neeleman 2018; Höhn 2020). If *nosotros/vosotros/ustedes los cocineros* ‘we/you (informal)/you the chefs’ constitute two DPs in apposition, as in (46a), the question is why **ellos los cocineros* ‘they the chefs’ cannot have this same structure, as in (46b).

- (46) a.  b. 

The third-person gap extends to singular pronoun-noun constructions in Spanish, which are less common but still grammatical. One context in which these expressions can be used felicitously is

to differentiate two individuals with respect to the roles that they perform, as shown in (47), where the third-person pronouns are the only forms that are disallowed.¹⁰

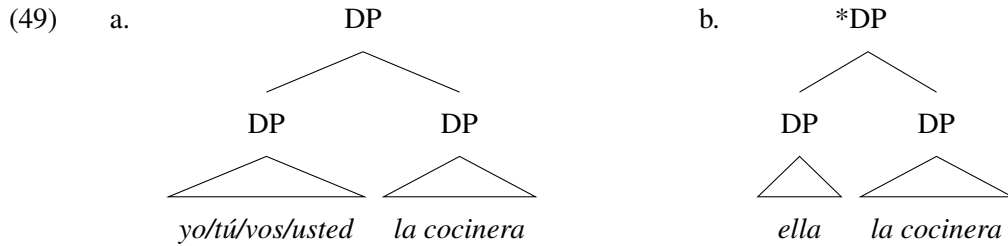
- (47) a. Yo el cocinero no tengo las mismas responsabilidades que {tú / vos / usted /
I the chef not have.1SG the same responsibilities than you you you
*él} el mesero, pero los dos desempeñamos un papel importante.
he the server but the two perform.1PL a role important
Literal: 'I the chef do not have the same responsibilities as you (informal)/you (informal)/you (formal) the server, but we both perform an important role.'
- b. Yo la cocinera no tengo las mismas responsabilidades que {tú / vos / usted /
I the chef not have.1SG the same responsibilities than you you you
*ella} la mesera, pero las dos desempeñamos un papel importante.
she the server but the two perform.1PL a role important
Literal: 'I the chef do not have the same responsibilities as you (informal)/you (informal)/you (formal) the server, but we both perform an important role.'

This observation is echoed by the data in (48), where the singular pronouns are followed by an epithet in the form of a definite description, one that I assume contains a covert noun: *el Ø distraído/la Ø distraída* 'the distracted one'.

- (48) a. Yo {el distraído / la distraída} perdí el tren.
I the distracted the distracted missed.1SG the train
Literal: 'I the distracted one missed the train.'
- b. Tú {el distraído / la distraída} perdiste el tren.
you the distracted the distracted missed.2SG the train
Literal: 'You (informal) the distracted one missed the train.'
- c. Vos {el distraído / la distraída} perdiste el tren.
you the distracted the distracted missed.2SG the train
Literal: 'You (informal) the distracted one missed the train.'
- d. Usted {el distraído / la distraída} perdió el tren.
you the distracted the distracted missed.3SG the train
Literal: 'You (formal) the distracted one missed the train.'
- e. *Él el distraído perdió el tren.
he the distracted missed.3SG the train
Literal: 'He the distracted one missed the train.'
- f. *Ella la distraída perdió el tren.
she the distracted missed.3SG the train
Literal: 'She the distracted one missed the train.'

As in the case of plural pronoun-noun constructions, the appositional structure of *yo/tú/vos/usted la cocinera* 'I/you (informal)/you (informal)/you (formal) the chef' in (49a) is undermined by the impossibility of *ella* 'she' occupying the first DP in (49b).

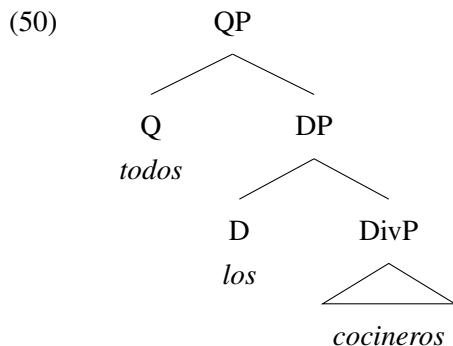
¹⁰I thank María Cristina Cuervo for calling my attention to the possibilities in (47) and (48).



Instead, it seems that pronoun-noun constructions in Spanish constitute a single extended nominal projection with two determiners, similar to *todos los cocineros* ‘all the chefs’. The issue is how to reconcile this conclusion with the view that D is the locus of person features, according to which pronouns should be mutually exclusive with the definite article (Ritter 1995; Lyons 1999; Panagiotidis 2002; Bernstein 2008a,b; Longobardi 2008; Danon 2011; Richards 2015). In the next section, I discuss work that addresses this problem by assigning person features to their own functional projection in the noun phrase.

4.4 The locus and representation of person

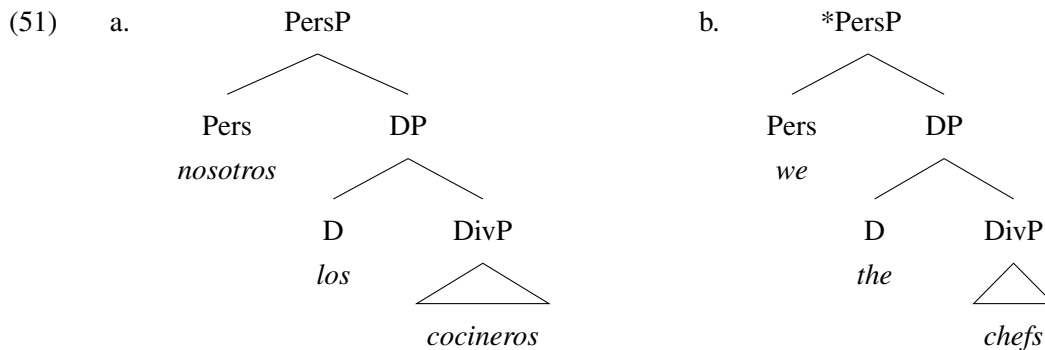
In section 4.3, I drew a comparison between pronoun-noun constructions like *nosotros los cocineros* ‘we the chefs’ and quantificational expressions like *todos los cocineros* ‘all the chefs’, both of which consist of a determiner that is followed by a definite description without an intervening prosodic break. In crosslinguistic research on quantifier float, many authors represent the equivalent of *todos los cocineros* as in (50), where *todos* ‘all’ takes *los cocineros* ‘the chefs’ as its complement (Shlonsky 1991, Merchant 1996, Benmamoun 1999, McCloskey 2000, Cirillo 2009, Lacerda 2016).



This structure could serve as a model for the analysis of *nosotros los cocineros*, though *nosotros* ‘we’ is not of category Q. Since pronouns are widely regarded to be a type of definite determiner, one could posit two DP layers in Spanish to account for the cooccurrence of *nosotros* and *los* ‘the’ in *nosotros los cocineros*, but there is another solution that proves more compelling.

Platzack (2004) is, to my knowledge, the first to propose a dedicated person phrase (PersP) that dominates DP, and although this author does not directly engage with the topic of pronoun-noun constructions, one can straightforwardly evaluate how this proposal fares in different languages. As

shown in (51), the separation of person and definiteness readily captures *nosotros los cocineros* ‘we the chefs’ in Spanish, but it fails to explain the ungrammaticality of **we the chefs* in English.



To resolve this issue in English, one could resort to fusion of Pers and D or a rule of contextual allomorphy where D is not pronounced in the environment of Pers to derive *we chefs* from the configuration in (51b). Alternatively, and ideally on the basis of independent evidence, one could parameterize PersP such that it projects in some languages but not in others.

As Choi (2013, 2014a,b) and Höhn (2016, 2017) maintain, there is empirical support for a language-specific approach to person features in the phenomenon of “unagreement”, or the use of first-person or second-person agreement morphology with a nonpronominal argument. Even though Spanish and Italian are closely related languages, only Spanish permits unagreement, as revealed by the contrast between (52) and (53) with *los cocineros/i cuochi* ‘the chefs’ in subject position (Bello 1847, Moravcsik 1978, Bosque & Moreno 1984, Jelinek 1984, Hurtado 1985, Torrego 1996, Ordóñez 1997, Olarrea 1998, Fernández Soriano 1999, Ordóñez & Treviño 1999, Martínez 1999, Rodríguez-Mondoñedo 2007, Rivero 2008, Saab 2008, Villa-García 2010, Saab 2013, Torrego & Laka 2015, Ackema & Neeleman 2018, Di Tullio et al. 2019).

- (52) a. Los cocineros tenemos muchas destrezas.
 the chefs have.1PL many skills
 Approximate: ‘We chefs have many skills.’
 b. Los cocineros tenéis muchas destrezas.
 the chefs have.2PL many skills
 Approximate: ‘You (informal) chefs have many skills.’

- (53) I cuochi {hanno / *abbiamo / *avete} molte competenze.
 the chefs have.3PL have.1PL have.2PL many skills
 ‘(The) chefs have many skills.’

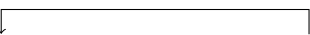
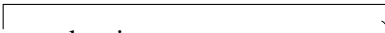
In section 4.1, I discussed the similarities between clitic doubling and object-verb agreement in Spanish, and based on these observations, it is evident from the examples in (54) that unagreement is possible in object position as well (Hurtado 1985, Ordóñez 1997, Fernández Soriano 1999, Ordóñez & Treviño 1999, Leonetti 2008, Saab 2013, Torrego & Laka 2015, Di Tullio et al. 2019).

- (54) a. Siempre nos subestiman a los cocineros.
 always 1PL underestimate.3PL DOM the chefs
 Approximate: ‘They always underestimate us chefs.’
- b. Siempre os subestiman a los cocineros.
 always 2PL underestimate.3PL DOM the chefs
 Approximate: ‘They always underestimate you (informal) chefs.’

According to Choi and Höhn, it is no coincidence that a pronoun can be inserted before a definite description in Spanish, as in (55), which results in a pronoun-noun construction that provides an identifiable first-person or second-person goal for agreement.

- (55) a.  Nosotros los cocineros tenemos muchas destrezas.
 we the chefs have.1PL many skills
 ‘We chefs have many skills.’
- b.  Siempre os subestiman a vosotros los cocineros.
 always 2PL underestimate.3PL DOM you.PL the chefs
 ‘They always underestimate you (informal) chefs.’

Assuming that the covert versions of these pronouns in (56) can also serve as goals, unagreement is no different than standard agreement.

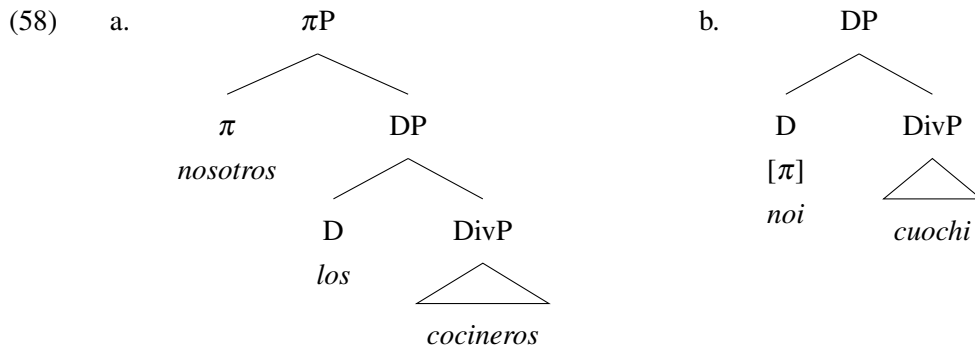
- (56) a.  $\emptyset_{[\pi:\text{PART}, \text{AUTH}]}$ los cocineros tenemos muchas destrezas.
 the chefs have.1PL many skills
 ‘We chefs have many skills.’
- b.  Siempre os subestiman a $\emptyset_{[\pi:\text{PART}]}$ los cocineros.
 always 2PL underestimate.3PL DOM the chefs
 ‘They always underestimate you (informal) chefs.’

As for Italian, the data in (57) indicate that pronouns are in complementary distribution with the definite article, so it follows that the verb forms in the first three sentences match the sole determiner in terms of person.

- (57) a. Noi cuochi abbiamo molte competenze.
 we chefs have.1PL many skills
 ‘We chefs have many skills.’
- b. Voi cuochi avete molte competenze.
 you.PL chefs have.2PL many skills
 ‘You chefs have many skills.’
- c. I cuochi hanno molte competenze.
 the chefs have.3PL many skills
 ‘(The) chefs have many skills.’

- d. *Noi i cuochi abbiamo molte competenze.
 we the chefs have.1PL many skills
 Literal: ‘We the chefs have many skills.’

For Höhn, the locus of person features is Pers in languages like Spanish that use the definite article in pronoun-noun constructions and D in languages like Italian that do not. The structures of *nosotros los cocineros* ‘we the chefs’ and *noi cuochi* ‘we chefs’ in (58) convey the essence of this proposal, but I use Béjar’s (2003) π notation instead of Pers to more transparently reflect the notion that person features can belong to their own functional projection or be subordinate to D.



In the next part of this section, I formalize the approach to person features that I adopt in this thesis and then build on Höhn’s analysis in (58a) to provide morphosyntactic representations and vocabulary items for all first-person and second-person pronouns in Spanish.

Ever since Forchheimer (1953) and Benveniste (1966) raised the possibility that third person is effectively the absence of person, there has been contention in the literature. In (59), I use Spanish to replicate a test by Ackema & Neeleman (2013b, 2018) that supports the claim that person features are exclusive to pronominal expressions, in which case third person must, in fact, be distinguished from Benveniste’s “non-personne”, or “no-person”.

- (59) a. Cualquiera que estudie gastronomía con los franceses aprenderá de
 anyone that studies.SBJV gastronomy with the French learn.FUT.3SG from
 {nosotros / vosotros / ustedes / ellos}.
 us you.PL you.PL them
 ‘Anyone who studies gastronomy with the French will learn from us/you all (informal)/you all/them.’
- b. Cualquiera que estudie gastronomía con ellos aprenderá de {*nosotros
 anyone that studies.SBJV gastronomy with them learn.FUT.3SG from us
 / *vosotros / *ustedes / ellos}.
 you.PL you.PL them
 ‘Anyone who studies gastronomy with them will learn from them.’

In the first sentence, *los franceses* ‘the French’ can be coreferential with any pronoun, but in the second, *ellos* ‘they’ cannot, suggesting that third-person pronouns are specified for person and that

definite descriptions are not. This conclusion echoes Ritter's (1995: 421) view that "person implies definiteness, but definiteness does not imply person" and has been independently reached in crosslinguistic work on pronouns (Bianchi 2006; Sichel & Wiltschko 2021; Toosarvandani 2023; Sichel & Toosarvandani to appear). As illustrated in (60), one can easily differentiate third person from no person in a privative system by using π in addition to [PART(ICIPANT)] and [AUTH(OR)] features (Béjar 2003; Béjar & Kahnemuyipour 2017; Sichel & Wiltschko 2021; Béjar & Kahnemuyipour 2023; Toosarvandani 2023; Alexiadou et al. 2024; Sichel & Toosarvandani to appear).¹¹

- (60) a. [π , PART, AUTH] First person
 b. [π , PART] Second person
 c. [π] Third person
 d. [] No person

Other authors who employ privative features either consider third person to be the lack of person, following Forchheimer and Benveniste, or do not take an explicit stance on this issue (Kayne 1993; Corver & Delfitto 1999; Lyons 1999; Harley & Ritter 2002; Panagiotidis 2002, 2003; McGinnis 2005; Adger & Harbour 2007; Béjar & Rezac 2009; Cowper & Hall 2009; Kratzer 2009; Moskal 2018; Bobaljik & Sauerland 2023; Deal to appear). One advantage of privative features is that they can be organized into geometries to capture any entailment relations among them, as in (61).

- (61) a. First person b. Second person c. Third person
- | | | |
|-------|-------|-------|
| π | π | π |
| | | |
| PART | PART | |
| | | |
| AUTH | | |

These dependencies are less clear in a binary system, which is the other common method of representing person features (Hale 1973; Silverstein 1976; Zwicky 1977; Ingram 1978; Greenberg 1993; Kerstens 1993; Halle 1997; Noyer 1997; Schlenker 2003; Bianchi 2006; Nevins 2007; Bernstein 2008a,b; Trommer 2008; Nevins 2011; Harbour 2016; Arregi & Nevins 2018; Grishin 2023). To distinguish third person and no person in this approach, one would need to maintain that [−PART, −AUTH] in (62c) is not the same as the complete absence of these features in (62d).

- (62) a. [+PART, +AUTH] First person
 b. [+PART, −AUTH] Second person
 c. [−PART, −AUTH] Third person
 d. [] No person

¹¹Ackema & Neeleman (2013a, 2018) use the features [PROXIMATE] and [DISTAL] in their theory of person, and there is a growing body of research that explores the potential connection between deixis and person (Gruber 2013; Choi 2014a,b; Harbour 2016; Lander & Haegeman 2018; Dékány 2021).

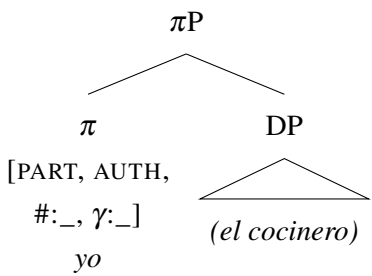
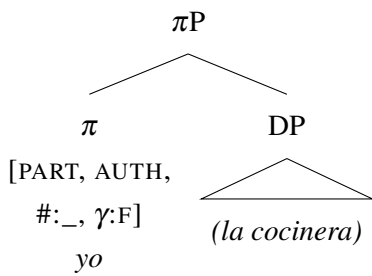
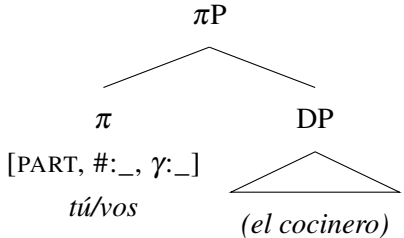
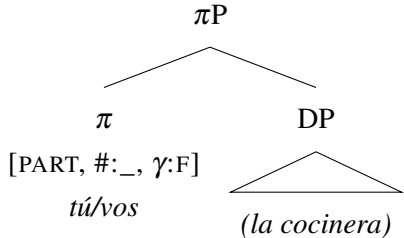
Due to these concerns, and because I use privative features in my analysis of gender and number in this thesis, I use the system in (60) in my analysis of person.

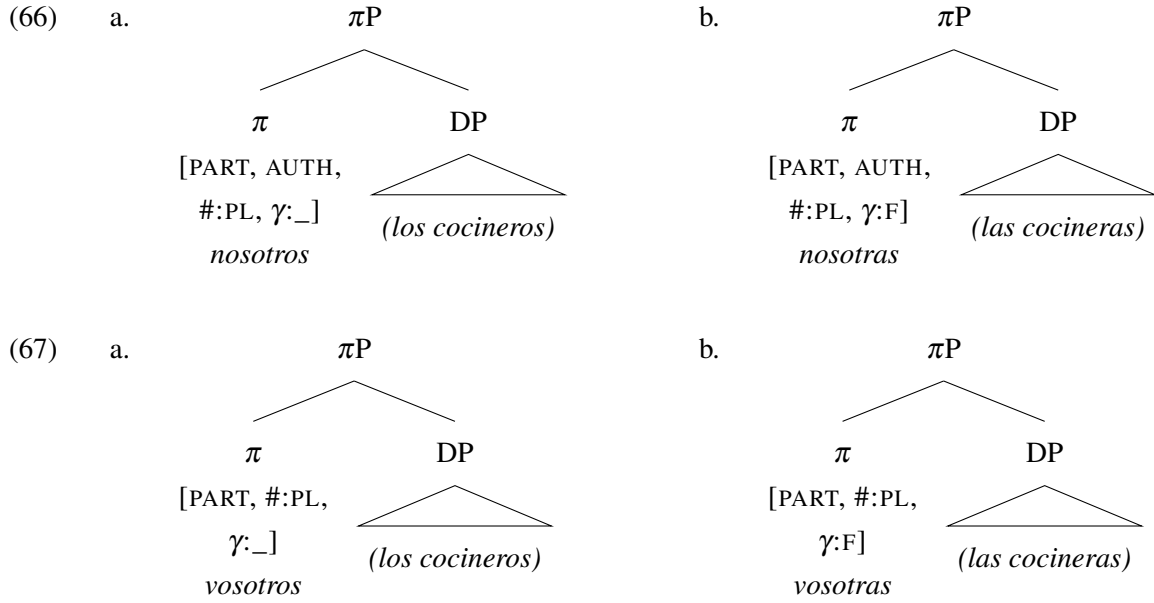
Having established that only pronominal expressions bear person features, it is necessary to address the person specification of what are traditionally described as third-person verb endings and clitics. There are at least two reasons to reject the view that these vocabulary items are specified as third person and instead represent them as elsewhere forms that are underspecified for this feature. First, the goal for subject-verb agreement in (63a) must be the personless definite description *los cocineros* ‘the chefs’ rather than a covert third-person pronoun since the version of this sentence in (63b) that includes the overt pronoun *ellos* ‘they’ is ungrammatical.

- (63) a. Los cocineros tienen muchas destrezas.
 the chefs have.3PL many skills
 ‘(The) chefs have many skills.’
 b. *Ellos los cocineros tienen muchas destrezas.
 the chefs have.3PL many skills
 Literal: ‘They they chefs have many skills.’

Second, if bare nouns are *nPs* or *DivPs*, as I argue in section 3.1, they lack πP and therefore cannot be third person. I assume that there is a failure of agreement with nonpronominal arguments and that default morphology is inserted as a result (Bianchi 2006, Preminger 2009, Arregi & Nevins 2012, Preminger 2014). In short, arguments can be first person, second person, third person, or personless, but agreement morphology can only be first person, second person, or personless.

It is now possible to finalize the morphosyntactic structures of the following first-person and second-person pronouns: *yo* ‘I’, *tú/vos* ‘you (informal)’, *nosotros/nosotras* ‘we’, and *vosotros/vosotras* ‘you all (informal)’. It is evident that these forms are generated in π because they can all precede a definite description, as indicated by the optionality of *el cocinero/la cocinera* ‘the chef’ and *los cocineros/las cocineras* ‘the chefs’ in (64) through (67).

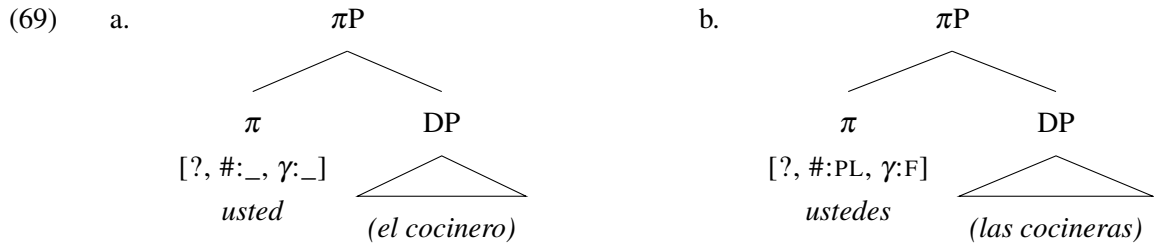
- (64) a.  b. 
- (65) a.  b. 



The corresponding vocabulary items in (68) capture the fact that, among the pronouns under consideration, only the plural forms overtly inflect for gender.¹²

- (68) a. /nosotras/ $\leftrightarrow \pi_{[AUTH, \# : PL, \gamma : F]}$
 b. /bosotras/ $\leftrightarrow \pi_{[PART, \# : PL, \gamma : F]}$
 c. /nosotros/ $\leftrightarrow \pi_{[AUTH, \# : PL]}$
 d. /bosotros/ $\leftrightarrow \pi_{[PART, \# : PL]}$
 e. /jo/ $\leftrightarrow \pi_{[AUTH]}$
 f. /tu/ or /bos/ $\leftrightarrow \pi_{[PART]}$

This analysis can be extended to *usted* ‘you (formal)’ and *ustedes* ‘you all’, as shown in (69), but their person specification remains to be determined.



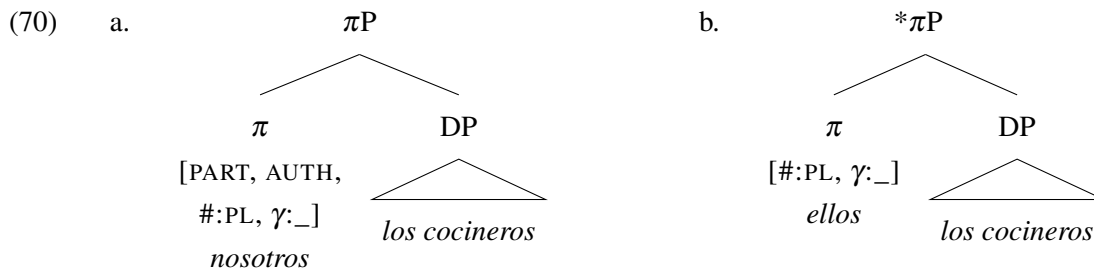
After identifying the semantic value of person features in section 5.4, I provide a complete account of *usted* and *ustedes* in section 5.5, where I argue that these pronouns are syntactically second person and lose their [PART] feature through a postsyntactic rule of impoverishment. In the remainder of

¹²See section 5.4 for discussion of the *otros/otras* ‘others’ component of *nosotros/nosotras* and *vosotros/vosotras* from a diachronic perspective. In principle, one could position *nos-/vos-* ‘we/you all (informal)’ in π and represent *otros/otras* ‘others’ as an adjunct to this head, a possibility that I leave to future research.

this chapter, I concentrate on the internal structure third-person pronouns, which necessarily contain π P because they are specified for person but do not seem to be confined to π in the way that the other stressed pronouns of Spanish are.

4.5 Deriving the third-person gap

In the previous section, I defended Höhn's (2016, 2017) claim that person and definiteness pertain to separate functional projections in Spanish and other languages that use the definite article in pronoun-noun constructions and permit so-called unagreement. Still, as Höhn (2020) acknowledges, the π P-DP split does not explain why Spanish displays the same third-person gap that English and many other languages do: *nosotros/vosotros/ustedes/*ellos los cocineros* 'we/you (informal)/you/they the chefs' and *we/you/*they chefs* (Postal 1966; Abney 1987; Cardinaletti 1994; Lyons 1999; Déchaine & Wiltschko 2002; Giusti 2002; Panagiotidis 2002; Roehrs 2005; Bernstein 2008a,b; Ackema & Neeleman 2018). If, for example, *nosotros* 'we' can spell out first-person π in (70a), the question is why *ellos* 'they' cannot spell out third-person π in (70b).



I consider the solution to this puzzle to lie in the morphological decomposition of third-person pronouns into markers of person, definiteness, class, and number.

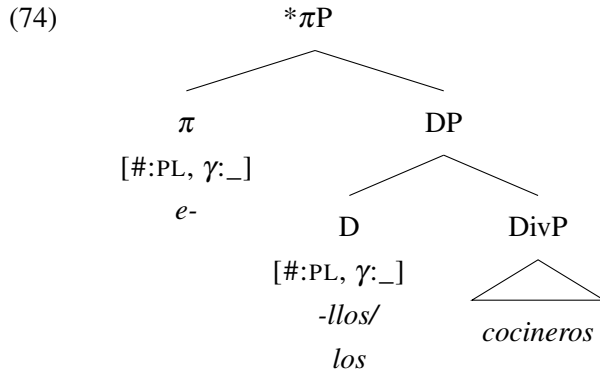
It has long been observed that third-person pronouns in Spanish bear a striking resemblance to the forms of the definite article (Bello 1847, Harris 1991, Schrotten 1992, Zwarts 1994, Uriagereka 1995, F. Roca 1996, Luján 2000, Fábregas 2018). There are, furthermore, morphological similarities between third-person pronouns and determiners in many Germanic and Romance languages (Postal 1966, Sommerstein 1972, Greenberg 1978, Cardinaletti 1994, Lyons 1999, Elbourne 2001, Kayne 2003, Elbourne 2005, Roehrs 2005, Elbourne 2013). Assuming that *e-* in the definite article *el* 'the' is epenthetic, as Harris (1991) remarks, the only substantial difference between the paradigms in (71) and (72) is the initial vowel of the pronominal series.

(71)	a.	él he	b.	ella she	c.	ellos they	d.	ellas they.F
(72)	a.	el the	b.	la the.F	c.	los the.PL	d.	las the.F.PL

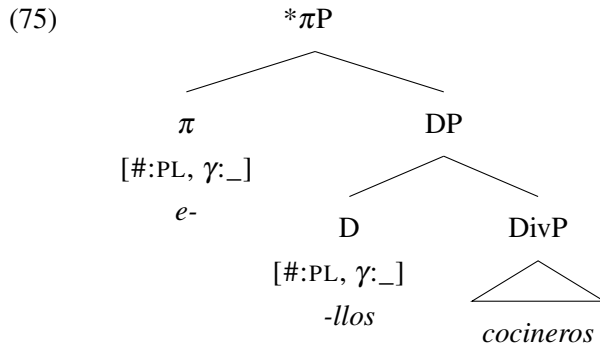
In section 4.4, I adapted a diagnostic by Ackema & Neeleman (2013b, 2018) to demonstrate that person features are exclusive to pronouns in Spanish, and I now propose that the nonepenthetic *é-/e-* that distinguishes the stressed third-person pronouns in (71) from the forms of the definite article in (72) is the morphological expression of third person. More precisely, I take *é-/e-* to be a vocabulary item that is specified for person but not gender or number, as formalized in (73).

$$(73) \quad /e/ \leftrightarrow \pi$$

If one were to divide *ellos* ‘they’ into *e-* in π and *-llos* in D, as in (74), the ungrammaticality of **ellos los cocineros* ‘they the chefs’ could be attributed to the inability of D to be spelled out twice.



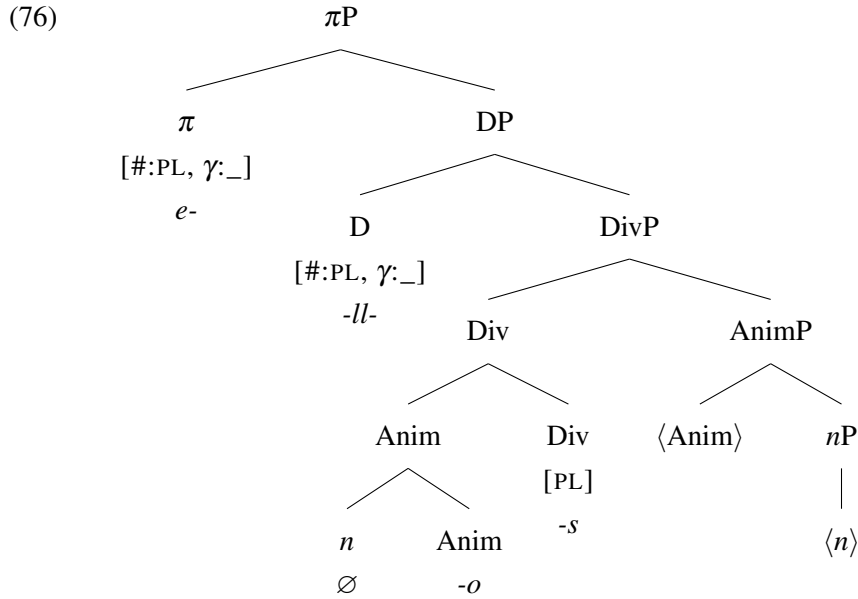
The problem with this analysis is that it does not explicitly rule out **ellos cocineros* ‘they chefs’ in (75), where D is only realized once as *-llos*.



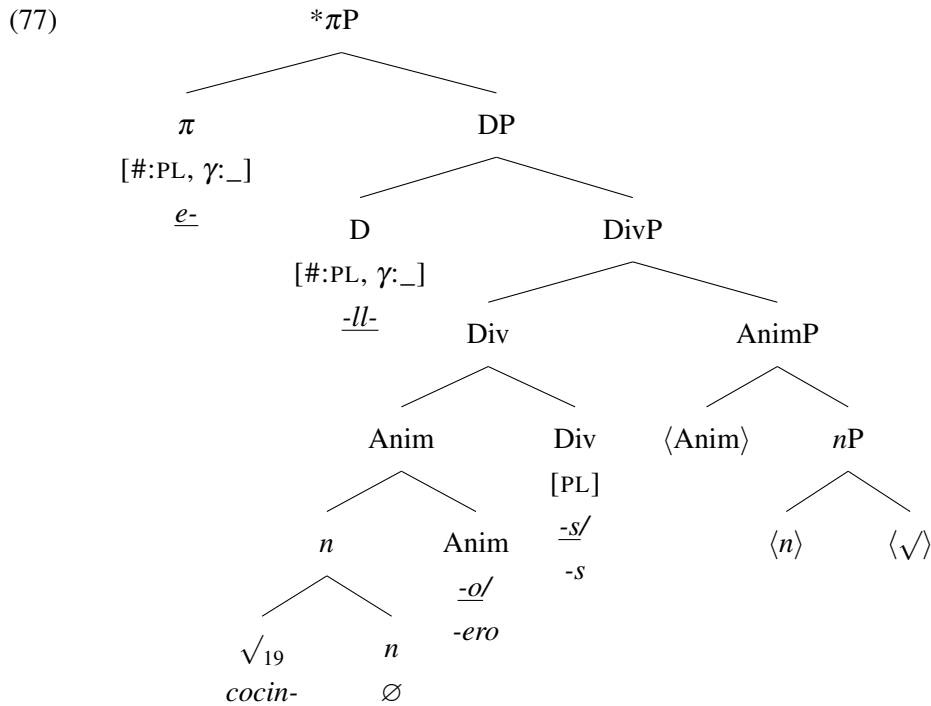
Although the *-llos* segment of *ellos* is nearly identical in form to the definite article *los* ‘the’, it is not necessarily the case that these elements map onto the same pieces of structure.

In the definite description *los cocineros* ‘the chefs’, the *-o* and *-s* of *los* ‘the’ both correspond to D, while the *-o* and *-s* of *cocineros* ‘chefs’ correspond to Anim and Div, respectively.¹³ In *ellos* ‘they’, on the other hand, the identity of *-o* and *-s* is less certain because this pronoun cannot be followed by a noun: **ellos cocineros* ‘they chefs’. The *-o* and *-s* of *ellos* could occupy D, as in (74) and (75) above, or they could occupy Anim and Div, as in (76).

¹³See sections 1.4 and 5.2 for arguments in favour of an animacy phrase (AnimP) above *nP* in Spanish.



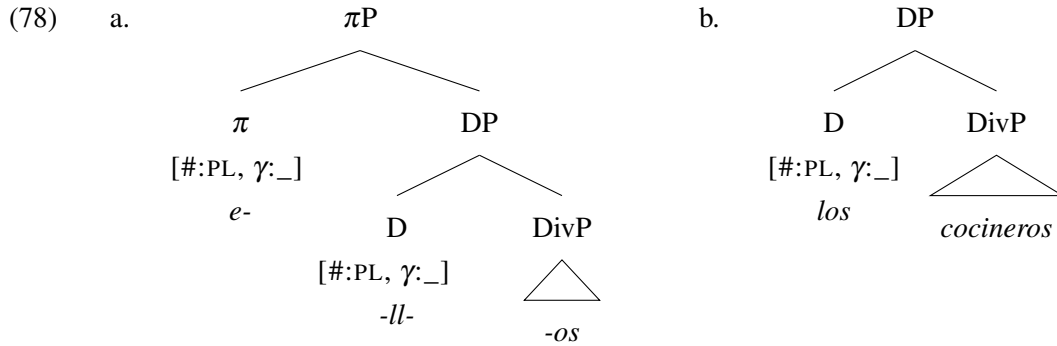
In the second approach, the *-o* and *-s* of *ellos* are identical to the *-o* and *-s* of *cocineros* (Jambrović to appear). From this perspective, the issue with **ellos cocineros* is the same as the issue with **los cocineros meseros* ‘the chefs servers’ or *Julia bailó cantó* ‘Julia danced sang’: only one vocabulary item can be inserted per node. This point is illustrated in (77), where I underline the components of *ellos* and attempt to include *cocineros* in the same extended nominal projection.



This account predicts the ungrammaticality of **ellos los cocineros* ‘they the chefs’ as well even before one factors in the clash between the *-ll-* of *ellos* and the definite article *los* in D. Any con-

struction that involves a third-person pronoun can, moreover, be ruled out on the same grounds, such as **él (el) cocinero* ‘he (the) chef’, **ella (la) cocinera* ‘she (the) chef’, and **ellas (las) cocineras* ‘they (the) chefs’. If, as I propose, third-person pronouns spell out both the functional and lexical layers of the noun phrase, it follows that they are mutually exclusive with other material.

This analysis of the third-person gap hinges on a plausible explanation for the unique realization of D in third-person pronouns. For instance, D has the same features, but not the same form, in the morphosyntactic structure of *ellos* ‘they’ in (78a) and that of *los cocineros* ‘the chefs’ in (78b).



Like Harris (1991) and Saab (2013), I consider the exponent of definiteness in Spanish to be a single consonant. In the case of the definite article *el/la/los/las* ‘the’ and the third-person clitics *lo/la/los/las*, this consonant is /l/, and in the case of the stressed third-person pronouns *él/ella/ellos/ellas* ‘he/she/they/they (feminine)’, this consonant varies from one variety to another but was diachronically /k/, which would neutralize to [l] in coda position, hence *él* instead of **ell*. No matter the consonant, it must be syllabified, which is the purpose of epenthetic *e-* in *el* and the vowels *-o* and *-a* in the other forms of the definite article as well as the clitics. As for the stressed pronouns, the syllabification of *-l/-ll-* in D is never a problem due to *é/-e-* in π , the vowel that is exclusive to this paradigm. The class marker *-a* in *ella* is not required phonologically, though it does serve to differentiate this pronoun from *él*, while the class markers *-o* in *ellos* and *-a* in *ellas* do perform a phonological role in that they syllabify the plural suffix *-s*.¹⁴ If one factors in the phonological effects that the initial vowel has in the stressed pronouns, the simpler form of D in this series is not as exceptional as it might otherwise seem.

Another way in which the third-person gap is manifested in Spanish involves pronouns that are modified by restrictive relative clauses. To demonstrate this phenomenon, it is first necessary to establish that the definite article must be present for a relative clause to be interpreted restrictively, as indicated in (79).

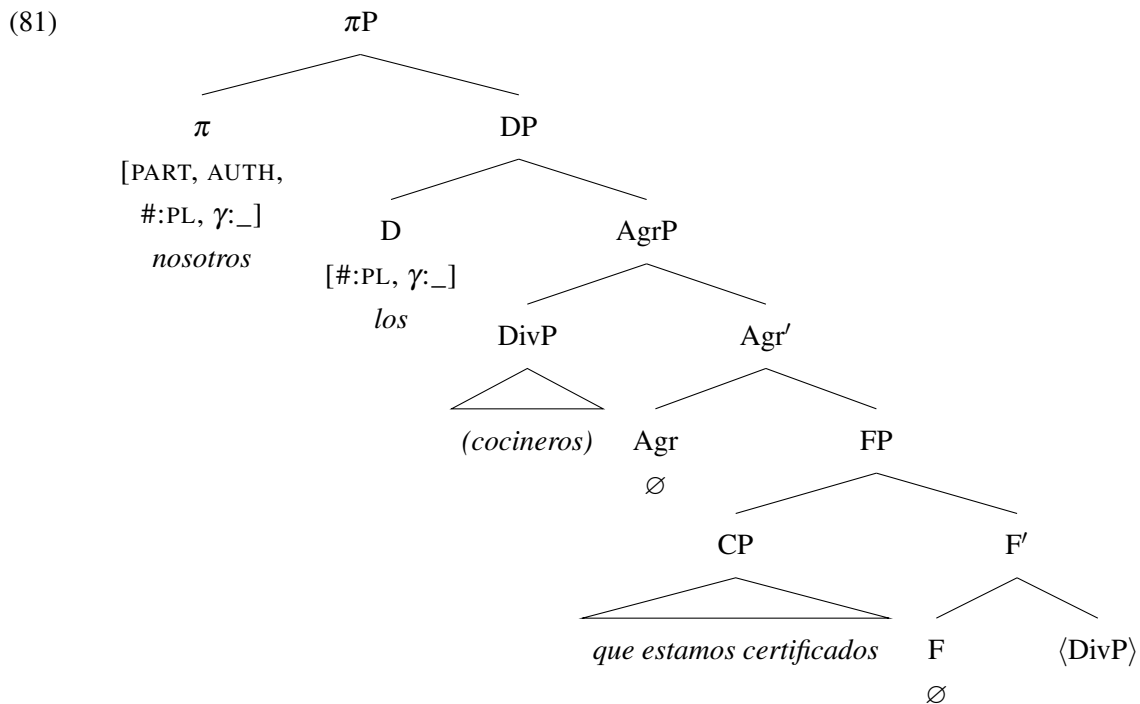
- (79) Solo nosotros **(los)* que estamos certificados podemos trabajar ahí.
 only we the that are.1PL certified can.1PL work there
 ‘Only we who are certified can work there.’

¹⁴Spanish does not permit lateral-fricative clusters apart from the noun *vals* ‘waltz’, which is Germanic in origin. Catalan does, as in the noun *cavalls* ‘horses’, and the counterpart of *ellos* in this language is, unsurprisingly, *ells* ‘they’ without an overt class marker.

The need for the definite article in this context suggests that it precedes a covert noun, whose presence is confirmed by the addition of *cocineros* ‘chefs’ in (80).

- (80) Solo nosotros los cocineros que estamos certificados podemos trabajar ahí.
 only we the chefs that are.1PL certified can.1PL work there
 ‘Only we chefs who are certified can work there.’

For Cinque (2010), restrictive relative clauses merge as specifiers of a functional phrase (FP) that is dominated by an agreement phrase (AgrP), just as adjectives do, and their postnominal order is the result of movement of the noun to the specifier of AgrP. Integrating this view of restrictive modification with the architecture of the noun phrase that I have motivated in this chapter yields the structure of *nosotros los (cocineros) que estamos certificados* ‘we (chefs) who are certified’ in (81).



I assume, following Heim & Kratzer (1998), that relative clause are property-denoting, or type $\langle e, t \rangle$, expressions that combine with nouns through predicate modification, the result of which is an expression of the same type. The relative clause has a restrictive reading in (81) because it introduces another property, and thus narrows down the set of possible referents, before D returns a unique maximal object.

As the data in (82) reveal, third-person pronouns are the only forms that cannot be restrictively modified by a relative clause (Jambrović to appear).

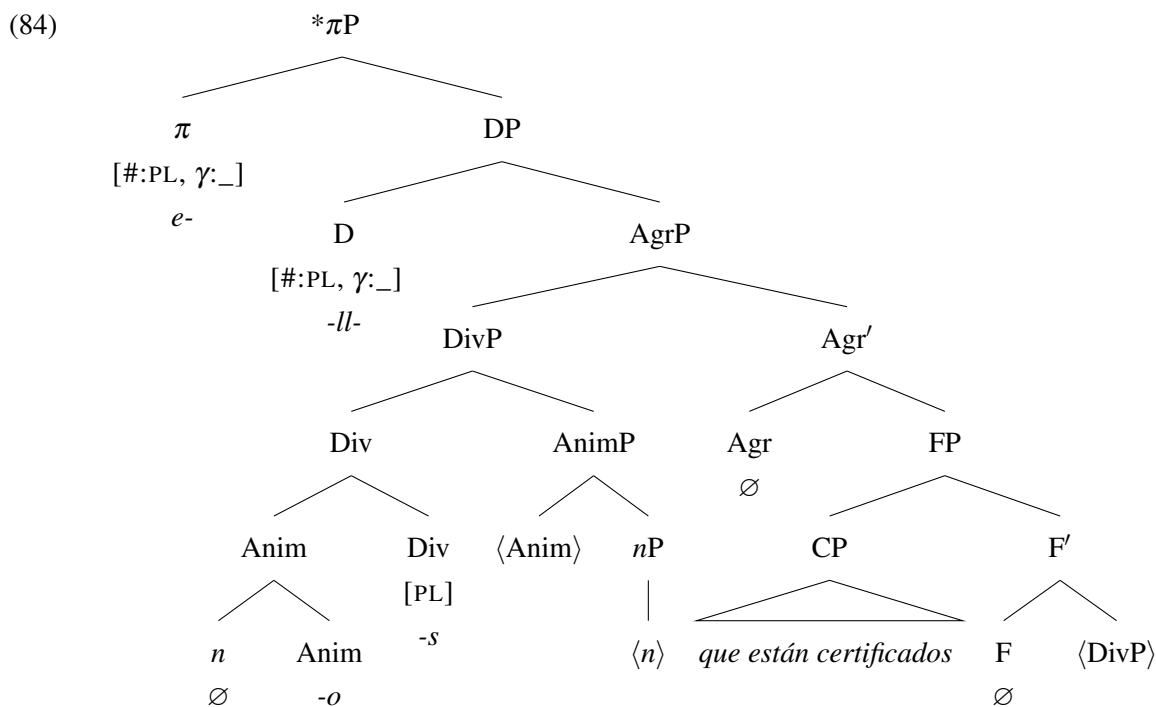
- (82) a. Solo nosotros los que estamos certificados podemos trabajar ahí.
 only we the that are.1PL certified can.1PL work there
 ‘Only we who are certified can work there.’

- b. Solo vosotros los que estáis certificados podéis trabajar ahí.
only you.PL the that are.2PL certified can.2PL work there
'Only you (informal) who are certified can work there.'
- c. Solo ustedes los que están certificados pueden trabajar ahí.
only you.PL the that are.3PL certified can.3PL work there
'Only you who are certified can work there.'
- d. *Solo ellos los que están certificados pueden trabajar ahí.
only they the that are.3PL certified can.3PL work there
Literal: 'Only they the ones who are certified can work there.'

In (75) above, I showed that the issue with **ellos los cocineros* 'they the chefs' cannot be reduced to the cooccurrence of the *-llos* component of *ellos* 'they' and *los* 'the' as the definite article since **ellos cocineros* 'they chefs' is just as ungrammatical. This observation extends to **ellos los que están certificados* 'they the ones who are certified' in (82d) based on the example in (83), which lacks the definite article and yet fares no better.

- (83) *Solo ellos que están certificados pueden trabajar ahí.
only they that are.3PL certified can.3PL work there
Literal: 'Only they who are certified can work there.'

The problem is evident from the structure of **ellos que están certificados* 'they who are certified' in (84), where it is the entire DivP, and not the complex head *n* + Anim + Div, that is closest to D.



This exploration of restrictive modification suggests that third-person pronouns require strict adjacency of the heads that they spell out, a condition that is met in (76) but not in (84).

- b. Vosotros dos (*cocineros) tenéis mucho que aprender.
you.PL two chefs have.2PL much that learn
'You (informal) two have much to learn.'
- c. Ustedes dos (*cocineros) tienen mucho que aprender.
you.PL two chefs have.3PL much that learn
'You two have much to learn.'
- d. Ellos dos (*cocineros) tienen mucho que aprender.
you.PL two chefs have.3PL much that learn
Literal: 'They two have much to learn.'

To include this noun, or any other, the definite article must intervene between the pronoun and the numeral, as in (89), in which case the third-person gap resurfaces.

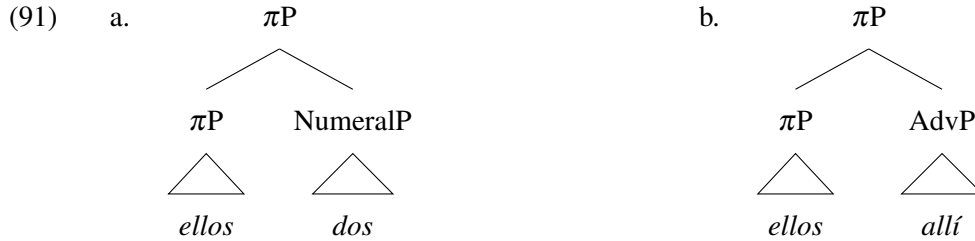
- (89) a. Nosotros los dos cocineros tenemos mucho que aprender.
we the two chefs have.1PL much that learn
'We two chefs have much to learn.'
- b. Vosotros los dos cocineros tenéis mucho que aprender.
you.PL the two chefs have.2PL much that learn
'You (informal) two chefs have much to learn.'
- c. Ustedes los dos cocineros tienen mucho que aprender.
you.PL the two chefs have.3PL much that learn
'You two chefs have much to learn.'
- d. *Ellos los dos cocineros tienen mucho que aprender.
they the two chefs have.3PL much that learn
Literal: 'They the two chefs have much to learn.'

Curiously, **nosotros los dos* 'we the two' does not exist as an alternative to *nosotros dos* 'we two' in Spanish, confirming that the article must be omitted whenever the noun is.

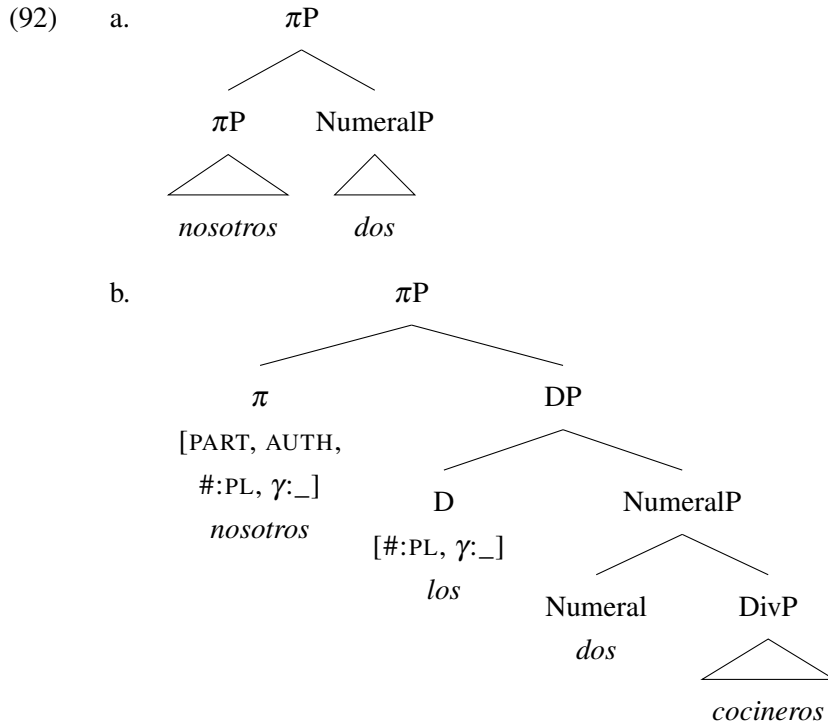
I propose that the numeral in pronoun-numeral constructions serves a deictic function similar to the adverbs that modify the pronouns in (90), none of which can be followed by a noun either.

- (90) a. Nosotros aquí (*cocineros) tenemos mucho que aprender.
we here chefs have.1PL much that learn
'We here have much to learn.'
- b. Vosotros ahí (*cocineros) tenéis mucho que aprender.
you.PL there chefs have.2PL much that learn
'You (informal) there have much to learn.'
- c. Ustedes ahí (*cocineros) tienen mucho que aprender.
you.PL there chefs have.3PL much that learn
'You there have much to learn.'
- d. Ellos allí (*cocineros) tienen mucho que aprender.
you.PL there chefs have.3PL much that learn
Literal: 'They there have much to learn.'

If numerals and adverbs adjoin to the highest level of the noun phrase, as shown in (91) with *ellos dos* ‘they two’ and *ellos allí* ‘they there’, they have no bearing on its internal configuration and, consequently, do not prevent a third-person pronoun from being spelled out.



As for the contrast between *nosotros dos* ‘we two’ and *nosotros los dos cocineros* ‘we the two chefs’, the structures in (92) capture the absence of the definite article in the first expression and the presence of the definite article in the second.



The fundamental point is that there is a principled explanation for *ellos dos*, so this construction does not undermine the analysis of the third-person gap that I have developed in this section.

4.6 Conclusion

The objective of this chapter has been to identify where stressed pronouns in Spanish are generated in the noun phrase. Since these forms cannot be preceded by determiners or modifiers in any productive sense, I rejected the possibility that they originate in *n*. Next, I demonstrated that pronoun-noun

constructions like *nosotros los cocineros* ‘we the chefs’ do not consist of two referential expressions in apposition, suggesting that the pronoun and the definite article occupy the same extended nominal projection. To account for the cooccurrence of these determiners, I adopted Höhn’s (2016, 2017) view that person and definiteness correspond to separate projections in languages like Spanish that use the definite article in pronoun-noun constructions and exhibit the phenomenon known as *una-agreement*. After motivating a distinction between third person and the lack of person and arguing that only pronominal expressions contain π P in Spanish, I assigned first-person and second-person pronouns to π and proposed that third-person pronouns realize π together with other heads in the noun phrase, namely D, Div, Anim, and *n*. The unique internal configuration of third-person pronouns renders them incompatible with nouns, restrictive relative clauses, and any other material that enters the derivation within the noun phrase, hence the third-person gap.

Having determined the general structure of pronouns in Spanish, the next task is to formalize the interpretation of φ -features in these expressions. In the first part of chapter 5, I focus on gender and animacy, and in the second part, I turn to number, person, and honorificity.

Chapter 5

A uniform account of φ -features

In chapter 4, I determined that first-person and second-person pronouns in Spanish are generated in a dedicated person phrase (π P) and that third-person pronouns are forms that spell out π along with other heads in the noun phrase. In this chapter, I provide morphosyntactic and interpretable structures for every pronoun in this language, including *ello* ‘it’, an endeavour that requires investigation into three issues. The first involves the representation of gender and animacy and the extent to which the analysis of pronouns can be subsumed under that of nouns. After exploring the difference between gender that has semantic effects and gender that is purely grammatical in nature, I confirm that pronouns only display a subset of the contrasts that are encoded by nouns. The second issue relates to the interpretation of number in pronouns like *nosotros* ‘we’, which denotes the speaker and at least one nonspeaker rather than a plurality of speakers. I reject the view that number is somehow different in these pronouns and instead maintain that their heterogeneous readings are due to the person features that they bear. The third issue concerns the syntactic identity of *usted* ‘you (formal)’ and *ustedes* ‘you all’, which behave semantically like second-person pronouns but appear with third-person verb forms and pronominal clitics. I establish on the basis of the Person-Case Constraint that these pronouns are syntactically second person and attribute their inability to trigger second-person agreement to a morphological process of impoverishment.

In section 5.1, I introduce the three dimensions of gender in Spanish and assign nouns and pronouns to six classes according to whether they denote animate or inanimate objects and presuppose maleness, femaleness, or neither. In section 5.2, I formally distinguish these six classes through the presence versus absence of an animacy phrase (AnimP) in their structure, privative gender features, and the head that hosts these features. In section 5.3, I revisit the topic of third-person pronouns and propose that *ello* ‘it’ is the only form that lacks AnimP. In section 5.4, I demonstrate how the interaction between person and number features accounts for the heterogeneity of *nosotros/nosotras* ‘we’ and *vosotros/vosotras* ‘you all (informal)’ and then finalize the interpretable structures of all first-person and second-person pronouns in Spanish. In section 5.5, I focus on *usted* ‘you (formal)’ and *ustedes* ‘you all’ and capture their disparate morphosyntactic and semantic properties by arguing that their structural representation varies from one component of the grammar to another. In section 5.6, I conclude with a summary of the main findings of this chapter.

5.1 Gender, animacy, and presupposition

As Harris (1991) remarks in work on Spanish, gender is a complex phenomenon that has syntactic, morphological, and semantic aspects. For instance, the nouns *cocinera* ‘(female) chef’ and *mano* ‘hand’ are both syntactically feminine, as evidenced by feminine concord in *la cocinera* ‘the (female) chef’ and *la mano* ‘the hand’, yet these nouns differ morphologically in their class markers, *cocinera* versus *mano*, and semantically in that *cocinera* conveys femaleness and *mano* does not. In section 4.1, I showed that stressed pronouns in Spanish can only refer to humans when they occur as the arguments of verbs, so I start by identifying the semantic contrasts that are expressed by human-denoting nouns and then consider whether pronouns make the same distinctions.

The first point to establish is that, in languages like Spanish that use a binary masculine-feminine system, masculine gender is semantically unmarked in comparison to feminine gender (Greenberg 1966; Echaide 1969; Jakobson 1971; Prado 1982; Klein 1989; I. Roca 1989; Harris 1991, 1996; Lloret & Viaplana 1997; I. Roca 2000; Alexiadou 2004; I. Roca 2005b; Sauerland 2008a; Bobaljik & Zocca 2011). As indicated in (1), the masculine noun *cocinero* ‘chef’ allows for the possibility of a female interviewee, but the feminine noun *cocinera* ‘(female) chef’ does not allow for the possibility of a male interviewee.

- (1) a. Si pudiera hablar con un cocinero famoso, sería {Jacques Pépin / Julia Child}.
 if could speak with a chef famous be.COND Jacques Pépin Julia Child
 ‘If I could speak with a famous chef, it would be Jacques Pépin/Julia Child.’
 b. Si pudiera hablar con una cocinera famosa, sería {#Jacques Pépin / Julia Child}.
 if could speak with a chef famous be.COND Jacques Pépin Julia Child
 ‘If I could speak with a famous (female) chef, it would be Julia Child.’

More precisely, *cocinera* presupposes femaleness because it introduces a gender-based inference that projects in the antecedent of a conditional, whereas *cocinero* does not presuppose maleness, at least not for the majority of speakers at the time of writing. Inclusive morphology is emerging in Spanish, and the class marker *-e* is gaining the most traction (Bonorino & Giammatteo 2018, Barrera Linares 2019, Martínez 2019, Gil 2020, Papadopoulos 2022, Fábregas 2024, Heap 2024, Papadopoulos 2024). For those speakers who have *cocinere* ‘chef’ as an inclusive alternative to *cocinero/cocinera*, the judgments may be as in (2) instead.

- (2) a. Si pudiera hablar con un cocinero famoso, sería {Jacques Pépin / #Julia Child}.
 if could speak with a chef famous be.COND Jacques Pépin Julia Child
 ‘If I could speak with a famous (male) chef, it would be Jacques Pépin.’
 b. Si pudiera hablar con una cocinera famosa, sería {#Jacques Pépin / Julia Child}.
 if could speak with a chef famous be.COND Jacques Pépin Julia Child
 ‘If I could speak with a famous (female) chef, it would be Julia Child.’
 c. Si pudiera hablar con une cocinere famose, sería {Jacques Pépin / Julia Child}.
 if could speak with a chef famous be.COND Jacques Pépin Julia Child
 ‘If I could speak with a famous chef, it would be Jacques Pépin/Julia Child.’

Until these judgments have been corroborated through corpus-based or experimental work, one can at most speculate how nouns like *cocinere* are interpreted or may affect the interpretation of their masculine and feminine counterparts. For this reason, I limit the empirical domain to binary pairs of nouns and pronouns and concentrate on the possible readings that a given form can have, not on the more complex question of whether a particular reading is more salient or preferable to another.

In addition to inflectional pairs like *cocinero/cocinera* ‘chef/(female) chef’ in Spanish, there are human-denoting nouns like *modelo* ‘model’ that do not overtly inflect for gender but nevertheless have masculine and feminine guises that are revealed through concord and agreement: *el modelo es alto* ‘the model is tall’ versus *la modelo es alta* ‘the (female) model is tall’. As the examples in (3) illustrate, the only difference between this type of noun and those that come in inflectional pairs is their invariable class marker.

- (3) a. Si pudiera hablar con un modelo famoso, sería {Alton Mason / Heidi Klum}.
 if could speak with a model famous be.COND Alton Mason Heidi Klum
 ‘If I could speak with famous model, it would be Alton Mason/Heidi Klum.’
 b. Si pudiera hablar con una modelo famosa, sería {#Alton Mason / Heidi Klum}.
 if could speak with a model famous be.COND Alton Mason Heidi Klum
 ‘If I could speak with famous (female) model, it would be Heidi Klum.’

That is, *modelo* presuppose femaleness in its feminine guise, but it does not presuppose maleness in its masculine guise.

Next, a select few human-denoting nouns come in derivational pairs, such as *rey/reina* ‘king/queen’, and suppletive pairs, such as *hombre/mujer* ‘man/woman’ (Lenz 1920; Echaide 1969; I. Roca 1989; Harris 1991; Ambadiang 1993, 1999; I. Roca 2000, 2005a,b; Saab 2008, 2010). The data sets in (4) and (5) show that both members of these pairs trigger gender-based presuppositions, distinguishing these masculine forms from other masculine nouns in Spanish.

- (4) a. Si pudiera hablar con un rey famoso, sería {Felipe VI / #Letizia Ortiz}.
 if could speak with a king famous be.COND Felipe VI Letizia Ortiz
 ‘If I could speak with famous king, it would be Felipe VI.’
 b. Si pudiera hablar con una reina famosa, sería {#Felipe VI / Letizia Ortiz}.
 if could speak with a queen famous be.COND Felipe VI / Letizia Ortiz
 ‘If I could speak with famous queen, it would be Letizia Ortiz.’
 (5) a. Si pudiera hablar con un hombre famoso, sería {Felipe VI / #Letizia Ortiz}.
 if could speak with a man famous be.COND Felipe VI Letizia Ortiz
 ‘If I could speak with famous man, it would be Felipe VI.’
 b. Si pudiera hablar con una mujer famosa, sería {#Felipe VI / Letizia Ortiz}.
 if could speak with a woman famous be.COND Felipe VI / Letizia Ortiz
 ‘If I could speak with famous woman, it would be Letizia Ortiz.’

There are, moreover, animal-denoting nouns that come in inflectional pairs (*cerdo/cerda* ‘pig/sow’), derivational pairs (*gallo/gallina* ‘rooster/hen’), and suppletive pairs (*toro/vaca* ‘bull/cow’), all of

which pattern identically to the corresponding human-denoting pairs in terms of whether a particular form presupposes maleness or femaleness.

Finally, there are epicene nouns like *personaje* ‘character, personality’ and *persona* ‘person’ that permit a single type of concord and agreement: *el/*la personaje es alto/*alta* ‘the character is tall’ and *la/*el persona es alta/*alto* ‘the person is tall’. These nouns can refer to individuals of any gender identity, so it is unsurprising that they do not introduce gender-based presuppositions in (6).

- (6) a. Si pudiera hablar con un personaje famoso, sería {Jacques Pépin / Julia Child}.
if could speak with a personality famous be.COND Jacques Pépin Julia Child
‘If I could speak with a famous personality, it would be Jacques Pépin/Julia Child.’
- b. Si pudiera hablar con una persona famosa, sería {Jacques Pépin / Julia Child}.
if could speak with a person famous be.COND Jacques Pépin Julia Child
‘If I could speak with a famous person, it would be Jacques Pépin/Julia Child.’

The fact that these human-denoting nouns must be differentially marked as objects in environments where inanimate nouns cannot be, as indicated in (7) and (8), confirms that animacy is distinct from gender that has semantic consequences.¹

- (7) a. Conozco *(a) ese personaje.
know.1SG DOM that character
‘I know that character.’
- b. Conozco (*a) ese libro.
know.1SG DOM that book
‘I know that book.’
- (8) a. Conozco *(a) esa persona.
know.1SG DOM that person
‘I know that person.’
- b. Conozco (*a) esa receta.
know.1SG DOM that recipe
‘I know that recipe.’

The use of differential object marking with animal-denoting nouns is more nuanced, and since my analysis of gender and animacy does not hinge on this issue, I leave it to future research.

Based on the previous discussion of the syntactic and semantic behaviour of nouns in Spanish, I organize them into the six classes in (9), each of which is accompanied by a representative example.

- (9) a. Animate masculine nouns that presuppose gender *rey* ‘king’
b. Animate feminine nouns that presuppose gender *reina* ‘queen’
c. Animate masculine nouns *cocinero* ‘chef’
d. Animate feminine nouns *persona* ‘person’
e. Inanimate masculine nouns *libro* ‘book’
f. Inanimate feminine nouns *receta* ‘recipe’

As for nouns like *modelo* ‘model’ that do not overtly inflect for gender, I assign their feminine guise to the second class in (9b), or animate feminine nouns that presuppose femaleness, and their masculine guise to the third class in (9c), or animate masculine nouns that do not presuppose maleness.

¹In Spanish, animacy, definiteness, and specificity all play a role in differential object marking (Jaeggli 1982, Kliffer 1984, Lois 1987, Masullo 1992, Brugè & Brugger 1996, Masullo 1996, Torrego 1998, Bleam 1999b, Torrego Salcedo 1999, Leonetti 2004, Bleam 2005, Laca 2006, Rodríguez-Mondoñedo 2007, Leonetti 2008, Fábregas 2013).

To evaluate stressed pronouns along the same dimensions as nouns, it is necessary to modify the diagnostic that I used in (1) through (6) to allow for a definite expression in the antecedent.² Not only do the sentences in (10) achieve this goal, but they also confirm that the masculine noun *cocinero* ‘chef’ does not presuppose maleness when *el cocinero* ‘the chef’ has an attributive reading, according to which the speaker does not have a particular referent in mind.

- (10) a. Si *el cocinero* es canadiense, no necesita visa.
 if the chef is Canadian not needs visa
 ‘If the chef is Canadian, he or she does not need a visa.’ (Attributive)
 ‘If the chef is Canadian, he does not need a visa.’ (Referential)
- b. Si *la cocinera* es canadiense, no necesita visa.
 if the chef is Canadian not needs visa
 ‘If the (female) chef is Canadian, she does not need a visa.’ (Attributive)
 ‘If the (female) chef is Canadian, she does not need a visa.’ (Referential)

If one replaces these definite descriptions with the third-person pronouns *él* ‘he’ and *ella* ‘she’, as in (11), the interpretation can only be referential, and a gender-based inference projects in the antecedent of both conditionals.

- (11) a. Si *él* es canadiense, no necesita visa.
 if he is Canadian not needs visa
 ‘If he is Canadian, he does not need a visa.’
- b. Si *ella* es canadiense, no necesita visa.
 if she is Canadian not needs visa
 ‘If she is Canadian, she does not need a visa.’

In other words, *él* presupposes maleness just as *ella* presupposes femaleness, and these judgments are likely to remain stable over time since neither pronoun has been identified as a viable candidate for an inclusive singular form in Spanish. Instead, *elle* ‘they (singular)’ seems poised to be adopted for this purpose (Barrera Linares 2019, Martínez 2019, Papadopoulos 2022, Fábregas 2024, Papadopoulos 2024). Although *él* is the masculine member of an inflectional pair, *él/ella*, this pronoun patterns semantically like the masculine members of derivational and suppletive pairs.

The remaining singular pronouns in Spanish are invariable in form but can appear with masculine or feminine agreement morphology, as shown in (12), so they are morphologically and syntactically comparable to nouns like *modelo* ‘noun’.

- (12) a. Yo soy {alto / alta}.
 I am tall tall
 ‘I am tall.’
- b. Tú eres {alto / alta}.
 you are.2SG tall tall
 ‘You (informal) are tall.’

²See Table 4.1 in section 4.1 for the complete paradigm of stressed pronouns in Spanish.

- c. Vos sos {alto / alta}.
you are.2SG tall tall
'You (informal) are tall.'
- d. Usted es {alto / alta}.
you is tall tall
'You (formal) are tall.'

The question is whether these pronouns are semantically comparable to *modelo* as well, in which case they would only trigger gender-based presuppositions in their feminine guise.

The facts in (13) are informative regarding this issue because the adjective *francés/francesa* 'French' inflects for gender but has a masculine form that lacks an overt class marker, leaving open the possibility that it could be used by speakers who identify as nonbinary.³

- (13) a. Si yo soy francés, ¿necesito visa?
if I am French need.1SG visa
'If I am French, do I need a visa?'
- b. Si yo soy francesa, ¿necesito visa?
if I am French need.1SG visa
'If I am French, do I need a visa?'

In (13b), it is unambiguously the case that agreement has taken place between *yo* 'I' and the adjective given that feminine gender is not the morphological default in Spanish, as demonstrated by Harris's (1991: 43) metalinguistic example in (14).

- (14) Tienes {demasiados / *demasiadas} "paras" en ese párrafo.
have.2SG too.many too.many "paras" in that paragraph
'You have too many "paras" in that paragraph.'

In (13a), on the other hand, one could argue that the pronoun is underspecified for gender and then attribute the masculine, or default, form *francés* to a failure of agreement (Bianchi 2006, Preminger 2009, Arregi & Nevins 2012, Preminger 2014). The same observations apply to the masculine and feminine guises of the singular pronouns in (15) through (17).

- (15) a. Si tú eres francés, no necesitas visa.
if you are.2SG French not need.2SG visa
'If you (informal) are French, you do not need a visa.'
- b. Si tú eres francesa, no necesitas visa.
if you are.2SG French not need.2SG visa
'If you (informal) are French, you do not need a visa.'
- (16) a. Si vos sos francés, no necesitás visa.
if you are.2SG French not need.2SG visa
'If you (informal) are French, you do not need a visa.'

³An explicitly inclusive form like *francese* 'French' is conceivable but has a morphophonological shape that could render it less preferable than *francés*.

- b. Si vos sos francesa, no necesitás visa.
if you are.2SG French, not need.2SG visa
'If you (informal) are French, you do not need a visa.'
- (17) a. Si usted es francés, no necesita visa.
if you is French not needs visa
'If you (formal) are French, you do not need a visa.'
- b. Si usted es francesa, no necesita visa.
if you is French not needs visa
'If you (formal) are French, you do not need a visa.'

Another reason to consider that *tú/vos* 'you (informal)' and *usted* 'you (formal)' do not presuppose maleness in their masculine guises is that these pronouns all permit the same generic reading that *uno* 'one' does in (18).

- (18) Si uno es francés, no necesita visa.
if one is French not needs visa
'If one is French, one does not need a visa.'

Due to these factors, I maintain that *yo*, *tú/vos*, and *usted* are indeed semantically akin to nouns like *modelo* 'model' in that it is only their feminine guises that introduce gender-based presuppositions.

The next topic to explore is the interpretation of gender in plural pronouns. I begin with the inflectional pairs *nosotros/nosotras* 'we', *vosotros/vosotras* 'you all (informal)', and *ellos/ellas* 'they' and once again employ the invariable adjective *canadiense* 'Canadian' so that any inferences that project can be tied to the forms of the pronouns themselves. In each pair of sentences in (19) through (21), the masculine pronouns do not presuppose that all of their referents identify as male, but the feminine pronouns do presuppose that all of their referents identify as female.

- (19) a. Si nosotros somos canadienses, ¿necesitamos visa?
if we are.1PL Canadian need.1PL visa
'If we are Canadian, do we need a visa?'
- b. Si nosotras somos canadienses, ¿necesitamos visa?
if we.F are.1PL Canadian need.1PL visa
'If we (feminine) are Canadian, do we need a visa?'
- (20) a. Si vosotros sois canadienses, no necesitáis visa.
if you.PL are.2PL Canadian not need.2PL visa
'If you all (informal) are Canadian, you do not need a visa.'
- b. Si vosotras sois canadienses, no necesitáis visa.
if you.F.PL are.2PL Canadian not need.2PL visa
'If you all (informal, feminine) are Canadian, you do not need a visa.'
- (21) a. Si ellos son canadienses, no necesitan visa.
if they are.3PL Canadian not need.3PL visa
'If they are Canadian, they do not need a visa.'

- b. Si ellas son canadienses, no necesitan visa.
if they.F are.3PL Canadian not need.3PL visa
'If they (feminine) are Canadian, they do not need a visa.'

Although *ustedes* 'you all' does not overtly inflect for gender in (22), hence the use of the adjective *francés/francesa* 'French', it exhibits the same behaviour as the other plural pronouns in that it only presupposes that all of its referents identify as female when it triggers feminine agreement.

- (22) a. Si ustedes son franceses, no necesitan visa.
if you.PL are.3PL French not need.3PL visa
'If you all are French, you do not need a visa.'
- b. Si ustedes son francesas, no necesitan visa.
if you.PL are.3PL French not need.3PL visa
'If you all are French, you do not need a visa.'

A related, but independent, question is whether the masculine forms in (19) through (21) presuppose that at least one of their referents identifies as male, and scenarios where the speaker has incomplete knowledge suggest that the answer is negative. If a woman is asking about a future situation that will involve a group of people and does not yet know the exact composition of the group, she could use the plural masculine pronoun *nosotros* 'we' in the antecedent of a conditional question, as in (23), without presupposing that the group will include at least one male individual.

- (23) Si nosotros queremos salir después de la cena, ¿hay bares en la zona?
if we want.1PL go.out after of the dinner has bars in the area
'If we want to go out after the dinner, are there bars in the area?'

Similarly, one could use *vosotros* 'you all (informal)' or *ellos* 'they' when discussing later plans, as in (24), without presupposing that there will be at least one male individual in attendance.

- (24) a. Si vosotros queréis salir después de la cena, hay muchos bares en la zona.
if you.PL want.2PL go.out after of the dinner has many bars in the area
'If you all (informal) want to go out after the dinner, there are many bars in the area.'
- b. Si ellos quieren salir después de la cena, hay muchos bares en la zona.
if they want.3PL go.out after of the dinner has many bars in the area
'If they want to go out after the dinner, there are many bars in the area.'

In summary, the feminine forms and guises of plural pronouns presuppose that all of their referents identify as female, but the masculine forms and guises of plural pronouns do not presuppose that all of their referents identify as male, nor that even one does.

The only stressed pronoun that remains to be addressed is the "neuter" third-person form *ello* 'it'. Outside of locutions like *debido a ello* 'because of this' and *por ello* 'therefore', this pronoun is limited to formal speech and literature and must have an antecedent in the discourse, which is typically a clause or an abstract notion. In (25), for example, *ello* occurs in subject position and is coreferential with *consolarte* 'to comfort you (informal)', the object of the verb *quiero* 'I want'.

- (25) No quiero hacerte llorar, sólo quiero consolarte, hasta donde ello es posible.
 not want.1SG make.2SG cry only want.1SG console.2SG until where it is possible
 ‘I do not want to make you (informal) cry, I only want to comfort you, to the extent that it
 [comforting you] is possible.’ (Elena Demuth by Plácido Bonmatí, 1993, page 36)

It is important to note that *ello* cannot refer to inanimate objects, even as the object of a preposition, and thus should not be classified as an inanimate pronoun. As shown in (26), *él* ‘it’ or *ella* ‘it’ must be used instead, and in this syntactic context, neither pronoun conveys gender or animacy.

- (26) a. El carro es tan viejo que no deberíamos invertir más dinero en {él / *ello}.
 the car is so old that not should.1PL invest more money in it it
 ‘The car is so old that we should not invest more money in it.’
 b. La casa es tan vieja que no deberíamos invertir más dinero en {ella / *ello}.
 the house is so old that not should.1PL invest more money in it it
 ‘The car is so old that we should not invest more money in it.’

If one were to claim that *él* and *ella* have animate and inanimate guises, it would then be necessary to explain why it is only the animate guise that can appear as the argument of a verb, a restriction that I first mentioned in section 4.1. Instead, I contend that *él*, *ella*, and their plural counterparts *ellos* ‘they’ and *ellas* ‘they (feminine)’ all have but one formal representation and that any presuppositions that these pronouns trigger can be cancelled when they occur as the object of a preposition, the sole environment where a covert form is not possible in Spanish: **no deberíamos invertir más dinero en Ø* ‘we should not invest more money in’ (Strozer 1976, Luján 1980, Montalbetti 1984, Schrotten 1992, Fernández Soriano 1999, Suñer 1999). After motivating a novel analysis of gender and animacy in section 5.2, I provide the complete morphosyntactic and interpretable structures of third-person pronouns in section 5.3.

Apart from *ello* ‘it’, the stressed pronouns of Spanish map onto the first three class of nouns in (9) above, as indicated by the addition of the third-person pronouns in (27).

- (27) a. Animate masculine (pro)nouns that presuppose gender *rey* ‘king’, *él* ‘he’
 b. Animate feminine (pro)nouns that presuppose gender *reina* ‘queen’, *ella* ‘she’
 c. Animate masculine (pro)nouns *cocinero* ‘chef’, *ellos* ‘they’
 d. Animate feminine nouns *persona* ‘person’
 e. Inanimate masculine nouns *libro* ‘book’
 f. Inanimate feminine nouns *receta* ‘recipe’

The plural feminine forms *nosotras* ‘we’, *vosotras* ‘you all (informal)’, and *ellas* ‘they’ also belong to the second class in (27b), as do the feminine guises of *yo* ‘I’, *tú/vos* ‘you (informal)’, *usted* ‘you (formal)’, and *ustedes* ‘you all’. The remaining pronouns, which include *nosotros* ‘we’, *vosotros* ‘you all (informal)’, and the masculine guises of *yo*, *tú/vos*, *usted*, and *ustedes*, all belong to the third class in (27c). Lastly, I consider *ello* to have the same lexical component as inanimate masculine nouns, a view that I elaborate on in section 5.3, but refrain from classifying this pronoun as

inanimate based on the data in (26), where *ello* cannot be coreferential with either of the inanimate nouns. Having identified the ways in which gender does, and does not, have semantic import in Spanish, the next step is to formalize a system that derives the contrasts in (27), ideally one that captures nominal and pronominal expressions using the same set of tools.

5.2 Two loci of gender features

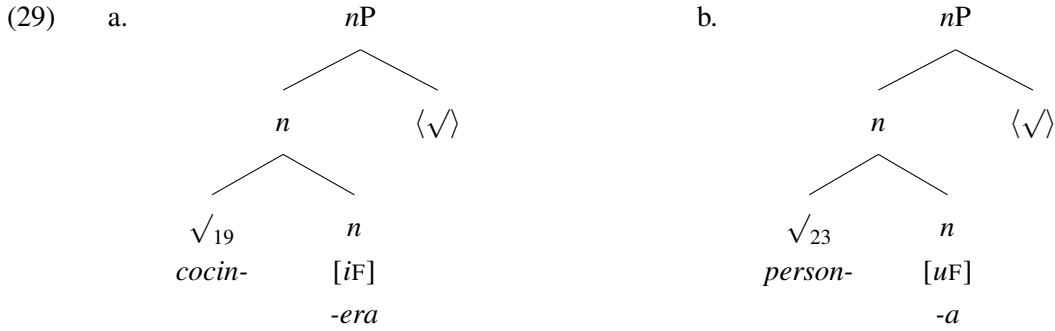
In the theoretical literature, gender is widely recognized to be intertwined with nominality (Corbett 1991, Harris 1991, Picallo 1991, Bernstein 1993, Harris 1996, Panagiotidis 2002, Ralli 2002, Panagiotidis 2003, Alexiadou 2004, Kornfeld & Saab 2004, Picallo 2008). Since the advent of lexical decomposition approaches to morphosyntax, particularly in the framework of Distributed Morphology, there have been numerous proposals that *n* is the locus of gender features (Lecarme 2002, Ferrari 2005, Kihm 2005, Ferrari-Bridgers 2008, Lowenstamm 2008, Saab 2008, Acquaviva 2009, Kramer 2009, Saab 2010).⁴ One reason to suspect that the source of gender is internal to the noun phrase even in the case of pronouns is that masculine and feminine forms can both be used in the absence of a linguistic antecedent (Tasmowski-De Ryck & Verluyten 1981, 1982; Dowty & Jacobson 1989; Sauerland 2007; Elbourne 2013; Patel-Grosz & Grosz 2017). With all three sentences in (28), neither *él* ‘he, him’ nor *ella* ‘she, her’ requires prior mention of the referent as long as the pronoun is accompanied by some type of identificatory gesture.

- (28) a. {Él / Ella} es canadiense.
 he she is Canadian
 ‘He/She is Canadian.’
- b. {Lo / La} conocí a {él / ella} en Francia.
 3SG.ACC 3SG.F.ACC met.1SG DOM him her in France
 ‘I met him/her in France.’
- c. Fui a París por primera vez con {él / ella}.
 went.1SG to Paris for first time with him her
 ‘I went to Paris for the first time with him/her.’

Even if one were to claim that *él* is underspecified for gender, it still must be the case that *ella* contains a feminine feature somewhere within its structure.

To differentiate gender that is relevant for interpretation from gender that is exclusively grammatical, many authors resort to “interpretable” versus “uninterpretable” versions of the same feature, such as [*i*F] and [*u*F] for feminine gender (Matushansky 2013; Kramer 2014, 2015; Anagnostopoulou 2017; Smith 2017; Wurmbrand 2017a,b; Panagiotidis 2019; Smith 2021; Steriopolo 2024; Adamson 2025; Adamson & Anagnostopoulou to appear). For instance, one could distinguish *cocinera* ‘(female) chef’, an animate feminine noun that presupposes femaleness, from *persona* ‘person’, an animate feminine noun that does not presuppose femaleness, through the use of [*i*F] versus [*u*F] features on *n*, as in (29).

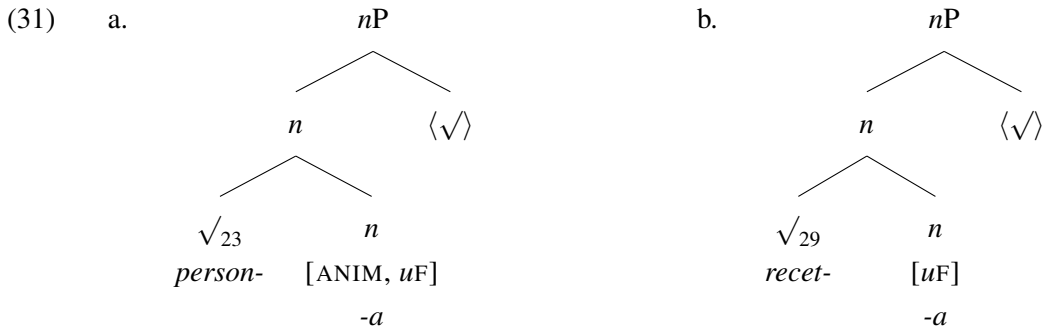
⁴See Fábregas 2022, 2024 for a nanosyntax-based analysis of gender and class markers in Spanish.



Recall from the minimal pair in (8) above, repeated in (30), that *persona* requires differential object marking in contexts where an inanimate feminine noun like *receta* ‘recipe’ does not, indicating that animacy must be syntactically represented as well.

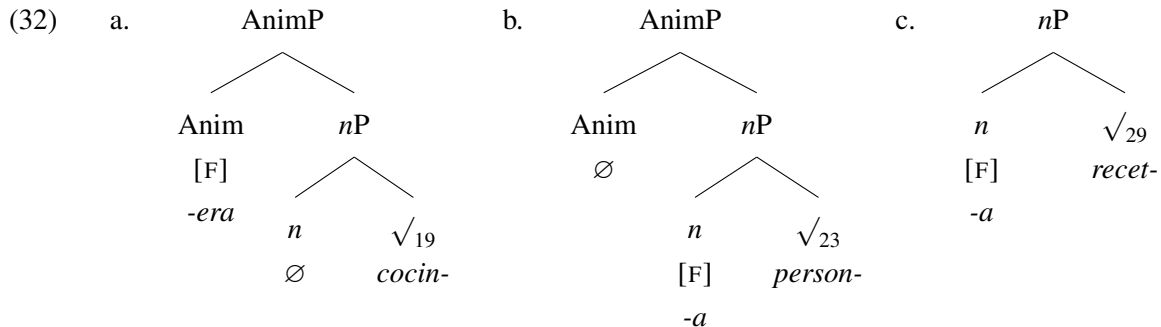
- (30) a. Conozco *(a) esa persona.
 know.1 SG DOM that person
 ‘I know that person.’
 b. Conozco (*a) esa receta.
 know.1 SG DOM that recipe
 ‘I know that recipe.’

One way to account for these facts would be to include an $[ANIM(ATE)]$ feature on *n* in the structure of *persona* but not in that of *receta*, as in (31).

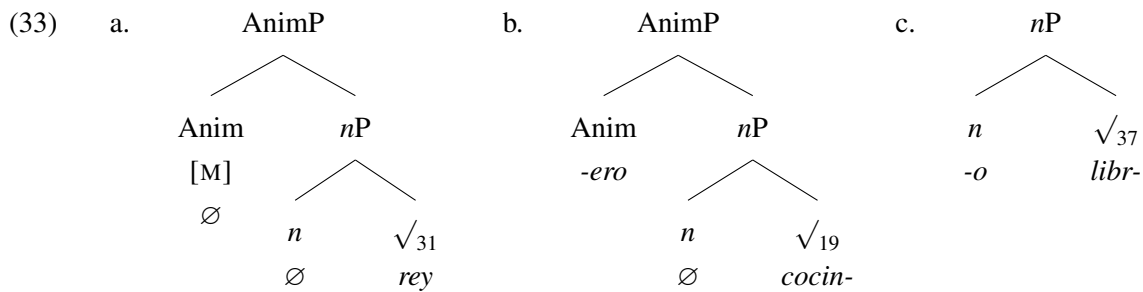


As for *cocinera*, one could either add $[ANIM]$ to *n* in (29a) or take animacy to be entailed by $[iF]$. A more pressing concern is whether the proliferation of feature values in the $[i\gamma]$ - $[u\gamma]$ approach to gender is justified. In section 1.4, I argued that Panagiotidis’s (2019) animacy phrase (AnimP) obviates the need for $[i\gamma]$ - $[u\gamma]$ notation because one can instead maintain that gender features have semantic effects when they are hosted by Anim but not by *n*. Later, in section 3.5, I extended this view to number features on Div versus *n* to explain the difference between canonical plurals like *gatos* ‘cats’ and lexical plurals like *tijeras* ‘scissors’. When gender and number features are present on *n*, a position where they do not “belong”, they are visible at logical form but do not enter into the computation of an interpretable structure.

By incorporating AnimP into the system, the three-way distinction between the feminine nouns *cocinera* ‘(female) chef’, *persona* ‘person’, and *receta* ‘recipe’ can be derived as in (32), where I omit head movement to facilitate the comparison.⁵



Masculine nouns exhibit the same number of contrasts that feminine nouns do: *rey* ‘king’ is animate and presupposes maleness, *cocinero* ‘chef’ is animate but does not presuppose maleness, and *libro* ‘book’ is inanimate. The corresponding structures follow in (33), where I use a privative [M] feature to differentiate *rey* from *cocinero* and *cocinera*.



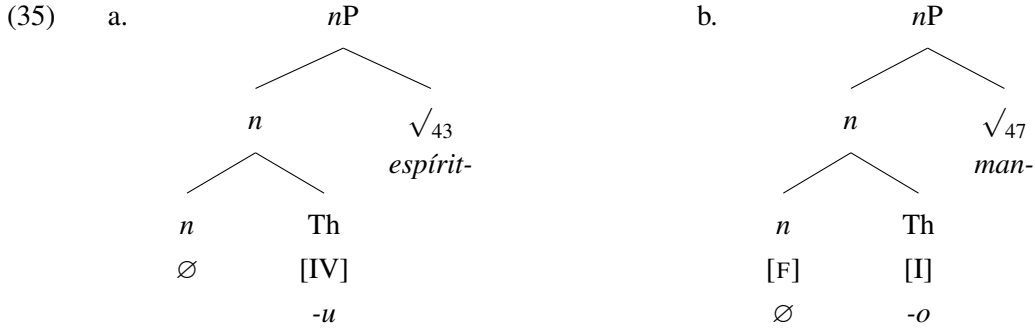
A further advantage of the AnimP-nP split is that it allows one to elegantly capture the rare cases of inanimate nouns that come in inflectional pairs, such as *cerezo/cereza* ‘cherry tree/cherry’ in (34).



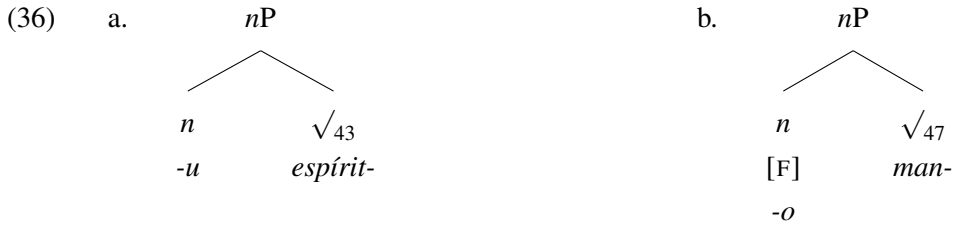
This proposal can be augmented by one’s preferred method of accounting for the various unpredictable forms that class markers can take in Spanish. For example, the masculine noun *espíritu* ‘spirit’ ends in *-u*, and the feminine noun *mano* ‘hand’ ends in *-o* (Harris 1991: 30–31). One option

⁵One could reduce the number of null exponents by positing fusion of *n* and Anim during morphological operations. In (32a) and (33b), I assign the suffix *-er-* to Anim rather than to *n* because there is a homophonous suffix *-er-* that is found in inanimate nouns like *llavero* ‘key ring’, from *llave* ‘key’, and *monedero* ‘coin purse’, from *moneda* ‘coin’.

is given in (35), where theme nodes and class features are inserted during morphological operations (Oltra-Massuet 1999, Embick & Halle 2005, Oltra-Massuet & Arregi 2005, Embick & Noyer 2007, Embick 2010, Lampitelli 2014, Kramer 2015).



Another option would be to appeal to root-conditioned allomorphy of the relevant head, Anim or *n*, allowing for the simpler structures in (36).



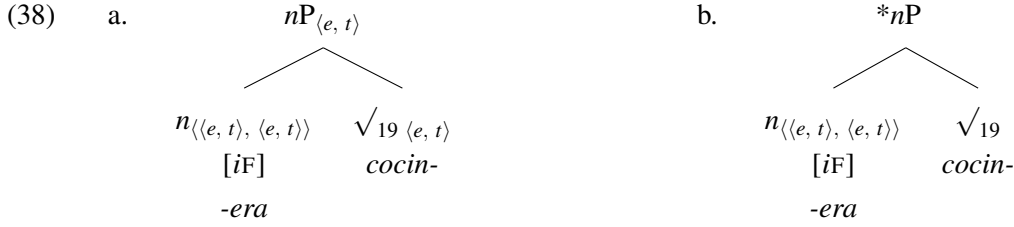
As mentioned in section 1.4, the inclusion of theme nodes is not essential to my analysis, so I abstract away from this issue in morphosyntactic representations of nouns and pronouns.

Not only does the use of AnimP simplify the inventory of gender features, but it also eliminates the need to assume that roots acquire a semantic type independently of their categorizers. To demonstrate this point, it is necessary to first discuss the presuppositional approach to gender that has repercussions at logical form. In (37), I provide a typical lexical entry for an *[F]* feature that applies to property-denoting, or type $\langle e, t \rangle$, expressions and returns a function of the same type that performs two operations: it first checks that for every object *x*, *x* is female, and if this condition is met, the function then maps *x* to the truth value 1 if and only if *x* has property *P* (Percus 2011, Matushansky 2013, Merchant 2014, Spathas & Sudo 2020, Sudo & Spathas 2020).⁶

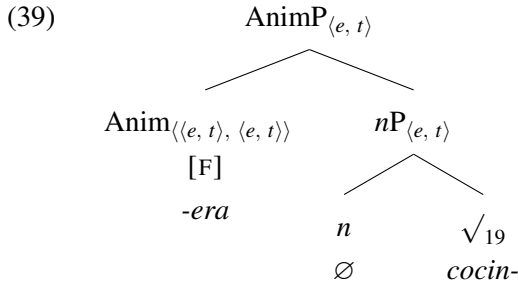
$$(37) \quad \llbracket [F] \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Female}(x) . P(x)$$

To implement this feature in the $[i\gamma]$ - $[u\gamma]$ treatment of gender, the root of *cocinera* ‘(female) chef’ would need to be of type $\langle e, t \rangle$, as in (38a), or else it would not belong to the domain of $n_{[iF]}$, as in (38b), in which case composition could not take place.

⁶Gender and other ϕ -features have long been characterized as presupposition triggers in work on pronouns (Cooper 1983, Dowty & Jacobson 1989, Heim & Kratzer 1998, Sauerland 2003, Schlenker 2003, Buring 2005, Heim 2008, Sauerland 2008a, Kratzer 2009, Jacobson 2012, Sudo 2012).



In the proposed system, where gender features introduce presuppositions of maleness or femaleness when they are subordinate to Anim but not to n , I maintain that it is only the combination of the root and the categorizer in (39) has an interpretation.



I consider Anim to make a semantic contribution even when it does not bear a gender feature and assign this head the lexical entry in (40), according to which it triggers a presupposition of animacy.

$$(40) \quad \llbracket \text{Anim} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Animate}(x) . P(x)$$

When Anim bears an [M] or [F] feature, the presuppositions of animacy and maleness or femaleness are conjoined as in (41).⁷

$$(41) \quad \begin{aligned} \text{a. } \llbracket \text{Anim}_{[M]} \rrbracket &= \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Animate}(x) \wedge \text{Male}(x) . P(x) \\ \text{b. } \llbracket \text{Anim}_{[F]} \rrbracket &= \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Animate}(x) \wedge \text{Female}(x) . P(x) \end{aligned}$$

For instance, the interpretable structure of *cocinera* in (39) has the semantic value in (42), where this noun denotes a function that first checks that for every object x , x is animate and female, and then maps x to the truth value 1 if and only if x is a chef.

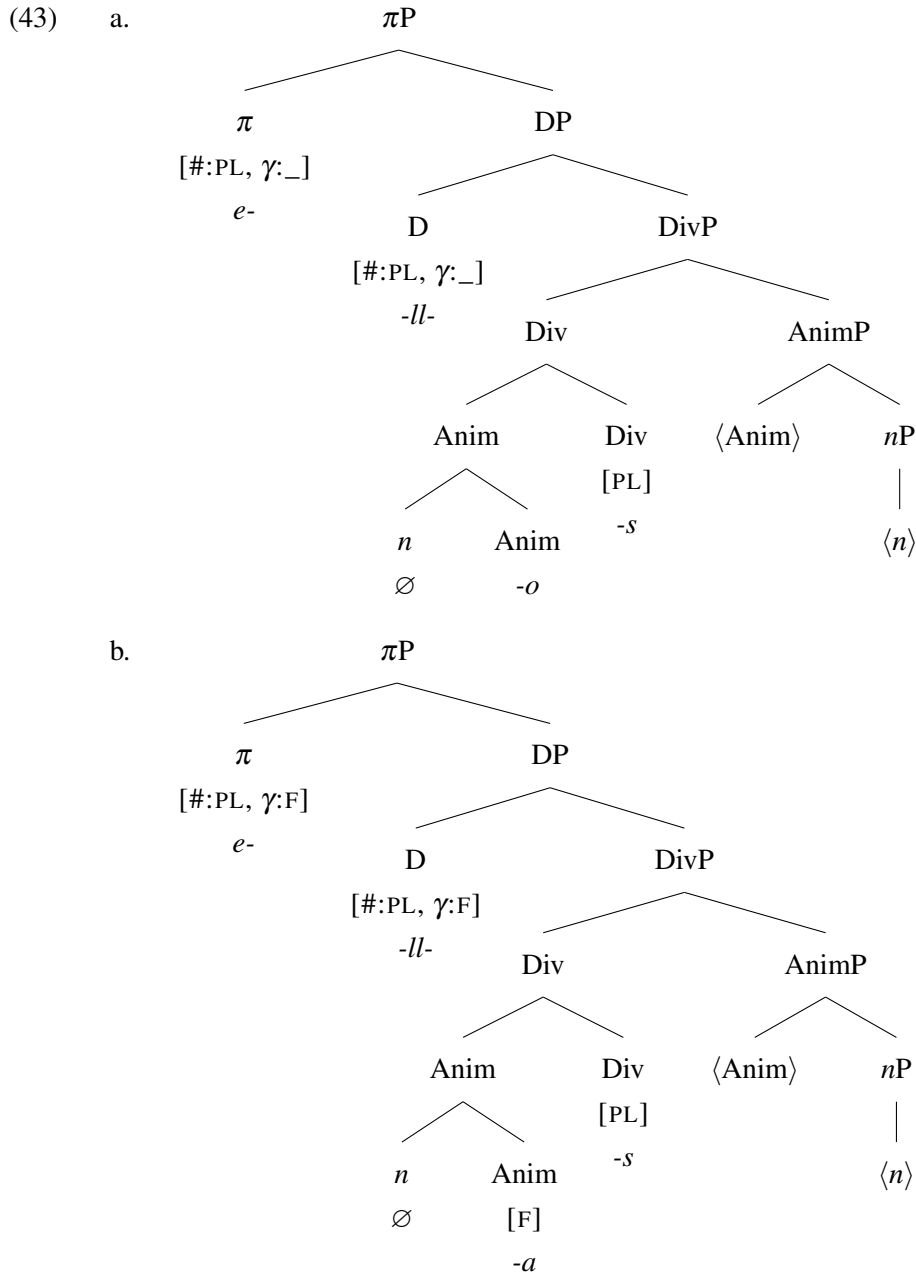
$$(42) \quad \llbracket \text{cocinera} \rrbracket = \lambda x_e : \text{Animate}(x) \wedge \text{Female}(x) . \text{Chef}(x)$$

In the next section, I employ this view of gender and animacy to differentiate the “neuter” pronoun *ello* ‘it’ from *él* ‘he’ and *ella* ‘she’ and then formalize the interpretation of all stressed third-person pronouns in Spanish.

⁷In research on Greek, a language that has masculine, feminine, and neuter gender, Spathas & Sudo (2020) and Sudo & Spathas (2020) argue that certain animate nouns both presuppose and assert maleness or femaleness. It is beyond the scope of this chapter to explore this topic in Spanish, but the lexical entries in (41) could easily be amended to reflect the notion of asserted gender.

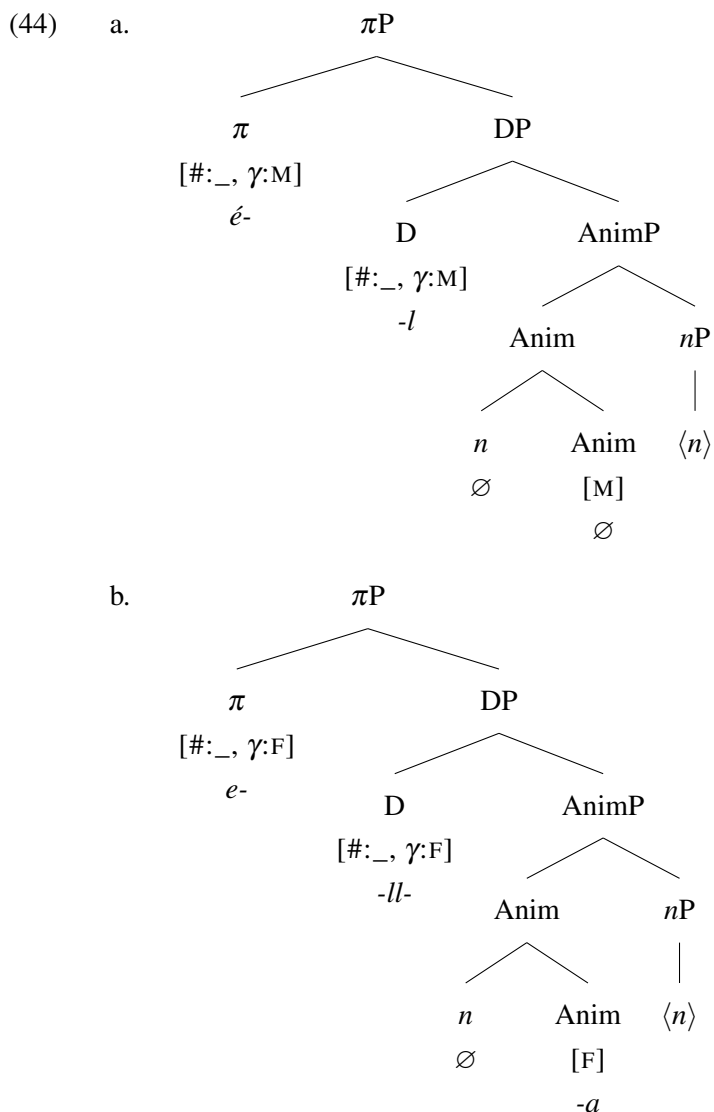
5.3 Third-person pronouns, revisited

Third-person pronouns differ from first-person and second-person pronouns in their intolerance of other material in the noun phrase, as evidenced by the contrast between *nosotros/vosotros/ustedes los cocineros* ‘we/you (informal)/you the chefs’ and **ellos los cocineros* ‘they the chefs’. In section 4.5, I proposed a novel account of this paradigmatic gap by arguing that third-person pronouns spell out the entirety of the noun phrase, as shown with *ellos* ‘they’ and *ellas* ‘they (feminine)’ in (43).



As discussed in section 5.1, *ellas* presupposes femaleness, but *ellos* does not presuppose maleness, so I only include a gender feature on Anim in the structure of *ellas*. The singular pronoun *él* ‘he’,

on the other hand, does presuppose maleness, just as its feminine counterpart *ella* ‘she’ presupposes femaleness, so there is empirical rationale for a gender feature on Anim in both of their morphosyntactic representations in (44).⁸



When it comes to agreement and concord, I assume that the feminine forms of adjectives and determiners are specified for gender and that the masculine, or default, forms are not. As illustrated with the adjective *francés/francesa* ‘French’ in (45), the masculine form can be predicated of *él* ‘he’, which bears an [M] feature, and of *el cocinero* ‘the chef’, which does not, but it cannot be predicated of *ella* ‘she’, which bears an [F] feature.

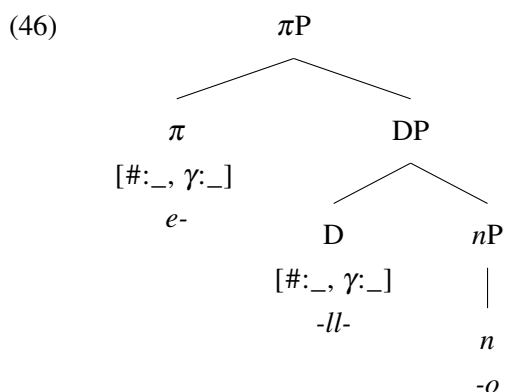
- (45) a. Si *él* es francés, ¿necesita visa?
 if he is French needs visa
 ‘If he is French, does he need a visa?’

⁸See section 3.3 for arguments that DivP does not project in nonplural definite expressions, including pronouns.

- b. Si el cocinero es francés, ¿necesita visa?
 if the chef is French needs visa
 ‘If the chef is French, does he (or she) need a visa?’
- c. Si ella es {francesa / *francés}, ¿necesita visa?
 if she is French French needs visa
 ‘If she is French, does she need a visa?’

In Spanish, arguments can be masculine, feminine, or genderless, but agreement and concord morphology can only be feminine or genderless. This characterization of gender reflects the view of person that I advanced in section 4.4, where I determined that arguments can be first person, second person, third person, or personless, but agreement morphology can only be first person, second person, or personless.

Turning to the “neuter” form *ello* ‘it’, I differentiate this pronoun from *él* ‘he’ and *ella* ‘she’ by omitting AnimP from its structure in (46).



The four other “neuter” determiners of Spanish all end in *-o* as well: *lo* ‘the’, *esto* ‘this’, *eso* ‘that’, and *aquello* ‘that (over there)’. One might ask whether their class markers also occupy *n* instead of D or Dem, but I reject this possibility based on the data in (47), where *ello* is the sole determiner that cannot be followed by a restrictive relative clause.

- (47)
- a. Lo que descubrimos significa que aún hay esperanza.
 the that discovered.1PL means that still has hope
 ‘What we discovered means that there is still hope.’
- b. Esto que descubrimos significa que aún hay esperanza.
 the that discovered.1PL means that still has hope
 Approximate: ‘This thing that we discovered means that there is still hope.’
- c. {Eso / Aquello} que descubrieron significa que aún hay esperanza.
 that that that discovered.3PL means that still has hope
 Approximate: ‘That thing that they discovered means that there is still hope.’
- d. Ello (*que descubrieron) significa que aún hay esperanza.
 it that discovered.3PL means that still has hope
 ‘It means that there is still hope.’

In section 4.5, I accounted for the contrast between *ustedes* ‘you all’ and *ellos* ‘they’ in (48) by arguing that the heads that are realized by third-person pronouns must be strictly adjacent, a configuration that is interrupted by the merge position of restrictive relative clauses below DP.

- (48) Solo {ustedes / *ellos} los que están certificados pueden trabajar ahí.
 only you.PL they the that are.3PL certified can.3PL work there
 ‘Only you who are certified can work there.’

If *lo*, *esto*, *eso*, and *aquello* all spelled out *n* together with D or Dem, they should not tolerate restrictive modification either, yet they do. One way to distinguish these forms from their masculine counterparts *el* ‘the’, *este* ‘this’, *ese* ‘that’, and *aquel* ‘that (over there)’ would be to posit that they surface in rootless structures, but I leave this possibility to future study.

The next step is to derive the semantics of third-person pronouns, and I use the interpretable structures of *él* ‘he’ and *ellas* ‘they (feminine)’ in (49) for this purpose.

- (49) a.
-
- b.
-

As proposed in section 3.1, *nPs* have the “count” denotation in (50a) in the presence of DivP and the “mass” denotation in (50b) in the absence of DivP, where *P* is contributed by the root.

- (50) a. $\llbracket nP \rrbracket = \lambda x_e : P(x) \wedge \text{ATOM}(x) . P(x) / \text{--- DivP}$
 b. $\llbracket nP \rrbracket = \lambda x_e . P(x)$

If there is no root, as in the case of pronouns, I consider *nPs* to have the bleached lexical entries in (51), the first of which is comparable to Kayne’s (2004, 2008) light noun **THING** and the second of which can be construed as **STUFF**.

- (51) a. $\llbracket nP \rrbracket = \lambda x_e : \text{Object}(x) \wedge \text{ATOM}(x) . \text{Object}(x) / \text{--- DivP}$
 b. $\llbracket nP \rrbracket = \lambda x_e . \text{Object}(x)$

Next, $\text{Div}_{[\text{PL}]}$ has the semantic value in (52a), where Link’s (1983) star operator (*) generates the algebraic closure of *P*, and when this head combines with an *nP*, the resulting DivP has the semantic value in (52b).

- (52) a. $\llbracket \text{Div}_{[\text{PL}]} \rrbracket = \lambda P_{\langle e, t \rangle} . \lambda x_e . *P(x)$
 b. $\llbracket \text{DivP} \rrbracket = \lambda x_e . *P(x)$, defined only if *P*(*x*) and $\text{ATOM}(x)$

In this system, *nPs* denote properties of unindividuated objects, and DivPs denote properties of individuated objects and their sums. For D, I adopt the lexical entry that Heim (2011: 998) assigns to the definite article in (53), according to which this determiner takes a property-denoting expression and returns a unique maximal object that has the property in question provided that one exists.

- (53) $\llbracket \text{the} \rrbracket = \lambda P_{\langle e, t \rangle} : \exists x_e \forall y_e [\text{MAX}(P)(y) \leftrightarrow x = y] . \iota x_e \text{MAX}(P)(x)$

Lastly, I model third-person π as the vacuous identity function in (54), which applies to an object-denoting expression and yields the same expression.

- (54) $\llbracket \pi \rrbracket = \lambda x_e . x$

Alternatively, this head could encode the “potential” to be a participant in the discourse (Sichel & Wiltschko 2021, Toosarvandani 2023, Sichel & Toosarvandani to appear). This notion has not yet been formalized in the literature, however, and I do not attempt to do so here.

It is now possible to calculate the denotations of the pronouns in (49) above. I start with *él* ‘he’ in (55), where “LE” stands for the lexical entries and “BR” stands for β -reduction, or simplification.

- (55) $\llbracket \acute{e} \rrbracket (\llbracket -I \rrbracket (\llbracket \emptyset \rrbracket (\llbracket \emptyset \rrbracket)))$
 $= \llbracket \acute{e} \rrbracket (\llbracket -I \rrbracket (\llbracket \lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Animate}(x) \wedge \text{Male}(x) . P(x) \rrbracket (\lambda y_e . \text{Object}(y))))$ LE
 $= \llbracket \acute{e} \rrbracket (\llbracket -I \rrbracket (\lambda x_e : \text{Animate}(x) \wedge \text{Male}(x) . \text{Object}(x)))$ BR
 $= \llbracket \acute{e} \rrbracket (\llbracket \lambda P_{\langle e, t \rangle} : \exists y_e \forall z_e [\text{MAX}(P)(z) \leftrightarrow y = z] . \iota y_e \text{MAX}(P)(y) \rrbracket (\lambda x_e : \text{Animate}(x) \wedge \text{Male}(x) . \text{Object}(x)))$ LE

$$\begin{aligned}
&= \llbracket \acute{e} \rrbracket (\iota y_e \text{MAX}(\text{Object})(y)) && \text{BR} \\
&\quad \text{defined only if } \text{Animate}(y), \text{Male}(y), \text{ and } \exists y_e \forall z_e [\text{MAX}(\text{Object})(z) \leftrightarrow y = z] \\
&= [\lambda x_e . x] (\iota y_e \text{MAX}(\text{Object})(y)) && \text{LE} \\
&\quad \text{defined only if } \text{Animate}(y), \text{Male}(y), \text{ and } \exists y_e \forall z_e [\text{MAX}(\text{Object})(z) \leftrightarrow y = z] \\
&= \iota y_e \text{MAX}(\text{Object})(y) && \text{BR} \\
&\quad \text{defined only if } \text{Animate}(y), \text{Male}(y), \text{ and } \exists y_e \forall z_e [\text{MAX}(\text{Object})(z) \leftrightarrow y = z]
\end{aligned}$$

In prose, *él* denotes a unique maximal object y that is defined only if y is animate, male, and such a unique maximal object exists. The only semantic difference between this pronoun and a definite description like *el rey* ‘the king’, whose denotation is given in (56), is that *él* involves a more abstract property than *el rey* because it lacks a root.

$$\begin{aligned}
(56) \quad \llbracket \text{el rey} \rrbracket &= \iota x_e \text{MAX}(\text{King})(x) \\
&\quad \text{defined only if } \text{Animate}(x), \text{Male}(x), \text{ and } \exists x_e \forall y_e [\text{MAX}(\text{King})(y) \leftrightarrow x = y]
\end{aligned}$$

The computation of *ellas* ‘they (feminine)’, which bears an [F] rather than [M] feature and contains DivP in its structure, follows in (57).

$$\begin{aligned}
(57) \quad &\llbracket \text{e} \rrbracket (\llbracket \text{II} \rrbracket (\llbracket \text{s} \rrbracket (\llbracket \text{a} \rrbracket (\llbracket \emptyset \rrbracket)))) \\
&= \llbracket \text{e} \rrbracket (\llbracket \text{II} \rrbracket (\llbracket \text{s} \rrbracket ([\lambda P_{\langle e, t \rangle} . \lambda x_e : \text{Animate}(x) \wedge \text{Female}(x) . P(x)](\lambda y_e : && \text{LE} \\
&\quad \text{Object}(y) \wedge \text{ATOM}(y) . \text{Object}(y)))))) \\
&= \llbracket \text{e} \rrbracket (\llbracket \text{II} \rrbracket (\llbracket \text{s} \rrbracket (\lambda x_e : \text{Animate}(x) \wedge \text{Female}(x) . \text{Object}(x)))), && \text{BR} \\
&\quad \text{defined only if } \text{Object}(x) \text{ and } \text{ATOM}(x) \\
&= \llbracket \text{e} \rrbracket (\llbracket \text{II} \rrbracket ([\lambda P_{\langle e, t \rangle} . \lambda y_e . *P(y)](\lambda x_e : \text{Animate}(x) \wedge \text{Female}(x) . && \text{LE} \\
&\quad \text{Object}(x))))), \\
&\quad \text{defined only if } \text{Object}(x) \text{ and } \text{ATOM}(x) \\
&= \llbracket \text{e} \rrbracket (\llbracket \text{II} \rrbracket (\lambda y_e . *\text{Object}(y))), && \text{BR} \\
&\quad \text{defined only if } \text{Object}(y), \text{ATOM}(y), \text{Animate}(y), \text{ and } \text{Female}(y) \\
&= \llbracket \text{e} \rrbracket ([\lambda P_{\langle e, t \rangle} : \exists x_e \forall z_e [\text{MAX}(P)(z) \leftrightarrow x = z] . \iota x_e \text{MAX}(P)(x)](\lambda y_e . && \text{LE} \\
&\quad *\text{Object}(y))) \\
&\quad \text{defined only if } \text{Object}(y), \text{ATOM}(y), \text{Animate}(y), \text{ and } \text{Female}(y) \\
&= \llbracket \text{e} \rrbracket (\iota x_e \text{MAX}(*\text{Object})(x)), && \text{BR} \\
&\quad \text{defined only if } \text{Object}(x), \text{ATOM}(x), \text{Animate}(x), \text{Female}(x), \\
&\quad \text{and } \exists x_e \forall z_e [\text{MAX}(*\text{Object})(z) \leftrightarrow x = z] \\
&= [\lambda y_e . y] (\iota x_e \text{MAX}(*\text{Object})(x)), && \text{LE} \\
&\quad \text{defined only if } \text{Object}(x), \text{ATOM}(x), \text{Animate}(x), \text{Female}(x), \\
&\quad \text{and } \exists x_e \forall z_e [\text{MAX}(*\text{Object})(z) \leftrightarrow x = z] \\
&= \iota x_e \text{MAX}(*\text{Object})(x), && \text{BR} \\
&\quad \text{defined only if } \text{Object}(x), \text{ATOM}(x), \text{Animate}(x), \text{Female}(x), \\
&\quad \text{and } \exists x_e \forall z_e [\text{MAX}(*\text{Object})(z) \leftrightarrow x = z]
\end{aligned}$$

That is, *ellas* denotes a unique maximal sum of objects x that is defined only if x is atomic, animate, female, and such a unique maximal sum of objects exists. Here as well, the only difference between *ellas* in (57) and a definite description like *las cocineras* ‘the (female) chefs’ in (58) is the property Object versus Chef.

$$(58) \quad \llbracket \text{las cocineras} \rrbracket = \iota x_e \text{MAX}(*\text{Chef})(x),$$

defined only if $\text{Chef}(x)$, $\text{ATOM}(x)$, $\text{Animate}(x)$, $\text{Female}(x)$,
and $\exists x_e \forall y_e [\text{MAX}(*\text{Chef})(y) \leftrightarrow x = y]$

Having provided interpretable structures and denotations for the third-person pronouns of Spanish, the objective is to now do the same for the first-person and second-person pronouns.

5.4 Heterogeneous plurality via person

The extension of a plural noun like *cocineros* ‘chefs’ is homogeneous in the sense that every object has the property of being a chef, whereas the extension of a plural pronoun like *nosotros* ‘we’ is heterogeneous in the sense that not every object has the property of being the speaker. In recent work, Ghomeshi & Massam (2020: 598) have attributed this contrast to the syntactic configurations in (59), where number features belong to their own functional projection in nouns but are “subordinated to person” in pronouns.

- (59) a. *Nominal structure*
 $[\text{DP D } [\text{NumP Num } [\text{nP } n \text{ } [\text{VP Root}]]]]$
- b. *Pronominal structure*
 $[\text{DP D } [\varphi\text{P } \varphi \text{ } [\text{nP } n \text{ } [\text{VP Root}]]]]$
 (φ includes #, π)

In this section, I dispute the claim that number is fundamentally different in pronominal expressions because it can be shown that the phenomenon of heterogeneous plurality is a natural consequence of the semantics of certain person features. To this end, I begin with the characterization of discourse roles in crosslinguistic research on person.

For Boas (1911: 39), “a true first person plural is impossible, because there can never be more than one self”. It turns out that there is typological support for the view that author, or speaker, is a unique discourse role, as is addressee. To date, no language has been found in which there is a pronominal form that solely denotes a plurality of authors or a plurality of addressees (Zwicky 1977, Greenberg 1988, McGregor 1989, Greenberg 1993, Noyer 1997, Cysouw 2003, Daniel 2005, Simon 2005, Bobaljik 2008, Harbour 2016, Ackema & Neeleman 2018). At most, languages morphologically encode four combinations of discourse roles in the forms of plural pronouns: the author and the addressee (inclusive first person), the author and one or more nonparticipants (exclusive first person), the addressee and one or more nonparticipants (second person), and two or more non-

participants (third person). As shown in (60), Spanish does not distinguish between inclusive and exclusive first person in its pronominal system.

- | | | | |
|------|----|-------------------------------|---|
| (60) | a. | author + addressee | <i>nosotros/nosotras</i> ‘we’ |
| | b. | author + nonparticipant(s) | <i>nosotros/nosotras</i> ‘we’ |
| | c. | addressee + nonparticipant(s) | <i>vosotros/vosotras</i> , ‘you all (informal)’,
<i>ustedes</i> ‘you all (formal only in Spain)’ |
| | d. | nonparticipants | <i>ellos/ellas</i> ‘they’ |

Still, the first-person pronouns *nosotros/nosotras* ‘we/we (feminine)’ decompose into *nos* ‘we’ and *otros/otras* ‘others/others (feminine)’, so it is worth exploring whether Spanish may have once differentiated inclusive *nos* from exclusive *nosotros*.

The history of *nosotros* ‘we’ begins with Old Spanish *vos* ‘you all’, a plural pronoun that came to be used for singular formal address, much like French *vous* ‘you all, you (formal)’ (Lapesa 1942, Páez Urdaneta 1981, Penny 1991, Rini 1999, Bentivoglio 2003, Pharies 2007). Old Spanish *vos* eventually lost its honorific status altogether, unlike French *vous*, and to disambiguate plural address from singular address, speakers began to add *otros* ‘others’ to *vos*, resulting in *vosotros* ‘you all (informal)’.⁹ Over time, *vosotros* replaced *vos* as the plural form, and *nosotros* replaced *nos* ‘we’ by analogy. There is, consequently, no diachronic evidence to suggest that *nos* and *nosotros* were ever used contrastively to expressclusivity.

From the form-meaning mappings in (60) emerge the two generalizations in (61), which are consistent with a large body of theoretical and typological work on plurality in pronominal systems (Boas 1911, Lenz 1920, Jespersen 1924, Benveniste 1966, Zwicky 1977, Greenberg 1993, Noyer 1997, Corbett 2000, Cysouw 2003, Moravcsik 2003, Siewierska 2004, Daniel 2005, Bobaljik 2008, Kratzer 2009, Ackema & Neeleman 2013a, Harbour 2016, Ackema & Neeleman 2018, Daniel 2020, Dékány 2021, Toosarvandani 2023).

- | | | |
|------|----|---|
| (61) | a. | The extensions of plural first-person and second-person pronouns are heterogeneous. |
| | b. | The extensions of plural third-person pronouns are homogeneous. |

Number is not different in all pronominal expressions, pace Ghomeshi & Massam (2020), but rather only in first-person and second-person forms.

In the interpretable structures of *él* ‘he’ and *ellas* ‘they (feminine)’ in (49) above, I assigned the semantic type $\langle e, e \rangle$ to π so that this head applies to and returns an object-denoting expression. As for the contribution of [PARTICIPANT] and [AUTH(OR)] features on π , both are usually taken to either presuppose or assert content that relates to discourse roles (Heim & Kratzer 1998, Sauerland 2003, Heim 2008, Sauerland 2008a, Bobaljik & Sauerland 2023, Toosarvandani 2023, Alexiadou et al. 2024, Deal to appear, Sichel & Toosarvandani to appear). In my account of these features, I adopt Harbour’s (2016) labels *i* (“I”) for the author and *u* (“you”) for the addressee along with

⁹In English, many speakers use *you all* or a similar collocation for plural reference. The equivalent in Spanish, *vos todos* ‘you all’, is attested in documents from the thirteenth and fourteenth centuries but ultimately fell out of use (Rini 1999).

Bobaljik & Sauerland’s (2023) use of the parthood relation $\leq (\sqsubseteq)$. As formalized in (62), [PART] presupposes that i_c (the author of context c) or u_c (the addressee of context c) is a part of object x , while [AUTH] presupposes that i_c is a part of object x .

$$(62) \quad \begin{aligned} \text{a. } \llbracket [\text{PART}] \rrbracket^c &= \lambda x_e : i_c \leq x \vee u_c \leq x . x \\ \text{b. } \llbracket [\text{AUTH}] \rrbracket^c &= \lambda x_e : i_c \leq x . x \end{aligned}$$

The disjunction in the lexical entry for [PART] results in an intuitive semantics for first-person $\pi_{[\text{PART}, \text{AUTH}]}$ in (63), where I conjoin and simplify the presuppositions of [PART] and [AUTH].

$$(63) \quad \begin{aligned} \llbracket \pi_{[\text{PART}, \text{AUTH}]} \rrbracket^c &= \lambda x_e : [i_c \leq x \vee u_c \leq x] \wedge i_c \leq x . x \\ &= \lambda x_e : i_c \leq x . x \end{aligned}$$

Second-person $\pi_{[\text{PART}]}$ is less straightforward in that one must effectively remove $i_c \leq x$ from its denotation. One option is to follow Deal (to appear) and invoke Heim’s (1991) competition principle Maximize Presupposition, as in (64), where the presupposed content of $\pi_{[\text{PART}]}$ strengthens to $u_c \leq x$ because $\pi_{[\text{PART}, \text{AUTH}]}$ already presupposes $i_c \leq x$.¹⁰

$$(64) \quad \begin{aligned} \llbracket \pi_{[\text{PART}]} \rrbracket^c &= \lambda x_e : i_c \leq x \vee u_c \leq x . x, \text{ which strengthens to} \\ &\lambda x_e : u_c \leq x . x, \text{ given that } \llbracket \pi_{[\text{PART}, \text{AUTH}]} \rrbracket^c = \lambda x_e : i_c \leq x . x \end{aligned}$$

I now demonstrate how the lexical entries for $\pi_{[\text{PART}, \text{AUTH}]}$ in (63) and $\pi_{[\text{PART}]}$ in (64) predict the heterogeneous plurality of first-person and second-person pronouns.

The interpretable structure of *nosotras* ‘we (feminine)’ in (65) has a DP component that is semantically, though not phonologically, identical to that of *ellas* ‘they (feminine)’ in (49b) above.

$$(65) \quad \begin{array}{c} \pi P_e \\ \swarrow \quad \searrow \\ \pi_{\langle e, e \rangle} \quad DP_e \\ [\text{PART}, \text{AUTH}] \quad \triangle \\ \text{nosotras} \quad \emptyset \end{array}$$

It is therefore possible to substitute the denotation of the DP from the calculation of *ellas* in (57) in the calculation of *nosotras* in (66).

$$(66) \quad \begin{aligned} \llbracket \text{nosotras} \rrbracket^c (\llbracket \emptyset \rrbracket^c) &= [\lambda x_e : i_c \leq x . x] (\iota y_e \text{ MAX}(*\text{Object})(y)), & \text{LE} \\ &\text{defined only if } \text{Object}(y), \text{ATOM}(y), \text{Animate}(y), \text{Female}(y), \\ &\text{and } \exists y_e \forall z_e [\text{MAX}(*\text{Object})(z) \leftrightarrow y = z] \end{aligned}$$

¹⁰Another option would be to follow Bobaljik & Sauerland (2023) and Alexiadou et al. (2024) and resort to exhaustification to derive the intended semantics of $\pi_{[\text{PART}]}$. I do not pursue this approach because it is at odds with the geometry of person features that I assume in this thesis.

$$\begin{aligned}
 &= \iota y_e \text{ MAX}(*\text{Object})(y), & \text{BR} \\
 &\text{defined only if Object}(y), \text{ATOM}(y), \text{Animate}(y), \text{Female}(y), \\
 &\exists y_e \forall z_e [\text{MAX}(*\text{Object})(z) \leftrightarrow y = z], \text{ and } i_c \leq \iota y_e \text{ MAX}(*\text{Object})(y)
 \end{aligned}$$

In other words, *nosotras* denotes a unique maximal sum of objects y that is defined only if y is atomic, animate, female, such a unique maximal sum of objects exists, and the speaker of the discourse context (i_c) is a part of this sum. The final presupposition is responsible for the heterogeneity of *nosotras* and is all that distinguishes this pronoun from *ellas*.

To capture the masculine pronoun *nosotros* ‘we’, all that is needed is to delete $\text{Female}(y)$ from the calculation of *nosotras* ‘we (feminine)’ in (66) since *nosotros* does not trigger a gender-based presupposition. The same reasoning applies to *vosotros* ‘you all (informal)’ and the masculine guises of *yo* ‘I’ and *tú/vos* ‘you (informal)’ below, so I focus on the more semantically complex feminine forms and guises of these pronouns in the final part of this section. Before doing so, it bears mentioning that one can derive the interpretation of pronoun-noun constructions like *nosotros los cocineros/nosotras las cocineras* ‘we the chefs’ by taking the denotation of *nosotros/nosotras* and replacing Object with Chef . After all, the only difference in meaning between *nosotros los cocineros* and *los cocineros* ‘the chefs’ is that *nosotros los cocineros* presupposes that the speaker is among the referents of the expression and *los cocineros* does not.

Next, in (67), I provide the interpretable structure and denotation of *vosotras* ‘you all (informal, feminine)’, which has the same DP as *nosotras* ‘we (feminine)’ and *ellas* ‘they (feminine)’.

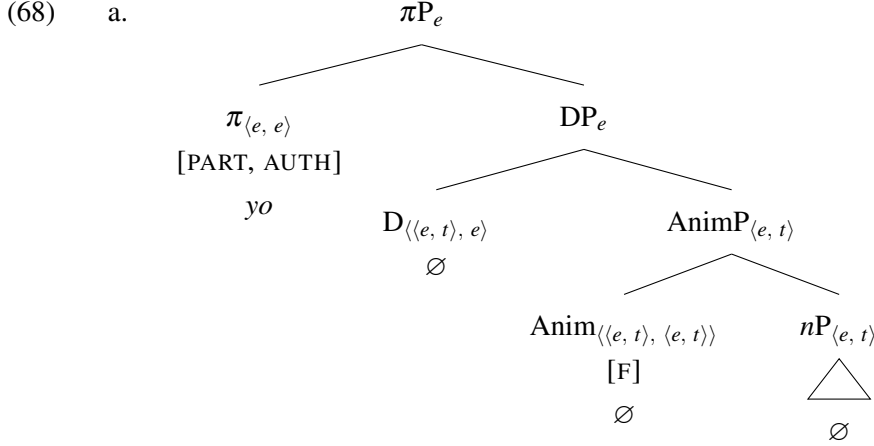
(67) a. πP_e

$$\begin{array}{cc}
 & \swarrow \quad \searrow \\
 \pi_{\langle e, e \rangle} & DP_e \\
 [\text{PART}] & \triangle \\
 \text{vosotras} & \emptyset
 \end{array}$$

b. $\llbracket \text{vosotras} \rrbracket^c = \iota x_e \text{ MAX}(*\text{Object})(x),$
defined only if $\text{Object}(x), \text{ATOM}(x), \text{Animate}(x),$
 $\text{Female}(x), \exists x_e \forall y_e [\text{MAX}(*\text{Object})(y) \leftrightarrow x = y],$
and $u_c \leq \iota x_e \text{ MAX}(*\text{Object})(x)$

This pronoun denotes a unique maximal sum of objects x that is defined only if x is atomic, animate, female, such a unique maximal sum of objects exists, and the addressee of the discourse context (u_c) is a part of this sum. In this case, the phenomenon of heterogeneous plurality results from the presupposition that the addressee is among the referents of *vosotras*.

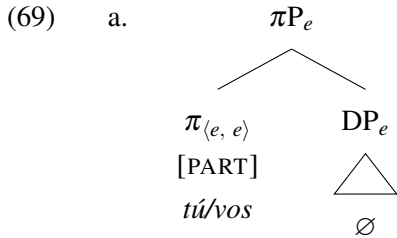
The final issue to explore is how the lexical entries for $\pi_{[\text{PART}, \text{AUTH}]}$ and $\pi_{[\text{PART}]}$ account for the singular pronouns *yo* ‘I’ and *tú/vos* ‘you (informal)’, which lack DivP , as do all nonplural definite expressions in the system that I have developed. The interpretable structure and semantic value of *yo* in its feminine guise are both given in (68).



- b. $\llbracket yo \rrbracket^c = \iota x_e \text{MAX}(\text{Object})(x)$,
 defined only if $\text{Animate}(x)$, $\text{Female}(x)$,
 $\exists x_e \forall y_e [\text{MAX}(\text{Object})(y) \leftrightarrow x = y]$, and $i_c \leq \iota x_e \text{MAX}(\text{Object})(x)$

That is, the feminine guise of *yo* denotes a unique maximal object x that is defined only if x is animate, female, such a unique maximal object exists, and the speaker of the discourse context (i_c) is a part of this object. Since the relation between i_c and x is one of parthood ($\leq$), and not of proper parthood ($<$), the speaker can be the same object as the referent of the DP. In fact, this outcome is inevitable due to the absence of DivP in the structure, which would be needed for the DP constituent of *yo* to denote a unique maximal sum of atomic objects instead of a unique maximal object.

Finally, *tú/vos* ‘you (informal)’ in their feminine guises have the person specification in (69a) and the denotation in (69b).



- b. $\llbracket tú/vos \rrbracket^c = \iota x_e \text{MAX}(\text{Object})(x)$,
 defined only if $\text{Animate}(x)$, $\text{Female}(x)$,
 $\exists x_e \forall y_e [\text{MAX}(\text{Object})(y) \leftrightarrow x = y]$, and $u_c \leq \iota x_e \text{MAX}(\text{Object})(x)$

Accordingly, these pronouns denote a unique maximal object x that is defined only if x is animate, female, such a unique maximal object exists, and the addressee of the discourse context (u_c) is a part of this object, which allows for the possibility that the addressee is the same as this object. Not only does this approach to person features capture the heterogeneity of plural first-person and second-person pronouns, but it also derives the correct semantics of their singular counterparts without any further assumptions or theoretical machinery.

5.5 Formality and impoverishment

The objective of this final section is to determine the syntactic identity of the pronouns *usted* ‘you (formal)’ and *ustedes* ‘you all’, the latter of which is formal only in Peninsular Spanish, and then provide their morphosyntactic and interpretable structures. As I first mentioned in section 4.1 and illustrate in (70) and (71), *usted* and *ustedes* are interpreted as second-person pronouns but occur with third-person verb forms and pronominal clitics.

- (70) a. *Usted* {es / *eres} brillante.
 you is are.2SG brilliant
 ‘You (formal) are brilliant.’
 b. *Ustedes* {son / *sois} brillantes.
 you.PL are.3PL are.2PL brilliant
 ‘You all are brilliant.’
- (71) a. {Lo / La / *Te} admiro a usted.
 3SG.ACC 3SG.F.ACC 2SG admire.1SG DOM you
 ‘I admire you (formal).’
 b. {Los / Las / *Os} admiro a ustedes.
 3PL.ACC 3PL.F.ACC 2PL admire.1SG DOM you.PL
 ‘I admire you all.’

Wang (2023) regards *usted* and *ustedes* as “lexicalized” honorifics because they evolved from the nominal expressions *vuestra merced* ‘your mercy’ and *vuestras mercedes* ‘your mercies’, respectively. Another common strategy that languages use to convey formal address is to recruit existing pronouns for this purpose, as is the case with French *vous* ‘you all, you (formal)’ and Italian *lei/Lei* ‘she/you (formal)’ and *loro/Loro* ‘they/you all (formal)’.

For *usted* ‘you (formal)’ and *ustedes* ‘you all’ to be interpreted as second person, they must either bear [PART] prior to logical form or acquire this feature at logical form. I evaluate these alternatives by investigating the behaviour of these pronouns with respect to the Person-Case Constraint, a well-studied phenomenon in languages that have clitics (Perlmutter 1970, 1971; Contreras & Rojas 1972; Kayne 1975; Bonet i Alsina 1991; Laenzlinger 1993; Bonet 1994, 1995; Parodi 1998; Anagnostopoulou 2003; Béjar 2003; Béjar & Rezac 2003, Anagnostopoulou 2005; Bianchi 2006; Adger & Harbour 2007; Nevins 2007; Ormazabal & Romero 2007; Béjar & Rezac 2009; Nevins 2011; Rezac 2011; Ormazabal & Romero 2013; Arregi & Nevins 2018; Toosarvandani 2023; Sichel & Toosarvandani to appear).¹¹ In constructions that involve accusative and dative clitics in Spanish, the accusative clitic cannot be first or second person if the dative clitic is third person, as in (72), where *te* ‘you (informal)’ can only be the dative clitic in the response to the question.

- (72) Julia, ¿te acuerdas de mis amigos?
 Julia 2SG remember.2SG of my friends
 ‘Julia, do you (informal) remember my friends?’

¹¹I thank Susana Béjar for the suggestion to explore clitic combinations in this endeavour.

- a. Te los presenté el año pasado.
2SG 3PL.ACC presented.1SG the year past
'I introduced them to you last year.'
- b. *Les te presenté el año pasado.
3PL.DAT 2SG presented.1SG the year past
Intended: 'I introduced you to them last year.'
- c. *Te les presenté el año pasado.
2SG 3PL.DAT presented.1SG the year past
Intended: 'I introduced you to them last year.'
- d. *Se te presenté el año pasado.
3 2SG presented.1SG the year past
Intended: 'I introduced you to them last year.'

If both clitics are third person, as in the answers to the question in (73), either form can occupy the accusative position.

- (73) ¿Ella se acuerda de mis amigos?
she 3 remembers of my friends
'Does she remember my friends?'

- a. Se los presenté el año pasado.
3 3PL.ACC presented.1SG the year past
'I introduced them to her last year.'
- b. Se la presenté el año pasado.
3 3SG.F.ACC presented.1SG the year past
'I introduced her to them last year.'

Given the argument structure of the question, the response in (73a) would be more natural, but that in (73b) is possible as well. The crucial point is that the only clitics that face distributional restrictions in Spanish are those that are coreferential with first-person and second-person arguments.

The next issue to consider is whether clitics that are coreferential with *usted* 'you (formal)' and *ustedes* 'you all' display rigid or flexible ordering in relation to clitics that are coreferential with third-person arguments. The data sets in (74) and (75) indicate that *usted* and *ustedes* both pattern like second-person pronouns in that the corresponding clitics must occur in dative position for the answers to be felicitous.¹²

- (74) Señora Child, ¿usted se acuerda de mis amigos?
Mrs. Child you 3 remembers of my friends
'Mrs. Child, do you (formal) remember my friends?'
- a. Se los presenté el año pasado.
3 3PL.ACC presented.1SG the year past
'I introduced them to you (formal) last year.'

¹²Rezac (2011) notes this effect with *usted* but takes this pronoun to be syntactically third person.

- b. #Se la presenté el año pasado.
 3 3SG.F.ACC presented.1SG the year past
 Intended: ‘I introduced you (formal) to them last year.’
- (75) Señora Child y señora Pépin, ¿ustedes se acuerdan de mis amigos?
 Mrs. Child and Mrs. Pépin you.PL 3 remember.3PL of my friends
 ‘Mrs. Child and Mrs. Pépin, do you remember my friends?’
- a. Se los presenté el año pasado.
 3 3PL.ACC presented.1SG the year past
 ‘I introduced them to you last year.’
- b. #Se las presenté el año pasado.
 3 3PL.F.ACC presented.1SG the year past
 Intended: ‘I introduced you to them last year.’

The contrast between the identical responses in (73b) and (74b) suggests that *usted* and *ustedes* originate with a [PART] feature in the syntax, which would not only explain their sensitivity to the Person-Case Constraint but also their interpretation as second rather than third person. It cannot, however, be the case that [PART] remains present when these pronouns serve as goals for agreement because neither one appears with second-person verb forms or clitics. In the next part of this section, I address the fate of their [PART] feature in different components of the grammar.

In Distributed Morphology, postsyntactic rules of impoverishment are often used to account for discrepancies between form and interpretation, but there must be an identifiable trigger. To capture the unique properties of *usted* ‘you (formal)’ and *ustedes* ‘you all’, I pursue an impoverishment-based analysis and argue that this trigger is an [HON(ORIFIC)], or [FORMAL], feature on π . During syntactic operations, *usted* and *ustedes* bear [HON] in addition to [PART], and during morphological operations, [HON] prompts the deletion of [PART] through the rule in (76).

- (76) [PART] $\rightarrow \emptyset / __$ [HON] Morphological operations

To reconcile this rule with the feature-geometric approach to person that I defended in section 4.4, I position [HON] on a separate node under π , as in (77a), instead of on one that is a dependent of [PART], as in (77b), so that [HON] does not bring about its own delinking.

- (77) a.
- b.

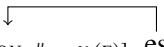
Since impoverishment and other morphological operations take place before lexical insertion, I do not include [PART] in the vocabulary items for *usted* and *ustedes* in (78).

- (78) a. /ustedes/ $\leftrightarrow \pi_{[\text{HON}, \#:\text{PL}]}$
 b. /usted/ $\leftrightarrow \pi_{[\text{HON}]}$

It is essential to note that [HON] is all that differentiates the vocabulary items in (78) from those that pertain to the other stressed pronouns in (79), so there is independent theoretical motivation for this feature in Spanish, pace Wang (2023).

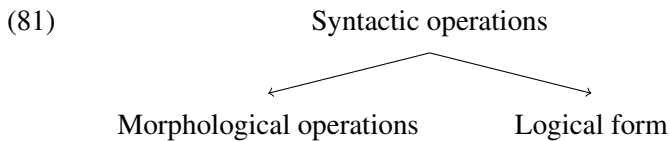
- (79) a. /nosotras/ $\leftrightarrow \pi_{[\text{AUTH}, \#:\text{PL}, \gamma:\text{F}]}$
 b. /vosotras/ $\leftrightarrow \pi_{[\text{PART}, \#:\text{PL}, \gamma:\text{F}]}$
 c. /nosotros/ $\leftrightarrow \pi_{[\text{AUTH}, \#:\text{PL}]}$
 d. /vosotros/ $\leftrightarrow \pi_{[\text{PART}, \#:\text{PL}]}$
 e. /yo/ $\leftrightarrow \pi_{[\text{AUTH}]}$
 f. /tu/ or /vos/ $\leftrightarrow \pi_{[\text{PART}]}$
 g. /e/ $\leftrightarrow \pi$

By the time *usted* and *ustedes* serve as goals for agreement in the clause, they have both lost their [PART] feature, so it follows that these pronouns occur with third-person rather than second-person verb forms and clitics in (80).

- (80) a.  Usted_[\pi:\text{HON}, \#:_, \gamma:(\text{F})] es brillante.
 you is brilliant
 ‘You (formal) are brilliant.’
 b.  Las admiro a ustedes_[\pi:\text{HON}, \#:\text{PL}, \gamma:\text{F}].
 3PL.F.ACC admire.1SG DOM you.PL
 ‘I admire you all (feminine).’

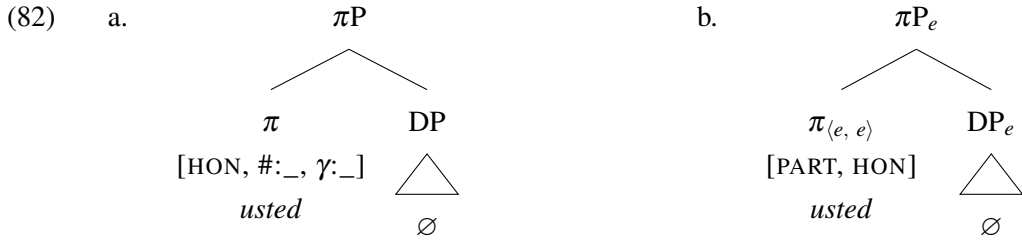
In previous work, the syncretism that is found with *usted* and *ustedes* has been attributed to impoverishment processes that target verbs and clitics, or the probes for agreement (Halle & Marantz 1994; Harris 1995, 1998; Arregi & Nevins 2018). In my account, on the other hand, this syncretism is the outcome of a rule that targets the pronouns themselves, or the goals for agreement.

In the architecture of the grammar that I adopted in section 1.3, part of which is reproduced in (81), the bifurcation after syntactic operations means that morphological operations like impoverishment have no bearing on logical form.



If, as I maintain, *usted* ‘you (formal)’ and *ustedes* ‘you all’ are generated with a [PART] feature in the syntax, they retain this feature at logical form even if they lose it along the route to phonological form. For example, the masculine guise of *usted* has the morphosyntactic structure in (82a), where

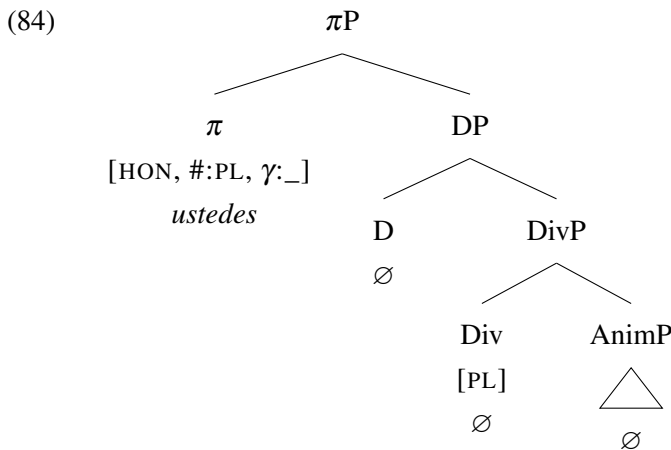
[PART] has been deleted in the context of [HON], and the interpretable structure in (82b), where both features remain present on π .



The situation is more complex with *ustedes* because Latin American Spanish does not distinguish between formal and informal address in plural pronouns. Consequently, there is no empirical motivation to include [HON] in the interpretable structure of *ustedes* in these varieties. To capture this difference between Latin American and Peninsular Spanish, I propose the rule of impoverishment at logical form in (83), which assumes my claim from section 3.3 that DivP only projects in plural definite expressions.

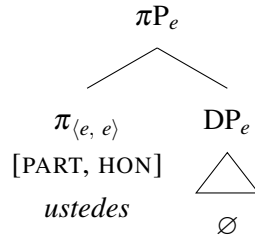
$$(83) \quad \pi_{[PART, HON]} \rightarrow \pi_{[PART]} / \text{--- Div}_{[PL]} \quad \text{Logical form (Latin American Spanish only)}$$

This rule is inspired by Ackema & Neeleman's (2018) analysis of honorific pronouns in several languages and is effectively a type of contextual allosemy in that the interpretation of π is conditioned by a head that is sufficiently local within the noun phase. In all varieties of Spanish, the masculine guise of *ustedes* has the morphosyntactic representation in (84), where [HON] is needed to differentiate *ustedes* as a vocabulary item from the second-person pronouns *vosotros/vosotras* 'you all (informal)' and the *e-* of the third-person pronouns *ellos/ellas* 'they'.

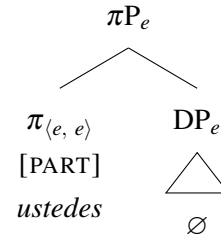


In Peninsular Spanish, the masculine guise of *ustedes* has the interpretable structure in (85a), where [HON] is preserved on π , but in Latin American Spanish, this same pronoun has the interpretable structure in (85b), where [HON] has been eliminated from π and does not factor into its denotation.

(85) a. Peninsular Spanish



b. Latin American Spanish



The next task is to develop a semantics of [HON] such that this feature makes the same contribution in *ustedes* when this pronoun is interpreted formally that it does in *usted*.

I begin with Ackema & Neeleman's (2013a, 2018) observation that the decision to employ a formal versus informal plural pronoun depends on the social status of the addressee, which can differ from the social status of the other referents of the pronoun. Consider a situation where there are four individuals who all speak Peninsular Spanish: a grandfather and grandmother, their granddaughter Carmen, and the granddaughter's friend Marta. If Marta wishes to ask the others about their holiday plans, she would likely formulate her question as in (86a) if speaking directly to Carmen and as in (86b) if speaking to either of the grandparents or to the room in general.

- (86) a. ¿Vosotros tenéis planes para las vacaciones?
 you.PL have.2PL plans for the holidays
 'Do you all (informal) have plans for the holidays?'
 b. ¿Ustedes tienen planes para las vacaciones?
 you.PL have.3PL plans for the holidays
 'Do you all (formal) have plans for the holidays?'

In other words, *vosotros* 'you all (informal)' does not necessarily convey that referents other than the addressee lack honorific status, just as *ustedes* 'you all (formal)' does not necessarily convey that referents other than the addressee have honorific status.¹³ This finding lends credit to the view from section 5.4 that addressee, like author, is a unique discourse role and that plural second-person pronouns have heterogeneous extensions.

Based on the previous discussion, I assign [HON] the lexical entry in (87), according to which this feature presupposes that the addressee of the discourse context (u_c) is honourable, a notion that is intentionally, and justifiably, vague.

$$(87) \quad \llbracket [HON] \rrbracket^c = \lambda x_e : \text{Honourable}(u_c) . x$$

For instance, a speaker may not personally consider someone to be honourable but may nevertheless address them as *usted* 'you (formal)' if the individual is of a certain age or holds a position with a high degree of social status. There are even varieties of Spanish where speakers use *usted* with immediate family and close friends, highlighting the need for a flexible semantics of honorificity.

¹³ See Morgan & Schwenter 2016, 2022 for a more nuanced picture of how *vosotros* and *ustedes* are used in Peninsular Spanish that is informed by corpus and elicitation data.

When π hosts [PART] as well as [HON], the presuppositions that are introduced by these features can be coordinated and simplified as in (88).

$$(88) \quad \llbracket \pi_{[\text{PART}, \text{HON}]} \rrbracket^c = \lambda x_e : u_c \leq x \wedge \text{Honourable}(u_c) . x \\ = \lambda x_e : \text{Honourable}(u_c) \leq x . x$$

In (89), I provide the semantic value of *ustedes* in Peninsular Spanish to show that it is only the addressee, and not all of the referents of the pronoun, who is presupposed to be honourable.

$$(89) \quad \llbracket \text{ustedes} \rrbracket^c = \iota x_e \text{ MAX}(*\text{Object})(x), \\ \text{defined only if } \text{Object}(x), \text{ATOM}(x), \text{Animate}(x), \\ \exists x_e \forall y_e [\text{MAX}(*\text{Object})(y) \leftrightarrow x = y], \\ \text{and } \text{Honourable}(u_c) \leq \iota x_e \text{ MAX}(*\text{Object})(x)$$

In this variety, *ustedes* denotes a unique maximal sum of objects x that is defined only if x is atomic, animate, such a unique maximal sum of objects exists, and the addressee of the discourse context (u_c) is both honourable and a part of this sum.

In a radically different approach to *usted* ‘you (formal)’ and *ustedes* ‘you all’, Collins & Ordóñez (2021: 250–251) argue that these forms have the “imposter” structures in (90), where a third-person DP contains a silent second-person pronoun that forms a constituent with the noun *usted*.

- (90) a. $[_{\text{DP}} \text{D} [\text{TÚ} \text{usted}]]$
b. $[_{\text{DP}} \text{D} [\text{TÚ} \text{usted}]\text{-s}]$

This analysis faces two problems that render it untenable. First, if *usted* were generated in the same position as nouns, *usted* and *ustedes* should be disallowed in pronoun-noun constructions, contrary to fact: *usted el cocinero* ‘you the chef’ and *ustedes las cocineras* ‘you the (female) chefs’. Second, if *usted* and *ustedes* are third-person DPs in the syntax, they should be immune to the Person-Case Constraint, but as I demonstrated in the first part of this section, these pronouns pattern identically to second-person pronouns in this regard.

Collins & Ordóñez (2021: 252–253) defend their characterization of *usted* ‘you (formal)’ on the basis of syntactic environments like those in (91) and (92), where *usted* need not be contrastive and *tú* ‘you (informal)’ is necessarily so.

- (91) a. ¡Coja {usted / USTED} el plato!
pick.up.SBJV you YOU the dish
‘Pick up/YOU (formal) pick up the dish!’
b. ¡Coge {*tú / TÚ} el plato!
pick.up.IMP you YOU the dish
‘YOU (informal) pick up the dish!’
- (92) a. Se irá {usted / USTED} acostumbrando al clima.
3 go.FUT.3SG you YOU getting.used to.the climate
‘You/YOU (formal) will get used to the climate.’

- b. Te irás { *tú / TÚ } acostumbrando al clima.
 2SG go.FUT.2SG you YOU getting.used to.the climate
 ‘YOU (informal) will get used to the climate.’

These facts build on observations by Fernández Soriano (1999), who remarks that the distinct behaviour of *usted* has a pragmatic or morphosyntactic explanation depending on the context. In some cases, speakers may use *usted* to emphasize a tone of respect, especially when giving a command, and in others, they may include *usted* to disambiguate the referent since this pronoun appears with the same agreement morphology as *él* ‘he’ and *ella* ‘she’. In light of these factors, the data in (91a) and (92a) cannot be taken as unassailable evidence that the internal structure of *usted* is different from that of other pronouns. Instead, I maintain that *usted* originates in π , just as *tú* does, and takes a covert DP complement that can be made overt, as in *usted el cocinero* ‘you (formal) the chef’.

5.6 Conclusion

In this chapter, I have provided morphosyntactic and interpretable structures for every stressed pronoun in Spanish. I started with an exploration of gender and animacy in this language and organized nouns and pronouns into six classes based on syntactic and semantic criteria: (i) animate masculine nouns and pronouns that presuppose maleness, (ii) animate feminine nouns and pronouns that presuppose femaleness, (iii) animate masculine nouns and pronouns that do not presuppose maleness, (iv) animate feminine nouns that do not presuppose femaleness, (v) inanimate masculine nouns, and (vi) inanimate feminine nouns. I then employed the AnimP-*n*P approach that I first defended in section 1.4, where gender features have semantic import on Anim but not on *n*, to propose a unique configuration of projections and features for each of these six classes. Next, I used this system to differentiate *él* ‘he’ from *ella* through an [M] versus [F] feature on Anim and both of these pronouns from *ello* ‘it’ through the presence versus absence of AnimP in the structure. I later argued that the heterogeneous plurality of first-person and second-person pronouns stems from the semantics of the person features [PART] and [AUTH], which presuppose that a discourse participant is among the referents of the pronoun, and not to a different type of number in pronominal expressions. Finally, I determined that *usted* ‘you (formal)’ and *ustedes* ‘you all’ are syntactically second person and attributed their unusual properties, some of which vary from one variety of Spanish to another, to impoverishment rules that apply during morphological operations and at logical form.

Chapter 6

Conclusion

The objective of this thesis has been to develop a theory of nominal and pronominal expressions in Spanish that captures their internal structure as well as their external distribution. To this end, I have explored the division of labour between syntactic operations, morphological operations, and logical form and proposed solutions to a variety of longstanding issues that eliminate unpronounced or uninterpreted pieces of structure, simplify the representation of $\varnothing$ -features, and unify the analysis of nouns and pronouns. In this final chapter, I return to the four research questions that I articulated in section 1.5 and summarize my findings before discussing their crosslinguistic implications.

The first research question concerns the role that determiners play in licensing arguments. In the literature on Spanish and other Romance languages, the most widely accepted explanation for the contrast in (1) is that bare nouns are introduced by a covert determiner that is subject to lexical government, meaning that it must be c-commanded by a verb or a preposition (Contreras 1986; Longobardi 1994; Contreras 1996; Longobardi 2000, 2001, 2005, 2008).

- (1) a. Llegaron cocineros.
arrived.3PL chefs
'Chefs arrived.'
- b. *Cocineros llegaron.
chefs arrived.3PL
Intended: 'Chefs arrived.'

In chapter 2, I identified several empirical and theoretical problems with the null determiner approach to bare nouns and reframed the issue in terms of composition at logical form. Uniting Kratzer's (1996) view of argument structure with Benedicto's (1998) claim that the domain of existential closure is determined by the position of the verb, I proposed a single type-shifting operation that targets verbs and prepositions and enables them to combine with property-denoting arguments that they c-command. Next, I built on insights by Demonte (1995) and Cuervo (2003a,b, 2008, 2010a,b) regarding the dative alternation in Spanish to extend my analysis to bare indirect objects, which are possible in ditransitive constructions but not in double-object constructions. Finally, I investigated three types of environments where bare nouns are disallowed even postverbally and demonstrated that semantic factors are responsible for the obligatory determiner in each case.

The second research question relates to the mass-count distinction and the ways in which morphological and syntactic complexity shape the interpretation of a noun, as in (2).

- (2) a. Quedaba tomate.
 remained.IMPF tomato
 ‘Tomato remained.’ (Mass only)
- b. Quedaban tomates.
 remain.IMPF.3PL tomatoes
 ‘Tomatoes remained.’ (Plural only)
- c. Quedaba un tomate.
 remained.IMPF a tomato
 ‘A tomato remained.’ (Singular only)
- d. Quedaba el tomate.
 remained.IMPF the tomato
 ‘The tomato remained.’ (Mass or singular)

In chapter 3, I argued that the absence versus presence of Borer’s (2005) division phrase (DivP) determines whether *n*Ps denote properties of unindividuated objects or properties of individuated objects, and I assigned Div the semantics of Link’s (1983) star operator (*), which yields the algebraic closure of a predicate, so that DivPs denote properties of individuated objects and their sums. After observing that nouns only have exclusively singular readings when they are preceded by a select few indefinite determiners (*cada/un/ningún tomate* ‘every/a/no tomato’), I amended the standard lexical entries for these determiners so that they take the extension of DivP and exclude the sums. Next, I showed that nonplural definite expressions are systematically ambiguous between denoting totalities of a substance and denoting atomic objects (*el/este/mi tomate* ‘the/this/my tomato’) and attributed this phenomenon to the lack of DivP in their structure and the maximality operator that is encoded by definite determiners. Later in the chapter, I addressed constructions where bare nonplural objects have number-neutral readings (*tengo carro* ‘I have a car/cars’) and defended Espinal & McNally’s (2011) view that these objects are interpreted as modifiers of the verb. Lastly, I proposed morphosyntactic and interpretable structures for lexical plurals (*los pantalones* ‘the pants’) and pluralia tantum (*las entrañas* ‘the entrails’) and discussed possible analyses of group nouns (*el equipo* ‘the team’) and aggregate nouns (*la cristalería* ‘the glassware’).

The third research question considered where pronouns are generated in the noun phrase. In English, there is empirical motivation to situate pronouns in D because they can be directly followed by a noun (*we/you chefs*), just like the definite article (*the chefs*). In Spanish, the situation is more complex because pronouns can only be followed by a definite description, as indicated by the need for the definite article in (3).

- (3) a. Llegamos nosotros *(los) cocineros.
 arrived.1PL we the chefs
 ‘We chefs arrived.’

- b. Llegasteis vosotros *(los) cocineros.
arrived.2PL you.PL the chefs
'You (informal) chefs arrived.'
- c. Llegaron ustedes *(los) cocineros.
arrived.3PL you.PL the chefs
'You chefs arrived.'

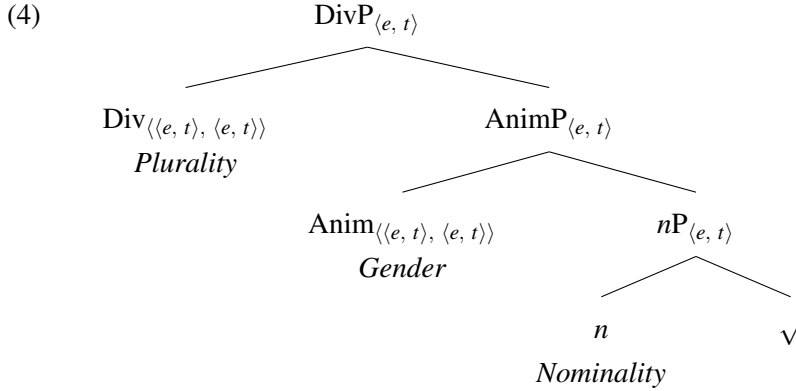
In chapter 4, I employed numerous diagnostic to rule out the possibility that these constructions involve a pronoun and a definite description in apposition and then argued in favour of Höhn's (2016, 2017) claim that person features belong to their own functional projection, person phrase (PersP), in languages like Spanish. I relabelled this projection as π P and, after formalizing a system of privative person features that differentiates third person from the absence of person, I provided morphosyntactic structures and vocabulary items for the first-person and second-person pronouns of Spanish, all of which are generated in π . To explain the inability of third-person pronouns to occur in pronoun-noun constructions (**ellos los cocineros* 'they the chefs') and be modified by restrictive relative clauses (**solo ellos que están certificados* 'only they who are certified'), I proposed that these forms realize multiple heads in the noun phrase (π , D, Div, Anim, and n) and can only do so when these heads are structurally adjacent.

The fourth research question pertains to ϕ -features and whether those that are common to nouns and pronouns, such as gender, animacy, and number, have the same representation in both types of expressions. In the first part of chapter 5, I focused on gender and animacy and sorted nouns and pronouns into six classes according to their syntactic and semantic behaviour: (i) animate masculine nouns and pronouns that presuppose maleness, (ii) animate feminine nouns and pronouns that presuppose femaleness, (iii) animate masculine nouns and pronouns that do not presuppose maleness, (iv) animate feminine nouns that do not presuppose femaleness, (v) inanimate masculine nouns, and (vi) inanimate feminine nouns. Next, I argued that gender features only have repercussions at logical form when they are hosted by Panagiotidis's (2019) animacy phrase (AnimP) and presented structural configurations for each of the six classes that I previously identified. For instance, I differentiated the animate feminine noun *cocinera* '(female) chef', which presupposes femaleness, from the animate feminine noun *persona* 'person', which does not, by positioning the [F] feature of *cocinera* on Anim and the [F] feature of *persona* on n . I then used the AnimP- n P split to distinguish the third-person pronouns *él* 'he' and *ella* 'she' from their "neuter" counterpart *ello* 'it'.

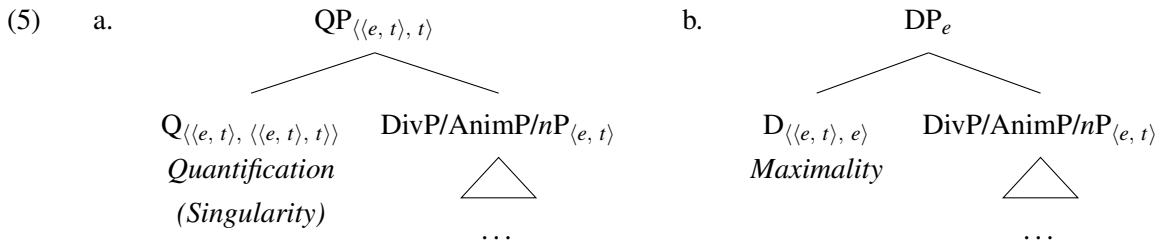
In the second part of chapter 5, I addressed the phenomenon of heterogeneous plurality in first-person and second-person pronouns and rejected the view that number has a different representation in pronominal expressions than it does in nominal expressions. I then showed that the heterogeneous plurality of *nosotros/nosotras* 'we', *vosotros/vosotras* 'you all (informal)', and *ustedes* 'you all' is predicted by the lexical entries that are standardly assigned to the person features [PARTICIPANT] and [AUTHOR]. Concretely, first-person and second-person pronouns presuppose that the speaker or addressee is among their referents, whereas third-person pronouns and definite descriptions do not. Finally, I determined on the basis of the Person-Case Constraint that *usted* 'you (formal)' and

ustedes are syntactically second person and proposed that they bear an [HON(ORIFIC)] feature that, postsyntactically, triggers the deletion of their [PART] feature. Due to this impoverishment process, *usted* and *ustedes* both appear with third-person rather than second-person agreement morphology.

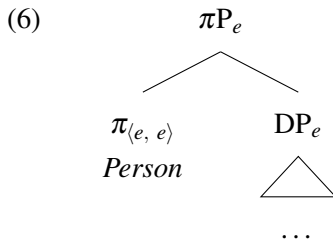
In (4), I provide the complete interpretable structure of the lower layers of the noun phrase and label the semantic contribution of each syntactic projection.



If Q combines with an *nP*, *AnimP*, or *DivP*, as in (5a), the result is a generalized quantifier, and if D combines with any of these constituents, as in (5b), the result is a nominal referential expression.



If π then combines with a *DP*, as in (6), the result is a pronominal referential expression, or one that is specified for person.



Having summarized the major findings of the thesis, I now discuss the implications of this work for crosslinguistic issues in syntactic, semantic, and morphological theory.

The account of bare nouns that I proposed in chapter 2 models existential closure as a general rescue operation, one that permits verbs and prepositions to apply directly to a property-denoting argument. This analysis is by no means specific to Spanish and can be extended to other languages

where bare nouns are confined to postverbal position and have obligatory existential readings, such as Catalan, Greek, and Italian. As for languages where bare nouns can appear preverbally, such as English, I hypothesize that covert type-shifting mechanisms are involved, a possibility that Chierchia (1998b), Van Geenhoven (2000), Cohen & Erteschik-Shir (2002), and Dayal (2004) consider at length, among other authors.

Many languages pattern identically to Spanish with respect to the mass-count distinction, so the structure-driven approach that I developed in chapter 3 has broad applications. For example, every aspect of the proposal makes the correct predictions for English: bare *n*Ps have mass readings (*tomato remained*), bare DivPs have plural readings (*tomatoes remained*), a limited number of indefinite determiners give rise to singular readings (*every/each/a/what/which/whichever tomato*), and the combination of a definite determiner and a nonplural noun permits both a mass and a singular reading (*the/this/my tomato remained*). Other languages have a different empirical landscape, including ones that are closely related to Spanish like Brazilian Portuguese, so further research is needed to determine the extent to which this account can be generalized.

English is one of many languages where third-person pronouns are disallowed in the same environments as in Spanish: *we/you/*they chefs* and *only we/you/*they who are certified*. It has, moreover, been noted that the pronoun *they* shares an initial element, *th-*, with the definite article *the* and the demonstratives *this* and *that* (Greenberg 1978, Bernstein 2008a, Kayne & Pollock 2010, Leu 2014). In chapter 4, I argued that the *-o* and *-s* of *ellos* ‘they’ spell out heads in the lowest layers of the noun phrase, and one could make the same claim regarding the *-ey* component of *they*. More broadly, my proposal that third-person pronouns realize the entirety of the noun phrase could explain the third-person gap in any language where there are morphological similarities between third-person pronouns and other determiners.

To close, all of the content in chapter 5 has crosslinguistic relevance as well. My view that gender features have semantic effects when they are hosted by Anim but not by *n* can be extended to other languages where some nouns introduce gender-based presuppositions and others do not, though it would likely be necessary to adjust the representation of gender features depending on the number or types of distinctions that are made in a given language. Likewise, the observation that the heterogeneous extensions of first-person and second-person pronouns stems from their person specification holds across languages. Lastly, my analysis of *usted* ‘you (formal)’ and *ustedes* ‘you all’ can serve as a blueprint for future work on the counterparts of these pronouns in Catalan, *vostè/vostès* ‘you (formal)/you all (formal)’, and in Portuguese, *você/vocês* ‘you/you all’.

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